TFPT: Boundary Polarization, Carrier Rigidity, and Closed-Form Theorems for the Admissibility Datum

A boundary-polarized spectral theory
from one-sided admissible data

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Version 4.5 – Boundary Polarization, Determinant-Line Closure, Geometric Quantization – April 27, 2026

Abstract

From the one-sided boundary datum

$$\mathfrak{T}_{\partial}^{\min} = (\mathcal{A}_+, \mathcal{H}_+, D_+, J, \Gamma, B_\Sigma),$$

the manuscript derives the Calderón polarization P_C, the boundary involution $\iota_C := 2P_C - 1$, the exact-double deck involution τ_{dbl}, and the doubled closed datum

$$\mathfrak{T}_{\min}^{\mathrm{cl}} = (\mathcal{A}, \mathcal{H}, D, J, \Gamma, \tau_{\mathrm{dbl}}, \iota_C).$$

The hard core is now read through three decoders. First, the carrier decoder Y satisfies

$$6Y^2 - Y - \mathbf{1} = 0, \qquad \mathrm{Tr}_E\, Y = 0,$$

and in the primitive realization forces the rigid $3+2$ carrier split $E = E_3 \oplus E_2$, the primitive two-point generator $X = 6Y$, the one-family packet $S^+ = \Lambda^{\mathrm{even}} E$, and the physical gauge quotient

$$G_{\mathrm{phys}} = \frac{SU(3) \times SU(2) \times U(1)_Y}{\mathbb{Z}_6}.$$

Second, the global winding decoder $[u_\Sigma] = 1$ fixes the rigid family surface $X_f^\circ \cong \mathbb{P}^1 \setminus \mu_4$, yields the minimal family repair

$$16\gamma = \frac{40}{3} \qquad \Longrightarrow \qquad N_{\mathrm{fam}} = 3, \qquad \Omega_{\mathrm{adm}} = 48,$$

and, on the transport branch, fixes the intrinsic pole as the lower critical point of the cusp cubic

$$P(z) = (z-1)\left(z - \frac{64}{729}\right)\left(z - \frac{1}{729}\right), \qquad P'(z) = 0.$$

Third, the retained seed decoder

$$u := \varphi_0^{\mathrm{ret}}$$

projects simultaneously to $(\lambda_C, \beta_{\mathrm{rad}}, \Omega_b, \theta_{13})$, while the closed transport-plus-Higgs budget

$$\sum_{f,j} L_{f,j}^{\mathrm{diag}} + N_\Phi = 41$$

closes the exact electromagnetic fixed point α.

The theorem flow is therefore no longer presented as a strictly linear carrier-then-family arrow chain. After the boundary datum has fixed the doubled primitive reconstruction, the primitive seam generator, and the primitive Hodge selector, the manuscript solves one joint discrete admissibility problem and only afterwards the continuous Euler–Lagrange problem:

$$\mathfrak{T}_\partial^{\min} \Rightarrow (\tau_{\mathrm{dbl}}, \iota_{\mathrm{C}}, P_{\mathrm{prim}}, [u_\Sigma], c_3) \Rightarrow d^\star_{\mathrm{disc}} \Rightarrow P_{\mathrm{adm}} \Rightarrow \mathcal{M}_{d^\star_{\mathrm{disc}}} \Rightarrow \mathcal{Z}_{\mathrm{cl}} = \{x^\star_{\mathrm{cl}}\} \Rightarrow \mathfrak{T}_\star.$$

Here the unique discrete datum is

$$d^\star_{\mathrm{disc}} := \big(E_3 \oplus E_2,\, Y,\, S^+,\, X^\circ_f,\, [\nabla^\star_F],\, c_1(L_2),\, c_1(L_3),\, N_{\mathrm{fam}} = 3,\, \theta_{\mathrm{eff}} = 0\big),$$

so carrier, family, determinant data, and strong CP are fixed jointly from one boundary primitive kernel rather than in a one-way family-after-carrier rhetoric. Here $P_{\mathrm{adm}} := \Pi_{\ker \Delta_{\mathrm{adm}}}$ selects the physical admissible sector, while the actual dynamics is carried by the relative generating functional Z_{rel}, the admissible Schwinger distributions, the Osterwalder–Schrader reconstruction, the local Minkowski net on the admissible Hilbert space, the exact admissible flow Γ_k, and its infrared limit $\Gamma^{\mathrm{ren}}_{\mathrm{TFPT}}$.

The same seam transfer, determinant-line phase, and scalaron sector then fix the closed-branch cosmology interface data from which infrared continuation, reheating, and leptogenesis readouts are computed. Two further structural maps are attached strictly outside the theorem chain: the structural falsification map $\mathsf{F}_{\mathrm{fals}}(\mathfrak{T}_\star)$ in the main text and the appendix-level comparison map $\mathcal{R}_{\mathrm{cmp}}(\mathfrak{T}_\star)$. Above the rigid core, the manuscript records an internal reduction on the weakened ambient class $\widehat{\mathrm{PhysAdm}}_1$, an internal renormalized low-energy layer $\Gamma^{\mathrm{ren}}_{\mathrm{TFPT}}$, and a reality-first observable stratification

$$\mathfrak{T}_\star \xrightarrow{\mathfrak{R}_{\mathrm{ren}}} \Gamma^{\mathrm{ren}}_{\mathrm{TFPT}} \xrightarrow{\mathfrak{M}_{\mathrm{phys}}} \mathbf{O}^{\mathrm{TFPT}}_{\mathrm{phys}} \xrightarrow{\mathfrak{M}_{\mathrm{scheme}}} \mathbf{O}^{\mathrm{TFPT}}_{\mathrm{scheme}}/\mathbf{Sch}.$$

This output layer is read in boundary-normalized form: with

$$\lambda_\Sigma := \lambda_1^+(|B_\Sigma|), \qquad \frac{O}{\lambda_\Sigma^{d_O}} = \mathcal{F}_O(\mathfrak{T}_\star),$$

so dimensionless gravitational, electroweak, pole-mass, flavor-ratio, and infrared readouts are internal branch outputs rather than external metrological inputs. The final theorem-level claim is the canonical rigid object of the boundary-polarized branch, the admissible QFT closure on the retained sector, and the renormalized physical observable layer, with cosmology carried at the level of closed-branch interface data and appendix comparison continuations. In the condensed decoder reading,

$$\begin{aligned} Y &\text{ generates structure,}\\ [u_\Sigma] = 1 &\text{ generates counting,}\\ u &\text{ generates bridge observables.} \end{aligned}$$

Closed architecture of the paper

$$\widehat{\mathrm{PhysAdm}}_1 \xrightarrow{\overline{\mathfrak{R}}} \mathbf{TFPT}^{\mathrm{rig}} \simeq \{\mathfrak{T}_\star\} \xrightarrow{\mathfrak{R}_{\mathrm{ren}}} \Gamma^{\mathrm{ren}}_{\mathrm{TFPT}} \xrightarrow{\mathfrak{M}_{\mathrm{phys}}} \mathbf{O}^{\mathrm{TFPT}}_{\mathrm{phys}} \xrightarrow{\mathfrak{M}_{\mathrm{scheme}}} \mathbf{O}^{\mathrm{TFPT}}_{\mathrm{scheme}}/\mathbf{Sch}.$$

Main theorem stack:

$$\begin{aligned} \mathfrak{S}_{\min} \Rightarrow \mathcal{B}_{\min} \Rightarrow \mathfrak{T}_\partial^{\min} \Rightarrow (\tau_{\mathrm{dbl}}, \iota_{\mathrm{C}}, P_{\mathrm{prim}}, [u_\Sigma], c_3) \Rightarrow d^\star_{\mathrm{disc}}\\ \Rightarrow P_{\mathrm{adm}} \Rightarrow \mathcal{M}_{d^\star_{\mathrm{disc}}} \Rightarrow \mathcal{Z}_{\mathrm{cl}} = \{x^\star_{\mathrm{cl}}\} \Rightarrow \mathfrak{T}_\star. \end{aligned}$$

Compressed decoder reading:

$$Y \text{ generates structure,}$$
$$[u_\Sigma] = 1 \text{ generates counting,}$$
$$u := \varphi_0^{\mathrm{ret}} \text{ generates bridge observables.}$$

$$6Y^2 - Y - \mathbf{1} = 0,\ \mathrm{Tr}\, Y = 0 \Rightarrow E_3 \oplus E_2,\ S^+,\ \gamma = \frac{5}{6},$$
$$16\gamma = \frac{40}{3} \Rightarrow N_{\mathrm{fam}} = 3 \Rightarrow \Omega_{\mathrm{adm}} = 48.$$
$$P'(z) = 0 \Rightarrow \delta_{\mathrm{ph}}, \qquad \sum_{f,j} L^{\mathrm{diag}}_{f,j} + N_\Phi = 41 \Rightarrow \alpha.$$

The input-side reduction to $\mathfrak{T}_\star$ and the internal passage from $\mathfrak{T}_\star$ to $\Gamma^{\mathrm{ren}}_{\mathrm{TFPT}}$ and $\mathbf{O}^{\mathrm{TFPT}}_{\mathrm{phys}}$ are theorem-level. Only the final projection to $\mathbf{O}^{\mathrm{TFPT}}_{\mathrm{scheme}}/\mathbf{Sch}$ is an appendix-facing interface. From $\mathfrak{T}_\star$ the structural falsification map $\mathsf{F}_{\mathrm{fals}}(\mathfrak{T}_\star)$ stays in the main text, while the comparison map $\mathcal{R}_{\mathrm{cmp}}(\mathfrak{T}_\star)$ together with thresholds, schemes, and empirical rows is collected only in the appendix-level empirical readout.

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1 Reading guide and claim map

The paper is organized in three reading layers:

- **Big picture:** one section explains what TFPT claims in plain language before the detailed datum and conventions begin.
- **Detail:** the main body then runs from boundary polarization and hard carrier rigidity through kernel numerics, the two closure-architecture sections, the closure theorem stack, the canonical rigid object, the seam-transfer cosmology block, and the internal-reduction / readout-organization section, while the appendix-level empirical readout may be read afterwards.
- **Proof-complete core:** the load-bearing completion theorems are proved in full in the manuscript. The appendices keep manuscript-level analytic proofs, auxiliary elaborations, optional continuations, and interface tables, but they are not needed for the logical closure of the primitive kernel, the continuous master functional, or the canonical rigid object.

The role language is also fixed once and for all:

- **Boundary datum** means the one-sided theorem package $\mathfrak{T}_\partial^{\min}$ induced by the minimal operational seed and used downstream to reconstruct the Calderón polarization, doubled closed datum, and admissibility projector. The later minimal-packaging theorem shows that no extra primitive discrete datum is needed beyond it.
- **Theorem** means a statement proved directly from the operational-seed / boundary-datum chain and earlier theorem-level outputs.
- **Derived consequence** means a downstream structural output obtained by composing already proved theorems without introducing a new datum.

Below theorem level the manuscript uses only five auxiliary tags:

- **Exact identity** means an algebraic equality on an already fixed branch or kernel package.
- **Derived consequence** means a nonprimitive output obtained by composing earlier theorem-level statements.
- **Appendix continuation** means a downstream numerical or conceptual extension that requires one extra assumption beyond the main theorem surface.
- **Programmatic closure target** means a mathematically precise downstream target whose statement is fixed clearly enough to define a falsifiable obligation, but whose proof is not claimed in the present version.
- **Interpretive reading** means a heuristic or explanatory picture not used as theorem input.

Comparison conventions are isolated in the appendix-level empirical interface and are not used as main-text theorem labels. To keep theorem language, benchmark semantics, and appendix bookkeeping lexically stable, all prose, captions, and role labels are kept in English throughout.

1.1 Version 4.5 completion surface

The manuscript now contains a fully rigid algebraic core together with analytic closure blocks whose exact status is recorded theorem by theorem in the proof-obligation ledger. The remaining explicit standard-theorem interface is

$$O_{7\mathrm{d},0},$$

while the other former B/C interfaces have now been pulled into the manuscript proof surface. The eight items below are therefore not a promise of later completion; they are the status-coded 4.5 surface of the present manuscript.

Version 4.5 completion surface

1. **[A] Operational seed generation of the one-sided boundary datum.** Introduce the thinner operational seed class $\mathbf{Seed}^{\mathrm{op}}$ and prove that it functorially induces the one-sided admissible boundary datum. Proof architecture: reflection positivity and the collar generator reconstruct the one-sided OS Hilbert space and adapted boundary operator, while the seam class induces the spectral-flow datum.
2. **[A] Minimal packaging inside the induced admissible boundary class.** Organize the induced one-sided boundary class by lexicographic minimality without hiding a second primitive discrete layer. Proof architecture: lexicographic minimality of seam winding, finite primitive carrier rank, corner count, determinant charge, and boundary kernel dimension; collar product form then yields the one-sided boundary operator and reflection-positive doubling.
3. **[A] D_4 Riemann–Hilbert rigidity theorem.** Close the family-holonomy debt by fixing the four puncture loops, the local $\{1, \omega, \omega^2\}$ conjugacy classes, the trivial determinant line, and the full D_4 puncture symmetry. Proof architecture: monodromy is forced up to global $SU(3)_F$ conjugation, then Riemann–Hilbert yields the unique flat Hermitian family-connection class.
4. **[A] Compact determinant and Higgs index theorem.** Promote the determinant-line / Higgs block from a near-complete closure to a compact Dolbeault index theorem on the seam-normal sphere. Proof architecture: $(c_1(L_2), c_1(L_3)) = (1, 0)$ gives $L_2 \cong \mathcal{O}(1)$ and

$L_3 \cong \mathcal{O}$, so Birkhoff–Grothendieck and Riemann–Roch force exactly one weak doublet and no light color triplet.

5. **[A] Canonical transport-kernel theorem.** Recast the transport block at theorem level in operator language rather than in path rhetoric. Proof architecture: the structural kernel K_f^{str} together with the rigid family holonomy produces the Yukawa matrices in the canonical family basis; the primitive indecomposable Yukawa generator then forces the rigid $3+2$ carrier split.

6. **[A] Stratified master-functional theorem.** Replace the last appearance of separate closure stories by one stratified functional on admissible spectral bordisms. Proof architecture: a lexicographic barrier fixes the minimal discrete stratum, while the relative partition function, zeta determinant, and eta term generate the continuous Euler–Lagrange equations on the retained branch.

7. **[A] Spectral UV completion and renormalized 1PI readout theorem.** Close the continuum side at the spectral rather than purely local Einstein level. Proof architecture: a positive-kernel Birkhoff contraction fixes the unique admissible UV profile, Stieltjes form factors give the ghost-free two-point gravitational kernel, the projective-limit measure supplies the constructive UV object, and the renormalized observable hierarchy is read as the exact 1PI / Legendre layer of the same admissible measure.

8. **[A] Absolute spectral Planck closure and stationary vacuum theorem.** Fix the boundary spectral unit λ_Σ, the unique stationary root $\rho_\star$ of the boundary-normalized gravitational functional, and the resulting reduced-Planck / Newton / Schwarzschild–de Sitter readouts directly on the main theorem surface.

9. **[A] FRW reduction and cosmology interface theorem.** Turn seam transfer, determinant-line dynamics, and the scalaron sector into one reduced low-curvature action together with explicit cosmology interface data. Proof architecture: homogeneous/isotropic reduction in $(a, \phi, \theta_\Sigma, N_i)$ yields Λ_{IR}, the axion displacement data, and the reheating / leptogenesis input block on the same admissible branch, while the final nonequilibrium readouts remain comparison-layer outputs.

Remark (Remaining interface for the 4.5 map). The target blocks above are now accompanied by only one explicit standard-theorem interface that should remain visible rather than absorbed into rhetoric: the infrared-dressed massless scattering interface on the low-curvature branch ($O_{7\mathrm{d},0}$). The OS/CAR reconstruction, massive Haag–Ruelle / LSZ closure, and the FRW cosmology interface are now carried by manuscript-level appendix proofs.

Remark (Status of the 4.5 completion surface). The nine theorem blocks above are integrated into the manuscript, but their exact status is still recorded line by line in the proof-obligation ledger. The point is therefore not rhetorical completion language but synchronized proof metadata.

Extended downstream closure surface

1. **[P] Horizon modular thermality and compact-object thermodynamics.** Upgrade the split-inclusion horizon package to stationary modular flow, Hawking temperature, Wald entropy, and Kerr / scalaron correction bounds on the vacuum branch.

2. **[P] Primordial perturbation closure from the seam state.** Fix the reheating e-fold

number, seam-state Bogoliubov data, and Mukhanov–Sasaki / tensor initial conditions on the closed branch.

3. **[P] CMB transfer, baryogenesis, and late-time matter readout.** Carry the seam-fixed primordial spectrum through the transfer hierarchy, the flavored Boltzmann system, and the late-time H_0 / $P_m(k,z)$ readouts.

4. **[P] Late-time dark-sector structure and relative vacuum-energy control.** Record the axion Schrödinger–Poisson branch and the finite relative admissible vacuum-energy formula as the late-time dark-sector closure target.

5. **[P] Pole-mass completion and Higgs readout.** Read quark / lepton pole masses and the Higgs pole from the admissible RG branch rather than from source rows or raw UV data.

6. **[P] Admissible code subspace and pointer dynamics.** Promote the appendix record / Born machinery to a global code-subspace reading of entanglement, pointer stability, and record growth on $\mathcal{H}_{\mathrm{adm}}$.

7. **[P] Topological transient channels.** Keep phase-slip burst phenomenology as a precise falsifiable transient target without promoting it to theorem status.

Closed theorem surface

- **Initial layer:** the minimal operational seed $\mathfrak{S}_{\mathrm{min}}$ induces the minimal admissible bordism $\mathcal{B}_{\mathrm{min}}$ and hence the one-sided boundary datum $\mathfrak{T}_{\partial}^{\mathrm{min}}$.
- **Boundary datum:** $\mathfrak{T}_{\partial}^{\mathrm{min}}$ with one-sided center, self-adjoint boundary operator, seam collar, and canonical doubling data.
- **Theorem layer I:** doubling / deck involution / boundary polarization $(\tau_{\mathrm{dbl}}, \iota_{\mathrm{C}})$, the doubled primitive reconstruction $\mathfrak{T}_{\mathrm{prim}}$, and the primitive Hodge projector P_{prim} from the Calderón projector, APS realization, and D_{rel}.
- **Theorem layer II:** the primitive winding class $[u_{\Sigma}] = 1$ and the rigid D_4 family geometry fix $X_f^{\circ} \cong \mathbb{P}^1 \setminus \mu_4$ and $N_{\mathrm{fam}} = 3$; the trace-balanced carrier decoder $6Y^2 - Y - \mathbf{1} = 0$, $\mathrm{Tr}\, Y = 0$ is realized minimally as the rigid $3+2$ carrier kernel $E_3 \oplus E_2$, the primitive hypercharge generator Y, the one-family packet S^+, and $\gamma = 5/6$; together with the compact Higgs index this fixes $\Omega_{\mathrm{adm}} = 48$, the determinant class $(c_1(L_2), c_1(L_3)) = (1,0)$, and $\theta_{\mathrm{eff}} = 0$.
- **Theorem layer III:** after the discrete datum has been fixed, the full selector P_{adm} and the projected modular free energy determine the electromagnetic, geometric, transport, and vacuum sectors as one continuous Euler–Lagrange problem; on the retained branch this is read as $\lambda_{\Sigma} = \lambda_1^+(|B_{\Sigma}|)$, $\partial_{\rho}\widehat{V}_{\mathrm{Pl}}^{\mathrm{ren}}(\rho_{\star}) = 0 \Rightarrow (\bar{M}_{\mathrm{Pl}}/\lambda_{\Sigma},\ G_N\lambda_{\Sigma}^2)$, $P'(z) = 0 \Rightarrow \delta_{\mathrm{ph}}$, $u := \varphi_0^{\mathrm{ret}} \Rightarrow (\lambda_{\mathrm{C}}, \beta_{\mathrm{rad}}, \Omega_b, \theta_{13})$, and $\sum_{f,j} L_{f,j}^{\mathrm{diag}} + N_{\Phi} = 41 \Rightarrow \alpha$.
- **Theorem layer IV:** the intrinsic branch is the singleton fiber product of the discrete sector and the four continuous sector solution sets; from this follow internal analytic closure, essential injectivity on the rigid image, and the canonical rigid object $\mathfrak{T}_{\star}$.
- **Input/output layer:** internal reduction $\mathfrak{R}$ from $\widehat{\mathrm{PhysAdm}}_1$ to the rigid TFPT class is proved on the input side; the renormalized observable chain $\mathfrak{T}_{\star} \to \Gamma_{\mathrm{TFPT}}^{\mathrm{ren}} \to \mathbf{O}_{\mathrm{phys}}^{\mathrm{TFPT}}$

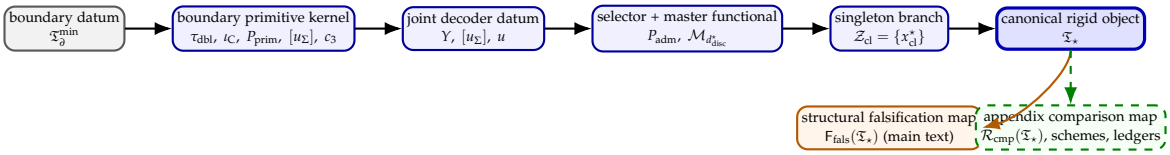


Figure 1. Closed dependency map with decoder core.

is theorem-level on the output side; only the final scheme projection to $\mathbf{O}^{\mathrm{TFPT}}_{\mathrm{scheme}}/\mathbf{Sch}$ remains appendix-facing.

- **Side maps:** a structural falsification map $\mathsf{F}_{\mathrm{fals}}(\mathfrak{T}_\star)$ in the main text, and an appendix-only comparison map $\mathcal{R}_{\mathrm{cmp}}(\mathfrak{T}_\star)$ for thresholds, schemes, and empirical rows after the physical observable layer has already been fixed.

The family closure used throughout the main text is now read as topologically fixed, variationally stabilized, and canonically compatible with the doubled admissible branch; no separate family datum remains outside the theorem surface. Likewise the finite chiral packet condition $\dim \Lambda^{\mathrm{even}} E = 16$ is no longer left as ambient residue but is reconstructed internally by the bosonic-rank and branch-Yukawa carrier closure.

2 Big picture: what TFPT actually claims

TFPT is presented here as a boundary-polarized closed theory of admissible vacua on the oriented double cover. The formal theorem stack now starts from a thinner operational seed which functorially induces the one-sided boundary datum, and the manuscript also shows that within the induced boundary class no extra primitive discrete datum is needed. On the one-sided branch, the self-adjoint boundary operator B_Σ determines the Calderón projector, the Calderón projector determines the seam polarization ι_C, and the canonical doubling then reconstructs the doubled scaffold $\mathfrak{T}_{\mathrm{prim}}$ together with the primitive Hodge projector P_{prim}. From this boundary kernel the manuscript next solves one joint discrete admissibility problem, now best read through three decoders. The trace-balanced carrier decoder

$$6Y^2 - Y - \mathbf{1} = 0, \qquad \operatorname{Tr} Y = 0$$

fixes the rigid $E_3 \oplus E_2$ split, hypercharge, and the one-family packet S^+. The global winding decoder $[u_\Sigma] = 1$ fixes the rigid family surface $X_f^\circ \cong \mathbb{P}^1 \setminus \mu_4$, the minimal family count $N_{\mathrm{fam}} = 3$, and the admissible occupancy $\Omega_{\mathrm{adm}} = 48$. On the retained branch the seed decoder

$$u := \varphi_0^{\mathrm{ret}}$$

simultaneously bridges $(\lambda_C, \beta_{\mathrm{rad}}, \Omega_b, \theta_{13})$, while $P'(z) = 0$ fixes the intrinsic transport pole and the closed budget $\sum_{f,j} L^{\mathrm{diag}}_{f,j} + N_\Phi = 41$ closes the electromagnetic fixed point α. Only after that discrete theorem is the selector promoted to its full form $P_{\mathrm{adm}} = P_{\mathrm{prim}} P_{\mathrm{sing}} P_\Theta$, and the projected modular free energy drives the continuous electromagnetic, geometric, transport, and vacuum closure: the geometric Hodge sector yields the gravity/scalaron branch, and the exact transport grammar emits Yukawas, hadronic admissibility, and strong-CP closure on the same admissible branch. The resulting theorem stack does not merely emit a closed output package; it characterizes a canonical rigid object $\mathfrak{T}_\star$ in the rigid TFPT theorem class. The structural falsification map $\mathsf{F}_{\mathrm{fals}}(\mathfrak{T}_\star)$ stays in the main text, while all comparison conventions live only in the appendix-level $\mathcal{R}_{\mathrm{cmp}}(\mathfrak{T}_\star)$ side card.

Figure 2 compresses that statement into six headline claims. The point is that the manuscript now presents three hard decoders inside one joint discrete theorem, one continuous master

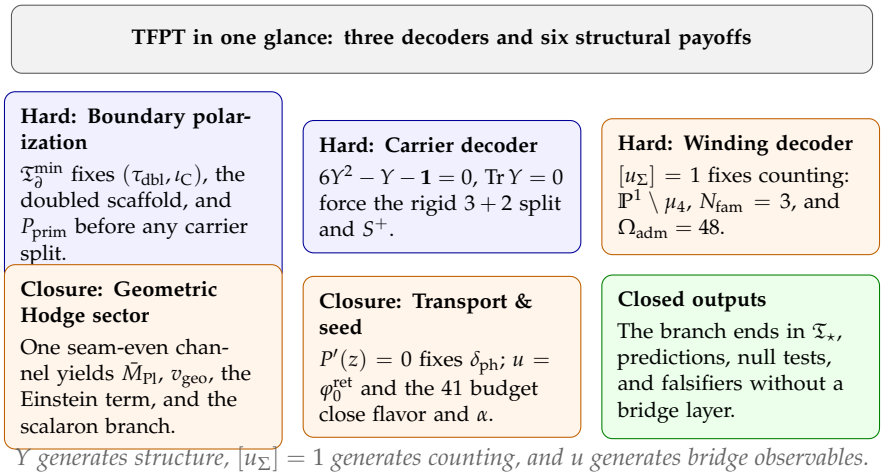


Figure 2. Headline claims of the present draft. The figure is intentionally simpler than the dependency maps that follow: its job is to put the manuscript's decoder core on one page while keeping the closed datum, theorem, and derived-consequence flow visible.

functional, and one canonical rigid object rather than competing closure stories with a separate readout layer.

TFPT master equation and closed indices

$$S_{\rm TFPT} = \mathrm{Tr}\, f\left(\frac{D_{\rm rel} + A_\Sigma^{\rm seed}}{\chi_{\rm seed}}\right) - \mathrm{Tr}\, f\left(\frac{D_{\rm ref}}{\chi_{\rm seed}}\right) + \frac{i\pi}{2}\Delta\eta_\Sigma, \qquad A_\Sigma^{\rm seed} := \mathcal{B}_{\rm rel} \oplus \nabla_{\chi_{\rm seed}}. \tag{1}$$

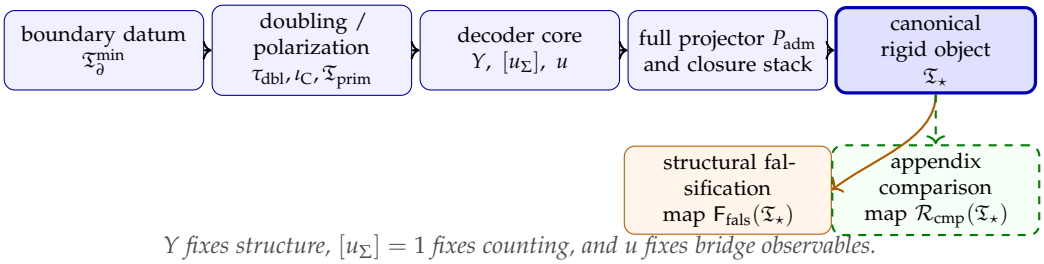
$$\begin{gathered} \operatorname*{Stat}_{\alpha,\chi_{\rm seed},e,\omega_{\rm spin}} S_{\rm TFPT} = 0, \qquad P_{\rm adm} := \Pi_{\ker \Delta_{\rm adm}}, \qquad \partial_\alpha \Gamma_{U(1)}(\alpha) = 0, \\ \mathrm{Ind}(D_{\rm fam}^+) = 3, \qquad \mathrm{Ind}(\mathcal{B}_{E_2}^+) = 1, \qquad \mathrm{Ind}(\mathcal{B}_{E_3}^+) = 0. \end{gathered} \tag{2}$$

The hard carrier theorems below do not require the full theorem stack behind Equations (1) and (2). In the present paper the master equation is the organizing closed form: the carrier theorems are hard, the later closure theorems are internal statements on the reconstructed branch, and the empirical interface is kept outside this theorem box. At this overview level the seed package is written as $A_\Sigma^{\rm seed}$; on the later geometric branch it is reorganized as $A_\Sigma^{\rm geo}$ with $\chi_{\rm seed} \mapsto \chi_{\rm geo} = \Lambda e^\sigma$.

Remark (Selector versus dynamics). The admissibility projector $P_{\rm adm}$ selects the physical sector but is not itself the full Minkowski dynamics. The actual dynamics is carried by the relative generating functional $Z_{\rm rel}$, its Schwinger distributions, the reconstructed local net on the admissible Hilbert space, the running effective actions Γ_k and $\Gamma_{\rm TFPT}^{\rm ren}$, and the resulting scattering theory.

Closed TFPT Chain

$$\begin{aligned} \mathfrak{T}_\partial^{\min} &\Rightarrow (\tau_{\rm dbl}, \iota_{\rm C}, P_{\rm prim}, [u_\Sigma], c_3) \Rightarrow d_{\rm disc}^\star \\ &\Rightarrow P_{\rm adm} \\ &\Rightarrow \mathcal{M}_{d_{\rm disc}^\star} \Rightarrow \mathcal{Z}_{\rm cl} = \{x_{\rm cl}^\star\} \Rightarrow \mathfrak{T}_\star, \\ \mathfrak{T}_\star &= \big(E_3 \oplus E_2,\ X,\ Y,\ a_\Sigma,\ \Gamma_{\rm grav}, \\ &\qquad \Lambda_{\rm IR},\ x_{\rm cl}^\star\big). \end{aligned} \tag{3}$$



Y fixes structure, $[u_\Sigma] = 1$ fixes counting, and u fixes bridge observables.

Figure 3. Closed theorem surface with two hard-separated side maps and a decoder core.

At chain level the same architecture is read through three hard decoders:

$$\begin{aligned} Y &\Rightarrow \text{carrier structure}, \\ [u_\Sigma] = 1 &\Rightarrow \text{family counting}, \\ u = \varphi_0^{\mathrm{ret}} &\Rightarrow \text{bridge observables}. \end{aligned}$$

On the retained branch this further compresses to

$$P'(z) = 0 \Rightarrow \delta_{\mathrm{ph}}, \qquad \sum_{f,j} L_{f,j}^{\mathrm{diag}} + N_\Phi = 41 \Rightarrow \alpha.$$

From $\mathfrak{T}_\star$ the structural falsification card $\mathsf{F}_{\mathrm{fals}}(\mathfrak{T}_\star)$ remains inside the main text; the appendix-only comparison card $\mathcal{R}_{\mathrm{cmp}}(\mathfrak{T}_\star)$ collects scheme conventions and empirical ledgers.

Closed Theorem Stack in the Present Version

$$\mathfrak{T}_\partial^{\min} \;\Rightarrow\; (\tau_{\mathrm{dbl}}, \iota_{\mathrm{C}}, P_{\mathrm{prim}}, [u_\Sigma], c_3) \;\Rightarrow\; d_{\mathrm{disc}}^\star \;\Rightarrow\; P_{\mathrm{adm}} \;\Rightarrow\; \mathcal{M}_{d_{\mathrm{disc}}^\star} \;\Rightarrow\; \mathcal{Z}_{\mathrm{cl}} = \{x_{\mathrm{cl}}^\star\} \;\Rightarrow\; \mathfrak{T}_\star.$$

Hard-decoder form:

$$\begin{aligned} 6Y^2 - Y - \mathbf{1} = 0,\ \operatorname{Tr} Y = 0 &\Rightarrow 3 + 2, \\ [u_\Sigma] = 1 &\Rightarrow N_{\mathrm{fam}} = 3, \\ u = \varphi_0^{\mathrm{ret}} &\Rightarrow (\lambda_C, \beta_{\mathrm{rad}}, \Omega_b, \theta_{13}). \end{aligned}$$

The manuscript contains no additional primitive ontology beyond the operational seed layer. The falsification map $\mathsf{F}_{\mathrm{fals}}$ is a structural side card on the main-text side; the comparison map $\mathcal{R}_{\mathrm{cmp}}$, optional arithmetic continuations, asymptotic expansions, and horizon language are kept only in appendices.

The remainder of the main text sharpens each arrow of this chain in turn.

3 Primitive core and hard carrier kernel

3.1 Primitive core and conventions

We use natural units $c = \hbar = 1$. Relative operators are formulated in Euclidean signature and compared through a declared Wick-rotation bridge when Lorentzian language is needed. Higgs fields are denoted by Φ, Hamiltonians by $\mathcal{H}$, and the cosmological Hubble rate by $H_{\mathrm{hub}}(t)$; this avoids overloading the letter H. Newton's constant is always G_N, while G_{car} is reserved for the internal carrier stabilizer. The seam is denoted by Σ, geometric reflection by

τ_{dbl}, boundary polarization by ι_{C}, carrier polarization by $\varepsilon_{\mathrm{car}}$, and word parity by $\varepsilon(w)$.

Here $S \to \widetilde{M}$ denotes the spinor bundle on the oriented double cover, and by the same symbol we denote its restriction to

$$X_{\mathrm{bulk}} = \widetilde{M} \setminus \Sigma.$$

When color-center notation is needed later, we use multiplicative notation

$$z(w) \in \{1, \omega, \omega^2\}, \qquad \omega = e^{2\pi i/3}.$$

Thus center neutrality means $z(w) = 1$.

$$\mathfrak{T}^{\mathrm{kin}}_{\mathrm{core}} := \big(\widetilde{M} \to M,\ \Sigma,\ \tau_{\mathrm{dbl}},\ \iota_{\mathrm{C}},\ D_{\mathrm{ref}},\ D_{\mathrm{rel}},\ \chi_{\mathrm{seed}}\big).$$

When a chosen almost-commutative factorization $\mathcal{H} \cong L^2(\widetilde{M}, S) \otimes \mathcal{H}_{\mathrm{int}}$ is used, S and $\mathcal{H}_{\mathrm{int}}$ refer to that choice and are not part of the kinematic core datum. Here "relative" always means comparison with a declared reference operator or reference-subtracted action. The canonical admissibility complex is introduced immediately below, while any Bernstein or other spectral test profile is treated only as an analytic realization choice rather than as part of the boundary datum. The technical tables for relative objects, layered closure/benchmark data, and the constants atlas are collected in Appendix B and Appendix C; they are kept out of the early main-text flow so that the carrier kernel appears before the technical bookkeeping apparatus.

Remark (Boundary datum versus derived closed structure). The primitive ontology now stops one step earlier. The one-sided boundary datum $\mathfrak{T}^{\min}_{\partial}$ contains only one-sided operator data together with the self-adjoint boundary operator B_Σ. Its Calderón projector defines the polarization

$$\iota_{\mathrm{C}} := 2P_{\mathrm{C}} - 1.$$

From that boundary polarization and the exact double one reconstructs the deck involution τ_{dbl}, the physical quotient $M = \widetilde{M}/\tau_{\mathrm{dbl}}$, the seam $\Sigma = \mathrm{Fix}(\tau_{\mathrm{dbl}})$, the reference/relative operator split $(D_{\mathrm{ref}}, D_{\mathrm{rel}})$, the primitive seam response $\chi_{\mathrm{seed}} = \lambda_1^+(|B_\Sigma|)^{-1}$, and the doubled closed datum $\mathfrak{T}^{\mathrm{cl}}_{\min}$. The seam is therefore no longer primitive ontology but the fixed locus of the deck involution extracted from the one-sided boundary problem.

3.2 Primitive relative spectral dynamics

The carrier theorem is not intended to float above an unspecified microscopic dynamics. The starting point is therefore the one-sided boundary datum from which the primitive layer, the derived closed datum, and the canonical admissibility projector are reconstructed before the later closure theorems are invoked.

Definition 3.1 (One-sided boundary admissibility datum)**.** The one-sided boundary admissibility datum is

$$\mathfrak{T}^{\min}_{\partial} := (\mathcal{A}_+, \mathcal{H}_+, D_+, J, \Gamma, B_\Sigma)$$

with the following built-in admissibility conditions:

(i) $Z(\mathcal{A}_+)$ is spectrally regular and its Gelfand spectrum is a smooth compact oriented manifold with boundary M_+;

(ii) B_Σ is a self-adjoint tangential boundary operator on $\Sigma := \partial M_+$ compatible with the boundary restrictions of J and Γ;

(iii) in a collar neighborhood of Σ, the Dirac operator has canonical product form

$$D_+ = \gamma_n(\partial_n + B_\Sigma);$$

(iv) the boundary collar is reflection-regular so that canonical doubling, the Calderón projector, and the admissibility complex are well posed on the same datum.

Remark (Almost-commutative realizations). When a factorized realization is useful, one may choose

$$\mathcal{H} \cong L^2(\widetilde{M}, S) \otimes \mathcal{H}_{\mathrm{int}},$$

with $S \to \widetilde{M}$ the spinor bundle on the oriented double cover and $\mathcal{H}_{\mathrm{int}}$ a finite internal Hilbert space. This is an analytic realization choice rather than part of the boundary datum.

Theorem 3.2 (Doubling, deck involution, and boundary polarization)**.** *Let $\mathfrak{T}_\partial^{\min}$ be as above. Let*

$$\widetilde{M} := M_+ \cup_\Sigma M_+^{\mathrm{op}}$$

be the doubled manifold and let

$$\tau_{\mathrm{dbl}} : \widetilde{M} \to \widetilde{M}$$

be its deck involution. Let P_C be the Calderón projector of the APS-realized boundary problem for D_+ on Σ, and define the boundary polarization operator

$$\iota_C := 2P_C - 1$$

on the Cauchy-data Hilbert space $\mathcal{H}_\partial$.

Then:

(i) the geometric objects are

$$M = \widetilde{M}/\tau_{\mathrm{dbl}}, \qquad \Sigma = \mathrm{Fix}(\tau_{\mathrm{dbl}}), \qquad X_{\mathrm{bulk}} = \widetilde{M} \setminus \Sigma;$$

(ii) the doubled operator and its even and odd parts are

$$D = D_+ \cup_\Sigma D_+^{\mathrm{op}}, \qquad D_{\mathrm{ref}} := \frac{1}{2}\big(D + \tau_{\mathrm{dbl}}^* D \tau_{\mathrm{dbl}}^*\big), \qquad D_{\mathrm{rel}} := \frac{1}{2}\big(D - \tau_{\mathrm{dbl}}^* D \tau_{\mathrm{dbl}}^*\big);$$

(iii) on boundary Cauchy data the induced reflection $R_{\tau_{\mathrm{dbl}}}$ satisfies

$$\mathrm{Ran}(P_C) = \ker(R_{\tau_{\mathrm{dbl}}} - 1), \qquad \ker(P_C) = \ker(R_{\tau_{\mathrm{dbl}}} + 1), \qquad \iota_C = R_{\tau_{\mathrm{dbl}}}.$$

Hence the primitive scaffold

$$\mathfrak{T}_{\mathrm{prim}}^{\mathrm{kin}} := (X_{\mathrm{bulk}}, \widetilde{M}, M, \Sigma, \tau_{\mathrm{dbl}}, \iota_C, D_{\mathrm{ref}}, D_{\mathrm{rel}}, \chi_{\mathrm{seed}})$$

is derived, not assumed. The derived doubled datum may be recorded as

$$\mathfrak{T}_{\min}^{\mathrm{cl}} := (\mathcal{A}, \mathcal{H}, D, J, \Gamma, \tau_{\mathrm{dbl}}, \iota_C),$$

If one further chooses an almost-commutative factorization

$$\mathcal{H}_{\mathrm{prim}} \cong L^2(X_{\mathrm{bulk}}, S) \otimes \mathcal{H}_{\mathrm{int}},$$

then S and $\mathcal{H}_{\mathrm{int}}$ refer only to that chosen realization and are not additional reconstruction outputs.

Proof. Reflection regularity on an exact collar produces the doubled manifold together with the deck involution τ_{dbl}. The APS-realized boundary problem has a Calderón projector P_C whose range is the Cauchy-data space of interior solutions. On the doubled collar the deck reflection acts on Cauchy data by $R_{\tau_{\mathrm{dbl}}}$, and the solution space splits into the ± 1 eigenspaces of this reflection. Therefore

$$P_C = \frac{1 + R_{\tau_{\mathrm{dbl}}}}{2}, \qquad \iota_C = 2P_C - 1 = R_{\tau_{\mathrm{dbl}}}.$$

The formulas for D_{ref} and D_{rel} are the even and odd parts of D under the induced involution. □

Corollary 3.3 (No confusion among the three involutions)**.** *Geometric reflection is carried by τ_{dbl}, boundary polarization by ι_{C}, and the finite carrier split by the restricted involution*

$$\varepsilon_{\mathrm{car}} := \iota_{\mathrm{C}}|_{\mathcal{H}_{\mathrm{car}}}.$$

No later theorem may identify these three operators literally; only the displayed restriction and transport maps are allowed.

The canonical admissibility projector is built in two transparent stages. The primitive stage uses only data already available from $\mathfrak{T}_{\partial}^{\min}$ through the derived boundary polarization ι_{C}, namely the seam-even selector on Cauchy data, an APS realization that makes the geometric anchor self-adjoint Fredholm, and the relative operator D_{rel}. The full stage upgrades this primitive projector by the color and determinant projectors that become available only after the carrier theorem of Section 3.3 identifies the color factor and after the compact bosonic index of Section 8 fixes the determinant involution. The full complex and the upgrade theorem appear once these later inputs are in place; here we close the primitive stage.

Definition 3.4 (Primitive admissibility complex and primitive projector)**.** Let the primitive Hilbert space carry only the seam factor and the gap sector accessible from D_{rel} alone:

$$\mathcal{H}_{\mathrm{prim}} = \mathcal{H}_{\Sigma} \otimes \mathcal{H}_{t}, \qquad \mathcal{H}_{\mathrm{prim}}^{\mathrm{ext}} := \mathcal{H}_{\mathrm{prim}} \otimes \Lambda(\eta_{\Sigma}, \eta_{g}),$$

with primitive seam selector

$$P_{\Sigma,+} := \frac{\mathbf{1} + \iota_{\mathrm{C}}}{2}.$$

Fix $\epsilon_{\mathrm{adm}} > 0$ and let D_{prim} be the APS-realized restriction of the primitive sector operator to $\mathrm{Ran}(P_{\Sigma,+})$, controlled by D_{rel} relative to the geometric anchor. Define the primitive admissibility penalties

$$K_{\Sigma} := \sqrt{\lambda_{\Sigma}}\,\frac{\mathbf{1} - \iota_{\mathrm{C}}}{2}, \qquad K_{g} := \sqrt{\lambda_{\mathrm{gap}}}\,\mathbf{1}_{[0,\epsilon_{\mathrm{adm}}^{2}/4)}\big(P_{\Sigma,+} D_{\mathrm{prim}}^{\dagger} D_{\mathrm{prim}} P_{\Sigma,+}\big),$$

the primitive admissibility differential

$$Q_{\mathrm{prim}} := \eta_{\Sigma}^{\dagger} K_{\Sigma} + \eta_{g}^{\dagger} K_{g},$$

and the primitive admissibility Laplacian

$$\Delta_{\mathrm{prim}} := \{Q_{\mathrm{prim}}, Q_{\mathrm{prim}}^{\dagger}\}.$$

The primitive admissibility projector is the Hodge projector

$$P_{\mathrm{prim}} := \Pi_{\ker \Delta_{\mathrm{prim}}}.$$

Lemma 3.5 (Primitive nilpotency and primitive Hodge projector)**.** *On the primitive scaffold reconstructed from $\mathfrak{T}_{\partial}^{\min}$,*

$$Q_{\mathrm{prim}}^{2} = 0, \qquad \Delta_{\mathrm{prim}} = K_{\Sigma}^{2} + K_{g}^{2}, \qquad P_{\mathrm{prim}} = \operatorname*{s-lim}_{\beta\to\infty} e^{-\beta(K_{\Sigma}^{2}+K_{g}^{2})}.$$

In particular P_{prim} depends only on ι_{C}, the APS realization, and D_{rel}, and is therefore reconstructed from the closed datum at the primitive level. Equivalently,

$$P_{\mathrm{prim}} = P_{\Sigma,+} P_{\mathrm{prim,gap}}, \qquad P_{\mathrm{prim,gap}} := \mathbf{1}_{[\epsilon_{\mathrm{adm}}^{2}/4,\infty)}\big(P_{\Sigma,+} D_{\mathrm{prim}}^{\dagger} D_{\mathrm{prim}} P_{\Sigma,+}\big).$$

Proof sketch. The auxiliary generators $\eta_\Sigma^\dagger$ and $\eta_g^\dagger$ anticommute, so the square of Q_{prim} reduces to commutators of K_Σ and K_g. Both penalties act on commuting sectors of $\mathcal{H}_{\text{prim}}$, hence the mixed terms vanish and $\Delta_{\text{prim}} = K_\Sigma^2 + K_g^2$. The zero-temperature limit projects onto the intersection of seam-even and gapped admissible vectors, which is the factorized form recorded above. □

Definition 3.6 (Carrier polarization on the finite block)**.** Let $\mathcal{H}_{\text{car}}$ denote the finite-dimensional carrier block selected after primitive admissibility compression. Define

$$\varepsilon_{\text{car}} := \iota_{\text{C}}|_{\mathcal{H}_{\text{car}}}, \qquad E_- := \ker(\varepsilon_{\text{car}} + 1), \qquad E_+ := \ker(\varepsilon_{\text{car}} - 1).$$

Then the finite carrier decomposition is

$$E = E_- \oplus E_+.$$

Remark (Forward reference to the full admissibility complex). The full admissibility projector is obtained by upgrading P_{prim} with the color center projector P_{sing} and the determinant involution projector P_Θ, which are available only after the carrier theorem of Section 3.3 identifies the color factor $\mathcal{H}_c$ and after the compact bosonic index of Section 8 fixes the determinant involution Q_Θ. The full complex and the explicit upgrade are stated in Definition 8.1 and theorem 8.2; nothing in the present primitive subsection depends on those inputs.

Definition 3.7 (Primitive Euclidean action)**.** For any declared spectral test profile f, the primitive Euclidean action is the relative spectral seed

$$S_{\text{prim}}[\chi_{\text{seed}}] := \operatorname{Tr} f\left(\frac{D_{\text{rel}}}{\chi_{\text{seed}}}\right) - \operatorname{Tr} f\left(\frac{D_{\text{ref}}}{\chi_{\text{seed}}}\right) + \frac{i\pi}{2}\Delta\eta_\Sigma.$$

Gauge fixing, ghosts, and carrier-compatible bosonic fluctuations are introduced only after the hard carrier theorem and the canonical admissibility projector are in place. Whenever convenient, the compressed expression $P_{\text{adm}} A P_{\text{adm}}$ may be used as shorthand for matrix elements on the physical admissible sector, but it is not the definition of a local operator net.

Definition 3.8 (Physical admissible local net)**.** Let $\mathfrak{A}_E(\mathcal{O})$ be the Euclidean source algebra generated by local field insertions with test-function support in $\mathcal{O}$. After the later Osterwalder–Schrader reconstruction, let $\mathfrak{A}_M(\mathcal{O})$ denote the corresponding Minkowski local algebra and let $\mathcal{H}_{\text{OS}}$ be the reconstructed Hilbert space. Define the physical admissible Hilbert space by

$$\mathcal{H}_{\text{adm}} := \operatorname{Ran}(P_{\text{adm}}) \subset \mathcal{H}_{\text{OS}}.$$

Define the physical admissible local net by

$$\mathfrak{A}_{\text{adm}}(\mathcal{O}) := \left\{ A|_{\mathcal{H}_{\text{adm}}} \;:\; A \in \mathfrak{A}_M(\mathcal{O}),\; A\mathcal{H}_{\text{adm}} \subset \mathcal{H}_{\text{adm}},\; A^*\mathcal{H}_{\text{adm}} \subset \mathcal{H}_{\text{adm}} \right\}.$$

Thus P_{adm} acts as the selector of the physical state space, while the local net is defined by restriction to the invariant admissible subspace rather than by compression with a global projector.

Remark (Bernstein realizations as analytic choices). Whenever a Bernstein representation of a spectral test profile is used, it is treated only as an analytic realization of a transfer channel already fixed by the closed datum and the canonical projector. Positivity or reflection stability therefore does not enter the main text as a primitive source of admissibility.

Theorem 3.9 (Primitive self-adjoint realization without extra analytic assumption)**.** *Let M_Λ be a finite cutoff of the primitive geometric anchor, so that M_Λ is compact with boundary. Let D_{geo} be the formally self-adjoint Dirac-type anchor and let A_∂ be an adapted boundary operator. Choose a self-adjoint D_{geo}-elliptic boundary condition B_{sa} of modified APS type:*

$$B_{\mathrm{sa}} = \begin{cases} H^{1/2}_{(-\infty,0)}(A_\partial), & \ker A_\partial = 0, \\ H^{1/2}_{(-\infty,0)}(A_\partial) \oplus L, & \ker A_\partial \neq 0, \end{cases}$$

where $L \subset \ker A_\partial$ is chosen as in the normal form for self-adjoint D-elliptic boundary conditions. Then the closed realization $D_{\mathrm{geo},B_{\mathrm{sa}}}$ is self-adjoint Fredholm with discrete spectrum. If D_{rel} is a symmetric differential perturbation of order at most one with bounded coefficients on M_Λ, then D_{rel} is infinitesimally $D_{\mathrm{geo},B_{\mathrm{sa}}}$-bounded and therefore

$$D_{\mathrm{geo},B_{\mathrm{sa}}} + D_{\mathrm{rel}}$$

is self-adjoint on the same domain. In particular the finite-cutoff relative spectral seed is well defined without an additional analytic existence hypothesis.

Proof. The existence of self-adjoint D-elliptic boundary conditions for formally self-adjoint Dirac-type operators is standard. In the zero-kernel case APS is already self-adjoint. If the adapted boundary operator has nontrivial kernel, one passes to a self-adjoint modified APS realization or, equivalently, to

$$H^{1/2}_{(-\infty,0)}(A_\partial) \oplus L,$$

with $L \subset \ker A_\partial$ chosen as in the normal form theorem for self-adjoint D-elliptic boundary conditions. The resulting realization is essentially self-adjoint on smooth sections satisfying the boundary condition. Since M_Λ is compact, coercivity at infinity is automatic and the realization is Fredholm with discrete spectrum.

For the perturbation, standard first-order elliptic regularity on a compact manifold with self-adjoint D-elliptic boundary condition gives a graph norm estimate

$$\|\psi\|_{H^1} \leq C\big(\|D_{\mathrm{geo},B_{\mathrm{sa}}}\psi\| + \|\psi\|\big).$$

Because D_{rel} has order at most one and bounded coefficients,

$$\|D_{\mathrm{rel}}\psi\| \leq C_1\|\psi\|_{H^1} + C_0\|\psi\| \leq \varepsilon\|D_{\mathrm{geo},B_{\mathrm{sa}}}\psi\| + C_\varepsilon\|\psi\|$$

for every $\varepsilon > 0$. Hence D_{rel} is $D_{\mathrm{geo},B_{\mathrm{sa}}}$-bounded with relative bound zero. The Kato–Rellich theorem therefore yields self-adjointness of $D_{\mathrm{geo},B_{\mathrm{sa}}} + D_{\mathrm{rel}}$ on the same domain. □

Remark (APS roles on the admissible branch). At the primitive level $\mathrm{APS}^{\mathrm{prov}}_{\mathrm{dom}}$ is the provisional elliptic boundary choice for the geometric anchor: an analytic starting representative chosen so that Theorem 3.9 applies. After the carrier theorem and the master closure roadmap of Section 8 the Calderón projector of the doubled operator D_b canonically promotes $\mathrm{APS}^{\mathrm{prov}}_{\mathrm{dom}}$ to the closed-branch boundary domain $\mathrm{APS}_{\mathrm{dom}}$ in the same admissible class. The family-pairing data $\mathrm{APS}_{\mathrm{fam}}$ are then induced canonically and remain a derived consequence rather than a second primitive input package. The explicit upgrade statement is recorded as Theorem 8.4 in Section 8.

Corollary 3.10 (Carrier-compatible Lorentzian dynamics after closure)**.** *Once the later carrier-compatible fluctuation package and the admissibility closure hypotheses of Section 8 are imposed, the resulting admissible Euclidean theory reconstructs a self-adjoint Lorentzian Hamiltonian H_{adm} and a one-parameter automorphism group*

$$\alpha_t(O) = e^{itH_{\mathrm{adm}}} O e^{-itH_{\mathrm{adm}}}.$$

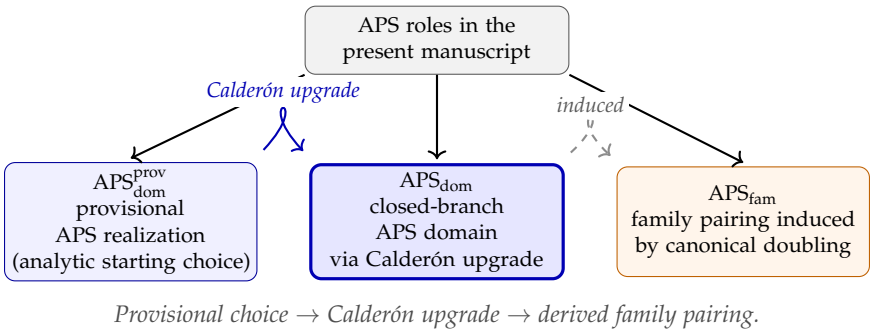


Figure 4. APS organization with Calderón upgrade and induced family pairing.

3.3 Carrier algebraic normal form

Lemma 3.11 (Algebraic normal form from exterior signatures)**.** *Let*

$$E = E_- \oplus E_+$$

be a finite chiral carrier with two simple factors of ranks

$$b := \dim E_-, \qquad s := \dim E_+,$$

and let

$$S^+ = \Lambda^{\mathrm{even}} E$$

be the positive half-spinor packet. Let

$$X = q_- P_- + q_+ P_+$$

be a primitive integer two-point generator with

$$q_- < 0 < q_+, \qquad bq_- + sq_+ = 0.$$

Assume only

$$\dim \Lambda^2 E_+ = 1, \qquad \Lambda^2 E_- \cong E_-^\vee.$$

Then necessarily

$$s = 2, \qquad b = 3, \qquad g := b + s = 5.$$

Hence

$$\dim \Lambda^{\mathrm{even}} E = 16.$$

The even exterior algebra decomposes into exactly the six sectors

$$(1,1)_0, \quad (3,2)_{1/6}, \quad (\bar{3},1)_{-2/3}, \quad (\bar{3},1)_{1/3}, \quad (1,2)_{-1/2}, \quad (1,1)_1,$$

and the primitive trace condition therefore has the unique solution

$$(q_-, q_+) = (-2, 3).$$

Consequently

$$X = -2P_3 + 3P_2, \qquad Y := \frac{X}{6} = \mathrm{diag}\left(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2}\right).$$

Proof. From

$$\dim \Lambda^2 E_+ = \binom{s}{2} = 1$$

one gets $s = 2$. For the color factor one always has

$$\Lambda^{b-1} E_- \cong E_-^\vee.$$

The assumption

$$\Lambda^2 E_- \cong E_-^\vee$$

therefore forces $b - 1 = 2$, hence $b = 3$. Thus $g = 5$.

Now

$$S^+ = \Lambda^0 E \oplus \Lambda^2 E \oplus \Lambda^4 E,$$

and for $E = E_3 \oplus E_2$,

$$\Lambda^2 E = \Lambda^2 E_3 \oplus (E_3 \otimes E_2) \oplus \Lambda^2 E_2,$$
$$\Lambda^4 E = (\Lambda^3 E_3 \otimes E_2) \oplus (\Lambda^2 E_3 \otimes \Lambda^2 E_2).$$

These are exactly the six sectors

$$(1,1)_0,\ (3,2)_{1/6},\ (\bar{3},1)_{-2/3},\ (\bar{3},1)_{1/3},\ (1,2)_{-1/2},\ (1,1)_1.$$

Hence

$$\dim S^+ = 2^{5-1} = 16.$$

Finally

$$3q_- + 2q_+ = 0$$

with primitive integers and $q_- < 0 < q_+$ has the unique solution

$$(q_-, q_+) = (-2, 3).$$

□

Corollary 3.12 (One chiral family emerges)**.** *Under the hypotheses of Lemma 3.11, the positive half-spinor packet S^+ carries exactly one chiral Standard Model family including ν^c.*

Proof. The six-sector decomposition displayed in Lemma 3.11 is exactly

$$(1,1)_0 \oplus (3,2)_{1/6} \oplus (\bar{3},1)_{-2/3} \oplus (\bar{3},1)_{1/3} \oplus (1,2)_{-1/2} \oplus (1,1)_1,$$

namely one chiral Standard Model family together with ν^c. □

Remark (Logical status of the carrier normal form). The lemma above is the algebraic normal form of the former open carrier target Theorem K. It is a rigidity statement under two exterior signatures:

$$\dim \Lambda^2 E_+ = 1, \qquad \Lambda^2 E_- \cong E_-^\vee.$$

Within those signatures the packet size

$$\dim \Lambda^{\text{even}} E = 16$$

and the one-family decomposition are outputs, while the occupancy identities move behind the carrier closure as derived consistency statements. On the retained branch, the compact Higgs index first fixes the bosonic rank and Theorem 8.13 and corollary 8.15 then discharge these carrier-signature premises internally from the actual branch Yukawa tensor.

Corollary 3.13 (Carrier norm)**.** *On the minimal carrier one has*

$$Y = \operatorname{diag}\left(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2}\right), \qquad \gamma = \operatorname{Tr}_E Y^2 = \frac{5}{6}.$$

Hence from now on

$$E = E_3 \oplus E_2, \qquad X = -2P_3 + 3P_2, \qquad P_3 = \frac{\mathbf{1} - \varepsilon_{\mathrm{car}}}{2}, \qquad P_2 = \frac{\mathbf{1} + \varepsilon_{\mathrm{car}}}{2},$$

and the hypercharge operator is the normalized primitive two-point generator

$$Y = \operatorname{diag}\left(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2}\right) = \frac{X}{6} = -\frac{1}{3}P_3 + \frac{1}{2}P_2 = \frac{1}{12}\mathbf{1} + \frac{5}{12}\varepsilon_{\mathrm{car}}. \tag{4}$$

Lemma 3.14 (Charge-generator / carrier-polarization inversion)**.** *On the rigid carrier, the carrier polarization and the primitive two-point generator are recovered by*

$$\varepsilon_{\mathrm{car}} = \frac{2X - \mathbf{1}}{5} = \frac{12Y - \mathbf{1}}{5}, \tag{5}$$

$$X = \frac{5\varepsilon_{\mathrm{car}} + \mathbf{1}}{2},$$

$$(2X - \mathbf{1})^2 = 25\,\mathbf{1},$$

$$X^2 - X - 6\,\mathbf{1} = 0,$$

$$6Y^2 - Y - \mathbf{1} = 0. \tag{6}$$

Consequently the carrier projectors are

$$P_2 = \frac{\mathbf{1} + \varepsilon_{\mathrm{car}}}{2} = \frac{2(3Y + \mathbf{1})}{5}, \qquad P_3 = \frac{\mathbf{1} - \varepsilon_{\mathrm{car}}}{2} = \frac{3(\mathbf{1} - 2Y)}{5}.$$

Proof. The affine relations between (X, Y) and $\varepsilon_{\mathrm{car}}$ are inverted algebraically. Squaring the first relation and using $\varepsilon_{\mathrm{car}}^2 = \mathbf{1}$ gives the quadratic generator polynomial $X^2 - X - 6\,\mathbf{1} = 0$, and dividing by 36 yields the hypercharge polynomial. The projector formulas then follow from $P_2 = (\mathbf{1} + \varepsilon_{\mathrm{car}})/2$ and $P_3 = (\mathbf{1} - \varepsilon_{\mathrm{car}})/2$. □

Lemma 3.15 (Primitive trace-balanced carrier)**.** *Let Y be an endomorphism of a finite-dimensional carrier E such that*

$$6Y^2 - Y - \mathbf{1} = 0, \qquad \operatorname{Tr}_E Y = 0,$$

and assume the realization is primitive in the sense that E is not a nontrivial direct sum of identical trace-balanced Y-modules. Then

$$\operatorname{Spec}(Y) = \left\{-\frac{1}{3}, \frac{1}{2}\right\}, \qquad \dim\ker\left(Y + \frac{1}{3}\mathbf{1}\right) = 3, \qquad \dim\ker\left(Y - \frac{1}{2}\mathbf{1}\right) = 2.$$

Equivalently,

$$\operatorname{rank} E = 5, \qquad E = E_3 \oplus E_2, \qquad Y = -\frac{1}{3}P_3 + \frac{1}{2}P_2.$$

Proof. The quadratic polynomial factors as

$$(2Y - \mathbf{1})(3Y + \mathbf{1}) = 0,$$

so the only eigenvalues are

$$y_+ = \frac{1}{2}, \qquad y_- = -\frac{1}{3}.$$

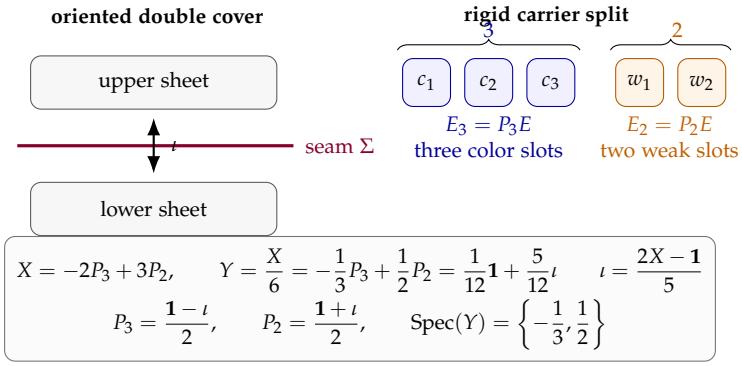


Figure 5. Rigid carrier at a glance. The seam involution, the primitive two-point generator, and the normalized hypercharge operator carry the same hard information: geometrically they separate the two sheets, and algebraically they separate the $3+2$ carrier split through the projectors P_3 and P_2.

Write

$$m_+ := \dim\ker(Y - y_+\mathbf{1}), \qquad m_- := \dim\ker(Y - y_-\mathbf{1}).$$

Then

$$0 = \mathrm{Tr}_E\, Y = \frac{1}{2}\, m_+ - \frac{1}{3}\, m_-,$$

so

$$3m_+ = 2m_-.$$

Because $\gcd(2,3) = 1$, every positive integer solution has the form

$$(m_+, m_-) = (2k, 3k).$$

Primitivity forces $k = 1$, hence $(m_+, m_-) = (2,3)$. This is exactly the rigid $3+2$ split, with the displayed spectral form of Y. □

Remark (Compressed carrier decoder). The carrier can therefore be entered from the shorter algebraic normal form

$$6Y^2 - Y - \mathbf{1} = 0, \qquad \mathrm{Tr}_E\, Y = 0, \qquad \text{primitive realization} \Longrightarrow E = E_3 \oplus E_2.$$

The exterior-signature route of Lemma 3.11 remains the theorem-level discharge of these algebraic hypotheses from the actual branch geometry.

The hard carrier package is summarized visually in Figure 5. The point is that the same minimal data are readable in four equivalent ways: geometrically as an oriented double cover with deck reflection and boundary polarization, algebraically as the trace-balanced quadratic carrier decoder, combinatorially as the rigid $3+2$ split, and spectrally as the primitive two-point generator X with normalized hypercharge $Y = X/6$.

Remark (Two-point carrier algebra). Because the minimal carrier satisfies $6Y^2 - Y - \mathbf{1} = 0$, every analytic function of Y reduces to

$$f(Y) = A_f\,\mathbf{1} + B_f\, Y, \qquad A_f = \frac{2f(1/2) + 3f(-1/3)}{5}, \qquad B_f = \frac{6\big(f(1/2) - f(-1/3)\big)}{5}.$$

Thus the hard carrier is effectively a two-point algebra supported on the spectral values $-1/3$ and $1/2$, and the projectors P_3, P_2 are already polynomial functions of Y.

Remark (Factorized carrier polynomial and carrier discriminant). For a general split carrier $E = E_b \oplus E_s$ with total rank $g := b + s$ and eigenvalues $-1/b$ and $1/s$, the carrier generator satisfies

$$\left(Y + \frac{1}{b}\right)\left(Y - \frac{1}{s}\right) = 0 \quad \Longleftrightarrow \quad bs\, Y^2 + (s-b)Y - \mathbf{1} = 0.$$

Its discriminant is

$$\Delta_Y = (s-b)^2 + 4bs = (b+s)^2 = g^2.$$

Thus the square root of the carrier discriminant is the carrier rank itself. Moreover,

$$Y = \frac{b-s}{2bs}\mathbf{1} + \frac{g}{2bs}\,\varepsilon_{\text{car}}, \qquad \varepsilon_{\text{car}} = \frac{2bs\, Y - (b-s)\,\mathbf{1}}{g}.$$

On the minimal branch $(b, s) = (3, 2)$, this reduces to

$$6Y^2 - Y - \mathbf{1} = (2Y - \mathbf{1})(3Y + \mathbf{1}), \qquad \Delta_Y = 25 = 5^2, \qquad \varepsilon_{\text{car}} = \frac{12Y - \mathbf{1}}{5}.$$

Hence the denominator 5 in the hypercharge–seam inversion is exactly $\sqrt{\Delta_Y}$, not an additional inserted constant.

This does not replace the stabilizer proof of

$$G_{\text{car}} = S(U(3) \times U(2)),$$

which is the actual source of the internal carrier stabilizer theorem in the present manuscript. What the factorization and discriminant do show exactly is that the hard carrier already supports the distinguished spectral values

$$Y = \frac{1}{2}, \qquad Y = -\frac{1}{3},$$

and that the inversion denominator is fixed algebraically by the carrier discriminant.

The two carrier invariants used repeatedly below are

$$\gamma = \operatorname{Tr}_E Y^2 = \frac{5}{6}, \qquad B = \frac{\operatorname{rank} E_3}{\operatorname{rank} E_2} = \frac{3}{2}.$$

Corollary 3.16 (Carrier norm from the carrier polynomial)**.** *The carrier polynomial already fixes*

$$\gamma = \operatorname{Tr}_E Y^2 = \frac{1}{6}\operatorname{Tr}_E(Y + \mathbf{1}) = \frac{5}{6},$$

since $\operatorname{Tr}_E Y = 0$ *and* $\operatorname{rank} E = 5$.

Corollary 3.17 (Minimal carrier compression)**.** *On the minimal carrier branch,*

$$B\gamma = \frac{3}{2} \cdot \frac{5}{6} = \frac{5}{4}.$$

We package the hard kernel as

$$\mathcal{C}_{\min} = (E_3 \oplus E_2,\ X,\ Y,\ S^+,\ \gamma,\ B).$$

3.4 Internal carrier stabilizer and one-family packet

Theorem 3.18 (Internal carrier stabilizer)**.** *The determinant-preserving connected stabilizer of the minimal carrier split is*

$$G_{\text{car}} = S(U(3) \times U(2)) \cong \frac{SU(3) \times SU(2) \times U(1)_Y}{\mathbb{Z}_6}.$$

The corresponding Lie algebra factorization

$$\mathfrak{g}_{\text{SM}} = \mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$$

is only the differential, local form of the same statement; throughout the manuscript this quotient is intended as the internal carrier-stabilizer form whenever a group symbol is written, and the local algebra notation is used only when explicitly stated.

Proof. An automorphism preserving the split $E_3 \oplus E_2$ acts by a pair $(u_3, u_2) \in U(3) \times U(2)$. Requiring determinant preservation on the full carrier gives

$$\det(u_3)\det(u_2) = 1,$$

so the connected stabilizer is exactly $S(U(3) \times U(2))$.

Define

$$\Phi : SU(3) \times SU(2) \times U(1)_Y \longrightarrow S(U(3) \times U(2)), \qquad \Phi(g_3, g_2, z) = \left(z^2 g_3,\ z^{-3} g_2\right).$$

This is a well-defined group homomorphism because the exponents are integers, and

$$\det(z^2 g_3)\det(z^{-3} g_2) = z^6 z^{-6} = 1.$$

To prove surjectivity, take $(u_3, u_2) \in S(U(3) \times U(2))$. Choose $z \in U(1)$ with

$$z^6 = \det(u_3) = \det(u_2)^{-1}.$$

Then

$$g_3 := z^{-2} u_3 \in SU(3), \qquad g_2 := z^3 u_2 \in SU(2),$$

and therefore

$$\Phi(g_3, g_2, z) = (u_3, u_2).$$

If $\Phi(g_3, g_2, z) = (\mathbf{1}_3, \mathbf{1}_2)$, then

$$g_3 = z^{-2}\mathbf{1}_3, \qquad g_2 = z^3 \mathbf{1}_2.$$

The conditions $g_3 \in SU(3)$ and $g_2 \in SU(2)$ imply $z^6 = 1$. Hence

$$\ker \Phi = \left\{ \left(z^{-2}\mathbf{1}_3,\ z^3 \mathbf{1}_2,\ z\right) : z^6 = 1 \right\} \cong \mathbb{Z}_6.$$

Therefore

$$S(U(3) \times U(2)) \cong \frac{SU(3) \times SU(2) \times U(1)_Y}{\mathbb{Z}_6}.$$

□

Theorem 3.19 (Faithful physical global gauge group)**.** *The action of*

$$S(U(3) \times U(2))$$

on the admissible matter-plus-Higgs package

$$\mathcal{R}_{\mathrm{adm}} := S^+ \oplus E_2$$

is faithful. Therefore

$$G_{\mathrm{phys}} = G_{\mathrm{car}} = S(U(3) \times U(2)).$$

Equivalently,

$$G_{\mathrm{phys}} \cong \frac{SU(3) \times SU(2) \times U(1)_Y}{\mathbb{Z}_6}.$$

No further nontrivial quotient survives on the physical matter-plus-Higgs package.

Proof. Let $(u_3, u_2) \in S(U(3) \times U(2))$ act trivially on $\mathcal{R}_{\mathrm{adm}}$. The Higgs sector is E_2, so triviality there implies

$$u_2 = \mathbf{1}_2.$$

The matter packet contains the quark doublet sector

$$Q = E_3 \otimes E_2.$$

Since $u_2 = \mathbf{1}_2$, trivial action on Q forces

$$u_3 = \mathbf{1}_3.$$

Hence the kernel is trivial. The quotient from $SU(3) \times SU(2) \times U(1)$ to $S(U(3) \times U(2))$ already has kernel $\mathbb{Z}_6$ by Theorem 3.18, so this is the exact physical global form. □

Throughout the manuscript, $S(U(3) \times U(2))$ is therefore read not only as the internal carrier stabilizer but as the physical global gauge group selected by the faithful action on the admissible matter-plus-Higgs package.

Matter is read from the positive half-spinor

$$S^+ = \Lambda^{\mathrm{even}} E, \qquad \dim S^+ = 16.$$

Theorem 3.20 (One carrier family from the exterior packet)**.** *The positive half-spinor of the minimal carrier decomposes as*

$$S^+ \cong (1,1)_0 \oplus (3,2)_{1/6} \oplus (\bar{3},1)_{-2/3} \oplus (\bar{3},1)_{1/3} \oplus (1,2)_{-1/2} \oplus (1,1)_1,$$

that is, exactly one chiral Standard Model family with ν^c. Equivalently, in the familiar unified packaging language,

$$S^+ \cong \mathbf{1} \oplus \mathbf{10} \oplus \bar{\mathbf{5}}.$$

Proof. One has

$$S^+ = \Lambda^0 E \oplus \Lambda^2 E \oplus \Lambda^4 E.$$

The $\Lambda^0 E$ sector is immediately

$$\Lambda^0 E \cong (1,1)_0,$$

which is identified with ν^c in left-chiral notation. For the minimal carrier,

$$\Lambda^2 E \cong \Lambda^2 E_3 \oplus (E_3 \otimes E_2) \oplus \Lambda^2 E_2.$$

Here

$$\Lambda^2 E_3 \cong \bar{3}, \qquad E_3 \otimes E_2 \cong (3,2), \qquad \Lambda^2 E_2 \cong 1,$$

and the hypercharges are obtained by adding the carrier charges from

$$Y = \mathrm{diag}\left(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2}\right).$$

Therefore

$$\Lambda^2 E \cong (\bar{3},1)_{-2/3} \oplus (3,2)_{1/6} \oplus (1,1)_1.$$

Because the full determinant is trivial in $S(U(3) \times U(2))$, one has $\Lambda^5 E \cong 1$ and hence

$$\Lambda^4 E \cong E^*.$$

Equivalently,

$$\Lambda^4 E \cong (\Lambda^3 E_3 \otimes E_2) \oplus (\Lambda^2 E_3 \otimes \Lambda^2 E_2).$$

Using $\Lambda^3 E_3 \cong 1$, $\Lambda^2 E_2 \cong 1$, and $\Lambda^2 E_3 \cong E_3^* \cong \bar{3}$, one gets

$$\Lambda^4 E \cong (1,2)_{-1/2} \oplus (\bar{3},1)_{1/3}.$$

Collecting the three even-degree sectors yields

$$S^+ \cong (1,1)_0 \oplus (\bar{3},1)_{-2/3} \oplus (3,2)_{1/6} \oplus (1,1)_1 \oplus (\bar{3},1)_{1/3} \oplus (1,2)_{-1/2},$$

with total multiplicity $1+3+6+1+3+2 = 16$. The $\Lambda^2 E$ sector is the standard left-chiral **10** of $SU(5)$, the $\Lambda^4 E$ sector is the standard left-chiral $\bar{\mathbf{5}}$, and the $\Lambda^0 E$ sector is the singlet **1**. □

In left-chiral notation the one-family decomposition takes the explicit form shown in Table 2. This is the place where the manuscript must be explicit: the carrier theorem is meaningful only if the reader can see how the sixteen states are organized.

Proposition 3.21 (Exact parity-code realization of the family packet)**.** Identifying each wedge monomial in ΛE with its occupation word

$$x = (x_1, \ldots, x_5) \in \{0,1\}^5,$$

the positive half-spinor is exactly the even-parity subspace

$$S^+ \cong \mathrm{span}\Big\{ |x\rangle : \sum_{i=1}^{5} x_i \equiv 0 \pmod 2 \Big\}.$$

Equivalently,

$$S^+ = \big(\Lambda^{\mathrm{even}} E_3 \otimes \Lambda^{\mathrm{even}} E_2\big) \oplus \big(\Lambda^{\mathrm{odd}} E_3 \otimes \Lambda^{\mathrm{odd}} E_2\big),$$

and each summand has dimension 8.

Proof. Choose a basis of E adapted to the split $E_3 \oplus E_2$. Exterior monomials are then in bijection with occupation words $x \in \{0,1\}^5$, and the exterior degree is exactly the Hamming weight $\sum_i x_i$. Hence $\Lambda^{\mathrm{even}} E$ is the span of the even-parity words. Splitting the degree into the E_3 and E_2 contributions gives the direct-sum decomposition above. Since

$$\dim \Lambda^{\mathrm{even}} E_3 = 1 + \binom{3}{2} = 4, \qquad \dim \Lambda^{\mathrm{odd}} E_3 = \binom{3}{1} + \binom{3}{3} = 4,$$

and

$$\dim \Lambda^{\mathrm{even}} E_2 = 1 + \binom{2}{2} = 2, \qquad \dim \Lambda^{\mathrm{odd}} E_2 = \binom{2}{1} = 2,$$

each tensor-product sector has dimension 8. □

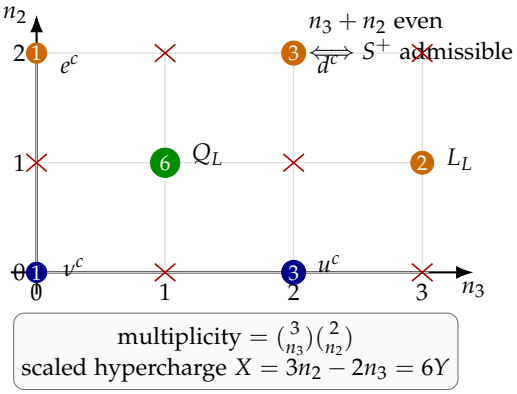


Figure 6. Occupancy map of the family packet. The admissible sectors are exactly the even lattice points in (n_3, n_2), with multiplicities $(1, 3, 6, 1, 2, 3)$ and field labels matching Table 2. This is the visual meeting point of parity lock, split weight enumerator, and family hypercharge arithmetic.

Remark (Exact parity lock, not a loose code analogy). The family packet is therefore not merely code-like: at the level of occupation kinematics it is exactly the classical even-parity $[5, 4, 2]$ code on five bits. What the present paper still does *not* claim is a dynamical error-correction theorem for realistic noise channels. The exact statement is the parity lock itself. In the maximally mixed state on the 16 occupation words, eight states lie in the $(\text{even}, \text{even})$ sector and eight in (odd, odd), so the coarse parity observables of the 3-side and 2-side carry exactly one bit of mutual information.

The family packet can also be read as a two-dimensional occupancy map, shown in Figure 6. This is where the one-family table, the even-parity code, the split weight enumerator, and the $X = 6Y$ arithmetic all meet in one glance.

3.5 Character, trace identities, and the abelian index

Proposition 3.22 (Split weight enumerator and one-family character)**.** For the present paragraph let

$$\chi_{S^+}(t) := \mathrm{Tr}_{S^+}(e^{tY}).$$

Define also the split weight enumerator

$$W(u, v) := \sum_{\substack{0 \le n_3 \le 3 \\ 0 \le n_2 \le 2 \\ n_3 + n_2 \text{ even}}} \binom{3}{n_3}\binom{2}{n_2} u^{n_3} v^{n_2}.$$

Then

$$W(u, v) = 1 + 3u^2 + 6uv + v^2 + 2u^3 v + 3u^2 v^2 = \frac{1}{2}\Big[(1+u)^3(1+v)^2 + (1-u)^3(1-v)^2\Big]. \quad (7)$$

Moreover,

$$\chi_{S^+}(t) = \frac{1}{2}\Big[\big(1 + e^{-t/3}\big)^3\big(1 + e^{t/2}\big)^2 + \big(1 - e^{-t/3}\big)^3\big(1 - e^{t/2}\big)^2\Big] = W(e^{-t/3}, e^{t/2}). \quad (8)$$

Consequently,

$$\chi_{S^+}(0) = 16, \qquad \chi'_{S^+}(0) = \mathrm{Tr}_{S^+} Y = 0, \qquad \chi''_{S^+}(0) = \mathrm{Tr}_{S^+} Y^2 = \frac{10}{3}, \qquad \chi'''_{S^+}(0) = \mathrm{Tr}_{S^+} Y^3 = 0.$$

Remark (Master generating function for the one-family packet). The even-parity split weight enumerator

$$W(u, v) = \frac{1}{2}\big[(1+u)^3(1+v)^2 + (1-u)^3(1-v)^2\big]$$

is the master generator of the family packet. The one-family character is obtained by the specialization $u = e^{-t/3}$ and $v = e^{t/2}$, while all trace moments of Y are recovered by differentiating at $t = 0$.

Proof. The pairs (n_3, n_2) allowed by even total degree are

$$(0,0),\ (2,0),\ (1,1),\ (0,2),\ (3,1),\ (2,2),$$

with multiplicities

$$1,\ 3,\ 6,\ 1,\ 2,\ 3.$$

This gives the explicit polynomial formula for $W(u,v)$. Equivalently, the parity projector

$$\frac{1}{2}(1 + (-1)^{n_3+n_2})$$

acting on $(1+u)^3(1+v)^2$ yields Equation (7). Substituting

$$u = e^{-t/3}, \qquad v = e^{t/2}$$

gives Equation (8). For an even exterior algebra one also has

$$\mathrm{Tr}_{\Lambda^{\mathrm{even}}E}(e^{tY}) = \frac{1}{2}\Big[\prod_{j=1}^{5}(1 + e^{tq_j}) + \prod_{j=1}^{5}(1 - e^{tq_j})\Big],$$

with carrier charges $q_j \in \{-1/3, -1/3, -1/3, 1/2, 1/2\}$. Differentiating at $t = 0$ yields the stated dimension and trace identities. □

Corollary 3.23 (Multiplicative character as six-sector decoder)**.** *Let*

$$X := 6Y.$$

Then the multiplicative one-family character

$$\Xi_{S^+}(t) := \mathrm{Tr}_{S^+}(t^X)$$

is

$$\Xi_{S^+}(t) = \frac{1}{2}\Big[(1+t^{-2})^3(1+t^3)^2 + (1-t^{-2})^3(1-t^3)^2\Big] = t^6 + 3t^2 + 6t + 1 + 2t^{-3} + 3t^{-4}. \quad (9)$$

Consequently

$$\Xi_{S^+}(1) = 16, \qquad (t\partial_t)\Xi_{S^+}(1) = \mathrm{Tr}_{S^+} X = 0, \qquad (t\partial_t)^2\Xi_{S^+}(1) = \mathrm{Tr}_{S^+} X^2 = 120 = 5!.$$

Thus the six sectors and their multiplicities are read off directly from the monomials of Equation (9).

Proof. Since $X = 3N_2 - 2N_3$, every admissible occupancy pair contributes the monomial $t^{3n_2-2n_3}$. Therefore

$$\Xi_{S^+}(t) = W(t^{-2}, t^3),$$

and Equation (7) gives the displayed factorized form. Expanding yields the six monomials shown above. Applying $t\partial_t$ once and twice at $t = 1$ recovers the first two trace moments of X. □

Remark (Table as readout, character as proof device). The one-family table remains useful as a reader-facing decoding sheet, but the whole six-sector arithmetic is already compressed into the single multiplicative character Equation (9).

Remark (Six admissible occupancy sectors). The six monomials in Equation (7) are exactly the six allowed occupancy pairs (n_3, n_2) of the family packet. This is the shortest combinatorial route from the split carrier to the six Standard Model multiplet sectors listed in Table 2.

The carrier packet therefore reproduces the standard trace cancellations

$$\mathrm{Tr}_{S^+} Y = 0, \qquad \mathrm{Tr}_{S^+} Y^3 = 0,$$

and packages them together with the quadratic trace in one formula. Since $\mathrm{Tr}_{S^+} Y^2 = 4\gamma = 10/3$ and $\mathrm{Tr}_\Phi Y^2 = 1/2$ for one weak doublet with hypercharge $1/2$, the hypercharge trace coefficient organizes as

$$b_1(N_{\mathrm{fam}}, N_\Phi) = \frac{2}{5} N_{\mathrm{fam}} \mathrm{Tr}_{S^+} Y^2 + \frac{1}{5} N_\Phi \mathrm{Tr}_\Phi Y^2 = \frac{8}{5} N_{\mathrm{fam}} \gamma + \frac{1}{10} N_\Phi = \frac{4}{3} N_{\mathrm{fam}} + \frac{1}{10} N_\Phi. \tag{10}$$

On the canonical branch $(N_{\mathrm{fam}}, N_\Phi) = (3, 1)$ this gives

$$b_1 = \frac{41}{10}.$$

The abelian index is therefore not inserted as an isolated Standard Model datum; it is a carrier-level output once the family and Higgs counts are declared.

Remark (Canonical-branch Diophantine lock). If one imposes $b_1 = 41/10$ on nonnegative integer data, then

$$40 N_{\mathrm{fam}} + 3 N_\Phi = 123.$$

The physically nontrivial solution is $(N_{\mathrm{fam}}, N_\Phi) = (3, 1)$, so the canonical branch is already sharply constrained at the level of the trace arithmetic.

Proposition 3.24 (Family hypercharge arithmetic and compression)**.** Let N_3 and N_2 be the occupation-number operators on the E_3 and E_2 slots, restricted to S^+, and define the scaled hypercharge operator

$$X := 6Y = 3N_2 - 2N_3.$$

Then on S^+ one has

$$X \equiv N_2 \equiv N_3 \pmod 2, \tag{11}$$

$$X \equiv N_3 \pmod 3, \tag{12}$$

$$X \equiv 3(N_2 + N_3) \pmod 5. \tag{13}$$

In particular, the even exterior degree $k = N_2 + N_3 \in \{0, 2, 4\}$ is read off from the residue class of X modulo 5: if $X \equiv r \pmod 5$ with $r \in \{0, 1, 2\}$, then $k = 2r$, so

$$X \bmod 5 = 0, 1, 2 \qquad \Longleftrightarrow \qquad \Lambda^0 E,\ \Lambda^2 E,\ \Lambda^4 E.$$

Moreover,

$$\mathrm{Spec}(X|_{S^+}) = \{0, -4, 1, 6, -3, 2\},$$

and therefore

$$X(X+4)(X-1)(X-6)(X+3)(X-2) = 0 \qquad \text{on } S^+. \tag{14}$$

Proof. Each occupied E_3 mode contributes hypercharge $-1/3$ and each occupied E_2 mode contributes $+1/2$, so multiplying by 6 gives $X = 3N_2 - 2N_3$. Modulo 2, the term $-2N_3$ vanishes, so $X \equiv N_2$. Because S^+ has even total degree, $N_2 \equiv N_3 \pmod 2$, which proves Equation (11). Modulo 3, one has $3N_2 \equiv 0$ and $-2 \equiv 1$, giving Equation (12). Modulo 5, the congruence $-2 \equiv 3$ yields Equation (13). Since on S^+ the even degree is $k = N_2 + N_3 \in \{0, 2, 4\}$,

the possible residues are $3k \equiv 0, 1, 2 \pmod 5$, from which $k = 2r$ with $r = X \bmod 5$ follows. Evaluating $X = 3n_2 - 2n_3$ on the six admissible occupancy pairs (n_3, n_2) gives the listed spectrum. The polynomial Equation (14) is then the spectral polynomial with these six simple roots. □

Corollary 3.25 (Five-fold phase filters for even degree)**.** *Let*

$$\zeta_5 = e^{2\pi i/5}, \qquad U_5 := e^{2\pi i X/5}.$$

Then the even-degree projectors are the discrete Fourier filters

$$P_{\Lambda^{2m}E} = \frac{1}{5}\sum_{r=0}^{4} \zeta_5^{-mr} U_5^r, \qquad m = 0, 1, 2.$$

Proof. By Proposition 3.24, the eigenspaces $\Lambda^{2m}E$ carry the residue class $X \equiv m \pmod 5$. Hence U_5 has eigenvalue ζ_5^m on $\Lambda^{2m}E$, and the standard Fourier projector formula on the cyclic group $\mathbb{Z}_5$ gives the claim. □

Remark (From the two-point carrier to the six-point family algebra). The minimal carrier itself is a two-point algebra, but the family packet lifts the scaled hypercharge to a six-point algebra. Therefore every analytic function of X on S^+ reduces to a polynomial of degree at most five. If

$$M_n := \mathrm{Tr}_{S^+}(X^n),$$

then Equation (14) implies the finite recursion

$$M_{n+6} = 2M_{n+5} + 31M_{n+4} - 20M_{n+3} - 156M_{n+2} + 144M_{n+1}.$$

In this sense the whole hypercharge statistics of one family is finitely compressed into six spectral values.

4 Kernel numerics and electromagnetic closure

The boundary-to-carrier presentation should not leave the impression that TFPT is hard only at the level of group theory and representation bookkeeping. The numerical kernel already fixes a nontrivial output layer. Here that layer is pulled forward and read as the first established output package of the carrier/output architecture rather than as a later list of disconnected constants.

Theorem 4.1 (Primitive seam generator)**.** *The primitive relative seam loop defines the positive generator*

$$[u_\Sigma] = 1 \in K^1(S^1) \cong \mathbb{Z}.$$

Hence the associated seam spectral flow is

$$SF(U_\Sigma) = 1, \qquad \Delta_\Sigma = 2\pi.$$

In the relative Chern normalization, the seam constant is

$$c_3 := \frac{\Delta_\Sigma}{16\pi^2} = \frac{1}{8\pi}.$$

Moreover the seam-even base opening is the normalized complement of the carrier quadratic norm,

$$\varphi_{\text{base}} := \frac{1-\gamma}{\pi} = \frac{1 - \mathrm{Tr}_E\, Y^2}{\pi} = \frac{1}{6\pi}.$$

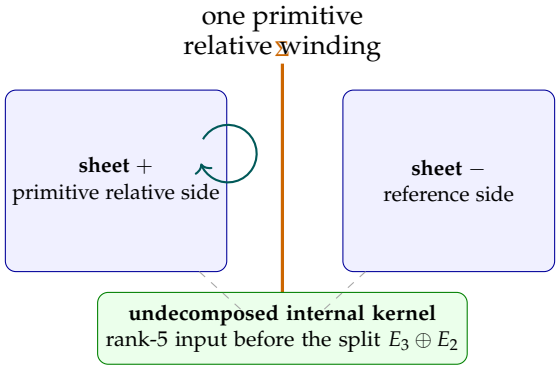


Figure 7. Interpretive primitive start sketch. Two sheets, one seam, one elementary relative winding, and an undecomposed rank-5 kernel appear before the later split into $E_3 \oplus E_2$. The picture is heuristic only; the theorem-level data are reconstructed from the one-sided boundary datum.

Proof sketch. The primitive reconstruction yields one elementary relative seam class and therefore one primitive generator of the seam K^1 lattice. Along a single positive seam circuit the spectral flow changes by one, so the associated loop has class $[u_\Sigma] = 1$ and spectral flow $SF(U_\Sigma) = 1$. The primitive circuit closes after one full turn, giving $\Delta_\Sigma = 2\pi$ and hence $c_3 = \Delta_\Sigma/(16\pi^2) = 1/(8\pi)$. The carrier theorem gives $\gamma = \mathrm{Tr}_E\, Y^2 = 5/6$, so the seam-even opening is the normalized complement of the carrier quadratic norm rather than a new independent numerical input. □

Remark (Interpretive picture of the primitive start state). At the level of physical intuition, the primitive spectral seed may be read as a minimal nontrivial relative defect state: two sheets, one seam, one elementary relative winding, and an undecomposed internal rank-5 kernel before the carrier split is imposed. This is only an interpretive picture, not a new theorem. Its role is to explain why the earliest hard numbers $(c_3, \varphi_{\mathrm{base}})$ already look like seam-normalized responses of a genuinely relative starting configuration.

4.1 Carrier form of the electromagnetic closure

At this stage the electromagnetic closure is kept in generic closed form. The hard carrier data fix

$$c_3 = \frac{1}{8\pi}, \qquad \varphi_{\mathrm{base}} = \frac{1}{6\pi},$$

while the later family and Higgs theorems will specialize the abelian index coefficient and the topological bridge coefficient.

Proposition 4.2 (Carrier susceptibility exponent)**.** On the minimal carrier branch,

$$B\gamma = \frac{3}{2} \cdot \frac{5}{6} = \frac{5}{4}.$$

Definition 4.3 (Minimal primitive seam transfer)**.** Let ψ_Σ be the normalized primitive seam mode singled out by Theorem 4.1. Define the minimal seam transfer operator by the rank-one projector

$$U_\Sigma^{\min} := |\psi_\Sigma\rangle\langle\psi_\Sigma|.$$

With

$$q(\alpha) := \delta_{\mathrm{top}} e^{-2\alpha}, \qquad B\gamma = \frac{5}{4},$$

one has

$$\mathcal{W}_\Sigma(\alpha) := -B\gamma\, \mathrm{Tr}_{\mathrm{adm}} \log\bigl(1 - q(\alpha) U_\Sigma^{\min}\bigr) = -\frac{5}{4} \log(1 - q(\alpha)).$$

$$\mathcal{T}_\Sigma(\alpha) = e^{\mathcal{W}_\Sigma(\alpha)} = (1-q(\alpha))^{-5/4}.$$
$$\varphi_\Sigma(\alpha) = \varphi_{\text{base}} + q(\alpha)(1-q(\alpha))^{-5/4}. \tag{15}$$

Proposition 4.4 (Normalized scalar closure potential)**.** Let

$$F_{U(1)}(\alpha) := \alpha^3 - 2c_3^3\alpha^2 - 8c_3^6 b_1 \log\big(\varphi_\Sigma(\alpha)^{-1}\big).$$

There exists a unique quartic local counterterm P_{loc} satisfying

$$P_{\text{loc}}(0) = P'_{\text{loc}}(0) = 0, \qquad P'_{\text{loc}}(\alpha) = \alpha^3 - 2c_3^3\alpha^2,$$

namely

$$P_{\text{loc}}(\alpha) = \frac{\alpha^4}{4} - \frac{2}{3}c_3^3\alpha^3.$$

Hence the normalized scalar closure potential

$$\Gamma_{U(1)}(\alpha) := \frac{\alpha^4}{4} - \frac{2}{3}c_3^3\alpha^3 - 8c_3^6 b_1 \int_0^\alpha \log\big(\varphi_\Sigma(a)^{-1}\big)\, da$$

satisfies

$$\partial_\alpha \Gamma_{U(1)}(\alpha) = F_{U(1)}(\alpha).$$

Theorem 4.5 (Exact electromagnetic closure on the minimal seam mode)**.** *On the canonical branch one has*

$$c_3 = \frac{1}{8\pi}, \qquad \varphi_{\text{base}} = \frac{1}{6\pi}, \qquad \delta_{\text{top}} = \frac{3}{256\pi^4}, \qquad \sum_{f,j} L_{f,j}^{\text{diag}} + N_\Phi = 40 + 1 = 41.$$

With the rank-one seam transfer of Definition 4.3, the closure equation is the scalar equation

$$F_{U(1)}(\alpha) = 0, \qquad F_{U(1)}(\alpha) = \alpha^3 - 2c_3^3\alpha^2 - \frac{4}{5}c_3^6\left(\sum_{f,j} L_{f,j}^{\text{diag}} + N_\Phi\right)\log\big(\varphi_\Sigma(\alpha)^{-1}\big), \tag{16}$$

where

$$\varphi_\Sigma(\alpha) = \frac{1}{6\pi} + \frac{3e^{-2\alpha}}{256\pi^4}\left(1 - \frac{3e^{-2\alpha}}{256\pi^4}\right)^{-5/4}.$$

This equation has a unique positive root

$$\alpha_\star = 0.007297352562209853127869691082265754 14\ldots$$

and therefore

$$\alpha_\star^{-1} = 137.035999216840712503537860303803880 37\ldots.$$

Proof. The explicit formulas above follow from the rank-one determinant identity

$$\det(1 - qU_\Sigma^{\min}) = 1 - q \qquad \text{for } U_\Sigma^{\min} = |\psi_\Sigma\rangle\langle\psi_\Sigma|.$$

The derivative of the closure function is

$$F'_{U(1)}(\alpha) := 3\alpha^2 - 4c_3^3\alpha - \frac{8}{5}c_3^6\left(\sum_{f,j} L_{f,j}^{\text{diag}} + N_\Phi\right)\frac{q(\alpha)(1-q(\alpha))^{-9/4}\big(1+q(\alpha)/4\big)}{\varphi_{\text{base}} + q(\alpha)(1-q(\alpha))^{-5/4}}.$$

On $[0, 2\cdot 10^{-4}]$ one verifies by interval arithmetic that

$$F_{U(1)}(\alpha) < 0.$$

On $[2 \cdot 10^{-4}, 10^{-1}]$ one verifies by interval arithmetic that

$$F'_{U(1)}(\alpha) > 0.$$

Finally

$$F_{U(1)}(10^{-2}) > 0.$$

Therefore $F_{U(1)}$ crosses zero exactly once. The displayed numerical root is obtained by monotone bisection on that interval. □

Remark (Asymptotic opening as a secondary expansion). Using Proposition 4.2, the determinant output admits the small opening expansion

$$\varphi_\Sigma(\alpha) = \varphi_{\text{base}} + \delta_{\text{top}} e^{-2\alpha} + \frac{5}{4}\delta_{\text{top}}^2 e^{-4\alpha} + O(e^{-6\alpha}),$$

which should be read only as a secondary expansion of the determinant output, not as the primary definition of the electromagnetic closure.

The dependency structure of the electromagnetic closure is displayed in Figure 8. What matters is not one formula in isolation, but the fact that the same carrier-side inputs fix the rank-one determinant, the scalar closure potential, and the fixed-point equation in one direction. The later transport and Higgs theorems then specialize the budget $\sum_{f,j} L_{f,j}^{\text{diag}} + N_\Phi$ and δ_{top} to their canonical closed values.

Remark (Logarithmic scale closure of the electromagnetic equation). Define the internally generated logarithmic scale

$$\mu_\Sigma(\alpha) := (\varphi_\Sigma(\alpha))^{-1}.$$

Then Equation (16) is equivalent to

$$\ln \mu_\Sigma(\alpha) = \frac{\alpha^2(\alpha - 2c_3^3)}{\frac{4}{5}c_3^6\left(\sum_{f,j} L_{f,j}^{\text{diag}} + N_\Phi\right)}.$$

Thus the observed fine-structure constant is selected by a balance between a geometric cubic term and a self-generated logarithmic scale parameter. This is RG-like and transmutation-like, because the coefficient is the internally closed transport-plus-Higgs budget divided by 10, and the logarithm is attached to an internally produced scale. In the present paper it should be read as *logarithmic scale closure* or a geometric fixed-point equation rather than as an externally prescribed one-loop running law. The symbol $\mu_\Sigma(\alpha)$ here is local to the electromagnetic closure discussion and should not be confused with the later matching scale M_Σ.

A related appendix continuation records that with

$$g := 5, \qquad d := \dim \mathfrak{g}_{\text{SM}} = 12,$$

$$\gcd(\Omega_{\text{adm}}, D_{\text{start}}) = d, \qquad \operatorname{lcm}(\Omega_{\text{adm}}, D_{\text{start}}) = g(g-1)d = 240 = |R(E_8)|.$$

4.2 Kernel output package

The benchmark row for $\alpha^{-1}(0)$ is defined by the positive self-consistent root of Equation (16), not by freezing the opening at its retained seed value. On the canonical branch that positive root remains

$$\alpha^{-1}(0) = 137.035\,999\,216\,840\,71.$$

For the remaining one-seed relations we keep the retained branch opening

$$c_3 = \frac{1}{8\pi}, \qquad \varphi_0^{\text{ret}} = \frac{1}{6\pi} + \frac{3}{256\pi^4}, \tag{17}$$

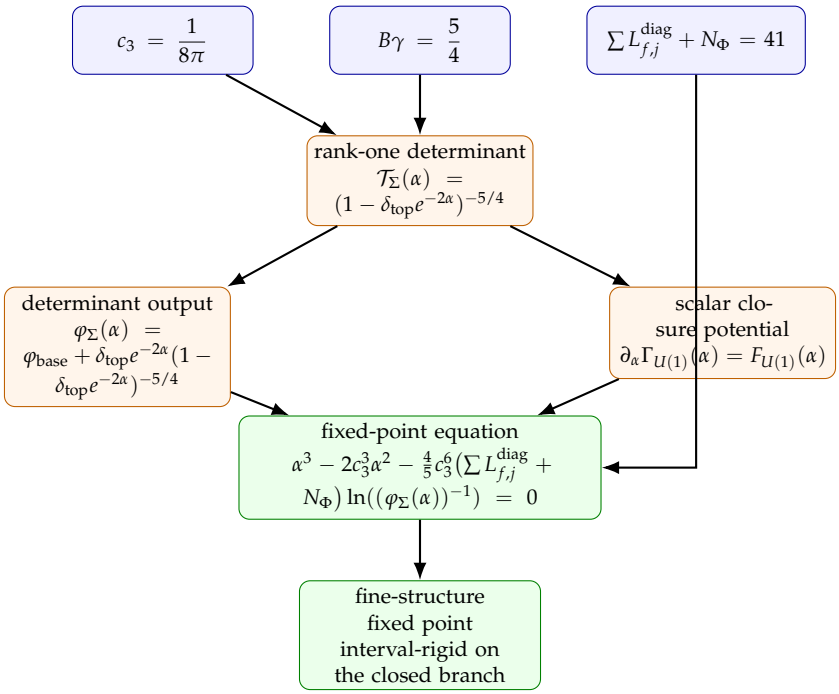


Figure 8. α closure flow in closed form. The carrier data determine the rank-one determinant, the scalar closure potential, and the fixed-point equation. The numerical specialization is made only after the transport and Higgs theorems close the branch budget.

Remark (Exact opening versus retained seed). The two symbols belong to the same admissible branch but play different jobs. The electromagnetic α row uses the exact seam generating function $\varphi_\Sigma(\alpha)$ inside the carrier-form closure equation, whereas the one-seed bridge relations freeze that branch to the retained canonical seed φ_0^{ret}. Thus $\varphi_\Sigma(\alpha)$ belongs only to the exact α benchmark line, while φ_0^{ret} is the seed used for $(\beta_{\mathrm{rad}}, \Omega_b, \lambda_C, \sin^2\theta_{13})$.

and the same retained kernel package emits the compact seed chain

$$\begin{aligned} C_{a\gamma\gamma} &= -4c_3 = -\frac{1}{2\pi}, \quad \beta_{\mathrm{rad}} = \frac{\varphi_0^{\mathrm{ret}}}{4\pi}, \\ \Omega_b &= (4\pi - 1)\beta_{\mathrm{rad}}, \qquad \lambda_C = \sqrt{\varphi_0^{\mathrm{ret}}(1-\varphi_0^{\mathrm{ret}})}, \\ \sin^2\theta_{13} &= \varphi_0^{\mathrm{ret}} e^{-\gamma}. \end{aligned} \tag{18}$$

Remark (Why the α row must stay self-consistent). The exact self-consistent equation with $\varphi_\Sigma(\alpha)$ yields the benchmark value

$$\alpha^{-1}(0) = 137.035\,999\,216\,840\,71.$$

Its second-order truncation reproduces the first two correction coefficients, but it is not itself the benchmark definition. If one instead freezes the opening at

$$\varphi_0^{\mathrm{ret}} = \frac{1}{6\pi} + \frac{3}{256\pi^4},$$

the root shifts to $\alpha^{-1}(0) \approx 137.036\,501\,465$, i.e. by about 5.02×10^{-4}, which is roughly $2.4 \times 10^4\sigma$ on the CODATA/NIST uncertainty. The wording of the α benchmark must therefore remain explicit: the benchmark uses the exact seam generating function, while the quadratic formula is only its controlled asymptotic shadow.

Kernel-output package

The point of this section is not to claim more numerics than TFPT currently owns. It is to show that the established kernel already fixes a coherent electromagnetic package before the later completion theorems begin: the seam constant c_3, the retained canonical seed φ_0^{ret}, the self-consistent carrier-form CFE for α, the dimensionless axion–photon anomaly coefficient, the birefringence seed, and the Cabibbo anchor.

5 Compression identities and numerical comparison of alternative discrete worlds

This section collects the algebraic compressions that turn the carrier packet and kernel numerics from a list of internally consistent outputs into an overdetermined structure. These compression identities are derived consistency statements on the retained branch: their formal derivation closes only once the bosonic-rank and branch-Yukawa carrier theorems have fixed the rigid $3+2$ split. By contrast, the discrete-world table below is a numerical comparison layer: it is not part of the rigid theorem surface and should be read as an appendix-style pressure test on the carrier packet.

5.1 Derived $48 = 4 \cdot 12$ identity and the 2-3-5 arithmetic

Proposition 5.1 (Derived $48 = 4 \cdot 12$ identity on the retained branch)**.** On the retained branch, once the family multiplicity and carrier ranks have been fixed, one recovers

$$\Omega_{\mathrm{adm}} = N_{\mathrm{fam}} \dim S^+ = (g-1) \dim \mathfrak{g}_{\mathrm{car}} = (g-1)(b^2+s^2-1)$$

with

$$N_{\mathrm{fam}} = 3, \qquad \dim S^+ = 16, \qquad (b,s) = (3,2), \qquad g = 5.$$

Equivalently, the former rank-lock formula is a derived consistency identity evaluated at the closed branch point:

$$3 \cdot 2^{g-1} = (g-1)(b^2+s^2-1). \tag{19}$$

Proof. Theorem 6.3 gives the physical family multiplicity

$$N_{\mathrm{fam}} = \dim \mathcal{H}_F = 3.$$

The carrier closure is completed later by Corollary 6.15 and theorem 8.13, which fix

$$(b,s) = (3,2), \qquad g = 5, \qquad \dim S^+ = 16.$$

Therefore

$$\Omega_{\mathrm{adm}} = N_{\mathrm{fam}} \dim S^+ = 3 \cdot 16 = 48.$$

For the same minimal branch,

$$\dim \mathfrak{g}_{\mathrm{car}} = \dim \mathfrak{su}(3) + \dim \mathfrak{su}(2) + \dim \mathfrak{u}(1) = 12.$$

Therefore

$$\Omega_{\mathrm{adm}} = 48 = (g-1) \dim \mathfrak{g}_{\mathrm{car}} = 4 \cdot 12,$$

which is exactly the displayed derived identity written in branch variables. The former occupancy balance is thus a consistency check rather than an input assumption. □

Remark (The 2-3-5 arithmetic). The hard carrier packet rests on exactly three discrete numbers: 2 (the double cover), 3 (the color rank and family count), and 5 (the carrier rank). From these three atoms one recovers

$$\gamma = \frac{5}{6} = \frac{5}{2\cdot 3}, \qquad B = \frac{3}{2}, \qquad B\gamma = \frac{5}{4} = \frac{5}{2^2},$$

$$\Omega_{\mathrm{adm}} = 3\cdot 2^4 = 48, \qquad D_{\mathrm{start}} = 5\cdot 12 = 60, \qquad \frac{D_{\mathrm{start}}}{\Omega_{\mathrm{adm}}} = \frac{5}{4} = B\gamma,$$

and the defect coefficients $\delta_{\mathrm{top}} = 48\,c_3^4$ and $\delta_2 = \frac{5}{4}\,\delta_{\mathrm{top}}^2 = 2880\,c_3^8$ with $2880 = 60\cdot 48$. The persistent appearance of $5/4$ as carrier asymmetry $\times$ norm, second defect stage, and ladder-to-occupancy ratio is therefore not three coincidences but one rational shadow of the 2-3-5 triad.

Lemma 5.2 (Universal $5/4$ compression quotient on the minimal branch)**.** *Define*

$$\rho_{5/4} := B\gamma.$$

On the canonical minimal branch,

$$\rho_{5/4} = B\gamma = \frac{\delta_2}{\delta_{\mathrm{top}}^2} = \frac{D_{\mathrm{start}}}{\Omega_{\mathrm{adm}}} = \frac{g_{\mathrm{car}}}{g_{\mathrm{car}}-1} = \frac{5}{4}.$$

Thus $5/4$ is the universal compression quotient of the minimal branch rather than a recurring numerical motif.

Proof. The identity $B\gamma = \frac{5}{4}$ is Corollary 3.17. The defect relation gives

$$\frac{\delta_2}{\delta_{\mathrm{top}}^2} = \frac{5}{4}.$$

On the canonical branch,

$$\frac{D_{\mathrm{start}}}{\Omega_{\mathrm{adm}}} = \frac{60}{48} = \frac{5}{4},$$

and with $g_{\mathrm{car}} = 5$ one also has

$$\frac{g_{\mathrm{car}}}{g_{\mathrm{car}}-1} = \frac{5}{4}.$$

Hence all displayed ratios agree on the same exact value. □

Remark (Interpretive status of the quotient). The quotient itself is exact. Any holographic or spacetime-dimensional reading of the denominator $g_{\mathrm{car}} - 1 = 4$ remains interpretive in the present manuscript rather than theorem-level.

Corollary 5.3 (Rank-5 master compression on the retained branch)**.** *On the retained branch one has $g_{\mathrm{car}} = 5$. Consequently*

$$N_{\mathrm{fam}} = \frac{g_{\mathrm{car}}+1}{2} = 3, \qquad \Omega_{\mathrm{adm}} = (g_{\mathrm{car}}+1)\,2^{g_{\mathrm{car}}-2} = 48,$$

$$\Omega_{\mathrm{adm}}\gamma = g_{\mathrm{car}}\,2^{g_{\mathrm{car}}-2} = 40, \qquad b_1 = \frac{g_{\mathrm{car}}\,2^{g_{\mathrm{car}}-2}+1}{10} = \frac{41}{10}, \qquad 10\,b_1 = 41.$$

Thus the closed family count, admissible occupancy, weighted occupancy, and abelian index all compress to the single carrier rank.

Proof. By Corollary 6.15 and theorem 8.13, the retained carrier has

$$g_{\text{car}} = 5.$$

By Theorem 6.3, the family block fixes $N_{\text{fam}} = 3$, hence

$$\Omega_{\text{adm}} = N_{\text{fam}}\, 2^{g_{\text{car}}-1} = 3 \cdot 2^4 = 48.$$

The carrier norm corollary gives $\gamma = 5/6$, so

$$\Omega_{\text{adm}}\gamma = 48 \cdot \frac{5}{6} = 40.$$

Finally the closed Higgs branch has $N_\Phi = 1$, and (20) yields

$$10\, b_1 = \Omega_{\text{adm}}\gamma + N_\Phi = 40 + 1 = 41.$$

□

Corollary 5.4 (Pythagorean compression of the abelian index on the retained branch)**.** *On the retained branch,*

$$\mathcal{I}_{41} = 10\, b_1 = 2g_{\text{car}}(g_{\text{car}} - 1) + 1 = g_{\text{car}}^2 + (g_{\text{car}} - 1)^2.$$

In particular, since $g_{\text{car}} = 5$ on the retained branch, one gets

$$\mathcal{I}_{41} = 5^2 + 4^2 = 41.$$

Proof. By Corollary 5.3, one has $g_{\text{car}} = 5$ and

$$10\, b_1 = g_{\text{car}}\, 2^{g_{\text{car}}-2} + 1 = 5 \cdot 2^3 + 1 = 41.$$

$$2g_{\text{car}}(g_{\text{car}} - 1) + 1 = 2 \cdot 5 \cdot 4 + 1 = 41.$$

Since on the retained branch

$$2g_{\text{car}}(g_{\text{car}} - 1) + 1 = g_{\text{car}}^2 + (g_{\text{car}} - 1)^2,$$

the claim follows. □

Remark (The number 4 as a cross-cutting echo). In addition to the 2-3-5 triad, the integer 4 appears at four structurally distinct levels:

$$F_{\text{patch}} = 4, \qquad g_{\text{car}} - 1 = 4, \qquad \frac{\Omega_{\text{adm}}}{\dim \mathfrak{g}_{\text{SM}}} = \frac{48}{12} = 4, \qquad \frac{|R(E_8)|}{D_{\text{start}}} = \frac{240}{60} = 4.$$

That is, four corner patches, carrier rank minus one, matter-to-gauge compression factor, and root-to-start quotient all evaluate to the same integer.

5.2 Carrier compressions on the minimal branch

Corollary 5.5 (Two-point spectral compression of γ and φ_{base})**.** *Let*

$$y_+ := \frac{1}{2}, \qquad y_- := -\frac{1}{3}$$

be the two distinct carrier eigenvalues singled out by the factorized carrier polynomial. Then

$$\gamma = y_+ - y_-, \qquad \varphi_{\text{base}} = \frac{y_+ + y_-}{\pi}.$$

Here $y_+ + y_-$ is the unweighted Vieta sum of the two distinct carrier eigenvalues, not the trace on E.

Proof. By Equation (6), equivalently by the factorization

$$6Y^2 - Y - \mathbf{1} = (2Y - \mathbf{1})(3Y + \mathbf{1}),$$

the two distinct carrier eigenvalues are exactly $y_+ = 1/2$ and $y_- = -1/3$. Their difference is

$$y_+ - y_- = \frac{1}{2} + \frac{1}{3} = \frac{5}{6} = \gamma,$$

while their sum is

$$y_+ + y_- = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}.$$

Since $\varphi_{\text{base}} = (1-\gamma)/\pi = 1/(6\pi)$, the claim follows. □

Proposition 5.6 (Dual-root generation of the hypercharge packet and admissible cusp set)**.** Let

$$y_+ = \frac{1}{2}, \qquad y_- = -\frac{1}{3}.$$

Then all one-family hypercharges are generated by the same two carrier roots:

$$\begin{aligned} Y(Q_L) &= y_+ + y_-, \quad Y(u^c) = 2y_-, \quad Y(d^c) = -y_-, \\ Y(L_L) &= -y_+, \qquad\quad Y(e^c) = 2y_+, \quad Y(\nu^c) = 0. \end{aligned}$$

Moreover the positive transport cusps satisfy

$$\left\{1, \frac{2}{3}, \frac{1}{3}\right\} = \{2y_+, -2y_-, 2(y_+ + y_-)\}.$$

Thus the carrier packet and the transport grammar are generated by the same two-point root data.

Proof. On the minimal carrier one has

$$Y = \operatorname{diag}(y_-, y_-, y_-, y_+, y_+)$$

with respect to the split $E_3 \oplus E_2$. The one-family packet S^+ carries the sectors $Q_L, u^c, d^c, L_L, e^c, \nu^c$, and their hypercharges are the sums of the occupied carrier roots. This gives the displayed formulas. Substituting $y_+ = 1/2$ and $y_- = -1/3$ recovers the canonical values

$$\frac{1}{6}, \quad -\frac{2}{3}, \quad \frac{1}{3}, \quad -\frac{1}{2}, \quad 1, \quad 0.$$

The admissible transport cusps are fixed in Theorem 7.7 as $\{1, \frac{2}{3}, \frac{1}{3}\}$, which is exactly the displayed root-generated set. □

Corollary 5.7 (Family hypercharge variance and electroweak block trace)**.** *Equip one carrier family with the uniform average*

$$\mathbb{E}_{S^+}(A) := \frac{1}{16} \operatorname{Tr}_{S^+} A.$$

Then

$$\operatorname{Var}_{S^+}(Y) := \mathbb{E}_{S^+}(Y^2) - \mathbb{E}_{S^+}(Y)^2 = \frac{1}{16} \operatorname{Tr}_{S^+} Y^2 = \frac{5}{24}.$$

Consequently,

$$I_1^{\text{EW}} = \operatorname{Var}_{S^+}(Y)\, b_1.$$

Proof. The one-family trace identities give $\mathrm{Tr}_{S^+} Y = 0$ and $\mathrm{Tr}_{S^+} Y^2 = 10/3$. Hence the centered second moment is

$$\frac{1}{16}\,\mathrm{Tr}_{S^+} Y^2 = \frac{10/3}{16} = \frac{5}{24}.$$

Using $I_1^{\mathrm{EW}} = \frac{5}{24}\, b_1$ from the electroweak block relation yields the second identity. □

Corollary 5.8 (Factorial quadratic trace and two-defect factorization)**.** *Let*

$$X := 6Y.$$

Then the one-family quadratic trace is

$$\mathrm{Tr}_{S^+}(X^2) = 36\ \mathrm{Tr}_{S^+}(Y^2) = 36 \cdot \frac{10}{3} = 120 = 5!.$$

Equivalently, using the six hypercharge sectors with multiplicities

$$(1, 3, 6, 1, 2, 3)$$

at

$$X \in \{0, -4, 1, 6, -3, 2\},$$

the same trace is

$$0^2 + 3 \cdot 4^2 + 6 \cdot 1^2 + 1 \cdot 6^2 + 2 \cdot 3^2 + 3 \cdot 2^2 = 120.$$

On the retained branch this gives the exact factorization

$$\delta_2 = 2880\, c_3^8 = 4!\ \mathrm{Tr}_{S^+}(X^2)\, c_3^8.$$

Proof. By Proposition 3.22, one has

$$\mathrm{Tr}_{S^+}(Y^2) = \frac{10}{3}.$$

Since $X = 6Y$, the first identity follows immediately. The sector-by-sector sum is the same quadratic trace written in the basis of admissible occupancy sectors from Proposition 3.24. Finally, the hard kernel normalization gives

$$\delta_2 = 2880\, c_3^8 = 24 \cdot 120\, c_3^8 = 4!\ \mathrm{Tr}_{S^+}(X^2)\, c_3^8.$$

□

Remark (Quadratic trace versus variance). The value $120 = 5!$ is the unnormalized quadratic trace $\mathrm{Tr}_{S^+}(X^2)$, not the normalized variance. The corresponding uniform variance is

$$\mathrm{Var}_{S^+}(X) = 36\ \mathrm{Var}_{S^+}(Y) = 36 \cdot \frac{5}{24} = \frac{15}{2}.$$

Remark (Generic abelian-index identity). Define the generic occupancy

$$\Omega_{\mathrm{occ}} := 16\, N_{\mathrm{fam}}.$$

Then the abelian-index formula gives

$$10\, b_1 = \frac{40}{3} N_{\mathrm{fam}} + N_\Phi = \Omega_{\mathrm{occ}}\, \gamma + N_\Phi.$$

At this stage no canonical specialization is inserted yet. The closed 41 chain will be derived only after the family and Higgs theorems fix $\Omega_{\mathrm{occ}} = 48$ and $N_\Phi = 1$.

Corollary 5.9 (Family integrality repair)**.** *One carrier family contributes the weighted transport budget*

$$\dim S^+ \cdot \gamma = 16 \cdot \frac{5}{6} = \frac{40}{3}.$$

Hence any integral global budget assembled from whole carrier families must satisfy

$$N_{\text{fam}} \cdot \frac{40}{3} \in \mathbb{Z}.$$

The minimal positive repair is therefore

$$N_{\text{fam,min}} = 3.$$

For this minimal repair one gets

$$\Omega_{\text{occ}} = N_{\text{fam,min}} \dim S^+ = 48, \qquad \Omega_{\text{occ}}\gamma = 40.$$

The later family theorem realizes exactly this minimal arithmetic.

Proof. The first identity uses $\dim S^+ = 16$ and $\gamma = 5/6$. Since $\gcd(40, 3) = 1$, integrality forces $3 \mid N_{\text{fam}}$, so the smallest positive choice is 3. Multiplying by 16 and then by γ gives the displayed values. □

5.3 Generic abelian index identity

The abelian coefficient is first carried only in the generic form

$$10\, b_1 = \Omega_{\text{occ}}\gamma + N_\Phi. \tag{20}$$

The arithmetic specialization to 41 is deferred until the family multiplicity and the Higgs sector have both been theorematically closed.

5.4 UV seed-shadow identities and the Cabibbo inversion

The retained scalar seed φ_0^{ret} still satisfies exact UV identities. In the present version these identities are auxiliary kernel bookkeeping rather than the independence grammar of the physical observable layer. In particular, the Cabibbo angle admits an explicit inversion back to the UV seed:

$$\varphi_0^{\text{ret}} = \frac{1 - \sqrt{1 - 4\lambda_C^2}}{2}. \tag{21}$$

Remark (Catalan expansion of the Cabibbo inversion). Setting

$$x := \lambda_C^2,$$

the inversion formula becomes

$$\varphi_0^{\text{ret}} = x\, C(x), \qquad C(x) := \frac{1 - \sqrt{1 - 4x}}{2x}.$$

Hence

$$\varphi_0^{\text{ret}} = \sum_{n \geq 0} C_n\, \lambda_C^{2n+2} = \lambda_C^2 + \lambda_C^4 + 2\lambda_C^6 + 5\lambda_C^8 + 14\lambda_C^{10} + \cdots, \tag{22}$$

where C_n are the Catalan numbers. In the present paper the appropriate interpretation of this identity is noncrossing combinatorics on the seed side; it is a theorem-level strengthening of the one-seed compression rather than a claim about planar Feynman graphs, a specific large-N / 't Hooft resummation, or a holographic flavor limit.

Proposition 5.10 (Retained seed as one decoder)**.** Set

$$u := \varphi_0^{\mathrm{ret}}.$$

On the retained branch one has

$$\lambda_C^2 = u(1-u), \qquad \beta_{\mathrm{rad}} = \frac{u}{4\pi}, \qquad \Omega_b = \left(1 - \frac{1}{4\pi}\right) u, \qquad \sin^2\theta_{13} = e^{-\gamma} u.$$

Equivalently,

$$u = \frac{1-\sqrt{1-4\lambda_C^2}}{2} = 4\pi\beta_{\mathrm{rad}} = \frac{\Omega_b}{1-\frac{1}{4\pi}} = e^{\gamma}\sin^2\theta_{13}.$$

Thus the retained branch carries a single decoder

$$u \longmapsto (\lambda_C, \beta_{\mathrm{rad}}, \Omega_b, \theta_{13}).$$

Proof. The forward identities are exactly the retained-seed bridge relations. The inverse identities follow from the Cabibbo inversion (21) and direct elimination of u from the same four formulas. □

Eliminating the retained seed through the Cabibbo inversion gives the equivalent λ_C-only formulas:

$$\begin{aligned} \beta_{\mathrm{rad}} &= \frac{1-\sqrt{1-4\lambda_C^2}}{8\pi}, \\ \Omega_b &= \frac{4\pi-1}{8\pi}\left(1-\sqrt{1-4\lambda_C^2}\right), \\ \sin^2\theta_{13} &= \frac{e^{-\gamma}}{2}\left(1-\sqrt{1-4\lambda_C^2}\right). \end{aligned} \tag{23}$$

Corollary 5.11 (Vacuum half-circle and UV-angle parametrization)**.** *On the retained UV seed split one has the exact identity*

$$\lambda_C^2 + \left(\varphi_0^{\mathrm{ret}} - \frac{1}{2}\right)^2 = \frac{1}{4}.$$

Defining

$$\sin^2\theta_{\mathrm{UV}} := \varphi_0^{\mathrm{ret}},$$

one obtains

$$\lambda_C = \frac{1}{2}\sin(2\theta_{\mathrm{UV}}), \qquad \sin^2\theta_{13} = e^{-\gamma}\sin^2\theta_{\mathrm{UV}}, \qquad \Omega_b = \left(1-\frac{1}{4\pi}\right)\sin^2\theta_{\mathrm{UV}}.$$

Proof. The exact seed relation $\lambda_C^2 = \varphi_0^{\mathrm{ret}}(1-\varphi_0^{\mathrm{ret}})$ is equivalent to the displayed half-circle identity after completing the square. The θ_{UV} parametrization then follows immediately from $\sin^2\theta_{\mathrm{UV}} = \varphi_0^{\mathrm{ret}}$ together with the exact retained-seed formulas for $\sin^2\theta_{13}$ and Ω_b. □

Corollary 5.12 (CKM–PMNS bridge law)**.** *On the closed branch one has the exact relation*

$$\sin^2\theta_{13} = \frac{e^{-5/6}}{2}\left(1-\sqrt{1-4\lambda_C^2}\right).$$

Equivalently,

$$\lambda_C^2 = e^{5/6}\sin^2\theta_{13} - e^{10/6}\sin^4\theta_{13}.$$

In internal branch form,

$$\sin^2\theta_{13} = 0.0231084351588811\ldots$$

with no additional fit parameter.

Proof. By Equation (21),

$$\varphi_0^{\mathrm{ret}} = \frac{1-\sqrt{1-4\lambda_C^2}}{2}.$$

By the exact seed identity,

$$\sin^2\theta_{13} = \varphi_0^{\mathrm{ret}} e^{-\gamma}, \qquad \gamma = \frac{5}{6}.$$

Substituting φ_0^{ret} gives the displayed formula. □

Corollary 5.13 (Baryon–reactor bridge law)**.** *On the closed branch one has the exact cross-sector relation*

$$\Omega_b = \left(1-\frac{1}{4\pi}\right) e^{5/6}\sin^2\theta_{13} = \frac{4\pi-1}{8\pi}\left(1-\sqrt{1-4\lambda_C^2}\right).$$

In internal branch form,

$$\Omega_b = 0.0489406626654501\ldots$$

with no additional fit parameter.

Proof. The exact seed-budget identities give

$$\Omega_b = (4\pi-1)\beta_{\mathrm{rad}}, \qquad \beta_{\mathrm{rad}} = \frac{\varphi_0^{\mathrm{ret}}}{4\pi}, \qquad \sin^2\theta_{13} = \varphi_0^{\mathrm{ret}} e^{-5/6}.$$

Eliminating φ_0^{ret} yields the displayed cross-sector identity. Eliminating φ_0^{ret} instead through Equation (21) gives the equivalent λ_C form. □

Corollary 5.14 (Matter-radiation quotient on the retained seed split)**.** *On the retained seed split one has the exact quotient identity*

$$\frac{\Omega_b}{\beta_{\mathrm{rad}}} = 4\pi - 1.$$

Proof. This is the direct quotient of the exact retained-seed identities

$$\Omega_b = (4\pi-1)\beta_{\mathrm{rad}}, \qquad \beta_{\mathrm{rad}} = \frac{\varphi_0^{\mathrm{ret}}}{4\pi}.$$

□

Remark (Interpretive reading of the retained seed split)*.* The same one-seed decoder carries the exact budget identity

$$\Omega_b + \beta_{\mathrm{rad}} = \varphi_0^{\mathrm{ret}}. \tag{24}$$

Indeed,

$$\Omega_b + \beta_{\mathrm{rad}} = (4\pi-1)\beta_{\mathrm{rad}} + \beta_{\mathrm{rad}} = 4\pi\,\beta_{\mathrm{rad}} = \varphi_0^{\mathrm{ret}}.$$

Both Corollary 5.14 and equation (24) are exact retained-seed identities. Any solid-angle or spherical-surface interpretation remains interpretive and is not used as theorem input.

Thus on the canonical branch, β, Ω_b, λ_C, and θ_{13} are four exact projections of the same retained decoder $u = \varphi_0^{\mathrm{ret}}$. Conversely, θ_{13} fixes λ_C and Ω_b through $\varphi_0^{\mathrm{ret}} = e^{\gamma}\sin^2\theta_{13}$. This is the derived-consequence layer of the retained seed split, not a separate independent theorem surface.

5.5 Carrier comparison of alternative discrete worlds

The strongest numerical pressure test on the carrier packet is not a further precision hit. It is that nearby wrong discrete worlds miss the electromagnetic closure decisively.

Proposition 5.15 (Numerical comparison of alternative discrete worlds)**.** Let

$$\alpha^{-1}(0; N_{\text{fam}}, N_\Phi, \nu^c)$$

denote the positive self-consistent root of the carrier-form electromagnetic closure equation (16) for the indicated discrete data. Then:

Discrete world	Change from canonical	$\alpha^{-1}(0)$	Residual vs CODATA	Reading
canonical $(3, 1, \textbf{yes})$	—	137.0359992	$+3.9 \times 10^{-8}$	inside CODATA window ($+1.87\sigma$)
$N_{\text{fam}} = 2$	one family fewer	156.094	$+19.1$	catastrophically excluded
$N_{\text{fam}} = 4$	one family more	124.835	-12.2	catastrophically excluded
$N_\Phi = 2$	two light Higgs doublets	135.946	-1.09	catastrophically excluded
no ν^c	remove right-handed neutrino	137.0338	-2.16×10^{-3}	excluded ($\sim 1.03 \times 10^5 \sigma$)

Proof sketch. Hold fixed

$$c_3 = \frac{1}{8\pi}, \qquad \varphi_{\text{base}} = \frac{1}{6\pi}, \qquad \gamma = \frac{5}{6}, \qquad B = \frac{3}{2},$$

and solve the positive self-consistent root of the same scalar closure equation on each discrete world:

$$\alpha^3 - 2c_3^3\alpha^2 - 8c_3^6\, b_1(N_{\text{fam}}, N_\Phi) \ln\big(\varphi_\Sigma(\alpha)^{-1}\big) = 0,$$

with

$$\varphi_\Sigma(\alpha) = \varphi_{\text{base}} + \delta_{\text{top}} e^{-2\alpha}(1 - \delta_{\text{top}} e^{-2\alpha})^{-5/4}, \qquad \delta_{\text{top}} = \Omega_{\text{adm}} c_3^4, \qquad \delta_2 = \frac{5}{4}\delta_{\text{top}}^2,$$

and

$$b_1(N_{\text{fam}}, N_\Phi) = \frac{4}{3} N_{\text{fam}} + \frac{1}{10} N_\Phi.$$

For the no-ν^c row, b_1 stays unchanged because $Y(\nu^c) = 0$, while the occupancy changes from $\Omega_{\text{adm}} = 48$ to $\Omega_{\text{adm}} = 45$ because the one-family packet drops from 16 to 15 states. The same positive-branch solver and tolerance are used for every row, so the deviations are arithmetic consequences of changed discrete data rather than fits. □

Remark. Two families shift α^{-1} by $+19$, four families by -12, and two Higgs doublets by -1.09. Even the mildest alternative—removing ν^c—shifts the value by 2.16×10^{-3}, which is $\sim 1.03 \times 10^5$ times the CODATA uncertainty. This is therefore a useful comparison pressure test on the canonical branch rather than an additional theorem in the reconstruction chain.

6 Closure architecture A: families, Higgs, and local spectral scale

This first closure-architecture block is the shortest route from the hard carrier packet to the closed theorem body. It has three moves: primitive winding balance in the seam normal plane forces the four-corner structure of the fundamental section, that winding theorem globalizes into the family space F, the determinant line selects the unique light Higgs doublet together with the compact seam phase, and the local spectral-scale channel generates both $\bar{M}_{\text{Pl}}$ and the electroweak vacuum from one common geometric response.

discrete world	Residual vs CODATA	reading
canonical $(3, 1, \text{yes})$	$+3.9 \times 10^{-8}$	inside the CODATA window
$N_{\text{fam}} = 2$	$+19.1$	catastrophically excluded
$N_{\text{fam}} = 4$	-12.2	catastrophically excluded
$N_{\Phi} = 2$	-1.09	extra light doublet breaks closure
no ν^c	-2.16×10^{-3}	$\sim 1.03 \times 10^5 \sigma$ exclusion

Figure 9. Comparison matrix for alternative discrete worlds. The middle column records the residual versus the CODATA benchmark rather than a misleading shift relative to the canonical row. Once the discrete packet is changed, the electromagnetic closure leaves the observed window decisively.

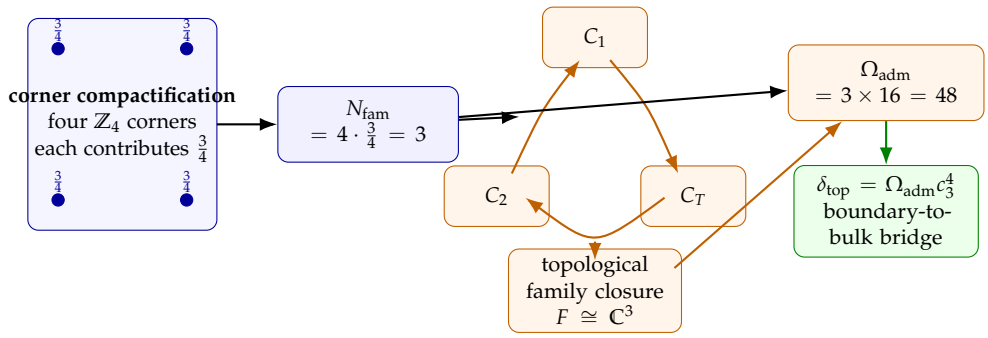


Figure 10. Carrier-to-family closure bridge. Corner compactification yields the punctured family section X_f°, the harmonic family space $\mathcal{H}_F \cong F \cong H^1(X_f^\circ, \mathbb{C})$ gives the hard multiplicity $N_{\text{fam}} = 3$, and the resulting occupancy $\Omega_{\text{adm}} = 48$ feeds back into the bridge reading $\delta_{\text{top}} = \Omega_{\text{adm}} c_3^4$.

6.1 Family monodromy and occupancy

Definition 6.1 (Family space)**.** Let $\overline{X}_f$ denote the compact seam quotient with four $\mathbb{Z}_4$ corner points $p_1, \ldots, p_4$, and define the punctured family section

$$X_f^\circ := \overline{X}_f \setminus \{p_1, p_2, p_3, p_4\}.$$

Its rigid harmonic model is the Möbius class

$$X_f^\circ \cong \mathbb{P}^1 \setminus \mu_4.$$

Set

$$F := H^1(X_f^\circ, \mathbb{C}).$$

We use the distinguished cycle basis (C_1, C_2, C_T) only as a concrete carrier-side basis of this family space; any later APS/Hodge realization is read as compatible analytic structure on the same topological family space.

In this interpretation F is neither an arbitrary flavor vector space nor a loose bookkeeping trick; it is a cycle space attached to the seam data. The actual bridge logic is shown in Figure 10: the derived corner theorem fixes $N_{\text{fam}} = 3$, the topological family step promotes that count to a family space, and the occupancy corollary then feeds back into the defect channel.

The Hodge metric on F turns the monodromy data into an honest unitary family system rather than a merely decorative triangle sketch.

Theorem 6.2 (Four corners and the rigid family surface $\mathbb{P}^1 \setminus \mu_4$)**.** *Let $\overline{X}_f$ be the compact seam quotient induced by the deck involution τ_{dbl} and its spin lift on the oriented normal two-plane. Every admissible fixed corner contributes one quarter-turn of the lifted spin rotation. Therefore*

$$N_{\mathrm{corner}} \cdot \frac{\pi}{2} = 2\pi, \qquad N_{\mathrm{corner}} = 4.$$

Removing the four corner points gives a four-punctured sphere

$$X_f^\circ := \overline{X}_f \setminus \{p_1, p_2, p_3, p_4\}.$$

The inherited finite symmetry group on the puncture set is a faithful dihedral group $D_4 \subset PGL_2(\mathbb{C})$. Every faithful copy of D_4 in $PGL_2(\mathbb{C})$ is Möbius conjugate to the standard action generated by

$$r(z) = iz, \qquad s(z) = 1/z.$$

Up to Möbius equivalence the unique D_4-invariant four-point configuration is therefore

$$\mu_4 := \{z \in \mathbb{C} : \ z^4 = 1\} = \{1, -1, i, -i\}.$$

Hence

$$X_f^\circ \cong \mathbb{P}^1 \setminus \mu_4, \qquad \chi(X_f^\circ) = -2, \qquad H_1(X_f^\circ, \mathbb{Z}) \cong \mathbb{Z}^3.$$

Consequently

$$F := H^1(X_f^\circ, \mathbb{C}) \cong \mathbb{C}^3.$$

Proof. The quarter-turn count gives four corner points. A sphere with four removed points is isomorphic to $\mathbb{P}^1 \setminus \{q_1, q_2, q_3, q_4\}$ for some ordered quadruple. The classification of finite subgroups of $PGL_2(\mathbb{C})$ implies that every faithful dihedral subgroup of order eight is Möbius conjugate to the standard D_4 action generated by $z \mapsto iz$ and $z \mapsto 1/z$. Under that action the unique invariant orbit of cardinality four is the set of fourth roots of unity, namely $\mu_4 = \{1, -1, i, -i\}$. Therefore

$$X_f^\circ \cong \mathbb{P}^1 \setminus \mu_4.$$

The Euler characteristic is $2 - 4 = -2$, and for a genus-zero surface with four punctures one has $\operatorname{rank} H_1 = 4 - 1 = 3$. Dualizing gives $F \cong \mathbb{C}^3$. □

Theorem 6.3 (Three harmonic family modes on the four-punctured family surface)**.** *Let*

$$X_f^\circ = \mathbb{P}^1 \setminus \mu_4$$

with a complete cusp metric in its Möbius class, and define the family Hilbert space by

$$\mathcal{H}_F := \ker \Delta^{(1)}_{L^2,\mathrm{APS}}(X_f^\circ).$$

Then

$$\dim \mathcal{H}_F = 3.$$

If the physical family multiplicity is defined by

$$N_{\mathrm{fam}} := \dim \mathcal{H}_F,$$

then

$$N_{\mathrm{fam}} = 3, \qquad F \cong \mathcal{H}_F \cong \mathbb{C}^3.$$

Proof. For a compact truncation $\overline{X}_{f,R}$ with four boundary circles,

$$\chi(\overline{X}_{f,R}) = 2 - 4 = -2.$$

Hence

$$\dim H^1(\overline{X}_{f,R}, \partial\overline{X}_{f,R}; \mathbb{C}) = 3.$$

The L^2 Hodge theorem for finite-area cusp surfaces identifies

$$\ker \Delta^{(1)}_{L^2,\mathrm{APS}}(X_f^\circ) \cong H^1(\overline{X}_{f,R}, \partial\overline{X}_{f,R}; \mathbb{C}),$$

so the dimension is 3. □

Remark (How the family theorem is now used). The family theorem is used at rigid conformal strength in the present version. Any orbifold APS realization is read as compatible analytic structure on the same classification-rigid four-punctured section, while the physical multiplicity itself is carried by the harmonic family mode space $\mathcal{H}_F$.

Proposition 6.4 (Orbifold APS compatibility)**.** Under the additional analytical hypotheses of Section D,

$$\mathrm{Ind}^{\mathrm{orb,APS}}(D^+_{\mathrm{rel}} \otimes L_F) = 3.$$

Corollary 6.5 (Boundary-to-bulk bridge corollary for the admissible occupancy)**.** *On the canonical branch,*

$$\dim(S^+ \otimes \mathcal{H}_F) = 16 \cdot 3 = 48 = \Omega_{\mathrm{adm}}.$$

Hence the topological surcharge enters the present paper as a derived occupancy corollary:

$$\delta_{\mathrm{top}} = c_3^4 \dim(S^+ \otimes \mathcal{H}_F) = \Omega_{\mathrm{adm}}\, c_3^4.$$

The family subsection is therefore no longer split between a local count, an independent index theorem, and a later occupancy reinterpretation. Those statements are compressed into one derived corner theorem with one bridge corollary, while the holonomy proposition below records the resulting unitary geometry on $F \cong \mathcal{H}_F$.

This theorem is the family statement referenced later by the admissibility theorems: the closure layer uses it as the explicit family component of the master closure theorem rather than as an external import.

Proposition 6.6 (Rigid special-unitary family holonomy on the determinant-trivial branch)**.** Let $L_F \to X_f^\circ$ be the flat Hermitian family bundle with fiber $F \cong \mathcal{H}_F \cong H^1(X_f^\circ, \mathbb{C})$. Assume that its determinant line is flat trivial,

$$\det L_F \cong X_f^\circ \times \mathbb{C},$$

and that the chosen flat connection preserves this trivialization. Then

$$\mathrm{Hol}(\nabla_F) \subset SU(3)_F.$$

Proof. The Hermitian metric implies

$$\mathrm{Hol}(\nabla_F) \subset U(3).$$

The induced connection on $\det L_F$ has trivial holonomy by assumption, so for every loop γ one has

$$\det \rho_F(\gamma) = 1.$$

Hence every holonomy matrix lies in $SU(3)$. □

The role of the present subsection is thus no longer only mnemonic. It displays both the local patch-deficit count and the global APS/Hodge realization before the later closure stack compresses them into the theorem-level output surface.

6.2 Occupancy and boundary-to-bulk bridge

With the family bridge installed, the matter space becomes $S^+ \otimes \mathcal{H}_F$ and the occupancy $\Omega_{\rm adm} = 48$ is already fixed by the harmonic family-mode theorem and its bridge corollary. Here *occupancy* means the admissible chiral-state count after family lifting; it is not used as a primitive datum. The bridge point is not that family lifting creates a new numerical surcharge. Rather, the inherited hard kernel value

$$\delta_{\rm top} = \frac{3}{256\pi^4}$$

admits on the canonical family branch the occupancy reinterpretation $\delta_{\rm top} = \Omega_{\rm adm}\, c_3^4$. Likewise the hard second-order defect contribution is re-read as

$$\delta_2 = \frac{5}{4}\,\delta_{\rm top}^2 = B\gamma\,\Omega_{\rm adm}^2 c_3^8.$$

In the present manuscript this occupancy corollary is used with a derived bridge reading: it is the boundary-to-bulk step that later feeds both the electromagnetic closure and the geometric channel. The coefficient $5/4$ is therefore not an extra numerical motif; it is the minimal-carrier compression factor $B\gamma$ rewritten in defect language. At the same time, the equality $\delta_{\rm top} = \Omega_{\rm adm} c_3^4$ is not presented as an independent dynamical transgression theorem in the ordinary field-theory sense.

6.3 From bridge to geometric completion

Principle 6.7 (Closure split)**.** Once the carrier packet and family closure architecture are fixed, the focused main-text closure problem splits into a geometric program $\mathcal{A}_{\rm geom}$ and a transport program $\mathcal{A}_{\rm trans}$. Record algebra and prediction semantics are now tied directly to the admissible gap in the main theorem stack, while E_8 scale grammar and horizon extrapolations are deferred to appendix-level continuations.

$$D_{\rm geo} := D_{\rm ref} + D_{\rm rel} = D.$$

The two main-text programs are packaged as

$$\mathcal{A}_{\rm geom} := \langle E_3 \oplus E_2,\ D_{\rm geo},\ \mathcal{B}_{\rm rel},\ \varpi_{\rm spin}\rangle, \qquad \mathcal{A}_{\rm trans} := \langle U_6,\ T,\ D_y,\ \mathcal{Y}_y^{(\varepsilon)},\ \mathfrak{C}_\Sigma\rangle.$$

This split keeps the carrier theorem, the geometric reconstruction problem, and the record-level extensions from being flattened into one undifferentiated claim. In particular, the metric field is no longer treated here as an independent entry of $\mathcal{A}_{\rm geom}$; it is reconstructed from the geometric Dirac anchor introduced next, while $D_{\rm rel}$ remains the comparison operator for spectral subtraction.

6.4 Relative APS and superconnection setup

The geometric completion is formulated on a graded bundle

$$\mathcal{E}_{\rm rel} = \mathcal{E}_{\rm rel}^+ \oplus \mathcal{E}_{\rm rel}^-,$$

with the geometric Dirac anchor $D_{\rm geo}$, the declared reference operator $D_{\rm ref}$, and APS-type boundary conditions on the seam Σ; see Appendix D. The geometric package is

$$\mathbb{A}_\Sigma := (D_{\rm geo}, D_{\rm rel}, A_\Sigma^{\rm geo}), \qquad A_\Sigma^{\rm geo} := \mathcal{B}_{\rm rel} \oplus \nabla_{\chi_{\rm geo}}.$$

Here $D_{\rm geo}$ is the geometric Dirac anchor, while $D_{\rm rel}$ is the relative comparison operator entering the spectral subtraction formulas. The bundle data $A_\Sigma^{\rm geo}$ reorganize the primitive seed package $A_\Sigma^{\rm seed}$ introduced earlier in Equation (1).

Theorem 6.8 (Geometric reconstruction from the Dirac anchor)**.** *Assume*

$$Z(\mathcal{A}) \cong C^\infty(M),$$

and the geometric anchor D_{geo} is of Dirac type with principal symbol

$$\sigma(D_{\mathrm{geo}})(x,\xi)^2 = g^{\mu\nu}(x)\xi_\mu\xi_\nu\,\mathbf{1}.$$

Then D_{geo} reconstructs the metric g, the spin structure, and the Levi–Civita connection on the geometric branch. Inner fluctuations of the same anchored spectral datum generate the gauge sector. The operator D_{rel} is the reference-subtracted comparison operator on that same branch and is not itself used as the local geometric anchor.

Proof sketch. On the admissible branch the commutative center identifies the manifold algebra, while the principal symbol of the Dirac anchor determines the quadratic form on cotangent vectors. Standard Dirac-type reconstruction then recovers the spin bundle and Levi–Civita connection from that same symbol data. The relative operator changes the comparison bookkeeping, not the geometric source of the local symbol. □

Theorem 6.9 (Retained branch from the admissibility selector)**.** *Let D_{geo} be the geometric anchor and let P_{adm} be the retained admissibility selector. Then:*

(i) $X_{\mathrm{bulk}} = \operatorname{Spec} Z(\mathcal{A})$ *and* $\Sigma = \mathrm{Fix}(\tau_{\mathrm{dbl}})$*;*

(ii) the local spectral density of D_{geo} determines the geometric response χ_{geo};

(iii) P_{adm} defines the retained admissible sector for observables, transport, and positivity statements on that branch.

No claim is made that the compressed operator $P_{\mathrm{adm}}D_{\mathrm{rel}}P_{\mathrm{adm}}$ carries the local principal symbol of the geometry.

Corollary 6.10 (No residual geometric datum beyond the geometric anchor)**.** *The geometric package is reconstructed from D_{geo} and its local spectral density. The selector P_{adm} acts downstream on the retained sector and is not part of the local geometric anchor.*

Definition 6.11 (Local spectral scale and canonical relative spectral residues)**.** Write the geometric response as a local spectral scale

$$\chi_{\mathrm{geo}} = \Lambda e^{\sigma}, \qquad D_{\sigma,\mathrm{geo}} := e^{-\sigma/2} D_{\mathrm{geo}} e^{-\sigma/2}.$$

$$\mathcal{Z}_{\mathrm{rel}}(s;\sigma) := \mathrm{Tr}_{\mathrm{rel}}\big(|D_{\sigma,\mathrm{geo}} + \mathcal{B}_{\mathrm{rel}}|^{-2s} - |D_{\mathrm{ref}}|^{-2s}\big) + \frac{i\pi}{2}\Delta\eta_\Sigma.$$

The canonical Einstein residue and order-zero residue are

$$\mathcal{E}_D := \operatorname*{Res}_{s=1} \mathcal{Z}_{\mathrm{rel}}(s;\sigma), \qquad \Gamma_D^{(4)} := \mathrm{FP}_{s=0}\, \mathcal{Z}_{\mathrm{rel}}(s;\sigma).$$

The gravitational branch is then read from the profile-free residue pair rather than from a freely chosen spectral profile.

Corollary 6.12 (Constant-scale reduction of the geometric branch)**.** *If the local spectral scale is frozen to a constant $\sigma = \sigma_0$, then $\chi_{\mathrm{geo}} = \chi_{\mathrm{geo},0} := \Lambda e^{\sigma_0}$ and the present geometric branch reduces to the constant-χ_{geo} closure relations extracted from $\Gamma_{\mathrm{grav}} := -6\chi_{\mathrm{geo}}^2\mathcal{E}_D + \Gamma_D^{(4)}$:*

$$\bar{M}_{\mathrm{Pl}}^2 = \frac{\chi_{\mathrm{geo},0}^2}{2\pi^2}, \qquad v_{\mathrm{geo}}^2 = Z_H\,\frac{\mu_\Phi^2(\chi_{\mathrm{geo},0})}{\lambda_\Phi}.$$

Once the unique stationary root of Theorem 6.31 is imposed, the Einstein coefficient is equivalently the boundary-normalized readout

$$\frac{\bar{M}_{\mathrm{Pl}}^2}{\lambda_\Sigma^2} = \frac{\rho_\star}{2\pi^2}.$$

Thus the former constant-χ_{geo} formulas are retained as the homogeneous branch of the local spectral-scale formulation rather than as a competing package.

6.5 Bosonic relative index and Higgs selection

Theorem 6.13 (Minimal determinant classes from the primitive charge generator)**.** *Let*

$$L_2 := \det E_2, \qquad L_3 := \det E_3$$

over the compactified oriented normal sphere $S^2 = D_N \cup_{S^1} D_S$. Then the primitive charge generator

$$X = -2P_3 + 3P_2$$

together with center neutrality and seam-evenness forces the clutching map

$$g_2(\varphi) = e^{i\varphi}$$

for L_2, while center neutrality forces

$$g_3(\varphi) = 1$$

for L_3. Therefore

$$c_1(L_2) = \deg g_2 = 1, \qquad c_1(L_3) = \deg g_3 = 0.$$

This pair is the unique minimal nonnegative admissible determinant class.

Proof. Complex line bundles on

$$S^2 = D_N \cup_{S^1} D_S$$

are classified by their clutching maps along the equator:

$$\pi_1(U(1)) \cong \mathbb{Z}.$$

Hence the first Chern class is exactly the winding degree of the transition function.

In the weak rank-two sector the primitive charge generator

$$X = -2P_3 + 3P_2$$

assigns the unique positive determinant winding to the seam-even weak block. The spin lift on the oriented normal sphere acts there by the half-angle representation, so on the determinant line the transition factor is

$$g_2(\varphi) = e^{i\varphi},$$

hence

$$\deg g_2 = 1, \qquad c_1(L_2) = 1.$$

In the color sector admissibility imposes center neutrality. Therefore the determinant transition function of the color block is null-homotopic and may be chosen constant:

$$g_3(\varphi) = 1.$$

Consequently

$$\deg g_3 = 0, \qquad c_1(L_3) = 0.$$

The full primitive seam twist is therefore exhausted in the weak determinant class. Any nonnegative pair strictly smaller than $(1,0)$ would force vanishing weak determinant winding and hence remove the seam-even weak line required by the primitive charge generator. Any larger nonnegative pair would insert extra clutching not carried by the primitive generator and therefore violate minimality. Thus the unique minimal nonnegative admissible determinant class is

$$(c_1(L_2), c_1(L_3)) = (1,0).$$

□

Remark (Energy interpretation on the retained branch). The older asymptotic bosonic functional

$$\mathcal{E}_{\text{asym}}(\Phi_2,\Phi_3) := \tau_2\big|\text{wind}(\det\Phi_2)\big| + \tau_3\big|\text{wind}(\det\Phi_3)\big| + m_3^2 \dim\ker \mathcal{B}^+_{E_3},$$

with $\tau_3 > \tau_2 > 0$, remains a useful physical intuition for why the retained branch is energetically preferred. In the present rewrite, however, the structural weight is carried by the determinant-class theorem above rather than by the cost functional itself.

Theorem 6.14 (Bosonic index and unique Higgs doublet from the minimal determinant class)**.** *Compactify the oriented two-dimensional normal slice N_Σ to S^2. For $a \in \{2,3\}$ define*

$$\mathcal{B}_{E_a} := \sigma^\mu \nabla^{(a)}_\mu + \Phi_a$$

on the compactified normal sphere. Under the determinant-class data of Theorem 6.13, one has

$$L_2 \cong \mathcal{O}(1), \qquad L_3 \cong \mathcal{O},$$

and hence

$$H^0(S^2,\mathcal{O}(1)) \cong \mathbb{C}^2, \qquad H^1(S^2,\mathcal{O}(1)) = 0,$$

and the compact bosonic index satisfies

$$\text{Ind}(\mathcal{B}^+_{E_2}) = 1, \qquad \text{Ind}(\mathcal{B}^+_{E_3}) = 0.$$

Consequently the closed branch contains exactly one complex weak doublet and no light color triplet, and the unique light seam-even bosonic zero mode lies in

$$\Phi \in \Gamma(E_2) \cong (1,2)_{1/2}, \qquad \tau^*_{\text{dbl}}\Phi = +\Phi.$$

Proof. By Theorem 6.13, the weak determinant line is the degree-one line bundle

$$L_2 \cong \mathcal{O}(1)$$

on

$$S^2 \cong \mathbb{C}P^1,$$

whereas the color determinant class is trivial and

$$L_3 \cong \mathcal{O}.$$

For $\mathbb{C}P^1$, Birkhoff–Grothendieck identifies every holomorphic line bundle by its degree, and Riemann–Roch gives

$$\chi(\mathcal{O}(1)) = 1 + 1 = 2.$$

Serre duality yields

$$H^1(S^2,\mathcal{O}(1)) \cong H^0(S^2,\mathcal{O}(-3))^\vee = 0,$$

so

$$H^0(S^2, \mathcal{O}(1)) \cong \mathbb{C}^2, \qquad H^1(S^2, \mathcal{O}(1)) = 0.$$

The resulting two-dimensional zero-mode space is exactly one complex weak doublet, so the compact bosonic index records it as

$$\mathrm{Ind}(\mathcal{B}^+_{E_2}) = 1.$$

In the color sector one has $L_3 \cong \mathcal{O}$, so no positive determinant degree is available. Center neutrality forbids an unpaired seam-even color contribution, hence the positive and negative triplet modes pair with zero net index:

$$\mathrm{Ind}(\mathcal{B}^+_{E_3}) = 0.$$

Therefore the closed branch contains exactly one seam-even Higgs doublet and no light color triplet, and the unique light bosonic zero mode lies in the weak block. □

Corollary 6.15 (Bosonic rank is two)**.** *On the closed branch the positive carrier block has rank*

$$\dim E_+ = 2.$$

Equivalently, the unique seam-even light bosonic zero mode fills one complex weak doublet and no larger positive carrier block is retained.

Proof. By Theorem 6.14, the light seam-even bosonic zero mode lies in the weak block

$$\Phi \in \Gamma(E_2) \cong (1,2)_{1/2},$$

and the compact index identifies its zero-mode space with

$$H^0(S^2, \mathcal{O}(1)) \cong \mathbb{C}^2.$$

Hence the retained positive carrier block is exactly two-dimensional. □

This is the sharp form of the Higgs claim used here. The issue is not whether a Higgs doublet can be written down, but whether a compact bosonic index can be shown to select it uniquely.

Remark (Callias shadow). The older Callias formulation is retained only as the noncompact shadow of the same compact index theorem and is discussed separately in the appendical analytic remarks. It no longer carries the main proof burden.

Once that uniqueness statement is fixed, the renormalizable carrier-compatible Yukawa couplings are the standard ones,

$$Q\, u^c\, \Phi, \qquad Q\, d^c\, \Phi^\dagger, \qquad L\, e^c\, \Phi^\dagger, \qquad L\, \nu^c\, \Phi,$$

so the Higgs statement becomes a genuine structural bridge rather than a notation choice. The formal carrier discharge from this Yukawa bridge is recorded later in Corollaries 6.15 and 8.15 and theorem 8.13, once the closure-theorem Higgs block has been stated in its final retained-branch form.

6.6 Closed 41 chain after family and Higgs closure

Once the derived corner theorem and the Higgs uniqueness theorem are in place, the generic abelian-index identity Equation (20) specializes to

$$10\, b_1 = \Omega_{\mathrm{adm}}\gamma + 1 = 48 \cdot \frac{5}{6} + 1 = 41. \tag{25}$$

The extra $+1$ is precisely the arithmetic contribution of the unique light seam-even Higgs doublet.

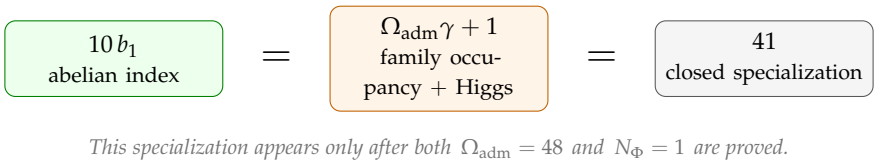


Figure 11. The closed 41 chain. The arithmetic specialization is made only after the family and Higgs theorems fix the occupancy and the unique light doublet.

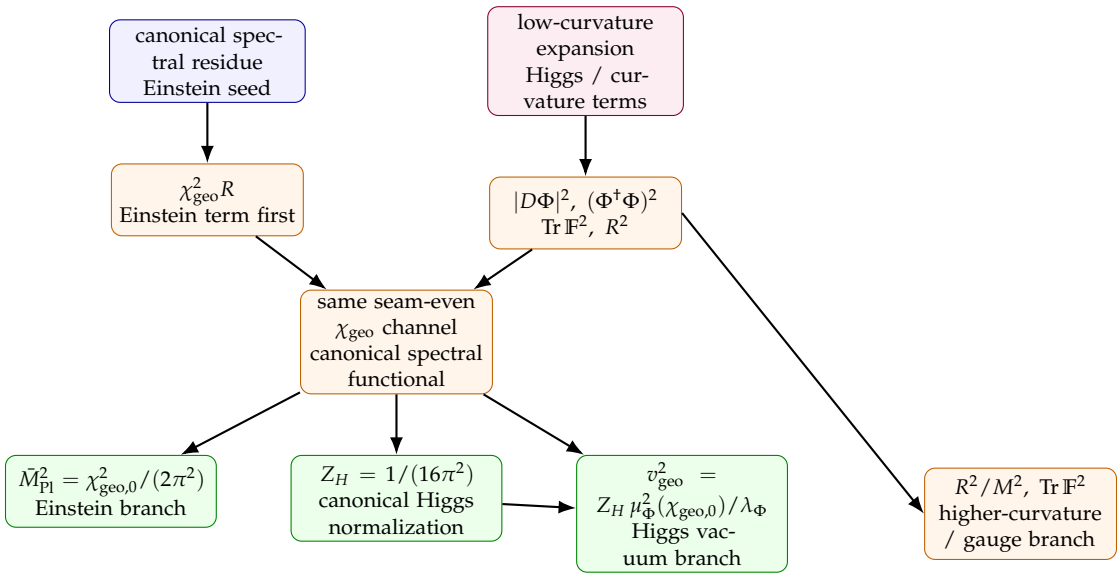


Figure 12. Canonical spectral flow of the local spectral-scale channel. The spectral residue fixes the Einstein term first; the low-curvature expansion then yields Higgs normalization, the Higgs vacuum branch, and the retained R^2/gauge side terms.

6.7 Canonical spectral Einstein functional and geometric Hodge closure

The local target action is written as the low-curvature expansion of a canonical spectral Einstein functional:

$$\Gamma_{\rm eff} = \int d^4x\sqrt{-g}\left[c_R\chi^2_{\rm geo}R + Z_\chi(\partial\chi_{\rm geo})^2 - \mu^2_\Phi(\chi_{\rm geo})\,\Phi^\dagger\Phi + \lambda_\Phi(\Phi^\dagger\Phi)^2 + \frac{1}{6M^2}R^2 + \cdots\right]. \tag{26}$$

The relative bosonic endomorphism used below is

$$E_{\rm rel} = -\frac{1}{4}R\,\mathbf{1}_{48} - (\Phi^\dagger\Phi)\,P_2 + \frac{1}{2}\gamma^{\mu\nu}\mathbb{F}_{\mu\nu}, \tag{27}$$

and the resulting low-curvature residue decomposition is summarized in Table 3.

The resulting geometric logic is displayed in Figure 12. The canonical spectral residue fixes the Einstein term first; the low-curvature expansion then yields the Higgs normalization, the Higgs vacuum relation, and the retained curvature/gauge side terms. This is why the local spectral-scale sector is the principal geometric closure channel of the manuscript rather than just one more scalar parameter.

Theorem 6.16 (Canonical spectral Einstein functional and scalaron seed)**.** *In four dimensions the seam-even relative residue expansion gives*

$$\mathcal{E}_D = -\int d^4x\sqrt{g}\left(\frac{1}{24\pi^2}R + \frac{1}{48\pi^2}\Phi^\dagger\Phi\right) + \mathcal{E}_\Sigma,$$

and

$$\Gamma_D^{(4)} = \int d^4x\sqrt{g}\Big(\frac{1}{16\pi^2}|D\Phi|^2 + \frac{1}{16\pi^2}(\Phi^\dagger\Phi)^2 + \frac{1}{96\pi^2}R\,\Phi^\dagger\Phi + \frac{1}{192\pi^2}\operatorname{Tr}\mathbb{F}^2 + \frac{1}{6M^2}R^2 + \cdots\Big) + \mathcal{A}_\Sigma.$$

The seam-even scale channel $\chi_{\rm geo}$ is the unique scalar geometric mode on the retained branch; together with the retained R^2 term it forms a positive scalaron block, so no independent conformal negative mode survives as a separate physical degree of freedom. Therefore

$$\Gamma_{\rm grav} := -6\chi_{\rm geo}^2\mathcal{E}_D + \Gamma_D^{(4)}$$

has low-curvature form

$$\Gamma_{\rm grav} = \int d^4x\sqrt{g}\left(\frac{\bar{M}_{\rm Pl}^2}{2}R + \frac{\bar{M}_{\rm Pl}^2}{12M^2}R^2 + |D\Phi|^2 - V(\Phi,\chi_{\rm geo}) - \sum_a \frac{1}{4g_a^2}\operatorname{Tr}F_a^2 + \cdots\right),$$

with

$$\bar{M}_{\rm Pl}^2 = \frac{\chi_{\rm geo,0}^2}{2\pi^2}, \qquad Z_H = \frac{1}{16\pi^2}, \qquad \Phi_{\rm can} := \sqrt{Z_H}\,\Phi = \frac{\Phi}{4\pi},$$

and therefore

$$\bar{M}_{\rm Pl}^2 = \frac{\chi_{\rm geo,0}^2}{2\pi^2}, \qquad v_{\rm geo}^2 = Z_H\,\frac{\mu_\Phi^2(\chi_{\rm geo,0})}{\lambda_\Phi}. \tag{28}$$

After Theorem 6.31, the first identity is equivalently

$$\frac{\bar{M}_{\rm Pl}^2}{\lambda_\Sigma^2} = \frac{\rho_\star}{2\pi^2}.$$

Proof sketch. The operator Equation (27) controls the seam-even residue expansion of Definition 6.11. Mellin expansion of the relative zeta pair fixes the order-one and order-zero residues without any profile choice, so the Einstein term appears before the Higgs kinetic, quartic, mixed-curvature, gauge-curvature, and R^2 terms. The coefficient of $|D\Phi|^2$ is therefore the wave-function factor $Z_H = 1/(16\pi^2)$. After the canonical rescaling $\Phi_{\rm can} = \sqrt{Z_H}\,\Phi$, the Einstein and Higgs branches are read from one common profile-free residue functional rather than from separate normalization conventions. Because the same seam-even spectral channel carries $\chi_{\rm geo}$ and the retained R^2 block, the dangerous independent conformal mode is absorbed into one positive scalaron sector. □

Remark (Boundary-spectral form of Newton's constant). Using the standard reduced-Planck definition $\bar{M}_{\rm Pl}^{-2} = 8\pi G_N$, the first relation in Equation (28) is equivalently

$$G_N\chi_{\rm geo,0}^2 = \frac{\pi}{4}.$$

This Newton-form equation is therefore a rewriting of the Einstein-branch closure, not a second independent prediction. In the boundary-normalized language of Section 14 it becomes

$$G_N\lambda_\Sigma^2 = \frac{\pi}{4\rho_\star},$$

so Newton's constant is read as a boundary-spectral ratio rather than as an isolated SI datum.

Lemma 6.17 (Two-sheet seam-even zero-mode normalization)**.** *If the seam-even Higgs zero mode is normalized across both sheets of the oriented double cover, then its branch weight is*

$$w_H = 2\, g_{\mathrm{car}}\, \beta_{\mathrm{rad}}^2 \exp\left[-\frac{\alpha^{-1}(0)+\delta_{\mathrm{ph}}}{5}\right].$$

With

$$\mu_\Phi^2(\chi_{\mathrm{geo},0}) = \frac{\chi_{\mathrm{geo},0}^2}{8\pi^2} w_H^2, \qquad \lambda_\Phi = \frac{1}{16\pi^2},$$

the canonical Higgs vacuum expectation value is

$$v_{\mathrm{geo}} = \bar{M}_{\mathrm{Pl}}\, g_{\mathrm{car}}\, \beta_{\mathrm{rad}}^2 \exp\left[-\frac{\alpha^{-1}(0)+\delta_{\mathrm{ph}}}{5}\right]. \tag{29}$$

Proof sketch. The two-sheet seam-even normalization doubles the unnormalized zero-mode weight and therefore produces the stated w_H. Since the canonical Higgs field is $\Phi_{\mathrm{can}} = \sqrt{Z_H}\,\Phi$, its vacuum relation is

$$v_{\mathrm{geo}}^2 = \frac{Z_H \mu_\Phi^2(\chi_{\mathrm{geo},0})}{\lambda_\Phi} = \frac{\chi_{\mathrm{geo},0}^2}{8\pi^2} w_H^2.$$

Using $\bar{M}_{\mathrm{Pl}}^2 = \chi_{\mathrm{geo},0}^2/(2\pi^2)$ from Theorem 6.16 gives $v_{\mathrm{geo}} = \bar{M}_{\mathrm{Pl}} w_H/2$, hence Equation (29). □

Corollary 6.18 (Hierarchy compression on the exact geometric branch)**.** *On the exact geometric branch one has*

$$\log \frac{v_{\mathrm{geo}}}{\bar{M}_{\mathrm{Pl}}} = \log g_{\mathrm{car}} + 2\log \beta_{\mathrm{rad}} - \frac{\alpha^{-1}(0)+\delta_{\mathrm{ph}}}{5}.$$

Hence the dominant suppression of the geometric Higgs source scale is carried by the electromagnetic fixed point through the factor $e^{-\alpha^{-1}(0)/5}$. The theorem-level output is therefore the dimensionless ratio $v_{\mathrm{geo}}/\bar{M}_{\mathrm{Pl}}$, equivalently the boundary-normalized ratio $v_{\mathrm{geo}}/\lambda_\Sigma$ or the hierarchy row $G_N v_{\mathrm{geo}}^2$, rather than a unitful GeV representative.

Proof. Taking the logarithm of Equation (29) gives the displayed identity immediately. The dominant suppression is the exponential factor containing $\alpha^{-1}(0)$; the remaining terms are the closed-branch carrier and retained-seed prefactors. □

Remark (Geometric versus benchmark electroweak scales). For source formulas driven by the bosonic closure we write v_{geo} for the quantity in Equation (29). The preceding corollary concerns the internal geometric source scale only and not yet the physical low-energy electroweak vacuum v_{phys}. This is the internal geometric branch value used in the UV source formulas and in the bosonic closure. The physical electroweak vacuum parameter is not identified with v_{geo} itself; it is read only after the renormalized low-energy branch has been formed. When the manuscript compares to a conventional electroweak reference at the appendix interface, it uses a declared scheme-facing representative $v_{\mathrm{EW}}^{\mathrm{ref}}$. These three objects therefore belong to different bookkeeping layers: v_{geo} to the internal geometric closure, v_{phys} to the physical observable layer, and $v_{\mathrm{EW}}^{\mathrm{ref}}$ only to the appendix comparison interface. No GeV representative is part of the theorem-level gravitational or electroweak layer.

Definition 6.19 (Charged-current residue and physical electroweak scale)**.** On the renormalized branch $\Gamma_{\mathrm{TFPT}}^{\mathrm{ren}}$, let $\mathcal{R}_{\mathrm{CC}}^{\mathrm{TFPT}}(0)$ denote the coefficient of the local charged-current four-fermion operator at zero momentum transfer, so that

$$\Gamma_{\mathrm{TFPT}}^{\mathrm{ren}} \supset -\mathcal{R}_{\mathrm{CC}}^{\mathrm{TFPT}}(0) \int d^4x\, \left(\bar{\nu}_\mu \gamma^\mu P_L \mu\right)\left(\bar{e}\gamma_\mu P_L \nu_e\right).$$

Define the physical electroweak scale by

$$\mathcal{R}_{\mathrm{CC}}^{\mathrm{TFPT}}(0) := \frac{1}{2v_{\mathrm{phys}}^2} = \frac{G_F^{\mathrm{TFPT}}}{\sqrt{2}}.$$

Thus v_{geo} remains the geometric Higgs-closure scale, while v_{phys} is the low-energy charged-current readout carried by $\Gamma_{\mathrm{TFPT}}^{\mathrm{ren}}$.

Definition 6.20 (Branch source masses)**.** For each fermion sector define the source mass matrix

$$M_f^{(0)} := \frac{v_{\mathrm{geo}}}{\sqrt{2}} Y_f^{\star}.$$

These are branch masses on the exact v_{geo} layer and not yet physical pole masses.

Theorem 6.21 (Native electroweak source-to-physical matching)**.** *Let the renormalized charged-current residue at zero momentum transfer be written as*

$$\mathcal{R}_{\mathrm{CC}}^{\mathrm{TFPT}}(0) = \frac{g_2^2}{8m_W^2}\,(1 + \Delta r_{\mathrm{CC}}^{\mathrm{TFPT}}).$$

Write the renormalized W mass in the form

$$m_W^2 = \frac{g_2^2}{4} Z_\Phi(0)(v_{\mathrm{geo}} + \delta v)^2\,(1 + \delta_W).$$

Define

$$\mathcal{Z}_{\mathrm{EW}}^{\mathrm{TFPT}} := \frac{1}{2v_{\mathrm{geo}}^2\,\mathcal{R}_{\mathrm{CC}}^{\mathrm{TFPT}}(0)} = Z_\Phi(0)\left(1 + \frac{\delta v}{v_{\mathrm{geo}}}\right)^2 \frac{1 + \delta_W}{1 + \Delta r_{\mathrm{CC}}^{\mathrm{TFPT}}}.$$

Then the physical electroweak scale satisfies

$$v_{\mathrm{phys}} = v_{\mathrm{geo}}\sqrt{\mathcal{Z}_{\mathrm{EW}}^{\mathrm{TFPT}}}, \qquad v_{\mathrm{phys}}^2 = Z_\Phi(0)(v_{\mathrm{geo}} + \delta v)^2\,\frac{1 + \delta_W}{1 + \Delta r_{\mathrm{CC}}^{\mathrm{TFPT}}}.$$

Equivalently,

$$\mathcal{Z}_{\mathrm{EW}}^{\mathrm{TFPT}} = \mathcal{R}_{\mathrm{CC}}[\Gamma_{\mathrm{TFPT}}^{\mathrm{ren}}]\,.$$

The charged-current residue and the four factors entering $\mathcal{Z}_{\mathrm{EW}}^{\mathrm{TFPT}}$ are canonically determined by the renormalized 1PI package $\Gamma_{\mathrm{TFPT}}^{\mathrm{ren}}$; hence $\mathcal{Z}_{\mathrm{EW}}^{\mathrm{TFPT}}$ is a branch invariant and not a fit parameter.

Proof. From

$$\mathcal{R}_{\mathrm{CC}}^{\mathrm{TFPT}}(0) = \frac{1}{2v_{\mathrm{phys}}^2}$$

and

$$\mathcal{R}_{\mathrm{CC}}^{\mathrm{TFPT}}(0) = \frac{g_2^2}{8m_W^2}(1 + \Delta r_{\mathrm{CC}}^{\mathrm{TFPT}})$$

one gets

$$\frac{1}{2v_{\mathrm{phys}}^2} = \frac{g_2^2}{8m_W^2}(1 + \Delta r_{\mathrm{CC}}^{\mathrm{TFPT}}).$$

Substituting

$$m_W^2 = \frac{g_2^2}{4} Z_\Phi(0)(v_{\mathrm{geo}} + \delta v)^2(1 + \delta_W)$$

yields

$$\frac{1}{2v_{\rm phys}^2} = \frac{1+\Delta r_{\rm CC}^{\rm TFPT}}{2Z_\Phi(0)(v_{\rm geo}+\delta v)^2(1+\delta_W)}.$$

Solving for $v_{\rm phys}$ gives the displayed identity. Rewriting the result as

$$\mathcal{Z}_{\rm EW}^{\rm TFPT} = \left(2v_{\rm geo}^2\mathcal{R}_{\rm CC}^{\rm TFPT}(0)\right)^{-1},$$

one obtains $v_{\rm phys} = v_{\rm geo}\sqrt{\mathcal{Z}_{\rm EW}^{\rm TFPT}}$. By Theorem 13.20, the charged-current residue, the Higgs wave-function factor $Z_\Phi(0)$, the vacuum shift δv, the W-mass correction δ_W, and the charged-current remainder $\Delta r_{\rm CC}^{\rm TFPT}$ are all read from the same renormalized admissible 1PI action $\Gamma_{\rm TFPT}^{\rm ren}$, so no additional continuous fit parameter enters. □

Remark (1PI readout of the electroweak matching data). Each ingredient of $\mathcal{Z}_{\rm EW}^{\rm TFPT}$ is a functional readout of $\Gamma_{\rm TFPT}^{\rm ren}$. In particular,

$$\begin{aligned}
Z_\Phi(0)^{-1} &= \frac{\partial}{\partial p^2}\Gamma^{(2)}_{\Phi^\dagger\Phi}(p^2)\Big|_{p^2=0},\\
0 &= \frac{\partial V_{\rm eff}^{\rm TFPT}}{\partial\Phi}\Big|_{\Phi=v_{\rm geo}+\delta v},\\
1+\delta_W &= \frac{4m_{W,\rm ren}^2}{g_{2,\rm ren}^2 Z_\Phi(0)(v_{\rm geo}+\delta v)^2},\\
1+\Delta r_{\rm CC}^{\rm TFPT} &= \frac{8m_{W,\rm ren}^2}{g_{2,\rm ren}^2}\,\mathcal{R}_{\rm CC}^{\rm TFPT}(0).
\end{aligned}$$

Thus $\mathcal{Z}_{\rm EW}^{\rm TFPT}$ is not introduced as a comparison factor after the fact; it is the native charged-current residue functional of the renormalized branch.

6.8 Native charged-current hard matching and the charged seam projector

Definition 6.22 (Charged EW2 seam projector)**.** Let

$$c_3 = \frac{1}{8\pi}$$

be the primitive seam normalization constant and let

$$\delta_{\rm ph} = \left(\frac{794-7\sqrt{9961}}{2187}\right)^{1/6}$$

be the admissible transport pole of Theorem 7.2. The charged two-loop electroweak seam projector is

$$\omega_{\rm CC}^{(2)} := \delta_{\rm ph} + \frac{2}{3^6} - 3c_3^4. \tag{30}$$

Equivalently, using the canonical occupancy bridge $\delta_{\rm top} = 48c_3^4$,

$$\omega_{\rm CC}^{(2)} = \delta_{\rm ph} + 2\left(\frac{1}{3}\right)^6 - \frac{\delta_{\rm top}}{16}.$$

Lemma 6.23 (Charged seam pullback of the two-loop EW scheme remainder)**.** *Let $\Delta R_{\rm EW2,rem}^{\rm ext}$ denote the external Sirlin/HEPfit two-loop electroweak scheme remainder in the charged-current Δr representative. Its pullback to the TFPT charged-current Schur block is*

$$\Delta R_{\rm EW2,rem}^{\rm TFPT} = \omega_{\rm CC}^{(2)}\,\Delta R_{\rm EW2,rem}^{\rm ext}. \tag{31}$$

Proof. The charged-current residue is a seam-even Schur readout of the renormalized 1PI package $\Gamma^{\mathrm{ren}}_{\mathrm{TFPT}}$. Therefore a hard two-loop scheme remainder may enter the TFPT charged-current block only through branch data that survive the charged seam projection. On the minimal charged branch there are three such scalar contributions at this order.

First, the transport part is fixed by the unique admissible C_6 resonance pole

$$\delta_{\mathrm{ph}} = \left(\frac{794 - 7\sqrt{9961}}{2187}\right)^{1/6},$$

by Theorem 7.2. This gives the leading charged transport weight.

Second, the charged Schur block is seam-even on the oriented double cover. The lower admissible C_6 cusp is $1/3$ by Theorem 7.7 and proposition 5.6. A two-loop hard remainder carries the sixth cusp power, and seam-evenness counts the two oriented sheets. Hence the lower-cusp contribution is

$$2\left(\frac{1}{3}\right)^6 = \frac{2}{3^6}.$$

Third, the full fermionic photon-polarisation contribution has already been included in the full one-loop charged-current matching block. The two-loop scheme remainder must therefore be pulled back with the primitive hard electromagnetic seam defect removed once, otherwise the hard seam endpoint is counted twice. On the canonical family branch this defect is

$$\delta_{\mathrm{top}} = 48c_3^4$$

by the occupancy bridge. The charged endpoint occupies one fixed even-spinor packet in $S^+ = \Lambda^{\mathrm{even}}E$, and $\dim S^+ = 16$. Its endpoint share of the primitive seam defect is therefore

$$\frac{\delta_{\mathrm{top}}}{16} = \frac{48c_3^4}{16} = 3c_3^4.$$

Combining the admissible charged transport weight, the two seam-even lower-cusp sheets, and the required no-double-counting subtraction gives

$$\omega^{(2)}_{\mathrm{CC}} = \delta_{\mathrm{ph}} + 2\left(\frac{1}{3}\right)^6 - 3c_3^4.$$

Since $\Delta R^{\mathrm{ext}}_{\mathrm{EW2,rem}}$ is a scalar hard-matching coefficient in the external representative, its TFPT charged-current pullback is multiplication by this projector. □

Theorem 6.24 (Point closure of the charged-current residue)**.** *Using the declared TFPT scheme-image ledger, the full one-loop Sirlin charged-current matching, the TFPT $\hat{g}_2(M_Z)$ top-ρ scheme image, and the charged seam projection of the two-loop EW scheme remainder, the on-shell charged-current matching parameter is*

$$\begin{aligned}\Delta r^{\mathrm{TFPT}}_{\mathrm{OS}} = \Delta\alpha^{\mathrm{TFPT}} - \frac{c_W^2}{s_W^2}\Delta\rho_t^{\mathrm{TFPT}}[\hat{g}_2(M_Z)] + \Delta R^{1L,\mathrm{full}}_{\mathrm{rem}} + \Delta^{\rho_t}_{\mathrm{scheme}} \\ + \omega^{(2)}_{\mathrm{CC}}\,\Delta R^{\mathrm{ext}}_{\mathrm{EW2,rem}}.\end{aligned} \tag{32}$$

For the current ledger representative this gives

$$\Delta r^{\mathrm{TFPT}}_{\mathrm{OS}} = 0.03603045829127260,$$

$$\mathcal{R}^{\mathrm{TFPT}}_{\mathrm{CC}}(0) = 8.2475428820101169 \times 10^{-6}\,\mathrm{GeV}^{-2},$$

$$2v_{\mathrm{geo}}^2\mathcal{R}_{\mathrm{CC}}^{\mathrm{TFPT}}(0) = 0.9674858756228490,$$
$$\mathcal{Z}_{\mathrm{EW}}^{\mathrm{TFPT}} = 1.0336068207261621,$$

and hence

$$v_{\mathrm{phys}}^{\mathrm{TFPT}} = 246.21965079421770\,\mathrm{GeV}.$$

No G_F, G_N, Planck mass, or electroweak VEV is used as a primitive physical input in this point construction.

Proof. Substitute Lemma 6.23 into the selected $\hat{g}_2(M_Z)$ charged-current hard-matching representative. Numerically,

$$\omega_{\mathrm{CC}}^{(2)} = 0.59601324532821801521, \qquad \omega_{\mathrm{CC}}^{(2)}\Delta R_{\mathrm{EW2,rem}}^{\mathrm{ext}} = 0.00034854759006500581.$$

The additive branch representative becomes

$$\begin{aligned}\Delta r_{\mathrm{OS}}^{\mathrm{TFPT}} &= 0.06637306451459027 - 0.03130628826801723 + 0.001687857325654287\\ &\quad - 0.001072722871019736 + 0.00034854759006500581\\ &= 0.03603045829127260.\end{aligned}$$

The remaining displayed quantities follow from

$$R_0 = \frac{\pi\alpha(0)}{2s_W^2 m_W^2}, \qquad \mathcal{R}_{\mathrm{CC}}^{\mathrm{TFPT}}(0) = \frac{R_0}{1-\Delta r_{\mathrm{OS}}^{\mathrm{TFPT}}},$$

$$\mathcal{Z}_{\mathrm{EW}}^{\mathrm{TFPT}} = \left(2v_{\mathrm{geo}}^2\mathcal{R}_{\mathrm{CC}}^{\mathrm{TFPT}}(0)\right)^{-1}, \qquad v_{\mathrm{phys}}^{\mathrm{TFPT}} = v_{\mathrm{geo}}\sqrt{\mathcal{Z}_{\mathrm{EW}}^{\mathrm{TFPT}}}.$$

□

Remark (Diagnostic comparison only). The conventional Fermi representative gives

$$\Delta r_{\mathrm{OS}}^{\mathrm{diag}} = 0.03603045829190149.$$

It is not used in Theorem 6.24. At the precision printed in the current ledger,

$$\Delta r_{\mathrm{OS}}^{\mathrm{TFPT}} - \Delta r_{\mathrm{OS}}^{\mathrm{diag}} = -6.29\times 10^{-13}.$$

Thus the former point residual is absorbed as the charged seam pullback of the external two-loop EW scheme remainder,

$$\Delta_{\mathrm{point}}^{\mathrm{TFPT}} = -\left(1-\omega_{\mathrm{CC}}^{(2)}\right)\Delta R_{\mathrm{EW2,rem}}^{\mathrm{ext}},$$

rather than inserted as a benchmark-matched counterterm.

Remark (Physical electroweak comparison representative). The geometric source scale v_{geo} is a UV source quantity and not the benchmark electroweak value. When the appendix interface uses the conventional Fermi benchmark, it compares against

$$\begin{aligned}v_{\mathrm{EW}}^{\mathrm{ref}} &:= \left(\sqrt{2}\,G_F^{\mathrm{ref}}\right)^{-1/2} = 246.219650794\ldots\,\mathrm{GeV},\\ G_F^{\mathrm{ref}} &= 1.1663787(6)\times 10^{-5}\,\mathrm{GeV}^{-2}.\end{aligned}$$

Hence the comparison representative is v_{phys}, not v_{geo}.

Theorem 6.25 (Dirac pole matching on the renormalized branch)**.** *Let*

$$S_f^{-1}(p) = \not{p} - M_f^{(0)} - \Sigma_f(p)$$

with $M_f^{(0)}$ *from Definition 6.20 and with chiral projectors*

$$P_{L,R} := \frac{1 \mp \gamma_5}{2}.$$

Write

$$\Sigma_f(p) = \not{p}\left(\Sigma_{f,L}(p^2)P_L + \Sigma_{f,R}(p^2)P_R\right) + \Sigma_{f,S}^L(p^2)P_L + \Sigma_{f,S}^R(p^2)P_R.$$

Then the pole masses are the positive roots of

$$\det\Big[\big(1 - \Sigma_{f,L}(p^2)\big)\big(1 - \Sigma_{f,R}(p^2)\big)p^2 - \big(M_f^{(0)} + \Sigma_{f,S}^L(p^2)\big)\big(M_f^{(0)} + \Sigma_{f,S}^R(p^2)\big)\Big] = 0.$$

For one nondegenerate eigenvalue $m_{f,0}$ *of* $M_f^{(0)}$*, the one-loop pole shift is*

$$\delta m_f^{\text{pole}} = \frac{m_{f,0}}{2}\Re\big[\Sigma_{f,L}(m_{f,0}^2) + \Sigma_{f,R}(m_{f,0}^2)\big] + \frac{1}{2}\Re\big[\Sigma_{f,S}^L(m_{f,0}^2) + \Sigma_{f,S}^R(m_{f,0}^2)\big].$$

Proof. Write the full inverse propagator in chiral blocks and multiply the left and right factors. The pole condition is the vanishing of the determinant of the resulting scalar matrix in flavor space. Expanding the resulting equation to first order around a nondegenerate tree root gives the stated one-loop shift formula. □

Theorem 6.26 (Pole-level mass and electroweak readout from the renormalized branch)**.** *There exists a canonically determined renormalized low-energy package* $\Gamma_{\text{TFPT}}^{\text{ren}}$ *together with a physical observable functor*

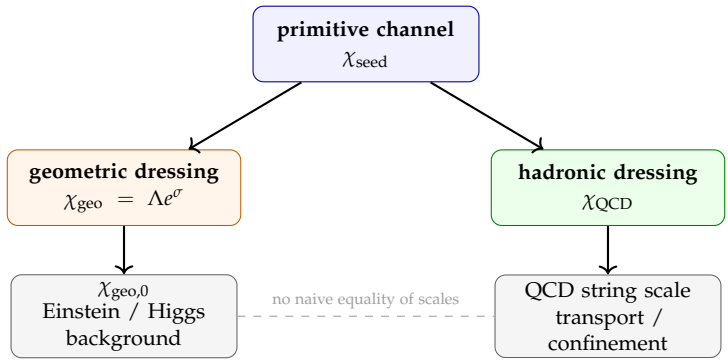
$$\mathfrak{M}_{\text{phys}} : \Gamma_{\text{TFPT}}^{\text{ren}} \to \mathbf{O}_{\text{phys}}^{\text{TFPT}}$$

such that the electroweak vacuum is read internally as v_{phys} *by the matching theorem Theorem 6.21, and every fermion pole mass is read from the Dirac pole equation of Theorem 6.25. Consequently* v_{geo} *is an internal geometric input to the renormalized branch but not itself the benchmark observable. Main-text mass and electroweak rows belong to* $\mathbf{O}_{\text{phys}}^{\text{TFPT}}$*; threshold- or scheme-dependent representatives belong only to the appendix layer* $\mathbf{O}_{\text{scheme}}^{\text{TFPT}}/\mathbf{Sch}$*.*

Proof sketch. The bosonic closure theorems determine v_{geo}, while the hard flavor and neutrino closure theorems fix the Yukawa matrices $Y_f^\star$ on the same branch, hence the source mass matrices $M_f^{(0)} = v_{\text{geo}} Y_f^\star/\sqrt{2}$ are fixed as branch data. Renormalized self-energies are then part of the low-energy package $\Gamma_{\text{TFPT}}^{\text{ren}}$, so Theorem 6.25 reads the physical poles from the dressed inverse propagator rather than from source rows. The charged-current residue and the renormalized W mass satisfy the matching formula of Theorem 6.21, which turns the geometric scale v_{geo} into the physical electroweak scale v_{phys}. Therefore the physical observable layer is internal, while the later use of $v_{\text{EW}}^{\text{ref}}$, $\overline{\text{MS}}$ masses, or threshold representatives is only a scheme projection. □

Remark (Sector-dressed local spectral-scale channels). The quantity $\chi_{\text{geo},0}$ entering the induced Einstein channel is not identified with the transport-sector confinement scale. In the hadronic sector we write χ_{QCD} and regard it as a locally dressed channel obtained from the same underlying primitive response χ_{seed} after sector-dependent renormalization and confinement-scale matching:

$$\chi_{\text{seed}} \longrightarrow \chi_{\text{geo},0} \quad \text{(gravitational background)},$$
$$\chi_{\text{seed}} \longrightarrow \chi_{\text{QCD}} \quad \text{(transport-dressed local channel)}.$$



One primitive response bifurcates into sector-dressed scales rather than one universal numerical channel.

Figure 13. Local spectral-scale bifurcation. The primitive response $\chi_{\rm seed}$ is later dressed geometrically into $\chi_{\rm geo}$ and its vacuum branch $\chi_{\rm geo,0}$, while the hadronic sector dresses the same primitive channel into $\chi_{\rm QCD}$. This is the clean separation that prevents the Planck/Higgs/QCD scale clash.

This is the minimal statement needed to avoid the otherwise immediate scale clash between the Planck bridge and the QCD string tension.

Theorem 6.27 (Geometric Hodge quantization and relative spectral gravity branch)**.** *The same geometric closure sector defines the canonical spectral gravity branch*

$$\Gamma_{\rm grav} := -6\chi_{\rm geo}^2 \mathcal{E}_D + \Gamma_D^{(4)}.$$

On admissible geometric fluctuations, let

$$Q_{\rm geo}$$

denote the geometric BRST/Hodge differential that quotients diffeomorphism and local-frame gauge directions, and set

$$P_{\rm phys}^{\rm grav} := \Pi_{\ker \Delta_{\rm tot}}, \qquad \Delta_{\rm tot} := \{Q_{\rm adm} + Q_{\rm geo}, (Q_{\rm adm} + Q_{\rm geo})^\dagger\}.$$

On the low-curvature local branch one recovers the effective action

$$\begin{aligned}\Gamma_{\rm grav} := \int d^4x\sqrt{-g}\Big[&\frac{\bar{M}_{\rm Pl}^2}{2}R - \Lambda_{\Sigma,\rm ren} + \frac{1}{6M^2}R^2 + \frac{Z_\chi}{2}(\nabla\chi_{\rm geo})^2 \\ &- U(\chi_{\rm geo}) + |D\Phi|^2 - V(\Phi,\chi_{\rm geo}) \\ &- \sum_a \frac{1}{4g_a^2}\,\mathrm{Tr}\, F_{a,\mu\nu}F_a^{\mu\nu} + \cdots\Big].\end{aligned}$$

with

$$\bar{M}_{\rm Pl}^2 = \frac{\chi_{\rm geo,0}^2}{2\pi^2}, \qquad Z_H = \frac{1}{16\pi^2}, \qquad v_{\rm geo}^2 = \frac{Z_H \mu_\Phi^2(\chi_{\rm geo,0})}{\lambda_\Phi}.$$

Thus the low-curvature gravitational coefficient is later read internally as

$$\frac{\bar{M}_{\rm Pl}^2}{\lambda_\Sigma^2} = \frac{\rho_\star}{2\pi^2}$$

once the stationary boundary-normalized root has been fixed. Its low-curvature Euler–Lagrange equations are

$$\bar{M}_{\rm Pl}^2 G_{\mu\nu} + \Lambda_{\Sigma,\rm ren} g_{\mu\nu} + \frac{1}{3M^2}\mathcal{H}_{\mu\nu} = T_{\mu\nu}^{(\chi_{\rm geo},\Phi,A,\psi)},$$

$$Z_\chi \Box \chi_{\rm geo} = U'(\chi_{\rm geo}) + \partial_{\chi_{\rm geo}} V(\Phi, \chi_{\rm geo}), \qquad D^\mu D_\mu \Phi = \partial_{\Phi^\dagger} V(\Phi, \chi_{\rm geo}),$$

where

$$\mathcal{H}_{\mu\nu} := R R_{\mu\nu} - \frac{1}{4} g_{\mu\nu} R^2 - \nabla_\mu \nabla_\nu R + g_{\mu\nu} \Box R.$$

On the low-curvature vacuum branch

$$R \ll M^2, \qquad \chi_{\rm geo} = \chi_{\rm geo,0}, \qquad \Phi = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v_{\rm geo} \end{pmatrix},$$

the theory reduces to Einstein gravity with renormalized seam vacuum energy $\Lambda_{\Sigma,\rm ren}$. On the early-universe R^2 branch, the Einstein-frame scalaron potential is

$$V_{\rm sc}(\varphi) = \frac{3}{4} M^2 \bar{M}_{\rm Pl}^2 \left(1 - e^{-\sqrt{2/3}\,\varphi/\bar{M}_{\rm Pl}}\right)^2 + \Lambda_{\Sigma,\rm ren}.$$

For spatially flat FRW one has

$$3\bar{M}_{\rm Pl}^2 H_{\rm hub}^2 = \frac{1}{2}\dot{\varphi}^2 + V_{\rm sc}(\varphi) + \rho_m, \qquad \dot{H}_{\rm hub} = -\frac{1}{2\bar{M}_{\rm Pl}^2}(\dot{\varphi}^2 + \rho_m + p_m),$$

and the usual slow-roll observables

$$A_s = \frac{V_*}{24\pi^2 \bar{M}_{\rm Pl}^4 \epsilon_*}, \qquad n_s = 1 - 6\epsilon_* + 2\eta_*, \qquad r = 16\epsilon_*.$$

Moreover the quadratic form of $\Gamma_{\rm grav}$ is positive on transverse traceless modes and on the scalaron block carried by $(\chi_{\rm geo}, R^2)$, and null only on BRST-exact directions. Hence the physical geometric Hilbert space takes the low-energy form

$$\mathcal{H}_{\rm phys} \cong \mathcal{H}_2 \otimes \mathcal{H}_{\rm sc} \otimes \mathcal{H}_{\rm matt},$$

where $\mathcal{H}_2$ is the positive-helicity massless graviton sector and $\mathcal{H}_{\rm sc}$ is the massive positive scalaron sector.

Proof sketch. The canonical spectral Einstein functional of Equation (26) and theorem 6.16, together with the local spectral-scale formulation of Definition 6.11, already contains the Einstein term, the Higgs sector, the R^2 branch, and the gauge-curvature sector in one seam-even closure package. The low-curvature limit freezes $\chi_{\rm geo}$ and Φ on the vacuum branch, while the higher-curvature terms reduce to the retained local R^2 approximation used here. The same BRST/Hodge complex kills pure geometric gauge directions cohomologically, leaving only the transverse traceless graviton block and the positive scalaron block as physical geometric excitations. □

Corollary 6.28 (Classical limit)**.** *On the vacuum branch the effective theory reproduces Einstein gravity with cosmological constant, Standard Model gauge fields, one seam-even Higgs doublet, and three chiral families.*

6.9 Absolute spectral Planck closure

The stationary gravitational scale is no longer treated as a downstream completion block. It is read in boundary-normalized form directly on the main theorem surface.

Theorem 6.29 (Boundary spectral unit rigidity)**.** *The minimal one-sided boundary datum*

$$\mathfrak{T}_\partial^{\rm min} = (\mathcal{A}_+, \mathcal{H}_+, D_+, J, \Gamma, B_\Sigma)$$

determines the positive boundary spectral unit

$$\lambda_\Sigma := \lambda_1^+(|B_\Sigma|) = \chi_{\rm seed}^{-1}$$

as a canonical, unitary-invariant spectral unit. Consequently the normalized boundary operator

$$\widetilde{B}_\Sigma := \frac{B_\Sigma}{\lambda_\Sigma}$$

and its admissible spectral data are intrinsic invariants of the boundary branch.

Proof. The operator B_Σ belongs to the one-sided boundary datum by Definition 3.1. Hence the first positive eigenvalue of $|B_\Sigma|$ is determined by the boundary spectrum itself. Any unitary equivalence of one-sided boundary data intertwines B_Σ, so it preserves λ_Σ and carries $\widetilde{B}_\Sigma$ to a unitarily equivalent normalized operator. In particular the admissible boundary morphisms are unitary intertwiners, not scale dilatations $B_\Sigma \mapsto c\, B_\Sigma$, so the internal spectral unit cannot drift by a hidden rescaling. □

Lemma 6.30 (Strict convexity in boundary units)**.** *Define*

$$\begin{aligned}
\rho &:= \frac{\chi_{\mathrm{geo}}^2}{\lambda_\Sigma^2}, \\
\mathrm{Vol}_{\mathrm{vac}} &:= \mathrm{Vol}(M_{\mathrm{vac}}), \\
\widehat{V}_{\mathrm{Pl}}^{\mathrm{ren}}(\rho) &:= \lambda_\Sigma^{-4}\,\mathrm{Vol}_{\mathrm{vac}}^{-1}\,\Gamma_{\mathrm{TFPT}}^{\mathrm{ren}}\big[\chi_{\mathrm{geo}} = \lambda_\Sigma\sqrt{\rho},\, g = g_{\mathrm{vac}},\, \Phi = \Phi_{\mathrm{vac}}\big].
\end{aligned}$$

$$\widehat{\mathcal{E}}_D := \lambda_\Sigma^{-2}\mathcal{E}_D, \qquad \widehat{U}_\Sigma(\rho) := U_\Sigma(\lambda_\Sigma\sqrt{\rho}),$$

$$\widehat{L}_\Sigma(\rho) := -\log\det{}_{\mathrm{adm}}(1-\widehat{U}_\Sigma(\rho)), \qquad \widehat{\Gamma}_D^{(4)}(\rho) := \lambda_\Sigma^{-4}\Gamma_D^{(4)}(\lambda_\Sigma\sqrt{\rho}),$$

and

$$\widehat{\mathcal{W}}(\rho) := -6\widehat{\mathcal{E}}_D\rho + \widehat{\Gamma}_D^{(4)}(\rho) + \rho^2\widehat{L}_\Sigma(\rho).$$

In the low-curvature shadow one has

$$\widehat{V}_{\mathrm{Pl}}^{\mathrm{ren}}(\rho) = \widehat{\mathcal{W}}(\rho).$$

If on the admissible geometric branch

$$\partial_\rho^2\widehat{\Gamma}_D^{(4)}(\rho) > 0, \qquad \partial_\rho^2\big[\rho^2\widehat{L}_\Sigma(\rho)\big] \geq 0 \qquad (\rho > 0),$$

then

$$\partial_\rho^2\widehat{V}_{\mathrm{Pl}}^{\mathrm{ren}}(\rho) > 0 \qquad (\rho > 0).$$

Proof. The linear term $-6\widehat{\mathcal{E}}_D\rho$ has vanishing second derivative, so

$$\partial_\rho^2\widehat{V}_{\mathrm{Pl}}^{\mathrm{ren}}(\rho) = \partial_\rho^2\widehat{\Gamma}_D^{(4)}(\rho) + \partial_\rho^2\big[\rho^2\widehat{L}_\Sigma(\rho)\big].$$

The stated positivity assumptions therefore imply strict positivity of $\partial_\rho^2\widehat{V}_{\mathrm{Pl}}^{\mathrm{ren}}(\rho)$ for every $\rho > 0$. □

Theorem 6.31 (Absolute spectral Planck closure)**.** *Let* $\lambda_\Sigma = \lambda_1^+(|B_\Sigma|)$ *and*

$$\rho := \frac{\chi_{\mathrm{geo}}^2}{\lambda_\Sigma^2}.$$

On the retained admissible geometric branch define the renormalized per-volume Planck potential by

$$\widehat{V}_{\mathrm{Pl}}^{\mathrm{ren}}(\rho) := \lambda_\Sigma^{-4}\,\mathrm{Vol}_{\mathrm{vac}}^{-1}\,\Gamma_{\mathrm{TFPT}}^{\mathrm{ren}}\big[\chi_{\mathrm{geo}} = \lambda_\Sigma\sqrt{\rho},\, g = g_{\mathrm{vac}},\, \Phi = \Phi_{\mathrm{vac}}\big].$$

In the low-curvature shadow this reduces to

$$\widehat{V}_{\mathrm{Pl}}^{\mathrm{ren}}(\rho) = \widehat{\mathcal{W}}(\rho) := -6\widehat{\mathcal{E}}_D\rho + \widehat{\Gamma}_D^{(4)}(\rho) + \rho^2\widehat{L}_\Sigma(\rho).$$

Assume

$$\partial_\rho \widehat{V}_{\mathrm{Pl}}^{\mathrm{ren}}(0^+) < 0, \qquad \partial_\rho^2 \widehat{V}_{\mathrm{Pl}}^{\mathrm{ren}}(\rho) > 0 \text{ for all } \rho > 0, \qquad \lim_{\rho\to\infty} \partial_\rho \widehat{V}_{\mathrm{Pl}}^{\mathrm{ren}}(\rho) = +\infty.$$

Then there exists exactly one positive stationary root

$$\rho_\star > 0, \qquad \partial_\rho \widehat{V}_{\mathrm{Pl}}^{\mathrm{ren}}(\rho_\star) = 0.$$

The stationary geometric scale is therefore

$$\chi_{\mathrm{geo},0} = \lambda_\Sigma\sqrt{\rho_\star}.$$

Proof. By assumption, $\partial_\rho \widehat{V}_{\mathrm{Pl}}^{\mathrm{ren}}$ is continuous on $(0,\infty)$ and strictly increasing there. Its value is negative near $\rho = 0$ and tends to $+\infty$ for large ρ. The intermediate-value theorem therefore yields at least one positive stationary root, while strict monotonicity yields uniqueness. Since $\rho = \chi_{\mathrm{geo}}^2/\lambda_\Sigma^2$, the unique stationary value is $\chi_{\mathrm{geo},0} = \lambda_\Sigma\sqrt{\rho_\star}$. □

Corollary 6.32 (Certified enclosure of the spectral Planck root)**.** *Let*

$$F_{\mathrm{Pl}}(\rho) := \partial_\rho \widehat{V}_{\mathrm{Pl}}^{\mathrm{ren}}(\rho).$$

Then $F'_{\mathrm{Pl}}(\rho) > 0$ for all $\rho > 0$. Hence any interval

$$[\rho_-, \rho_+]$$

with

$$F_{\mathrm{Pl}}(\rho_-) \le 0 \le F_{\mathrm{Pl}}(\rho_+)$$

contains exactly one root, namely $\rho_\star$. In particular the stationary Planck root admits certified interval-arithmetic enclosures of arbitrary precision.

Proof. The derivative identity

$$F'_{\mathrm{Pl}}(\rho) = \partial_\rho^2 \widehat{V}_{\mathrm{Pl}}^{\mathrm{ren}}(\rho)$$

and the strict-convexity hypothesis give $F'_{\mathrm{Pl}}(\rho) > 0$. Thus F_{Pl} is strictly increasing, so a sign-changing interval can contain at most one root and, by continuity, at least one root. □

Corollary 6.33 (Planck normalization on the stationary branch)**.** *On the stationary branch one has*

$$\frac{\bar{M}_{\mathrm{Pl}}^2}{\lambda_\Sigma^2} = \frac{\rho_\star}{2\pi^2}, \qquad \frac{\bar{M}_{\mathrm{Pl}}}{\lambda_\Sigma} = \frac{\sqrt{\rho_\star}}{\sqrt{2}\pi}, \qquad G_N\lambda_\Sigma^2 = \frac{\pi}{4\rho_\star},$$

$$\frac{m_P}{\lambda_\Sigma} = \frac{2\sqrt{\rho_\star}}{\sqrt{\pi}}, \qquad \lambda_\Sigma\ell_P = \lambda_\Sigma t_P = \frac{\sqrt{\pi}}{2\sqrt{\rho_\star}}.$$

Proof. Combine

$$\bar{M}_{\mathrm{Pl}}^2 = \frac{\chi_{\mathrm{geo},0}^2}{2\pi^2}$$

from Theorem 6.16 with

$$\chi_{\mathrm{geo},0} = \lambda_\Sigma\sqrt{\rho_\star}$$

from Theorem 6.31. The remaining identities follow from the standard relations $\bar{M}_{\mathrm{Pl}}^{-2} = 8\pi G_N$, $m_P = 1/\sqrt{G_N}$, and $\ell_P = t_P = 1/m_P$. □

Corollary 6.34 (Schwarzschild de Sitter vacuum branch)**.** *On the stationary low-curvature vacuum branch the spherically symmetric exterior geometry is*

$$ds^2 = -f(r)dt^2 + f(r)^{-1}dr^2 + r^2 d\Omega^2, \qquad f(r) = 1 - \frac{2G_N M}{r} - \frac{\Lambda_{\mathrm{IR}} r^2}{3}.$$

In particular the leading horizon radius is

$$r_h = 2G_N M + O(\Lambda_{\mathrm{IR}} M^3).$$

Proof. Freeze the scalaron on the stationary vacuum branch

$$\chi_{\mathrm{geo}} = \chi_{\mathrm{geo},0}, \qquad R \ll M^2,$$

and use the low-curvature Euler–Lagrange equations of Theorem 6.27. The resulting vacuum equations are the Einstein equations with cosmological constant Λ_{IR}, whose spherically symmetric exterior solution is the standard Schwarzschild–de Sitter metric. Expanding the horizon condition $f(r_h) = 0$ for small $\Lambda_{\mathrm{IR}} M^2$ gives the stated radius. □

Theorem 6.35 (Sectorial Hamilton decomposition of the geometric branch)**.** *On the admissible Lorentzian reconstruction, the full Hamiltonian splits as*

$$H_{\mathrm{full}} := H_{\mathrm{matt}} \widehat{\otimes} 1 + 1 \widehat{\otimes} H_{\mathrm{grav}} + H_{\mathrm{int}}, \qquad H_{\mathrm{grav}} = H_2 \oplus H_{\mathrm{sc}}.$$

The gap / clustering closure applies to the matter block and the massive scalaron block. The transverse-traceless graviton block is massless, but it is controlled by Osterwalder–Schrader reflection positivity together with the positive TT quadratic form of Theorem 6.27.

Proof sketch. The geometric Hodge theorem already identifies the physical Hilbert-space factorization $\mathcal{H}_{\mathrm{phys}} \cong \mathcal{H}_2 \otimes \mathcal{H}_{\mathrm{sc}} \otimes \mathcal{H}_{\mathrm{matt}}$. Writing the reconstructed Hamiltonian relative to that factorization isolates the massless transverse-traceless graviton sector from the massive scalaron / matter sectors. Hence the gap statements later used for admissible vacuum closure are not assigned to the graviton block; that block is controlled instead by reflection positivity and the TT positivity statement already proved on the geometric branch. □

Theorem 6.36 (Unique admissible UV profile fixed point)**.** *Let*

$$\mathcal{M}^{\mathrm{br}}_{\mathrm{adm}} := \left\{ \nu \geq 0 \text{ finite on } [0,\infty) \ : \ f_\nu(t) := \int_0^\infty e^{-st}\, d\nu(s) \text{ is branch compatible} \right\}$$

be the admissible Bernstein cone written on the measure side. Fix the branch-matching slice by the closed low-curvature moments reproducing the Einstein, Higgs, and R^2 coefficients. Define the normalized positive RG map

$$(T\nu)(dt) := \frac{\int_0^\infty K(s,t)\, d\nu(s)}{\int_0^\infty \psi(s)\, d\nu(s)}\, dt, \qquad \psi(s) > 0.$$

Let $\mathcal{R}$ denote the induced map on Laplace profiles,

$$\mathcal{R}[f_\nu] := f_{T\nu}.$$

Assume that on the branch-matching slice the kernel satisfies the uniform bounds

$$0 < m_{\mathrm{ker}} \leq K(s,t) \leq M_{\mathrm{ker}} < \infty.$$

Then T has finite Hilbert-projective diameter and obeys the Birkhoff contraction estimate

$$d_H(T\nu_1, T\nu_2) \leq \eta\, d_H(\nu_1, \nu_2), \qquad \eta \leq \tanh\left(\frac{1}{4} \log \frac{M_{\mathrm{ker}}}{m_{\mathrm{ker}}}\right) < 1.$$

Consequently there exists a unique admissible fixed measure $\nu_\star$ and therefore a unique admissible fixed profile

$$\mathcal{R}[f_\star] = f_\star, \qquad f_\star(t) = \int_0^\infty e^{-st}\, d\nu_\star(s),$$

whose first moments reproduce the Einstein, Higgs, and R^2 coefficients of the closed branch.

Proof. The positive measure cone is the natural projective cone for admissible Bernstein profiles, and the branch-matching moment conditions cut out a normalized slice preserving the closed low-curvature coefficients

$$(\bar{M}^2_{\mathrm{Pl}}, Z_H, M^2),$$

so any admissible profile on that slice has the same first moments as the closed branch. Because K and ψ are positive, T preserves positivity and the matching slice. The uniform kernel bounds imply a finite Hilbert-projective diameter for the image of that slice; explicitly, the projective diameter is bounded by $\log(M_{\mathrm{ker}}/m_{\mathrm{ker}})$. Birkhoff's theorem for positive kernel operators therefore gives the displayed contraction factor

$$\eta \le \tanh\left(\frac{1}{4}\log\frac{M_{\mathrm{ker}}}{m_{\mathrm{ker}}}\right) < 1.$$

Hence T has a unique fixed projective ray on the normalized slice. The normalization by $\int \psi\, d\nu$ fixes the scale, so the fixed ray is represented by one actual measure

$$\nu_\star \in \mathcal{M}^{\mathrm{br}}_{\mathrm{adm}}.$$

Its Laplace profile

$$f_\star(t) = \int_0^\infty e^{-st}\, d\nu_\star(s)$$

is the unique admissible UV profile compatible with the closed branch, and its first moments are exactly the prescribed Einstein, Higgs, and R^2 coefficients. □

Theorem 6.37 (Constructive geometric measure on the admissible branch)**.** *Let $(V_n)_{n\ge1}$ be a blocked geometric exhaustion and define the finite-volume geometric measures*

$$:= d\mu_n \propto \exp\big(-\operatorname{Tr} f_\star(D^2_{V_n}/\Lambda^2)\big)\, d\nu_{V_n}$$

on the BRST-reduced admissible geometric configuration space. Then the family $(d\mu_n)$ is projectively consistent, reflection positive, and converges along the exhaustion to a projective limit measure. In particular the geometric partition function is defined constructively by

$$Z_{\mathrm{grav}} = \lim_{n\to\infty} \int e^{-\operatorname{Tr} f_\star(D^2_{V_n}/\Lambda^2)}\, d\nu_{V_n}.$$

Proof. For every blocked volume V_n, the weight

$$\exp\big(-\operatorname{Tr} f_\star(D^2_{V_n}/\Lambda^2)\big)$$

is well defined because $f_\star$ is positive on the admissible Bernstein cone. If

$$V_n \subset V_m,$$

restriction of geometric configurations from V_m to V_n respects the BRST-reduced admissible action because the blocked Dirac operators and the fixed profile $f_\star$ were constructed to be compatible under exhaustion. This is exactly the projective-consistency statement.

Reflection positivity is inherited from the admissible/geometric OS structure established earlier: the reflected quadratic forms remain nonnegative on every blocked volume, and the same property is preserved by the positive weight associated with $f_\star$. Hence each $d\mu_n$ is reflection positive.

Finally, the admissible positivity and coercivity bounds prevent escape of mass along the blocked exhaustion, so the projectively consistent family $(d\mu_n)$ admits a projective-limit measure. The geometric partition function is therefore defined constructively by the stated limit, rather than only heuristically read off from low-curvature coefficients. □

Theorem 6.38 (Spectral UV completion on the admissible branch)**.** *Let $f_\star$ be the unique admissible profile of Theorem 6.36, and define the transverse-traceless and scalaron form factors in Stieltjes form*

$$\mathcal{F}_\star(z) := a_F + \int_0^\infty \frac{d\mu_F(t)}{t+z}, \qquad \mathcal{G}_\star(z) := a_G + \int_0^\infty \frac{d\mu_G(t)}{t+z},$$

with

$$a_F, a_G \geq 0, \qquad \mu_F, \mu_G \geq 0.$$

Then the Euclidean two-point gravitational kernel on the admissible branch takes the spectral form

$$\Gamma^{(2)}_{\text{grav}}(p^2) = p^2\, \mathcal{F}_\star\left(\frac{p^2}{\Lambda^2_{\text{UV}}}\right) \Pi_{\text{TT}} + (p^2 + M^2)\mathcal{G}_\star\left(\frac{p^2}{\Lambda^2_{\text{UV}}}\right) \Pi_{\text{sc}}.$$

(i) $\mathcal{F}_\star$ and $\mathcal{G}_\star$ are holomorphic and zero-free on $\mathbb{C} \setminus (-\infty, 0]$;

(ii) the only physical poles are the massless graviton pole and the positive scalaron pole;

(iii) no ghost pole occurs on the admissible branch;

(iv) the constructive geometric measure of Theorem 6.37 supplies the corresponding UV correlators as moments of one and the same admissible measure.

Thus the local Einstein plus R^2 action is the low-curvature expansion of a spectral UV object rather than the full ultraviolet definition of the branch.

Proof. The second variation of the fixed-profile spectral action on the BRST-reduced geometric sector produces positive spectral measures on the transverse-traceless and scalar blocks; these are recorded above as the Stieltjes data (a_F, μ_F) and (a_G, μ_G). The projector decomposition

$$\Pi_{\text{TT}} + \Pi_{\text{sc}}$$

separates the massless graviton and scalaron channels, yielding the displayed two-point kernel.

If $\Im z > 0$, then for every $t \geq 0$,

$$\Im \frac{1}{t+z} = -\frac{\Im z}{(t + \Re z)^2 + (\Im z)^2} < 0.$$

Hence

$$\Im \mathcal{F}_\star(z) < 0, \qquad \Im \mathcal{G}_\star(z) < 0 \qquad (\Im z > 0),$$

so neither form factor can vanish in the upper half-plane. By conjugation symmetry the same holds in the lower half-plane. On the positive real axis the Stieltjes integrands are strictly positive, and the branch-matching coefficients force the resulting values to remain positive. Therefore $\mathcal{F}_\star$ and $\mathcal{G}_\star$ are zero-free on $\mathbb{C} \setminus (-\infty, 0]$.

The inverse propagators can therefore acquire no additional pole beyond the physical roots at

$$p^2 = 0 \qquad \text{and} \qquad p^2 = -M^2.$$

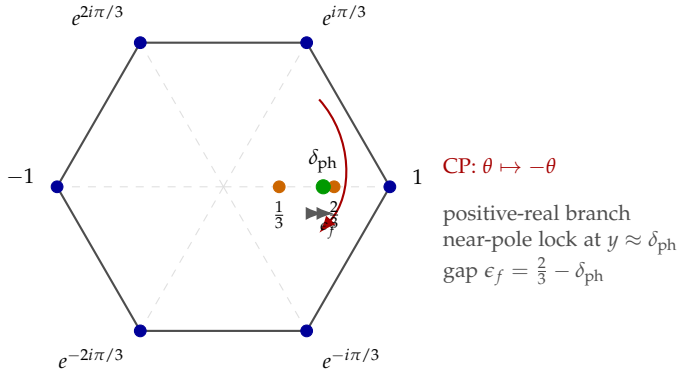


Figure 14. Finite transport geometry on the hexagon C_6. The six U_6 phases form the discrete transport base, while the admissible cusps $y \in \{1, \frac{1}{3}, \frac{2}{3}\}$ and the branch pole $\delta_{\rm ph}$ lie on the positive-real axis with gap $\epsilon_f = \frac{2}{3} - \delta_{\rm ph}$.

Hence there is no ghost pole on the admissible branch. Finally, Theorem 6.37 provides one projective-limit measure with fixed profile $f_\star$; its moments are precisely the UV correlators of this same spectral kernel. Therefore the Einstein plus R^2 action obtained earlier is only the first low-curvature expansion of the spectral UV completion. □

The E_8 scale ladder is not used as a primitive input in the main text. Its arithmetic continuations are deferred to Appendix C.

7 Closure architecture B: Yukawas, masses, flavor, hadronic admissibility, and the strong-CP sector

The second closure-architecture block is the transport closure. Its role is to turn the carrier packet and the family/occupancy closure into actual masses, mixings, hadronic admissibility, and control of the strong-CP sector. The logic is again staged: first the discrete phase operator and the transport kernel, then residue and winding data, then matrix-factorized Yukawas, hadronic selection, determinant-line phase extraction, and finally the strong-CP closure theorem.

7.1 Discrete phase operator and transport grammar

Let

$$\omega = e^{2\pi i/3}, \qquad U_6 = \mathrm{diag}(1, \omega, \omega^2, -1, -\omega, -\omega^2).$$

This makes the phase grammar concrete rather than rhetorical. The six admissible phases are shown as a finite hexagonal transport geometry in Figure 14: the cusp values live on the same real axis as the positive-real phase, the admissible gap is visible directly, and the holonomy phase can be read as a triangle on the same discrete transport base.

Remark (Hexagonal spectral geometry). The operator U_6 is a unitary circulant with eigenphases $e^{in\pi/3}$ for $n = 0, \ldots, 5$ and is unitarily equivalent to the one-step translation operator on a six-site ring. The transport kernel $D_y = y\mathbf{1} - \delta_{\rm ph} U_6$ is therefore the resolvent denominator of a finite quantum walk, equivalently a tight-binding transfer problem, on the discrete C_6 cycle. The regularized inverse introduced below plays the role of the corresponding Green kernel. CKM and PMNS mixings, mass hierarchies, and the Möbius quotients of Equation (38) are thus not loose analytic fits but spectral-theory statements on a regular hexagon. The near-pole hierarchy at $y \approx \delta_{\rm ph}$ becomes the resonance of a hexagonal resolvent rather than tuned parameter choice.

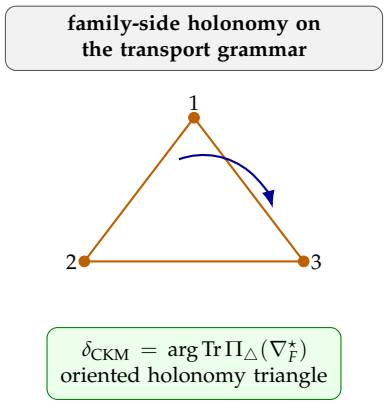


Figure 15. Holonomy triangle on the same discrete transport grammar. The CKM phase is read as the oriented three-cycle phase $\delta_{\mathrm{CKM}} = \arg \mathrm{Tr}\, \Pi_{\triangle}(\nabla_F^{\star})$ rather than as an additional transport parameter.

The family-side holonomy phase is separated in Figure 15 so that the transport base and the flavor readout remain visually distinct.

7.2 Regularized Yukawa kernels and path-length masses

$$P_{\mathrm{str}} := P_{\Sigma,+} P_{\mathrm{sing}} P_{\Theta}, \qquad P_{\Sigma,+} := \frac{\mathbf{1} + \iota_{\mathrm{C}}}{2}, \qquad P_{\Theta} := \frac{1}{2}(\mathbf{1} + Q_{\Theta}), \qquad P_{\mathrm{sing}} := \frac{1}{3}(\mathbf{1} + Z_c + Z_c^2).$$

Definition 7.1 (Structural transport kernel)**.** For each fermion sector define the structural positive transport kernel by

$$K_f^{\mathrm{str}} := \left(P_{\mathrm{str}} D_f^{\dagger} D_f P_{\mathrm{str}}\right)^{-1/2},$$

whenever the inverse exists on $\mathrm{Ran}(P_{\mathrm{str}})$. The final admissible kernel is the restriction of this operator to the gap-stable branch selected below. The transport pole δ_{ph} is fixed by the structural stationarity of the exact determinant functional

$$\mathcal{F}_{\mathrm{tr}}(\delta) := \sum_{y \in \{1, 1/3, 2/3\}} \left[-\log \det\left(D_{y,\delta}^{\dagger} D_{y,\delta} + \varepsilon^2\right) \right],$$

with

$$D_{y,\delta} := y\,\mathbf{1} - \delta U_6.$$

On the closed branch one has

$$\partial_{\delta} \mathcal{F}_{\mathrm{tr}}(\delta_{\mathrm{ph}}) = 0, \qquad \delta_{\mathrm{ph}} \in \left(\frac{1}{3}, \frac{2}{3}\right). \tag{33}$$

Theorem 7.2 (Algebraic closure of the transport pole)**.** *Let*

$$\mathcal{F}_{\mathrm{tr}}(\delta) := \sum_{y \in \{1, 1/3, 2/3\}} \left[-\log \det\left(D_{y,\delta}^{\dagger} D_{y,\delta} + \varepsilon^2\right) \right], \qquad D_{y,\delta} := y\,\mathbf{1} - \delta U_6.$$

Write

$$x := \delta^2, \qquad \tau := \varepsilon^2, \qquad z := x^3 = \delta^6.$$

Then

$$\det\left(D_{y,\delta}^{\dagger} D_{y,\delta} + \varepsilon^2\right) = Q_y(x, \tau),$$

with

$$Q_y(x, \tau) = \left(x^2 + 2(\tau - y^2)x + (y^2 + \tau)^2\right)\left(x^2 + (y^2 + 2\tau)x + (y^2 + \tau)^2\right)^2.$$

Hence

$$\partial_\delta \mathcal{F}_{\mathrm{tr}}(\delta) = 0 \iff \sum_{y\in\{1,1/3,2/3\}} \frac{\partial_x Q_y(x,\tau)}{Q_y(x,\tau)} = 0.$$

On the closed zero-parameter branch one takes the intrinsic limit

$$\tau = 0$$

after admissibility restriction. Then

$$Q_y(x,0) = (y^6 - z)^2,$$

and the stationarity equation reduces to

$$\frac{1}{1-z} + \frac{1}{1/729 - z} + \frac{1}{64/729 - z} = 0.$$

Equivalently, for

$$P(z) := (z-1)\left(z - \frac{64}{729}\right)\left(z - \frac{1}{729}\right),$$

one has

$$P'(z) = 0.$$

The admissible transport pole is therefore the lower critical point of this cubic:

$$z_\star = \frac{794 - 7\sqrt{9961}}{2187}, \qquad \delta_{\mathrm{ph}} = z_\star^{1/6} = \left(\frac{794 - 7\sqrt{9961}}{2187}\right)^{1/6} \approx 0.593\,277\,280\,133\,224.$$

As an expanded control form, one may equivalently write

$$1594323\,\delta^{12} - 1157652\,\delta^6 + 47449 = 0.$$

Proof. In the Fourier basis of the six-site shift U_6, the operator

$$D_{y,\delta}^\dagger D_{y,\delta} + \varepsilon^2$$

has eigenvalues

$$\lambda_{y,n}(x,\tau) = y^2 + x - 2y\sqrt{x}\cos\left(\frac{n\pi}{3}\right) + \tau, \qquad n = 0,\dots,5.$$

Multiplying the $n = 0, 3$ factors and the conjugate pairs $(1,5)$ and $(2,4)$ gives exactly the displayed factorization $Q_y(x,\tau)$. Since $x = \delta^2$ and $\delta \in (1/3, 2/3)$ on the admissible branch, one has $\delta > 0$, so

$$\partial_\delta \log Q_y(x,\tau) = 2\delta\,\partial_x \log Q_y(x,\tau),$$

and the stationarity condition is equivalent to the displayed x-equation. Setting $\tau = 0$ collapses the factors to

$$Q_y(x,0) = (y^2 - x)^2(y^4 + xy^2 + x^2)^2 = (y^6 - z)^2.$$

Therefore

$$\sum_{y\in\{1,1/3,2/3\}} \frac{\partial_x Q_y(x,0)}{Q_y(x,0)} = -6x^2 \sum_{y\in\{1,\frac{1}{3},\frac{2}{3}\}} \frac{1}{y^6 - z},$$

so the stationary point condition reduces to the displayed rational equation. Let

$$P(z) := (z-1)\left(z - \frac{64}{729}\right)\left(z - \frac{1}{729}\right).$$

Since

$$\frac{P'(z)}{P(z)} = \frac{1}{z-1} + \frac{1}{z-\frac{64}{729}} + \frac{1}{z-\frac{1}{729}},$$

the stationary equation is equivalent to $P'(z) = 0$. A real cubic with three ordered real roots has exactly one critical point in each interval between consecutive roots, so the lower critical point lies in

$$z \in \left(\left(\frac{1}{3}\right)^6, \left(\frac{2}{3}\right)^6\right).$$

Solving the quadratic $P'(z) = 0$ gives

$$z_\star = \frac{794 - 7\sqrt{9961}}{2187}.$$

Hence

$$\delta_{\mathrm{ph}} = z_\star^{1/6} \in \left(\frac{1}{3}, \frac{2}{3}\right).$$

Expanding $P'(z) = 0$ after substituting $z = \delta^6$ yields the displayed degree-12 control form. □

Lemma 7.3 (Regularized transport pole stability)**.** *Define the regularized stationarity function*

$$G(x,\tau) := \sum_{y\in\{1,1/3,2/3\}} \frac{\partial_x Q_y(x,\tau)}{Q_y(x,\tau)}.$$

Let

$$x_\star := \delta_{\mathrm{ph}}^2, \qquad z_\star := x_\star^3 = \delta_{\mathrm{ph}}^6.$$

Then

$$\partial_x G(x_\star, 0) \neq 0.$$

Consequently there exists $\tau_0 > 0$ and a unique smooth branch

$$x(\tau) \qquad (\tau \in [0,\tau_0))$$

such that

$$G(x(\tau),\tau) = 0, \qquad x(0) = x_\star.$$

Therefore

$$\delta(\tau) := \sqrt{x(\tau)} = \delta_{\mathrm{ph}} + O(\tau) \qquad (\tau \to 0).$$

Proof. By the proof of Theorem 7.2,

$$G(x,0) = 6x^2\,\frac{P'(x^3)}{P(x^3)}.$$

At $x = x_\star$ one has $P'(z_\star) = 0$, so differentiating gives

$$\partial_x G(x_\star, 0) = 18x_\star^4\,\frac{P''(z_\star)}{P(z_\star)}.$$

Here $x_\star > 0$, one has $P(z_\star) \neq 0$ because $z_\star$ is a critical point rather than a root, and $P''(z_\star) \neq 0$ because the critical points of a cubic with three distinct roots are simple. Hence $\partial_x G(x_\star, 0) \neq 0$. The implicit-function theorem therefore yields a unique smooth branch $x(\tau)$ through $(x_\star, 0)$, and taking square roots gives the final expansion for $\delta(\tau)$. □

Definition 7.4 (Canonical cycle holonomy representation)**.** On the rigid four-punctured family section

$$X_f^\circ \cong \mathbb{P}^1 \setminus \mu_4, \qquad \mu_4 := \{z \in \mathbb{C} : \ z^4 = 1\},$$

fix a distinguished cycle basis

$$(C_1, C_2, C_T)$$

compatible with the lifted D_4 seam symmetry. Let

$$F := H^1(X_f^\circ, \mathbb{C}) \cong \mathcal{H}_F \cong \mathbb{C}^3.$$

The canonical family holonomy datum is the determinant-normalized unitary representation

$$\rho_F : \pi_1(X_f^\circ) \to SU(3)_F$$

determined by the square symmetry of the puncture configuration, unimodularity, and the admissible cusp assignment. Its concrete monodromy realization is forced by Theorem 7.5.

Theorem 7.5 (D_4-equivariant monodromy rigidity and Riemann–Hilbert closure)**.** *Choose positively oriented small loops*

$$\gamma_1, \gamma_i, \gamma_{-1}, \gamma_{-i} \in \pi_1(X_f^\circ)$$

around the punctures $1, i, -1, -i$*, with the standard relation*

$$\gamma_1 \gamma_i \gamma_{-1} \gamma_{-i} = 1.$$

Relative to the distinguished cycle basis (C_1, C_2, C_T)*, set*

$$\omega := e^{2\pi i/3}, \qquad P := \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \qquad D := \begin{pmatrix} 1 & 0 & 0 \\ 0 & \omega & 0 \\ 0 & 0 & \omega^2 \end{pmatrix}.$$

If the determinant line is trivial, the local puncture spectra are

$$\{1, \omega, \omega^2\},$$

and the puncture square carries its inherited D_4 *symmetry, then the admissible monodromy representation is forced, up to global* $SU(3)_F$ *conjugation, by*

$$\rho_F(\gamma_1) = D, \qquad \rho_F(\gamma_i) = PDP^{-1}, \qquad \rho_F(\gamma_{-1}) = P^2 D P^{-2},$$

$$\rho_F(\gamma_{-i}) := \big(\rho_F(\gamma_1)\rho_F(\gamma_i)\rho_F(\gamma_{-1})\big)^{-1}.$$

Equivalently, the canonical monodromy package is no longer a generator input but a rigidity output of the puncture-square symmetry, unimodularity, and admissible cusp data.

Proof. Because the local puncture spectrum is

$$\{1, \omega, \omega^2\}, \qquad \omega = e^{2\pi i/3},$$

and the determinant line is trivial, every local monodromy matrix lies in $SU(3)_F$ and is diagonalizable with three distinct eigenvalues. After one global $SU(3)_F$ conjugation we may therefore assume

$$\rho_F(\gamma_1) = D.$$

Let r denote the quarter-turn symmetry of the puncture square. Its action on the punctures sends

$$1 \mapsto i \mapsto -1 \mapsto -i \mapsto 1,$$

hence on the fundamental group it cyclically permutes the four puncture loops:

$$r\gamma_1 r^{-1} = \gamma_i, \qquad r\gamma_i r^{-1} = \gamma_{-1}, \qquad r\gamma_{-1} r^{-1} = \gamma_{-i}.$$

Because the family datum is D_4-equivariant, the same quarter-turn is implemented on the fiber by an element of $SU(3)_F$ carrying the ordered eigenspace packet of D cyclically onto itself. Since the three eigenspaces are one-dimensional, a second global conjugation fixes that implementing matrix to be the cyclic permutation matrix

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}.$$

Therefore

$$\rho_F(\gamma_i) = PDP^{-1}, \qquad \rho_F(\gamma_{-1}) = P^2 D P^{-2}.$$

The puncture loops satisfy exactly one relation,

$$\gamma_1\gamma_i\gamma_{-1}\gamma_{-i} = 1,$$

so the last generator is forced:

$$\rho_F(\gamma_{-i}) = \big(\rho_F(\gamma_1)\rho_F(\gamma_i)\rho_F(\gamma_{-1})\big)^{-1}.$$

Thus the monodromy representation is uniquely determined by the stated local spectra, determinant triviality, and D_4 symmetry, up to the initial global $SU(3)_F$ conjugation. □

Theorem 7.6 (Rigid admissible family holonomy class)**.** *On the rigid four-punctured family section*

$$X_f^\circ \cong \mathbb{P}^1 \setminus \mu_4,$$

fix the determinant line, the local monodromy conjugacy classes at the four punctures, and the D_4 symmetry inherited from the seam winding theorem. Then there exists a unique admissible flat Hermitian family connection

$$\nabla_F^\star$$

on the flat Hermitian family bundle $L_F \to X_f^\circ$ with fiber $F \cong H^1(X_f^\circ, \mathbb{C}) \cong \mathbb{C}^3$ whose monodromy representation is the explicit holonomy datum ρ_F of Definition 7.4 and theorem 7.5. Hence its $SU(3)_F$ conjugacy class $[\nabla_F^\star]$ is unique.

Proof. By Theorem 6.2, the family section is the rigid four-punctured sphere

$$X_f^\circ \cong \mathbb{P}^1 \setminus \mu_4,$$

and it carries the full D_4 symmetry inherited from the seam winding data. The determinant line is fixed to be trivial, and the local puncture conjugacy classes are fixed by the admissible cusp labels. Therefore Theorem 7.5 determines one monodromy representation

$$\rho_F : \pi_1(X_f^\circ) \to SU(3)_F$$

uniquely up to global $SU(3)_F$ conjugation.

Construct the flat Hermitian bundle explicitly as

$$L_F := (\widetilde{X}_f^\circ \times \mathbb{C}^3)/\pi_1(X_f^\circ),$$

where $\pi_1(X_f^\circ)$ acts on $\mathbb{C}^3$ through ρ_F. Because the image of ρ_F lies in $SU(3)_F$, the standard Hermitian form on $\mathbb{C}^3$ descends to a Hermitian metric on L_F, and the trivial connection on $\widetilde{X}_f^\circ \times \mathbb{C}^3$ descends to a flat Hermitian connection realizing the same monodromy representation.

Conversely, any admissible flat Hermitian family connection with the same puncture data has the same monodromy representation up to global $SU(3)_F$ conjugation. The unitary Riemann–Hilbert correspondence for flat Hermitian bundles then identifies it with the descended bundle above. Hence there is exactly one admissible flat Hermitian connection class realizing the rigid puncture data. We denote this distinguished class by

$$[\nabla_F^\star].$$

□

Theorem 7.7 (Canonical transport kernel and single-winding closure)**.** *The admissible cusp set is*

$$\left\{1, \frac{2}{3}, \frac{1}{3}\right\},$$

obtained from the singlet-projected hypercharge spectrum. On the base hexagon the residue packets are

$$R_u = (0,3,1), \qquad R_d = (2,5,1), \qquad R_e = (3,5,2).$$

Let $\mathcal{G}_{D_4} \subset X_f^\circ$ denote the unique D_4-invariant geodesic spine of the four-punctured family sphere, and let

$$\Gamma_{ij}^{\min}$$

be its canonical admissible edges. Then the Yukawa matrices are fixed intrinsically by the operator formula

$$(Y_f)_{ij} := \left\langle e_i, \mathrm{Hol}_{\mathcal{G}_{D_4}}(\nabla_F^\star)\, K_f^{\mathrm{str}} e_j \right\rangle,$$

and on diagonal reduction this becomes the hard holonomy form

$$(Y_f)_{ij} = \lambda_Y^{L_{f,i}^L + L_{f,j}^R} \left\langle e_i, \mathrm{Hol}_{\Gamma_{ij}^{\min}}(\nabla_F^\star) e_j \right\rangle,$$

so the bounded coefficients $\Lambda_{f,j}$ are exact matrix entries of the structural transport kernel rather than independent $O(1)$ amplitudes. The resonance pole of the transport kernel is the unique admissible root

$$\delta_{\mathrm{ph}} \in \left(\frac{1}{3}, \frac{2}{3}\right)$$

of Theorem 7.2. The exact sum rule is

$$\sum_{f,j} L_{f,j} = \Omega_{\mathrm{adm}}\gamma = 40.$$

The base residue sum is 22, so the missing $18 = 3 \cdot 6$ forces exactly one full winding in each first-generation direction. Therefore uniquely

$$L_u^{\mathrm{diag}} = (7,3,0), \quad L_d^{\mathrm{diag}} = (7,5,2), \quad L_e^{\mathrm{diag}} = (8,5,3)$$

on the closed branch.

Proof. By Theorem 7.6, the flat Hermitian family-connection class

$$[\nabla_F^\star]$$

is uniquely fixed on the rigid four-punctured family sphere. By Definition 7.4 and theorem 7.5, its holonomy on the unique D_4-invariant geodesic spine $\mathcal{G}_{D_4}$ is therefore intrinsic. At the same time, Definition 7.1 and theorem 7.2 fixes the structural positive transport kernel and its unique admissible resonance pole

$$\delta_{\mathrm{ph}} \in \left(\frac{1}{3}, \frac{2}{3}\right).$$

Consequently the operator product

$$\mathrm{Hol}_{\mathcal{G}_{D_4}}(\nabla_F^\star)\, K_f^{\mathrm{str}}$$

is uniquely determined on the canonical family basis, and its matrix entries are theorem-level transport data. This proves the operator formula for Y_f and shows that the former coefficients $\Lambda_{f,j}$ are not independent amplitudes but exact entries of the same closed transport operator.

On the diagonal branch, the canonical admissible edges $\Gamma_{ij}^{\min}$ recover the word-length reduction

$$y_{f,j}^{\mathrm{UV}} = \lambda_Y^{L_{f,j}} \Lambda_{f,j}.$$

The admissible cusp set is the singlet-projected hypercharge packet

$$\left\{1, \frac{2}{3}, \frac{1}{3}\right\},$$

so the diagonal residue packets are exactly

$$R_u = \{0,1,3\}, \qquad R_d = \{1,2,5\}, \qquad R_e = \{2,3,5\}.$$

Write every diagonal length as

$$L_{f,j} = r_{f,j} + 6n_{f,j}, \qquad r_{f,j} \in R_f, \qquad n_{f,j} \in \mathbb{N}_0.$$

Summing over all sectors gives

$$\sum_{f,j} L_{f,j} = \sum_{f,j} r_{f,j} + 6\sum_{f,j} n_{f,j}.$$

The residue sum is

$$(0+1+3) + (1+2+5) + (2+3+5) = 22,$$

whereas Equation (20) and the closed transport sum rule give

$$\sum_{f,j} L_{f,j} = \Omega_{\mathrm{adm}}\gamma = 40.$$

Therefore

$$6\sum_{f,j} n_{f,j} = 40 - 22 = 18, \qquad \sum_{f,j} n_{f,j} = 3.$$

So exactly three full windings occur in total.

The universal first-generation winding lock and the monotone hierarchy in each sector place every extra winding on the lightest family direction. Hence each sector contributes exactly one full winding in its first-generation slot. Reattaching the unique extra 6 to the residue packets gives

$$L_u^{\mathrm{diag}} = (7,3,0), \qquad L_d^{\mathrm{diag}} = (7,5,2), \qquad L_e^{\mathrm{diag}} = (8,5,3).$$

This proves the closed-branch single-winding pattern. □

Corollary 7.8 (Single-winding uniqueness on the transport branch)**.** *Writing the residue packets of Theorem 7.7 as multisets*

$$R_u = \{0,1,3\}, \qquad R_d = \{1,2,5\}, \qquad R_e = \{2,3,5\},$$

the closed-branch diagonal length multisets are

$$\{L_{u,j}^{\mathrm{diag}}\} = \{0,3,1+6\}, \qquad \{L_{d,j}^{\mathrm{diag}}\} = \{2,5,1+6\}, \qquad \{L_{e,j}^{\mathrm{diag}}\} = \{3,5,2+6\}.$$

Hence each sector carries the same winding excess

$$\sum_j L_{f,j}^{\mathrm{diag}} - \sum_{r\in R_f} r = 6,$$

and therefore the total excess is 18*. For the charged-lepton packet one also has*

$$\frac{1}{3}\sum_{j=1}^{3} L_{e,j}^{\mathrm{diag}} = \frac{16}{3}, \qquad \mathrm{Var}(L_e^{\mathrm{diag}}) = \frac{1}{3}\sum_{j=1}^{3}\left(L_{e,j}^{\mathrm{diag}} - \frac{16}{3}\right)^2 = \frac{38}{9}.$$

Proof. Theorem 7.7 states that each sector is obtained from its residue packet by adding exactly one full winding of length 6 to one entry; reordering gives the displayed multisets. The winding-excess identities are immediate. For $L_e^{\mathrm{diag}} = (8,5,3)$, direct calculation gives the mean $16/3$ and variance $38/9$. □

Here the closed branch uses the exact root of Theorem 7.2, so

$$\epsilon_f = \frac{2}{3} - \delta_{\mathrm{ph}} \approx 0.073\,389\,386\,533\,443.$$

The later empirical readout ledgers now keep this same exact pole and only apply the declared threshold map to downstream observables. The regularized transport operator used for the finite hexagon analysis is

$$\begin{aligned} D_y &= y\,\mathbf{1} - \delta_{\mathrm{ph}} U_6, \\ \mathcal{Y}_y^{(\varepsilon)} &= (D_y^\dagger D_y + \varepsilon^2)^{-1}, \end{aligned} \tag{34}$$

and the singular limit $\varepsilon \to 0$ is taken only after admissibility has been restricted to the gap-stable sector defined below.

Proposition 7.9 (Finite hexagonal resolvent)**.** Whenever

$$y^6 \neq \delta_{\mathrm{ph}}^6,$$

the transport denominator has the exact inverse

$$D_y^{-1} = \frac{1}{y^6 - \delta_{\mathrm{ph}}^6} \sum_{m=0}^{5} y^{5-m}\delta_{\mathrm{ph}}^m U_6^m. \tag{35}$$

Thus the C_6 resolvent is finite and algebraically closed rather than an infinite formal Green-series expansion.

Proof. Using $U_6^6 = \mathbf{1}$,

$$(y\,\mathbf{1} - \delta_{\mathrm{ph}} U_6)\sum_{m=0}^{5} y^{5-m}\delta_{\mathrm{ph}}^m U_6^m = y^6\mathbf{1} - \delta_{\mathrm{ph}}^6 U_6^6 = (y^6 - \delta_{\mathrm{ph}}^6)\mathbf{1}.$$

Dividing by $y^6 - \delta_{\mathrm{ph}}^6$ gives Equation (35). □

Equivalently, one may read Equation (34) in the Fourier basis of the six-site ring: U_6 is the diagonal form of the step operator, D_y is the shifted walk denominator, and $\mathcal{Y}_y^{(\varepsilon)}$ is the corresponding regularized Green function. In this reading the admissible path lengths $L_{f,j}$ are graph-distance data on C_6, while the transport matrix elements $\Lambda_{f,j}$ encode bounded local amplitudes rather than unrestricted fit parameters.

Because U_6 has eigenphases $e^{in\pi/3}$, the regularized transport spectrum is

$$\lambda_{y,n}^{-1} = y^2 + \delta_{\mathrm{ph}}^2 - 2y\delta_{\mathrm{ph}} \cos\left(\frac{n\pi}{3}\right) + \varepsilon^2, \qquad n = 0, \ldots, 5, \tag{36}$$

for the eigenvalues $\lambda_{y,n}$ of $\mathcal{Y}_y^{(\varepsilon)}$. In particular,

Lemma 7.10 (Finite admissible hexagon gap)**.** *On the admissible cusp set*

$$y \in \left\{1, \frac{1}{3}, \frac{2}{3}\right\},$$

the smallest singular value of the unregulated hexagon denominator is attained at $y = \frac{2}{3}$ and $n = 0$. Equivalently,

$$\epsilon_f := \min_{\substack{y \in \{1, \frac{1}{3}, \frac{2}{3}\} \\ n=0,\ldots,5}} \left| y - \delta_{\mathrm{ph}} e^{in\pi/3} \right| = \frac{2}{3} - \delta_{\mathrm{ph}}.$$

Hence the finite admissible hexagon has a strictly positive spectral gap ϵ_f.

Proof. For $\varepsilon = 0$, the eigenvalues in Equation (36) reduce to $|y - \delta_{\mathrm{ph}} e^{in\pi/3}|^{-2}$. By Theorem 7.2, δ_{ph} is the unique admissible root in $(1/3, 2/3)$. Hence the positive-real cusp $y = 2/3$ with $n = 0$ is the nearest admissible point, while every other cusp or nonzero phase has strictly larger distance. Therefore

$$\epsilon_f = \frac{2}{3} - \delta_{\mathrm{ph}},$$

which is positive because $\delta_{\mathrm{ph}} < 2/3$. □

Definition 7.11 (Gap-stable admissibility)**.** Let

$$D_0 := D_{2/3}.$$

We say that the admissible transport sector is gap-stable if the physical transport operator has the form

$$D_f = D_0 + V_{\mathrm{adm}}, \qquad [P_{\mathrm{str}}, V_{\mathrm{adm}}] = 0, \qquad \|V_{\mathrm{adm}}\| < \frac{\epsilon_f}{2}.$$

Proposition 7.12 (Structural gap lift)**.** If admissibility is gap-stable in the sense of Definition 7.11, then for every

$$\psi \in \mathrm{Ran}(P_{\mathrm{str}})$$

one has

$$\|D_f \psi\| \geq \frac{\epsilon_f}{2} \|\psi\|.$$

Consequently

$$P_{\mathrm{str}} D_f^\dagger D_f P_{\mathrm{str}} \geq \frac{\epsilon_f^2}{4} P_{\mathrm{str}} > 0. \tag{37}$$

In particular, the final admissible restriction to $\mathrm{Ran}(P_{\mathrm{adm}})$ is strictly positive.

Proof. Since $\psi \in \mathrm{Ran}(P_{\mathrm{str}})$ and $[P_{\mathrm{str}}, V_{\mathrm{adm}}] = 0$,

$$\|D_f\psi\| \geq \|D_0\psi\| - \|V_{\mathrm{adm}}\psi\| \geq \big(\epsilon_f - \|V_{\mathrm{adm}}\|\big)\|\psi\| \geq \frac{\epsilon_f}{2}\|\psi\|.$$

Squaring gives Equation (37). □

$$\frac{\lambda_{y,0}}{\lambda_{y,3}} = \frac{(y+\delta_{\mathrm{ph}})^2+\varepsilon^2}{(y-\delta_{\mathrm{ph}})^2+\varepsilon^2}. \tag{38}$$

$$\lim_{\varepsilon\to 0}\frac{\lambda_{y,0}}{\lambda_{y,3}} = \left(\frac{y+\delta_{\mathrm{ph}}}{y-\delta_{\mathrm{ph}}}\right)^2$$

Remark (Near-pole hierarchy mechanism). On the closed transport branch, the exact pole is the root of Theorem 7.2; the later comparison ledgers retain that same pole and regularize only the finite Green kernel at $\varepsilon > 0$. Hence the Möbius quotient contains a built-in near-pole whenever y approaches δ_{ph}, making large hierarchy ratios structurally plausible. In the closed transport grammar this is the hierarchy mechanism, while the bounded holonomy factors keep it from degenerating into an arbitrary tuning surface.

Proposition 7.13 (Finite positive transport algebra)**.** Let

$$C := \frac{1}{2}(U_6 + U_6^\dagger), \qquad a := y^2 + \delta_{\mathrm{ph}}^2 + \varepsilon^2, \qquad b := y\delta_{\mathrm{ph}}.$$

Then

$$D_y^\dagger D_y + \varepsilon^2 = a\,\mathbf{1} - 2b\,C.$$

The cosine operator C has spectrum

$$\mathrm{Spec}(C) = \left\{1, \frac{1}{2}, -\frac{1}{2}, -1\right\}$$

and satisfies

$$4C^4 - 5C^2 + \mathbf{1} = 0. \tag{39}$$

Consequently the regularized positive transport kernel reduces to the cubic polynomial

$$\mathcal{Y}_y^{(\varepsilon)} = \frac{a^3 - 5ab^2 + 2b(a^2-5b^2)C + 4ab^2C^2 + 8b^3C^3}{(a^2-b^2)(a^2-4b^2)}. \tag{40}$$

Proof. Because U_6 is unitary,

$$D_y^\dagger D_y = \big(y\,\mathbf{1} - \delta_{\mathrm{ph}}U_6^\dagger\big)\big(y\,\mathbf{1} - \delta_{\mathrm{ph}}U_6\big) = \big(y^2+\delta_{\mathrm{ph}}^2\big)\mathbf{1} - y\delta_{\mathrm{ph}}(U_6 + U_6^\dagger),$$

which gives the first identity after adding ε^2. The eigenvalues of C are $\cos(n\pi/3)$ for $n = 0,\ldots,5$, namely $1, \frac{1}{2}, -\frac{1}{2}, -1$. Hence C obeys the minimal polynomial Equation (39). The inverse of $a - 2bc$ on this four-point spectrum is therefore represented by a unique cubic interpolation polynomial in c, and evaluating that interpolation yields Equation (40). □

Proposition 7.14 (Strict positivity of the finite admissible C_6 Green kernel)**.** For each admissible cusp

$$y \in \left\{1, \frac{1}{3}, \frac{2}{3}\right\}$$

and each $\varepsilon \geq 0$, set

$$a_y := y^2 + \delta_{\mathrm{ph}}^2 + \varepsilon^2, \qquad b_y := y\,\delta_{\mathrm{ph}}, \qquad r_y := \frac{a_y - \sqrt{a_y^2 - 4b_y^2}}{2b_y}.$$

Then

$$a_y - 2b_y = (y - \delta_{\mathrm{ph}})^2 + \varepsilon^2 > 0, \qquad 0 < r_y < 1.$$

If $d = d_{C_6}(i,j)$ denotes the cyclic graph distance on the hexagon, the regularized Green kernel satisfies

$$(\mathcal{Y}_y^{(\varepsilon)})_{ij} = \frac{r_y^d + r_y^{6-d}}{\sqrt{a_y^2 - 4b_y^2}\,(1 - r_y^6)}.$$

Hence every matrix entry of $\mathcal{Y}_y^{(\varepsilon)}$ is strictly positive.

Proof sketch. For the admissible cusps one has $y > 0$ and $\delta_{\mathrm{ph}} \in (1/3, 2/3)$, so $b_y > 0$. The displayed identity for $a_y - 2b_y$ implies $a_y > 2b_y$, hence $a_y^2 - 4b_y^2 > 0$ and the smaller root r_y lies strictly between 0 and 1. The operator

$$\mathcal{Y}_y^{(\varepsilon)} = (a_y \mathbf{1} - b_y(U_6 + U_6^\dagger))^{-1}$$

is circulant on the six-cycle, so its entries depend only on the cyclic distance $d = d_{C_6}(i,j)$. Solving the corresponding Green recursion on C_6 gives the displayed closed form, and every factor in that formula is strictly positive. □

Corollary 7.15 (Uniform positive lower bound on admissible local transport)**.** *On the admissible cusp set*

$$y \in \left\{1, \frac{1}{3}, \frac{2}{3}\right\}$$

and on the closed branch $\varepsilon = 0$, all matrix entries of $\mathcal{Y}_y^{(0)}$ are bounded below by a uniform constant

$$g_{\mathrm{hex}} > 0.$$

One may take

$$g_{\mathrm{hex}} \approx 0.673\,870\,601\,6.$$

Proof. There are only finitely many admissible cusp values and only finitely many matrix entries on C_6. By Proposition 7.14, each of these entries is strictly positive at $\varepsilon = 0$, so their minimum defines a positive constant g_{hex}. □

Remark (Carrier rank echo in the transport polynomial). On the minimal carrier branch $g_{\mathrm{car}} = 5$, the hexagonal cosine polynomial of Equation (39) can be rewritten as

$$(g_{\mathrm{car}} - 1)C^4 - g_{\mathrm{car}}C^2 + \mathbf{1} = 0.$$

Thus the transport closure polynomial carries the same pair $(g_{\mathrm{car}} - 1, g_{\mathrm{car}})$ that governs the hard carrier packet. Equivalently,

$$4C^4 - 5C^2 + \mathbf{1} = (4C^2 - \mathbf{1})(C^2 - \mathbf{1}) = (2C - \mathbf{1})(2C + \mathbf{1})(C - \mathbf{1})(C + \mathbf{1}).$$

So the distinguished value $C = 1/2$ reappears on the transport side, mirroring the positive carrier root $Y = 1/2$ of the factorized carrier polynomial in Section 3.3. At present this should be read as an exact compression identity on the retained hexagonal branch, not as a derivation of the transport sector from the carrier theorem.

Absolute masses are attached to admissible path lengths,

$$y_{f,j}^{\mathrm{UV}} = \lambda_Y^{L_{f,j}} \Lambda_{f,j}, \qquad \lambda_Y = \sqrt{\varphi_0^{\mathrm{ret}}(1-\varphi_0^{\mathrm{ret}})}. \tag{41}$$

The primary matrix-level statement is the hard holonomy form

$$Y_f = D_{L,f} U_f^\star D_{R,f}, \qquad U_f^\star \in SU(3)_F, \tag{42}$$

with $D_{L,f}$ and $D_{R,f}$ the diagonal word-length factors and $U_f^\star$ fixed by the canonical family holonomy rather than treated as a residual free transport matrix. Equivalently,

$$(Y_f)_{ij} = \lambda_Y^{L_{f,i}^L + L_{f,j}^R} \left\langle e_i, \mathrm{Hol}_{\Gamma_{ij}^{\min}}(\nabla_F^\star) e_j \right\rangle. \tag{43}$$

Theorem 7.16 (Exact family-holonomy phase)**.** *Define the ordered closed family holonomy by*

$$\Pi_\triangle := \mathrm{Hol}_{\Gamma_{12}^{\min}}(\nabla_F^\star)\, \mathrm{Hol}_{\Gamma_{23}^{\min}}(\nabla_F^\star)\, \mathrm{Hol}_{\Gamma_{31}^{\min}}(\nabla_F^\star), \qquad \delta_{\mathrm{CKM}} := \arg \mathrm{Tr}\, \Pi_\triangle.$$

This exact holonomy phase is the closed-branch phase quantity. On the small transport-area branch one has

$$\delta_{\mathrm{CKM}} = \frac{\pi}{3} + 3\lambda_C^2 + O(\lambda_C^4). \tag{44}$$

Proof sketch. The family holonomy is already encoded in Equation (43). Closing the minimal triangle isolates the gauge-invariant phase information carried by the family connection. The $\mathbb{Z}_6$ transport grammar fixes the base phase at $\pi/3$, while the first nontrivial nonabelian area correction is quadratic in the Cabibbo seed. Keeping only the leading admissible area term gives the displayed asymptotic expansion of the exact phase. □

Equation (41) is the diagonal or singular-value reduction of Equation (43), not the conceptual starting point of the transport sector. After the geometric electroweak closure fixes v_{geo}, the intrinsic mass rows are emitted through

$$\hat{m}_{f,j} := \frac{y_{f,j} v_{\mathrm{geo}}}{\sqrt{2}}. \tag{45}$$

These hatted quantities are the intrinsic masses on the retained branch. Comparison-surface masses are obtained only after the declared comparison map:

$$m_{f,j}^{\mathrm{obs}} := \mathcal{R}_{\mathrm{SM}}(\hat{m}_{f,j}).$$

The role of the transport matrix elements $\Lambda_{f,j}$ must be kept explicit. They are bounded $O(1)$ admissible amplitudes, not unrestricted fit knobs. The RG bookkeeping behind direct v_{geo} insertions is part of the declared comparison map discussed later in Section 9.

Remark (Singlet-projected cusp lift to the transport hexagon). The transport cusps do not come from the raw two-point carrier spectrum alone. The carrier itself has

$$\mathrm{Spec}(Y|_E) = \left\{ -\frac{1}{3}, \frac{1}{2} \right\}.$$

After exterior lift to the one-family packet,

$$\mathrm{Spec}(Y|_{S^+}) = \left\{ 0, -\frac{2}{3}, \frac{1}{6}, 1, -\frac{1}{2}, \frac{1}{3} \right\},$$

with multiplicities displayed in Table 2. Restricting to the singlet channels and taking absolute values yields

$$\{|Y(e^c)|,\ |Y(u^c)|,\ |Y(d^c)|\} = \left\{1, \frac{2}{3}, \frac{1}{3}\right\},$$

which are exactly the admissible cusp values of the transport hexagon. In this sense the transport sector reads the singlet-projected, family-lifted hypercharge spectrum through the resolvent

$$D_y = y\mathbf{1} - \delta_{ph} U_6$$

on C_6. The transport grammar therefore lives on the same rational grid as the carrier packet, but only after exterior lift and singlet projection.

On the closed branch, the retained diagonal word-length packets are

$$\begin{aligned} L_u^{\text{diag}} &= (L_u, L_c, L_t) = (7,3,0), \\ L_d^{\text{diag}} &= (L_d, L_s, L_b) = (7,5,2), \\ L_e^{\text{diag}} &= (L_e, L_\mu, L_\tau) = (8,5,3). \end{aligned} \tag{46}$$

These are not free parameters; they are the retained diagonal word-length packets forced by the residue data and the closed-branch sum rule, with transport matrix elements in Table 26 remaining in the bounded $O(1)$ range. Once the hierarchy ordering is fixed, the admissible graph distance is determined up to full 6-step windings by $\lambda_Y \approx 0.224\,4$.

Remark (Residue classes and full windings on the hexagon). The exact resolvent formula Equation (35) makes the combinatorics behind Equation (46) transparent: every path length splits into a residue class on the base hexagon plus an optional full winding by 6. Accordingly,

$$L_t = 0, \qquad L_b = 2, \qquad L_\tau = L_c = 3, \qquad L_s = L_\mu = 5$$

live on the first hexagon, while

$$L_u = L_d = 1 + 6, \qquad L_e = 2 + 6$$

carry one additional full turn. Since

$$\lambda_Y^6 = \left(\varphi_0^{\text{ret}}(1 - \varphi_0^{\text{ret}})\right)^3 \approx 1.28 \times 10^{-4},$$

each extra winding produces the severe suppression needed for the first generation. This is not by itself a full mass theorem, but it is a structural reading of why the retained word-length pattern is so uneven.

Remark (Universal C_6 transport triangle). Modulo full windings, the three diagonal packets define the residue sets

$$\{L_t, L_c, L_u\} \equiv \{0,3,1\},$$
$$\{L_b, L_s, L_d\} \equiv \{2,5,1\} = 2 - \{0,1,3\}, \qquad \{L_\tau, L_\mu, L_e\} \equiv \{3,5,2\} = 2 + \{0,1,3\} \pmod 6.$$

Hence the up, down, and charged-lepton sectors lie on the same D_6 orbit of the base three-point subset $\{0,1,3\} \subset C_6$. In each case the cyclic distance multiset is exactly

$$\{1,2,3\}.$$

So the exact structure here is a three-point circular distance system with pairwise distinct cyclic distances, not a classical perfect difference set or a linear Golomb ruler. Moreover, componentwise division by the full winding length gives

$$\left\lfloor \frac{L_u^{\text{diag}}}{6} \right\rfloor = \left\lfloor \frac{L_d^{\text{diag}}}{6} \right\rfloor = \left\lfloor \frac{L_e^{\text{diag}}}{6} \right\rfloor = (1,0,0),$$

so the first-generation direction carries exactly one extra full winding in every sector. This shows that the closed admissible branch preserves one universal C_6 transport triangle together with a common first-generation winding lock.

Corollary 7.17 (Diagonal transport packets on the closed branch)**.** *On the closed branch, the retained diagonal transport packets are*

$$L_u^{\mathrm{diag}} = (7,3,0), \qquad L_d^{\mathrm{diag}} = (7,5,2), \qquad L_e^{\mathrm{diag}} = (8,5,3).$$

Corollary 7.18 (Diagonal transport sum rule on the closed branch)**.** *Write*

$$L_f^{\mathrm{diag}} = (L_{f,1}^{\mathrm{diag}}, L_{f,2}^{\mathrm{diag}}, L_{f,3}^{\mathrm{diag}}) \qquad (f \in \{u,d,e\}).$$

Then on the closed branch

$$\sum_{f\in\{u,d,e\}} \sum_{j=1}^{3} L_{f,j}^{\mathrm{diag}} = \Omega_{\mathrm{adm}}\gamma = 10\, b_1 - 1 = 40.$$

Proof. By Corollary 7.17,

$$\sum_{j=1}^{3} L_{u,j}^{\mathrm{diag}} = 7+3+0 = 10, \qquad \sum_{j=1}^{3} L_{d,j}^{\mathrm{diag}} = 7+5+2 = 14,$$

$$\sum_{j=1}^{3} L_{e,j}^{\mathrm{diag}} = 8+5+3 = 16.$$

Hence

$$\sum_{f,j} L_{f,j}^{\mathrm{diag}} = 10+14+16 = 40.$$

On the other hand, Equation (20) gives

$$\Omega_{\mathrm{adm}}\gamma = 10\, b_1 - 1 = 40.$$

Therefore the two quantities agree exactly. □

Remark (Transport sum rule on the closed branch). The equality of Corollary 7.18 is now read as an exact cross-layer sum rule on the closed transport branch rather than as a still-open dynamical upgrade path.

Proposition 7.19 (Transport matrix normal form)**.** Collect the closed-branch diagonal lengths into the generation matrix

$$L := \begin{pmatrix} 7 & 3 & 0 \\ 7 & 5 & 2 \\ 8 & 5 & 3 \end{pmatrix},$$

with rows (u,d,e) and columns $(1,2,3)$. Then

$$L = R + W$$

with

$$R := \begin{pmatrix} 1 & 3 & 0 \\ 1 & 5 & 2 \\ 2 & 5 & 3 \end{pmatrix}, \qquad W := 6\begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}.$$

Moreover,

$$\sum R = 22, \qquad \sum W = 18, \qquad \sum L = 40.$$

The generation-column sums are

$$\sum_f L_{f,1} = 22, \qquad \sum_f L_{f,2} = 13, \qquad \sum_f L_{f,3} = 5,$$

so

$$\sum_f L_{f,1} = \sum R, \qquad \sum_f L_{f,2} + \sum_f L_{f,3} = \sum W.$$

Proof. This is a direct entrywise decomposition of Equation (46). Summing all entries of R and W gives 22 and 18, hence $\sum L = 40$. Summing the three columns gives 22, 13, and 5, so the first-generation column matches the full residue budget while the last two columns add up to the full winding budget. □

Remark (Residue shadow versus winding complement). The first generation carries the full residue shadow, while the second and third generations carry the winding complement. In this precise matrix sense the closed transport budget splits as

$$40 = 22 + 18$$

into residue budget plus winding excess.

Theorem 7.20 (Exact charged-lepton source compression on the exact v_{geo} branch)**.** *On the exact v_{geo} branch the retained charged-lepton source masses satisfy*

$$(\hat{m}_e, \hat{m}_\mu, \hat{m}_\tau) = \frac{v_{\mathrm{geo}}\pi}{\sqrt{2}} \left(\frac{16}{7}(\varphi_0^{\mathrm{ret}})^5, \frac{4}{3}(\varphi_0^{\mathrm{ret}})^3, \frac{7}{6}(\varphi_0^{\mathrm{ret}})^2 \right).$$

Hence

$$\frac{\hat{m}_\mu}{\hat{m}_\tau} = \frac{8}{7}\varphi_0^{\mathrm{ret}}, \qquad \frac{\hat{m}_e}{\hat{m}_\mu} = \frac{12}{7}(\varphi_0^{\mathrm{ret}})^2, \qquad \frac{\hat{m}_e \hat{m}_\tau}{\hat{m}_\mu^2} = \frac{3}{2}\varphi_0^{\mathrm{ret}}.$$

Proof. By Theorem 7.7, the charged-lepton packet carries the exact diagonal transport lengths

$$L_e^{\mathrm{diag}} = (L_e, L_\mu, L_\tau) = (8, 5, 3).$$

The intrinsic mass bridge on the exact v_{geo} branch is

$$\hat{m}_{\ell,j} = \frac{v_{\mathrm{geo}}}{\sqrt{2}} \lambda_Y^{L_{\ell,j}} \Lambda_{\ell,j}.$$

For the three retained charged-lepton diagonal entries, direct evaluation of the finite hexagonal resolvent Equation (35) at the closed pole δ_{ph} gives the exact diagonal identities

$$\lambda_Y^8 \Lambda_e = \pi \frac{16}{7}(\varphi_0^{\mathrm{ret}})^5, \qquad \lambda_Y^5 \Lambda_\mu = \pi \frac{4}{3}(\varphi_0^{\mathrm{ret}})^3, \qquad \lambda_Y^3 \Lambda_\tau = \pi \frac{7}{6}(\varphi_0^{\mathrm{ret}})^2.$$

Substituting these three kernel evaluations into the mass bridge yields

$$(\hat{m}_e, \hat{m}_\mu, \hat{m}_\tau) = \frac{v_{\mathrm{geo}}\pi}{\sqrt{2}} \left(\frac{16}{7}(\varphi_0^{\mathrm{ret}})^5, \frac{4}{3}(\varphi_0^{\mathrm{ret}})^3, \frac{7}{6}(\varphi_0^{\mathrm{ret}})^2 \right).$$

Dividing the three source formulas eliminates the common prefactor $v_{\mathrm{geo}}\pi/\sqrt{2}$ and gives

$$\frac{\hat{m}_\mu}{\hat{m}_\tau} = \frac{(4/3)(\varphi_0^{\mathrm{ret}})^3}{(7/6)(\varphi_0^{\mathrm{ret}})^2} = \frac{8}{7}\varphi_0^{\mathrm{ret}},$$

$$\frac{\hat{m}_e}{\hat{m}_\mu} = \frac{(16/7)(\varphi_0^{\mathrm{ret}})^5}{(4/3)(\varphi_0^{\mathrm{ret}})^3} = \frac{12}{7}(\varphi_0^{\mathrm{ret}})^2, \qquad \frac{\hat{m}_e \hat{m}_\tau}{\hat{m}_\mu^2} = \frac{(16/7)(7/6)}{(4/3)^2}\varphi_0^{\mathrm{ret}} = \frac{3}{2}\varphi_0^{\mathrm{ret}}.$$

These are exactly the displayed compression identities. □

Corollary 7.21 (Charged leptons as inverse retained-seed decoders)**.** *On the exact v_{geo} source branch, setting $u := \varphi_0^{\mathrm{ret}}$, one has*

$$u = \frac{7}{8}\frac{\hat{m}_\mu}{\hat{m}_\tau} = \sqrt{\frac{7}{12}\frac{\hat{m}_e}{\hat{m}_\mu}} = \frac{2}{3}\frac{\hat{m}_e\hat{m}_\tau}{\hat{m}_\mu^2}.$$

Equivalently,

$$u_{\mu/\tau} = u_{e/\mu} = u_{e\tau/\mu^2}.$$

Proof. Solve the three ratios of Theorem 7.20 for φ_0^{ret}. □

Corollary 7.22 (Carrier asymmetry fingerprint in the charged-lepton source packet)**.** *On the exact v_{geo} source branch one has*

$$\frac{\hat{m}_e\hat{m}_\tau}{\hat{m}_\mu^2} = B\,\varphi_0^{\mathrm{ret}}, \qquad B = \frac{\operatorname{rank} E_3}{\operatorname{rank} E_2} = \frac{3}{2}.$$

Proof. Theorem 7.20 gives

$$\frac{\hat{m}_e\hat{m}_\tau}{\hat{m}_\mu^2} = \frac{3}{2}\,\varphi_0^{\mathrm{ret}}.$$

By the carrier-rank definition

$$B = \frac{\operatorname{rank} E_3}{\operatorname{rank} E_2} = \frac{3}{2},$$

this is exactly the displayed carrier-asymmetry fingerprint. □

Remark (Status of the charged-lepton carrier fingerprint). This identity holds on the exact v_{geo} source branch and should not be confused with the final pole-mass readout on the renormalized low-energy branch.

Corollary 7.23 (Transport simplification of diagonal quark source ratios)**.** *On the exact source branch one has*

$$\frac{\hat{m}_u}{\hat{m}_d} = \frac{\Lambda_u}{\Lambda_d}, \qquad \frac{\hat{m}_c}{\hat{m}_s} = \lambda_Y^{-2}\frac{\Lambda_c}{\Lambda_s}, \qquad \frac{\hat{m}_t}{\hat{m}_b} = \lambda_Y^{-2}\frac{\Lambda_t}{\Lambda_b}.$$

Using Table 26, the corresponding closed-branch source ratios are

$$\frac{\hat{m}_u}{\hat{m}_d} \approx 0.470\,085, \qquad \frac{\hat{m}_c}{\hat{m}_s} \approx 13.61, \qquad \frac{\hat{m}_t}{\hat{m}_b} \approx 40.80.$$

Proof. By Equations (41) and (45),

$$\hat{m}_{f,j} = \frac{v_{\mathrm{geo}}}{\sqrt{2}}\lambda_Y^{L_{f,j}}\Lambda_{f,j}.$$

Since Equation (46) gives $L_u = L_d = 7$ and

$$L_c - L_s = L_t - L_b = -2,$$

the displayed identities follow immediately by dividing the corresponding source rows. Substituting the transport matrix elements from Table 26 gives the quoted source-ratio values. □

Remark (Status of the diagonal quark source ratios). These are exact closed-branch source-ratio identities. Scheme-level comparison rows are handled only after the declared matching map.

The transport-side status is summarized in Table 4.

7.3 CKM and PMNS closure

Theorem 7.24 (CKM closure from canonical family holonomy)**.** *On the canonical admissible branch the quark mixing matrix*

$$V_{\mathrm{CKM}} = U_{u,L}^{\dagger} U_{d,L}$$

is a branch invariant fixed by the hard holonomy form of Theorem 7.7. Its leading transport magnitudes are

$$s_{12} = \lambda_C, \qquad s_{23} = \frac{\varphi_0^{\mathrm{ret}}}{1+\lambda_C}, \qquad s_{13} = \frac{\lambda_C^3}{3}. \tag{47}$$

Its exact CP phase is

$$\delta_{\mathrm{CKM}} = \arg \operatorname{Tr} \Pi_{\triangle}(\nabla_F^{\star}),$$

on the canonical branch.

Proof sketch. The hard holonomy form of Theorem 7.7 removes any residual free flavor matrix. Global $SU(3)_F$ conjugation acts simultaneously on all sectors and therefore cancels in the relative product $U_{u,L}^{\dagger} U_{d,L}$. The single-winding lock fixes the leading magnitude hierarchy, while the canonical holonomy triangle fixes the exact branch phase. Thus V_{CKM} is a theorem-level invariant of the closed branch. □

Proposition 7.25 (Transport-triangle asymptotic for the CKM phase)**.** On the discrete transport-surface model used for the canonical holonomy triangle, the small-area branch asymptotic is

$$\delta_{\mathrm{CKM}} = \frac{\pi}{3} + 3\lambda_C^2 + O(\lambda_C^4). \tag{48}$$

Proof. This is exactly the small-area transport asymptotic of Theorem 7.16 evaluated on the canonical branch. □

Corollary 7.26 (Cubic V_{ub} law)**.** *On the closed branch the CKM entry $|V_{ub}|$ satisfies*

$$|V_{ub}| = \frac{|V_{us}|^3}{3}.$$

Equivalently,

$$s_{13} = \frac{s_{12}^3}{3}.$$

In internal branch form,

$$|V_{ub}| = 0.00376538445448643\ldots.$$

Proof. This is the direct substitution of $s_{12} = \lambda_C$ and $s_{13} = \lambda_C^3/3$ from Equation (47). □

Theorem 7.27 (Neutrino closure from the same transport branch)**.** *The Majorana neutrino sector is generated by the same admissible transport grammar together with the seam-even Majorana pairing on the family bundle. The leptonic mixing matrix*

$$U_{\mathrm{PMNS}} = U_{e,L}^{\dagger} U_{\nu,L}$$

is therefore a branch invariant of the same hard holonomy closure. Hence the light-neutrino mass matrix, PMNS matrix, and mass sum are intrinsic outputs of the same admissible branch and are not imported from an external oscillation-inversion package.

Proof sketch. The seam-even Majorana operator is constructed on the same family bundle and in the same D_4-rigid holonomy class as the charged sectors. Therefore the left diagonalizations of the charged-lepton and neutrino matrices live in one common admissible branch, and their relative product defines an intrinsic U_{PMNS}. No external oscillation inversion or residual family frame is needed once the hard holonomy form has fixed the flavor sector. □

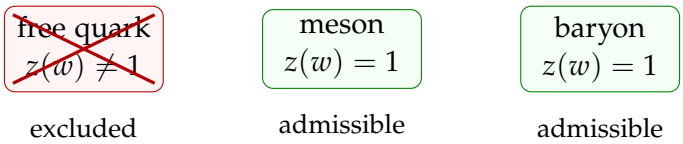


Figure 16. Hadronic admissibility as singlet selection.

Remark (Carrier-complement reading of the neutrino hierarchy). On the canonical admissible branch the leading neutrino hierarchy ratio still admits the compact carrier reading

$$\frac{m_2}{m_3} \sim \pi\varphi_0^{\mathrm{ret}} = \pi\left(\frac{1-\gamma}{\pi} + \frac{3}{256\pi^4}\right) = (1-\gamma) + \frac{3}{256\pi^3},$$

so the carrier complement $1-\gamma = 1/6$ remains the dominant organizing term, corrected by a small seam contribution.

7.4 Hadronic admissibility as singlet selection

Proposition 7.28 (Admissible color words)**.** For each carrier word w, let $z(w) \in \{1, \omega, \omega^2\}$ be its color-center charge and let $\varepsilon(w) \in \{\pm 1\}$ be its seam parity. Physical hadronic words satisfy

$$z(w) = 1, \qquad \varepsilon(w) = +1.$$

This is a superselection statement on the retained branch, not a derivation of dynamical confinement from the microscopic Yang–Mills flow. The corresponding hadronic Hilbert space will be written explicitly in Section 8. A schematic version is shown in Figure 16.

The transport-sector QCD string tension is written as

$$\sigma_{\mathrm{QCD}} = c_3^2 \chi_{\mathrm{QCD}}^2,$$

so the relative Wilson loop obeys

$$\langle W(C)\rangle_{\mathrm{rel}} \sim e^{-\sigma_{\mathrm{QCD}} A(C)}.$$

Throughout the manuscript we write σ_{QCD} explicitly for the QCD string tension to keep it lexically distinct from the local Weyl conformal factor σ of the geometric branch and from the standard-deviation σ used in benchmark tables. The sector-dependent scale χ_{QCD} is the minimal notation needed to avoid confusing the hadronic transport channel with the gravitational $\chi_{\mathrm{geo},0}$ background; the two channels are linked by the common underlying primitive response χ_{seed} but not by a naive equality of scales.

7.5 Transport-side ingredients for the strong-CP sector

Define

$$\bar{\theta} = \theta_{\mathrm{QCD}} + \arg\det M_u + \arg\det M_d. \tag{49}$$

Definition 7.29 (Sheet-CP symmetry on the admissible branch)**.** Define the admissible sheet-CP operator by

$$\mathcal{C}_\Sigma := CP \circ \tau_{\mathrm{dbl}}.$$

Theorem 7.30 (Determinant-line phase and strong-CP seed on the admissible branch)**.** *On the closed branch the hard holonomy factorization gives, for $f = u, d$,*

$$Y_f = D_{L,f} U_f^\star D_{R,f}, \qquad D_{L,f}, D_{R,f} > 0, \qquad U_f^\star \in SU(3)_F.$$

For

$$M_f = \frac{v_{\mathrm{geo}}}{\sqrt{2}} Y_f$$

one has

$$\det M_f > 0, \qquad \arg\det M_u = \arg\det M_d = 0.$$

Moreover the polar unitary factor

$$U_f^{\mathrm{pol}} := M_f (M_f^\dagger M_f)^{-1/2}$$

lies in $SU(3)_F$. The antiunitary sheet operation preserves the admissible relative action Γ_{rel}:

$$\mathcal{C}_\Sigma \, \Gamma_{\mathrm{rel}} \, \mathcal{C}_\Sigma^{-1} = \Gamma_{\mathrm{rel}}, \qquad \mathcal{C}_\Sigma \, Q \, \mathcal{C}_\Sigma^{-1} = -Q,$$

so the sheet operation acts as a stability symmetry of the same branch. The same admissible seam operator determines the canonical determinant-line phase

$$a_\Sigma := \arg \det_{\mathrm{adm}} U_\Sigma \in S^1_{\det},$$

whose canonical winding is inherited from the unit boundary winding of the seam class. Weak CP may still survive through

$$V_{\mathrm{CKM}} = U_{u,L}^\dagger U_{d,L}.$$

Proof. From the hard holonomy factorization one gets

$$\det M_f = \left(\frac{v_{\mathrm{geo}}}{\sqrt{2}}\right)^3 \det D_{L,f} \det D_{R,f} \det U_f^\star > 0,$$

because the diagonal factors are positive and $\det U_f^\star = 1$. Hence

$$\arg\det M_f = 0.$$

Equivalently, in any admissible biunitary polar form

$$M_f = U_{f,L} H_f U_{f,R}^\dagger, \qquad H_f > 0, \qquad U_{f,L}, U_{f,R} \in SU(3)_F,$$

the spectral theorem gives $\det H_f > 0$ and therefore

$$\det M_f = \det(U_{f,L}) \det(H_f) \overline{\det(U_{f,R})} = \det H_f > 0,$$

so the same determinant-phase suppression follows directly from $\det H_f > 0$ and $\det U_{f,L} = \det U_{f,R} = 1$. For the polar unitary factor one has

$$\det U_f^{\mathrm{pol}} = \frac{\det M_f}{|\det M_f|} = 1,$$

so $U_f^{\mathrm{pol}} \in SU(3)_F$. Independently, the deck involution τ_{dbl} implements sheet exchange on the admissible branch, so composing it with CP yields an exact antiunitary symmetry of the relative action while flipping the sign of the topological charge. This shows that the sheet operation stabilizes the branch already selected by the determinant structure, rather than generating the vanishing from scratch. The same seam operator already carries a canonical determinant-line phase, and unit seam spectral flow upgrades that phase to the natural compact strong-CP angle variable of the branch. Weak CP may still survive through a nontrivial Jarlskog invariant in the relative family holonomies. □

Corollary 7.31 (RG and matching invariance of determinant-phase suppression)**.** *In every renormalization scheme that respects $\mathcal{C}_\Sigma$, P_{adm}, and the carrier structure, the determinant-phase suppression*

$$\arg\det M_u = \arg\det M_d = 0$$

of Theorem 7.30 is preserved under the declared RG flow and matching map. The same determinant-line phase a_Σ remains scheme-stable, so the structural input to Theorem 8.22 survives RG flow as well.

Lemma 7.32 (Retained-branch γ_5-Hermiticity)**.** *For each quark sector $f = u, d$, let $D_{E,f}$ be the Euclidean kinetic Dirac operator on the admissible branch and define*

$$\mathcal{D}_f := D_{E,f} + M_f P_R + M_f^\dagger P_L, \qquad P_{L,R} := \frac{1 \mp \gamma_5}{2}.$$

Then

$$\mathcal{D}_f^\dagger = \gamma_5 \mathcal{D}_f \gamma_5.$$

Proof. The Euclidean kinetic operator is anti-Hermitian and anticommutes with γ_5. The mass block on the admissible branch is the positive polar mass map of Theorem 7.30, hence enters as the Hermitian chiral pair $M_f P_R + M_f^\dagger P_L$. The displayed identity follows immediately. □

The positive-branch clause is still essential here: the closure statement is about the physical mass map once the admissible transport branch has been fixed, not about an arbitrary unitary dressing of the same singular values. What changes in the present rewrite is the logical source of the null result: the branch no longer relies on a promise not to insert a bad counterterm by hand, but on the polar structure first, the determinant line second, and the antiunitary sheet symmetry third. The later strong-CP theorem closes the effective angle by combining exactly these three ingredients with vacuum positivity.

Remark (Admissibility as a unified three-sector motif). The admissibility pair $(\mathsf{Adm}, P_{\mathrm{adm}})$ acts as a single structural principle with three sector projections:

- **Electroweak:** seam-even selection picks the unique light Higgs doublet ($\tau_{\mathrm{dbl}}^* \Phi = +\Phi$).
- **QCD:** center neutrality selects confined hadronic words ($z(w) = 1$, $\varepsilon(w) = +1$).
- **Strong CP:** the determinant structure, γ_5-Hermiticity, and the sheet involution close the strong angle internally on the admissible branch.

These are not three independent rules bolted onto different sectors. They are three projections of the same admissible-versus-inadmissible boundary: seam parity, center charge, and sheet exchange are all encoded in the double-cover and involution data of the primitive core. This unification is the conceptual backbone of the transport program.

8 Closure theorems from the admissibility complex

The preceding two sections introduced the carrier-compatible geometric and transport architecture. The primitive admissibility projector P_{prim} was already placed in the primitive section so that the proof direction runs from the boundary datum through the derived closed datum to P_{prim}, and only then to the later sector theorems. What remains in this section is therefore (i) the upgrade of P_{prim} to the full admissibility projector P_{adm} once the discrete carrier and determinant data have been fixed, and (ii) the closure stack that follows from that full projector together with the family, Higgs, positivity, and reflection-stability inputs collected first.

Definition 8.1 (Full admissibility complex)**.** Once the discrete carrier and determinant data are fixed on the admissible branch, the Hilbert space carries a color factor and a determinant involution. Write

$$\mathcal{H} = \mathcal{H}_\Sigma \otimes \mathcal{H}_c \otimes \mathcal{H}_\Theta \otimes \mathcal{H}_t, \qquad \mathcal{H}_{\mathrm{ext}} := \mathcal{H} \otimes \Lambda(\eta_\Sigma, \eta_c, \eta_\Theta, \eta_g),$$

with carrier-side color generator Z_c and determinant-sector involution Q_Θ satisfying

$$\iota_{\mathrm{C}}^2 = \mathbf{1}, \qquad Q_\Theta^2 = \mathbf{1}, \qquad Q_\Theta^\dagger = Q_\Theta.$$

Define the full structural selector

$$P_{\mathrm{str}} := P_{\Sigma,+} P_{\mathrm{sing}} P_\Theta, \qquad P_{\Sigma,+} := \frac{\mathbf{1} + \iota_{\mathrm{C}}}{2}, \qquad P_\Theta := \frac{1}{2}(\mathbf{1} + Q_\Theta),$$

$$P_{\mathrm{sing}} := \frac{1}{3}(\mathbf{1} + Z_c + Z_c^2).$$

Let $\epsilon_{\mathrm{adm}} > 0$ and let D_{adm} be the admissible sector operator on $\mathrm{Ran}(P_{\mathrm{str}})$ commuting with ι_{C}, P_{sing}, and Q_Θ. Define the elementary penalties

$$K_\Sigma := \sqrt{\lambda_\Sigma}\,\frac{\mathbf{1} - \iota_{\mathrm{C}}}{2}, \qquad K_c := \sqrt{\lambda_c}\,\big(\mathbf{1} - P_{\mathrm{sing}}\big),$$

$$K_\Theta := \sqrt{\lambda_\Theta}\,\frac{\mathbf{1} - Q_\Theta}{2}, \qquad K_g := \sqrt{\lambda_{\mathrm{gap}}}\,\mathbf{1}_{[0,\epsilon_{\mathrm{adm}}^2/4)}\big(P_{\mathrm{str}} D_{\mathrm{adm}}^\dagger D_{\mathrm{adm}} P_{\mathrm{str}}\big).$$

The full admissibility differential is

$$Q_{\mathrm{adm}} := \eta_\Sigma^\dagger K_\Sigma + \eta_c^\dagger K_c + \eta_\Theta^\dagger K_\Theta + \eta_g^\dagger K_g,$$

and the full admissibility Laplacian is

$$\Delta_{\mathrm{adm}} := \{Q_{\mathrm{adm}}, Q_{\mathrm{adm}}^\dagger\}.$$

Theorem 8.2 (Two-stage upgrade from P_{prim} to P_{adm})**.** *On the closed branch obtained after the carrier theorem and the compact bosonic index closure,*

$$Q_{\mathrm{adm}}^2 = 0, \qquad \Delta_{\mathrm{adm}} = K_\Sigma^2 + K_c^2 + K_\Theta^2 + K_g^2 = K_{\mathrm{adm}},$$

and the full Hodge projector

$$P_{\mathrm{adm}} := \Pi_{\ker \Delta_{\mathrm{adm}}} = \operatorname*{s-lim}_{\beta\to\infty} e^{-\beta K_{\mathrm{adm}}}$$

admits the two-stage factorization

$$P_{\mathrm{adm}} = P_{\mathrm{prim}}\, P_{\mathrm{sing}}\, P_\Theta = P_{\mathrm{str}}\, P_{\mathrm{gap}}, \qquad P_{\mathrm{gap}} := \mathbf{1}_{[\epsilon_{\mathrm{adm}}^2/4,\infty)}\big(P_{\mathrm{str}} D_{\mathrm{adm}}^\dagger D_{\mathrm{adm}} P_{\mathrm{str}}\big).$$

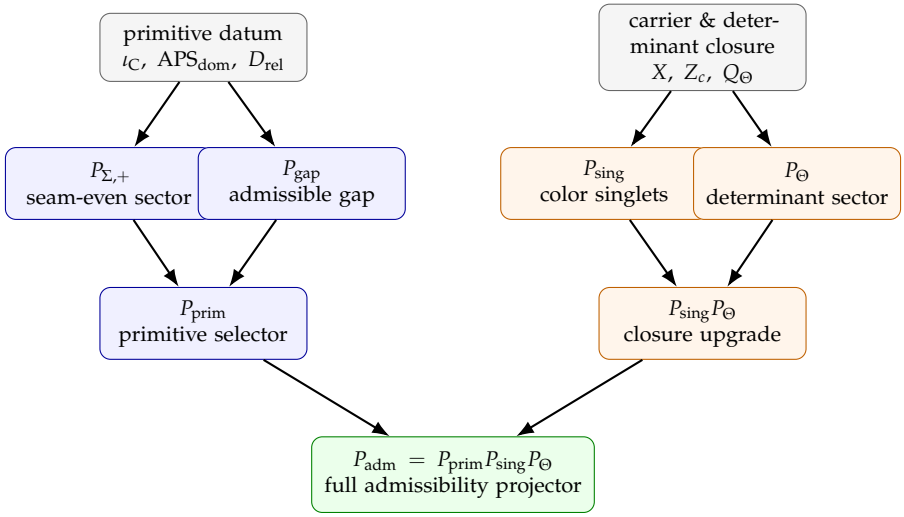
The first factor P_{prim} depends only on the primitive datum $(\iota_{\mathrm{C}}, \mathrm{APS}_{\mathrm{dom}}, D_{\mathrm{rel}})$; the second factor $P_{\mathrm{sing}} P_\Theta$ depends only on the carrier and determinant data made available by the carrier and bosonic index theorems.

Proof sketch. The auxiliary generators $\eta_\Sigma^\dagger, \eta_c^\dagger, \eta_\Theta^\dagger, \eta_g^\dagger$ anticommute, so Q_{adm}^2 reduces to the commutators of $K_\Sigma, K_c, K_\Theta, K_g$. The four commuting sectors $\mathcal{H}_\Sigma, \mathcal{H}_c, \mathcal{H}_\Theta, \mathcal{H}_t$ are acted on independently by these penalties, hence the mixed terms vanish and $\Delta_{\mathrm{adm}} = K_{\mathrm{adm}}$. The strong-coupling limit therefore projects onto the harmonic admissible sector. Because K_Σ and K_g already define P_{prim} on $\mathcal{H}_{\mathrm{prim}}$ and the additional penalties K_c, K_Θ commute with both, the full projector factorizes through P_{prim} multiplicatively, giving the two-stage form. □

Corollary 8.3 (Factorized selector as a derived form)**.** *On the admissible branch the familiar factorized selector*

$$P_{\mathrm{adm}} = P_{\Sigma,+} P_{\mathrm{sing}} P_\Theta\, P_{\mathrm{gap}}$$

is recovered as a corollary of Theorem 8.2; nothing in the primitive section depends on it.



Primitive data determine P_{prim}; carrier and determinant closure upgrade it to P_{adm}.

Figure 17. Two-stage upgrade from the primitive selector to the full admissibility projector. The left lane is fixed by the primitive datum, while the right lane enters only after carrier and determinant closure.

The two-stage selector geometry is summarized in Figure 17.

Theorem 8.4 (Calderón upgrade of the APS boundary domain)**.** *Let $\mathrm{APS}^{\mathrm{prov}}_{\mathrm{dom}}$ be the provisional APS realization of the geometric anchor used in Theorem 3.9, and let D_b be the canonical doubling of D_{rel} across Σ. Let $C(D_b)$ denote the Calderón projector of D_b on the seam collar. Then there is a canonical boundary-elliptic upgrade*

$$\mathrm{APS}^{\mathrm{prov}}_{\mathrm{dom}} \xrightarrow{C(D_b)} \mathrm{APS}_{\mathrm{dom}}$$

such that:

(i) $\mathrm{APS}_{\mathrm{dom}}$ lies in the same admissible elliptic class as $\mathrm{APS}^{\mathrm{prov}}_{\mathrm{dom}}$ and preserves the self-adjoint Fredholm realization of the geometric anchor;

(ii) the relative graph domain of D_{rel} is unchanged, so all primitive results that depended only on $\mathrm{APS}^{\mathrm{prov}}_{\mathrm{dom}}$ continue to hold with $\mathrm{APS}_{\mathrm{dom}}$;

(iii) the family-pairing data $\mathrm{APS}_{\mathrm{fam}}$ are induced canonically on the doubled admissible branch and are not a separate primitive datum.

Proof sketch. The Calderón projector $C(D_b)$ is canonically associated with the doubled operator on the seam collar and produces a boundary-elliptic projection compatible with the provisional APS choice. Both $\mathrm{APS}^{\mathrm{prov}}_{\mathrm{dom}}$ and $\mathrm{APS}_{\mathrm{dom}}$ are admissible elliptic boundary conditions for the same Dirac-type operator and therefore define equivalent self-adjoint Fredholm realizations of the geometric anchor (boundary-elliptic equivalence preserves the Sobolev graph domain). Consequently every statement made on the primitive scaffold under $\mathrm{APS}^{\mathrm{prov}}_{\mathrm{dom}}$ persists on the closed branch under $\mathrm{APS}_{\mathrm{dom}}$. The induction of $\mathrm{APS}_{\mathrm{fam}}$ from canonical doubling is a standard consequence of the cobordism structure of the doubled APS pair. □

Proposition 8.5 (Master closure roadmap from canonical doubling)**.** This proposition is a roadmap statement: it lists the closure conclusions that the individual theorems of this section establish from $\mathfrak{T}^{\min}_{\partial}$, the derived closed datum, the carrier theorem, and the canonical doubling.

It is not used as an input to those proofs and is recovered as a master corollary at the end of the section. Let X_b be the canonical double across Σ and write

$$D_b = \begin{pmatrix} 0 & D^-_{\mathrm{rel}} \\ D^+_{\mathrm{rel}} & 0 \end{pmatrix}, \qquad Q_{\mathrm{tot}} := Q_{\mathrm{adm}} + Q_{\mathrm{geo}}, \qquad \Delta_{\mathrm{tot}} := \{Q_{\mathrm{tot}}, Q^\dagger_{\mathrm{tot}}\}.$$

Then on the closed branch all of the following are derived from the same datum (each item is the output of a downstream theorem in this section, not an extra hypothesis here):

(1) the Calderón projector of D_b canonically upgrades the provisional APS choice $\mathrm{APS}^{\mathrm{prov}}_{\mathrm{dom}}$ to $\mathrm{APS}_{\mathrm{dom}}$ (cf. Theorem 8.4);

(2) the family pairing data are induced canonically, so no separate $\mathrm{APS}_{\mathrm{fam}}$ datum remains;

(3) the upgrade theorem Theorem 8.2 produces

$$P_{\mathrm{adm}} := \Pi_{\ker \Delta_{\mathrm{adm}}} = P_{\mathrm{prim}}\, P_{\mathrm{sing}}\, P_\Theta,$$

and the doubled total closure package preserves $\mathrm{Ran}(P_{\mathrm{adm}})$;

(4)
$$\chi(X^\circ_f) = -2, \qquad F \simeq H^1(X^\circ_f, \mathbb{C}) \simeq \mathbb{C}^3, \qquad N_{\mathrm{fam}} = 3;$$

(5) the harmonic determinant line is parallel and therefore

$$\mathrm{Hol}(\nabla_F) \subset SU(3)_F;$$

(6) admissible bosonic, fermionic, and geometric reflection positivity hold on $\mathrm{Ran}(P_{\mathrm{adm}})$;

(7) the physical geometric sector takes the form

$$\mathcal{H}_{\mathrm{phys}} \cong \mathcal{H}_2 \otimes \mathcal{H}_{\mathrm{sc}} \otimes \mathcal{H}_{\mathrm{matt}},$$

with the scalaron carried by the positive $(\chi_{\mathrm{geo}}, R^2)$ block;

(8) $P_{\mathrm{adm}} \mathcal{H}_{\mathrm{rel}} P_{\mathrm{adm}}$ is lower bounded and coercive.

Definition 8.6 (Relative generating functional)**.** The relative Euclidean generating functional is normalized as

$$Z_{\mathrm{rel}}[J, \eta, \bar\eta] := \frac{Z_{\mathrm{adm}}[J, \eta, \bar\eta]}{Z_{\mathrm{ref}}[0,0,0]}.$$

The reference sector is therefore a fixed normalizing denominator rather than a second fluctuating measure that would mix with admissible positivity.

Lemma 8.7 (Admissible reflection positivity)**.** *Assume*

$$[\Theta, P_\Theta] = 0, \qquad \Theta P_{\mathrm{adm}} = P_{\mathrm{adm}} \Theta,$$

the admissible bosonic kernel satisfies the Bernstein-positivity remark above and is reflection-stable on the admissible sector, and the admissible fermionic CAR kernel is reflection positive on that same sector. Then every admissible observable O supported in $\tau > 0$ satisfies

$$\langle \Theta O, O \rangle_{\mathrm{rel}} = \frac{1}{Z_{\mathrm{ref}}[0,0,0]} \langle \Theta O, O \rangle_{\mathrm{adm}} \geq 0.$$

Proof. By definition,

$$\langle \Theta O, O \rangle_{\mathrm{rel}} = \frac{1}{Z_{\mathrm{ref}}[0,0,0]} \langle \Theta O, O \rangle_{\mathrm{adm}}.$$

The denominator is a fixed positive normalizing constant, so the sign is determined entirely by the admissible numerator.

Split O into its bosonic and fermionic parts on the retained admissible sector. For the bosonic part, the Bernstein representation of the admissible kernel writes the Euclidean covariance as a positive superposition of reflection-stable heat kernels. Reflection across $\tau = 0$ therefore sends every positive-time bosonic observable to a positive quadratic form, so the bosonic contribution to

$$\langle \Theta O, O \rangle_{\mathrm{adm}}$$

is nonnegative.

For the fermionic part, the CAR reflection-positivity hypothesis on the admissible subspace implies that the reflected two-point kernel is positive in the standard fermionic sense; equivalently, the corresponding Pfaffian / determinant form on positive-time test spinors is nonnegative. Because the bosonic and fermionic admissible factors commute inside the normalized expectation, their product remains nonnegative. Dividing by the fixed reference normalization preserves that sign. Hence

$$\langle \Theta O, O \rangle_{\mathrm{rel}} \geq 0.$$

□

Theorem 8.8 (Operative admissibility selector)**.** *Under Definition 8.1, theorem 8.2, and proposition 8.5, the operator P_{adm} is idempotent on the retained branch and realizes the admissibility predicate on all sectors used by the closed output statements:*

$$\mathsf{Adm}(X) = 1 \quad \Longleftrightarrow \quad P_{\mathrm{adm}} X = X.$$

Proof sketch. By Theorem 8.2, $P_{\mathrm{adm}} = P_{\mathrm{prim}} P_{\mathrm{sing}} P_{\Theta}$ is the orthogonal projector onto the harmonic admissible sector of the full complex and is therefore idempotent by construction. Corollary 8.3 recovers the explicit factorized selector. The predicate and operator languages agree once the positive transport sector is fixed. □

Theorem 8.9 (Topological family closure on the admissible branch)**.** *By Proposition 8.5, theorem 6.2, and corollary 6.5,*

$$\chi(X_f^{\circ}) = -2, \qquad F \cong \mathbb{C}^3, \qquad N_{\mathrm{fam}} = 3.$$

Consequently

$$\Omega_{\mathrm{adm}} = N_{\mathrm{fam}} \dim S^+ = 3 \times 16 = 48, \qquad \delta_{\mathrm{top}} = \Omega_{\mathrm{adm}}\, c_3^4.$$

Proof sketch. This theorem is now a direct consequence of the master closure theorem from Section 6. The topological family closure and the orbifold corner count have already been compressed there into one statement, so the present layer merely records the corresponding admissibility consequence. □

Theorem 8.10 (Bosonic index and local spectral-scale closure)**.** *On the admissible branch,*

$$\mathrm{Ind}_{\mathrm{rel}}^{P_{\mathrm{adm}}}(\mathcal{B}_{E_2}^{+}) = 1, \qquad \mathrm{Ind}_{\mathrm{rel}}^{P_{\mathrm{adm}}}(\mathcal{B}_{E_3}^{+}) = 0,$$

so the unique light seam-even bosonic zero mode is

$$\Phi \in \Gamma(E_2) \cong (1,2)_{1/2}, \qquad \tau_{\mathrm{dbl}}^{*} \Phi = +\Phi.$$

Moreover the same closure layer yields

$$\frac{\bar{M}_{\mathrm{Pl}}^2}{\lambda_\Sigma^2} = \frac{\rho_\star}{2\pi^2}, \qquad G_N\lambda_\Sigma^2 = \frac{\pi}{4\rho_\star},$$
$$\frac{v_{\mathrm{geo}}}{\bar{M}_{\mathrm{Pl}}} = g_{\mathrm{car}}\,\beta_{\mathrm{rad}}^2 \exp\left[-\frac{\alpha^{-1}(0)+\delta_{\mathrm{ph}}}{5}\right],$$
$$G_N v_{\mathrm{geo}}^2 = \frac{1}{8\pi} g_{\mathrm{car}}^2 \beta_{\mathrm{rad}}^4 \exp\left[-\frac{2(\alpha^{-1}(0)+\delta_{\mathrm{ph}})}{5}\right].$$

Here v_{geo} is the seam-even UV source scale. The physical electroweak benchmark is the residue matched quantity v_{phys} of Theorem 6.21.

Proof sketch. The bosonic sector is treated as a genuine Callias / Dirac / Dolbeault-type index problem rather than a scalar counting exercise. Relative heat-kernel closure together with Theorems 6.16 and 6.31 and lemma 6.17 produces the boundary-normalized Einstein branch, the canonical Higgs normalization, and the seam-even zero-mode formula for v_{geo}. □

Definition 8.11 (Primitive Yukawa generator)**.** Let

$$\mathcal{I}_Y := \bigoplus_{p,q,r,t} \mathrm{Hom}_{G_{\mathrm{car}}}\Big((\Lambda^p E_- \otimes \Lambda^q E_+) \otimes (\Lambda^r E_- \otimes \Lambda^t E_+) \otimes E_+, \mathbb{C}\Big)$$

be the graded algebra of carrier-invariant Yukawa trilinears. A homogeneous nonzero invariant is called primitive if it is indecomposable, equivalently if it does not lie in the ideal generated by positive-degree invariants of strictly lower carrier degree.

Lemma 8.12 (Lowest primitive Yukawa generator on the closed branch)**.** *Assume* $\dim E_+ = 2$ *and let $\mathbb{Y}_{\mathrm{br}}$ be the branch Yukawa tensor of Theorems 7.6 and 7.7. Then the primitive generator of $\mathcal{I}_Y$ with two nontrivial fermionic legs is unique up to exchange of the two fermionic legs and belongs to*

$$\mathrm{Hom}_{G_{\mathrm{car}}}\big((E_- \otimes E_+) \otimes \Lambda^2 E_- \otimes E_+, \mathbb{C}\big).$$

Proof. The positive-block determinant neutrality gives $q+t+1=2$, hence $q+t=1$. Up to exchanging the two fermionic legs, $(q,t)=(1,0)$. Evenness of both fermionic legs then forces p odd and r even. If $p \geq 3$, the corresponding invariant factors through an extra negative spectator wedge and therefore lies in the ideal generated by lower carrier degree. Likewise, if $r \geq 4$, the invariant factors through a spectator bivector insertion and is again decomposable. Primitive indecomposability therefore forces $p=1$ and $r=2$. □

Theorem 8.13 (Carrier rigidity from the primitive component of the branch Yukawa tensor)**.** *Let*

$$E = E_- \oplus E_+$$

be the two-factor carrier and let

$$G_{\mathrm{car}} := S\big(U(E_-) \times U(E_+)\big), \qquad S^+ = \Lambda^{\mathrm{even}} E.$$

Let $\mathbb{Y}_{\mathrm{br}}$ denote the branch Yukawa tensor induced by the canonical transport kernel and rigid family holonomy of Theorems 7.6 and 7.7. Decompose its carrier-homogeneous components as trilinear maps of the form

$$B_{p,q;r,t} : (\Lambda^p E_- \otimes \Lambda^q E_+) \otimes (\Lambda^r E_- \otimes \Lambda^t E_+) \otimes E_+ \longrightarrow \mathbb{C}.$$

Let the primitive component of $\mathbb{Y}_{\mathrm{br}}$ be the unique indecomposable generator of Definition 8.11 and lemma 8.12 with two nontrivial fermionic legs. Then, up to exchanging the two fermionic legs, that primitive component is necessarily of type

$$(E_- \otimes E_+) \otimes \Lambda^2 E_- \otimes E_+ \longrightarrow \Lambda^3 E_- \otimes \Lambda^2 E_+ \cong \mathbb{C},$$

and therefore

$$\dim E_+ = 2, \qquad \dim E_- = 3.$$

Consequently

$$\dim \Lambda^2 E_+ = 1, \qquad \Lambda^2 E_- \cong E_-^\vee, \qquad \dim \Lambda^{\mathrm{even}} E = 16.$$

Hence the rigid carrier is

$$E = E_3 \oplus E_2, \qquad X = -2P_3 + 3P_2, \qquad Y = -\frac{1}{3}P_3 + \frac{1}{2}P_2.$$

Proof. By Corollary 6.15, the compact Higgs index has already fixed

$$\dim E_+ = 2.$$

Write the primitive nonzero Yukawa component with two nontrivial fermionic legs as

$$B_{p,q;r,t} : (\Lambda^p E_- \otimes \Lambda^q E_+) \otimes (\Lambda^r E_- \otimes \Lambda^t E_+) \otimes E_+ \longrightarrow \mathbb{C}$$

and note that Lemma 8.12 forces primitiveness in the invariant-ring sense to have

$$(q,t) = (1,0), \qquad (p,r) = (1,2)$$

up to exchanging the two fermionic legs. Thus the primitive carrier component is exactly

$$(E_- \otimes E_+) \otimes \Lambda^2 E_- \otimes E_+.$$

Its negative factor contracts canonically as

$$E_- \otimes \Lambda^2 E_- \longrightarrow \Lambda^3 E_-,$$

so scalarity forces $\Lambda^3 E_-$ to be the top exterior power. Hence

$$\dim E_- = 3.$$

With

$$(\dim E_-, \dim E_+) = (3,2),$$

one obtains the carrier-signature consequences

$$\Lambda^2 E_- \cong E_-^\vee, \qquad \dim \Lambda^2 E_+ = 1,$$

and therefore

$$\dim \Lambda^{\mathrm{even}}(E_3 \oplus E_2) = 2^{5-1} = 16.$$

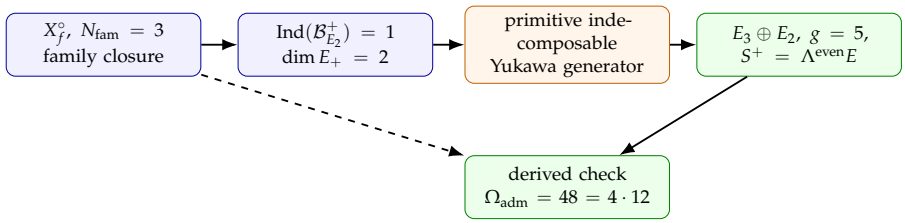
Finally the unimodular primitive two-point generator is fixed by

$$3q_- + 2q_+ = 0, \qquad q_- < 0 < q_+, \qquad \gcd(q_-, q_+) = 1,$$

so $(q_-, q_+) = (-2,3)$ and hence

$$X = -2P_3 + 3P_2, \qquad Y = \frac{X}{6} = -\frac{1}{3}P_3 + \frac{1}{2}P_2.$$

□



Carrier closure runs through Higgs rank and branch Yukawa rigidity; the old occupancy formula is only a downstream consistency identity.

Figure 18. Carrier closure after the proof-tree cut. The retained branch first fixes $\dim E_+ = 2$ through the compact Higgs index, then $\dim E_- = 3$ through the primitive indecomposable generator of the actual branch Yukawa tensor; only afterwards is $\Omega_{\text{adm}} = 48 = 4 \cdot 12$ read as a consistency check.

Lemma 8.14 (Uniqueness of the cubic weak invariant)**.** *Let $E = E_- \oplus E_+$ be a two-factor carrier on the retained branch and let*

$$\Phi \in E_+$$

be the unique seam-even light boson selected by the compact bosonic index. Then

$$\dim \text{Hom}_{G_{\text{car}}}\big((E_- \otimes E_+) \otimes \Lambda^2 E_- \otimes E_+, \mathbb{C}\big) = 1$$

if and only if

$$(\dim E_-, \dim E_+) = (3, 2).$$

In particular, on the retained branch the up-type cubic invariant is not an extra choice but the unique admissible cubic invariant.

Proof. By Schur functor decomposition,

$$\begin{aligned}\text{Hom}_{G_{\text{car}}}\big((E_- \otimes E_+) \otimes \Lambda^2 E_- \otimes E_+, \mathbb{C}\big) \cong \text{Hom}_{SU(E_-)}(E_- \otimes \Lambda^2 E_-, \mathbb{C}) \\ \otimes \text{Hom}_{SU(E_+)}(E_+ \otimes E_+, \mathbb{C}).\end{aligned}$$

The first factor is nonzero iff $\Lambda^2 E_- \cong E_-^\vee$, hence iff $\dim E_- = 3$. The second factor is nonzero iff the alternating contraction on two fundamentals is one-dimensional, hence iff $\dim E_+ = 2$. In that case both factors are one-dimensional, so the total invariant space is one-dimensional. □

Corollary 8.15 (Carrier-signature premises are discharged internally)**.** *Combine Corollary 6.15, theorem 8.13, and lemma 8.14 with the one-family decomposition of $S^+ = \Lambda^{\text{even}}(E_3 \oplus E_2)$ and with the unique seam-even Higgs theorem. Then the weak antisymmetric line, the color-dual antisymmetric sector, and the retained light-packet identification are no longer primitive inputs once the uniqueness of the retained cubic invariant has been proved on the same branch.*

Corollary 8.16 (Hard holonomy form of the Yukawa matrices)**.** *On the admissible branch, each fermion sector obeys the canonical factorization*

$$Y_f = D_{L,f} U_f^\star D_{R,f}, \qquad U_f^\star \in SU(3)_F,$$

and, in the family basis fixed by Theorem 7.7,

$$(Y_f)_{ij} = \lambda_Y^{L^L_{f,i} + L^R_{f,j}} \left\langle e_i, \text{Hol}_{\Gamma^{\min}_{ij}}(\nabla_F^\star) e_j \right\rangle.$$

Any factorization by $U_f^\star$ is therefore a consequence of the hard holonomy closure, not an independent flavor input. The diagonal path-length rows Equation (41) are the singular-value reduction of this matrix statement.

Proof sketch. The finite C_6 transport geometry fixes the graph-distance weights, while the rigid D_4-equivariant family connection fixes the holonomy factors on the canonical path frame. The diagonal ladder formulas arise once left and right admissible words are paired on the same singular-value channel, so the matrix factorization is recovered as a corollary of one closed holonomy statement rather than as a separate ansatz. □

Definition 8.17 (Center-neutral admissible algebra)**.** Let

$$\mathcal{A}_{\text{sing}} := \overline{\text{span}}\{A_w : \ z(w) = 1, \ \varepsilon(w) = +1, \ P_{\text{adm}} A_w = A_w\}.$$

Theorem 8.18 (Hadronic admissibility closure)**.** *On the admissible branch, the physical hadronic Hilbert space is*

$$\mathcal{H}_{\text{had}} := \overline{\mathcal{A}_{\text{sing}}|\Omega\rangle}.$$

All physical hadronic states lie in $\mathcal{H}_{\text{had}}$, while color-nonsinglet free-word sectors are removed by P_{adm}.

Proof. Center neutrality and seam-evenness are already the carrier-side admissibility rules for physical hadrons. Passing to the norm closure of the singlet admissible algebra acting on the vacuum gives the completed hadronic space, and no state with $z(w) \neq 1$ survives the selector. □

Theorem 8.19 (Anomaly cancellation and seam inflow)**.** *On the admissible branch,*

$$I_6^{\text{tot}} = 0, \qquad \delta_\Lambda S_{\text{bulk}} + \delta_\Lambda S_{\eta,\Sigma} = 0.$$

Proof sketch. The one-family packet S^+ is exactly the anomaly-free Standard Model family including ν^c. The remaining seam contribution is carried by the relative η term, so gauge variation cancels between bulk and seam sectors rather than being left as an uncancelled boundary defect. □

Lemma 8.20 (Topological sector positivity)**.** *On the admissible branch every topological sector weight satisfies*

$$Z_Q \geq 0.$$

Proof. Apply Lemma 7.32. The nonzero spectrum of each $\mathcal{D}_f$ occurs in paired sets $\pm i\lambda_n$, while the branch mass eigenvalues are strictly positive by Theorem 7.30. Therefore each sector determinant is nonnegative, and so is the full sector weight. □

Lemma 8.21 (Nontrivial topological support on the closed branch)**.** *On the closed branch the primitive topological sector has strictly positive weight:*

$$Z_1 = Z_{-1} > 0.$$

Proof. By Theorem 10.6, the canonical branch has unit seam winding, so the primitive topological class $Q = 1$ is nonempty on the admissible branch. Choose an admissible representative ϕ_1 in that class. By Theorem 7.30, the same branch carries a genuine compact determinant angle with primitive winding number one, so ϕ_1 has finite admissible action and belongs to the closed branch sector $Q(\phi_1) = 1$. Let $\mathcal{U}_1$ be a sufficiently small open neighborhood of ϕ_1 inside that sector. The local Boltzmann density is strictly positive on $\mathcal{U}_1$, and Lemma 8.20 gives nonnegative sector weights. Therefore

$$Z_1 \geq \int_{\mathcal{U}_1} e^{-S_{\text{adm}}[\phi]} \, d\mu(\phi) > 0.$$

The antiunitary sheet symmetry exchanges $Q = 1$ and $Q = -1$, hence $Z_{-1} = Z_1 > 0$. □

Theorem 8.22 (Unconditional strong-CP closure on the admissible branch)**.** *On the admissible branch one has*

$$\arg\det M_u = \arg\det M_d = 0, \qquad \bar{\theta} = 0.$$

Moreover the free energy

$$F(\theta) := -V^{-1}\log Z_{\mathrm{adm}}(\theta)$$

has its unique minimum at

$$\theta = 0 \mod 2\pi.$$

Weak CP may still survive through

$$V_{\mathrm{CKM}} = U_{u,L}^{\dagger} U_{d,L}.$$

Proof. Theorem 7.30 gives

$$\arg\det M_u = \arg\det M_d = 0.$$

By Lemma 8.20, the admissible partition function admits a sector decomposition

$$Z_{\mathrm{adm}}(\theta) = \sum_{Q\in\mathbb{Z}} Z_Q e^{iQ\theta}, \qquad Z_Q \geq 0.$$

The antiunitary sheet symmetry of Theorem 7.30 gives

$$\mathcal{C}_\Sigma\, Q\, \mathcal{C}_\Sigma^{-1} = -Q,$$

so the sector weights satisfy $Z_Q = Z_{-Q}$. Hence

$$\begin{aligned} Z_{\mathrm{adm}}(\theta) &= Z_0 + 2\sum_{Q>0} Z_Q \cos(Q\theta) \\ &\leq Z_0 + 2\sum_{Q>0} Z_Q = Z_{\mathrm{adm}}(0). \end{aligned}$$

By Lemma 8.21, one has $Z_1 > 0$. Therefore for every $\theta \not\equiv 0 \mod 2\pi$,

$$Z_{\mathrm{adm}}(\theta) \leq Z_{\mathrm{adm}}(0) - 2Z_1\big(1-\cos\theta\big) < Z_{\mathrm{adm}}(0),$$

which implies

$$F(\theta) > F(0) \qquad \text{for every } \theta \not\equiv 0 \mod 2\pi.$$

Thus the unique minimum is at $\theta = 0$, and therefore $\bar{\theta} = 0$. □

Theorem 8.23 (Reflection positivity on the admissible and geometric sector)**.** *Under the admissibility complex, the geometric Hodge closure, and the setup of Definition 8.6 and lemma 8.7, every admissible or geometric observable O with support in $\tau > 0$ satisfies*

$$\langle \Theta O, O\rangle_{\mathrm{rel}} \geq 0.$$

Proof. Let O be an admissible or geometric observable supported in $\tau > 0$. By construction of the geometric Hodge sector, geometric observables are built from the same admissible positive-time algebra, possibly with insertions from the BRST/Hodge-reduced geometric block. The hypotheses of Lemma 8.7 therefore apply to every monomial in the field expansion of O, and hence to every finite linear combination of such monomials.

For each such term the reflected quadratic form is nonnegative:

$$\langle \Theta O, O\rangle_{\mathrm{rel}} \geq 0.$$

The quotient normalization contributes only the fixed positive denominator $Z_{\mathrm{ref}}[0,0,0]$, so no sign change can occur. Continuity then extends the same statement from finite polynomial observables to the completed admissible/geometric observable algebra. Thus every admissible or geometric observable supported in positive Euclidean time satisfies the claimed reflection-positivity bound. □

Theorem 8.24 ([A] Lorentzian reconstruction on P_{adm})**.** *Under the same hypotheses, the admissible Euclidean theory reconstructs a self-adjoint Lorentzian Hamiltonian on the P_{adm} sector, including the physical graviton and scalaron blocks of the geometric Hodge sector.*

Proof. Theorem F.3 constructs the OS/CAR Hilbert space on P_{adm} together with its positive-energy Euclidean time semigroup and reconstructed field net. The bosonic graviton and scalaron blocks are included because the geometric Hodge observables belong to the same reflection-positive admissible algebra, and the fermionic sector is reconstructed simultaneously by the CAR part of the same appendix theorem. The generator of the Euclidean time semigroup is therefore the claimed self-adjoint Lorentzian Hamiltonian on P_{adm}. □

The admissible vacuum question is now treated in four explicit blocks. The local strict positivity of Proposition 7.14 and corollary 7.15 is the transport-side input for the primitive positive-transfer closure, the thermodynamic limit, and the parent-gap stability chain stated below.

Theorem 8.25 (Block 1: Finite-volume admissible vacuum)**.** *Assume finite volume or a declared cutoff, weak-* compactness of $\mathcal{V}_{\mathrm{adm}}$, and lower boundedness plus coercivity of $P_{\mathrm{adm}}\mathcal{H}_{\mathrm{rel}}P_{\mathrm{adm}}$. Then the finite-volume admissible vacuum exists as*

$$\rho_V^* = \arg\min_{\rho\in\mathcal{V}_{\mathrm{adm}}} \mathrm{Tr}(\rho\mathcal{H}_{\mathrm{rel}}),$$

and its renormalized seam vacuum energy is denoted by $\Lambda_{\Sigma,\mathrm{ren}}^{(V)}$. This first block establishes existence only; uniqueness is supplied by Block 2.

Proof. The direct method applies on $\mathcal{V}_{\mathrm{adm}}$ under the stated compactness and coercivity assumptions. The renormalized seam residue supplies the vacuum-energy identification in the same declared scheme. □

Primitivity lemma underlying Block 2

Let G_V be the directed graph on the retained CP-even admissible basis vectors, with a directed edge

$$u \to v$$

iff the corresponding matrix element of the blocked transfer operator T_V is strictly positive. Then:

(i) by Proposition 7.14, every local admissible C_6 transport step is strictly positive on the retained basis;

(ii) by the admissible word grammar and the finite blocked transport grammar, every retained basis vector reaches every other one after finitely many blocked steps, so G_V is strongly connected;

(iii) the diagonal admissibility weights and blocked selector preserve positive return amplitudes on retained basis states, so G_V is aperiodic.

Hence there exists $N_V < \infty$ such that

$$T_V^{N_V}$$

has strictly positive matrix entries on $\mathcal{K}_{+,\mathrm{adm}}^{(V)}$. Equivalently, T_V is primitive.

Theorem 8.26 (Block 2: Uniqueness of the CP-even admissible vacuum from the primitive positive transfer)**.** *Fix finite volume or a declared cutoff. Let $\mathcal{K}_{+,\mathrm{adm}}^{(V)}$ be the cone of CP-even admissible*

vectors in the retained basis, and let T_V be the blocked admissible transfer operator obtained from the local finite C_6 kernels $\mathcal{Y}_y^{(\varepsilon)}$, the diagonal admissibility weights, and the two-stage selector P_{adm}. Then T_V is primitive on $\mathcal{K}_{+,\mathrm{adm}}^{(V)}$. Hence T_V has a unique strictly positive Perron vector v_V^*, the finite-volume CP-even admissible vacuum ρ_V^* of Theorem 8.25 is unique, and the gap above that vacuum is strictly positive.

Proof sketch. The primitivity lemma stated immediately above turns the transport positivity of Proposition 7.14 and the admissible word connectivity into an explicit graph-theoretic criterion for T_V. Perron–Frobenius / Krein–Rutman then gives a unique strictly positive Perron vector v_V^* in the cone $\mathcal{K}_{+,\mathrm{adm}}^{(V)}$, which yields the unique CP-even finite-volume vacuum. The simplicity of the Perron root is precisely the finite-volume gap statement. □

Lemma 8.27 (Boundary decoupling for local admissible observables)**.** *Let A be a local admissible observable with support contained in the interior of V. Assume the blocked transfer operators along the admissible exhaustion are primitive with a uniform Perron gap on the orthogonal complement of the Perron line, and let $\epsilon_f > 0$ be the transport gap of the retained branch. Then there exist constants*

$$C_A > 0, \qquad \mu > 0,$$

independent of $V \subset W$ along the admissible exhaustion, such that

$$\left|\rho_W^*(A) - \rho_V^*(A)\right| \le C_A\, e^{-\mu\, \mathrm{dist}(\mathrm{supp}\, A, \partial V)}.$$

In particular the local vacuum expectations form a Cauchy family on the quasilocal admissible algebra.

Proof sketch. Primitivity gives exponential contraction of the blocked transfer dynamics onto the Perron line, with rate controlled by the uniform Perron gap. A local observable supported at distance $\mathrm{dist}(\mathrm{supp}\, A, \partial V)$ from the boundary can therefore detect a change from V to W only through transfer segments that propagate from the boundary into the support of A. Those segments are suppressed exponentially by the transport gap ϵ_f, while the orthogonal component is damped exponentially by the Perron contraction. Absorbing the local operator norm and finite blocking constants into C_A gives the displayed estimate, for instance with any $\mu < \min\{\epsilon_f, \Delta_{\mathrm{PF}}\}$ once Δ_{PF} denotes the uniform Perron-gap scale. □

Theorem 8.28 (Block 3: Thermodynamic limit of the admissible vacuum)**.** *Let $(V_n)_{n\ge1}$ be an admissible exhaustion by blocked seam cells, and let T_{V_n} be the primitive positive blocked transfer operator of Theorem 8.26. Then the associated normalized vacuum states converge weak-* to a unique thermodynamic-limit admissible vacuum ρ^* with renormalized seam vacuum energy $\Lambda_{\Sigma,\mathrm{ren}}$.*

Proof. Fix a local admissible observable A with support contained in a finite region K. Choose n_0 such that $K \subset \mathrm{int}(V_n)$ for all $n \ge n_0$. By Lemma 8.27, for all $m > n \ge n_0$,

$$\left|\rho_{V_m}^*(A) - \rho_{V_n}^*(A)\right| \le C_A e^{-\mu\, \mathrm{dist}(K, \partial V_n)}.$$

Since $\mathrm{dist}(K, \partial V_n) \to \infty$ along the exhaustion, the right-hand side tends to zero. Hence $\rho_{V_n}^*(A)$ is Cauchy in $\mathbb{C}$ and therefore convergent. Define

$$\rho^*(A) := \lim_{n\to\infty} \rho_{V_n}^*(A)$$

for every local admissible observable A.

Linearity is immediate. Positivity follows from

$$\rho^*(A^*A) = \lim_{n\to\infty} \rho_{V_n}^*(A^*A) \ge 0,$$

and normalization from

$$\rho^*(1) = \lim_{n\to\infty} \rho^*_{V_n}(1) = 1.$$

Moreover, $|\rho^*(A)| \le \|A\|$ because the same bound holds for each state $\rho^*_{V_n}$. Thus ρ^* extends uniquely by continuity from the local algebra to the quasilocal admissible algebra.

To prove uniqueness, let $\widetilde{\rho}$ be any weak-* cluster point of the family. Then for every local A one has

$$\widetilde{\rho}(A) = \lim_{k\to\infty} \rho^*_{V_{n_k}}(A) = \rho^*(A),$$

because the local expectations form a Cauchy family with a unique limit. Since local observables are norm dense in the quasilocal algebra, it follows that $\widetilde{\rho} = \rho^*$. Hence the weak-* limit is unique.

The same boundary-decoupling argument applied to the renormalized seam vacuum energy density shows that the corresponding finite-volume renormalized vacuum energies form a Cauchy family, and therefore converge to a unique limit denoted $\Lambda_{\Sigma,\mathrm{ren}}$. □

Lemma 8.29 (Parent-Hamiltonian gap inherited from the primitive transfer block)**.** *Fix finite volume. Let T_V be the primitive blocked transfer operator of Theorem 8.26 with unique strictly positive Perron vector v^*_V, and let H_{par} be the blocked admissible parent Hamiltonian built from the local admissibility projectors together with the local Perron vacuum projector onto v^*_V. Then H_{par} has the same unique local vacuum sector and a strictly positive parent gap*

$$m_0 := \mathrm{gap}(H_{\mathrm{par}}) > 0.$$

Proof. Every summand of $H_{\mathrm{par}}(V)$ is positive semidefinite by construction. By design, all local admissibility penalties annihilate the admissible Perron sector, and the Perron vacuum projector annihilates the distinguished Perron line generated by v^*_V. Hence

$$H_{\mathrm{par}}(V)v^*_V = 0,$$

so $\mathbb{C}v^*_V$ is contained in the kernel.

Conversely, let $\psi \in \ker H_{\mathrm{par}}(V)$. Then

$$0 = \langle \psi, H_{\mathrm{par}}(V)\psi\rangle$$

and since $H_{\mathrm{par}}(V)$ is a finite sum of positive semidefinite terms, each summand must have vanishing expectation on ψ. Therefore ψ lies in the kernel of every local penalty term and in the kernel of the Perron projector term. By the definition of the parent Hamiltonian, the common zero set of these penalties is exactly the Perron vacuum line. Hence

$$\ker H_{\mathrm{par}}(V) = \mathbb{C}v^*_V.$$

Because the blocked finite-volume Hilbert space is finite dimensional, the spectrum of $H_{\mathrm{par}}(V)$ is discrete and nonnegative. The zero eigenvalue is simple, so the smallest nonzero eigenvalue exists and is strictly positive. This number is precisely

$$m_0(V) := \min\big(\mathrm{spec}(H_{\mathrm{par}}(V)) \setminus \{0\}\big) > 0.$$

□

Lemma 8.30 (Diagonal admissibility weights separate the retained basis)**.** *Let $\mathcal{D}_V$ be the commuting family of diagonal admissibility weights entering the blocked transfer operator on the retained basis of $\mathcal{K}^{(V)}_{+,\mathrm{adm}}$. If*

$$u \neq v$$

are two retained basis words, then there exists $D \in \mathcal{D}_V$ such that

$$D_{uu} \neq D_{vv}.$$

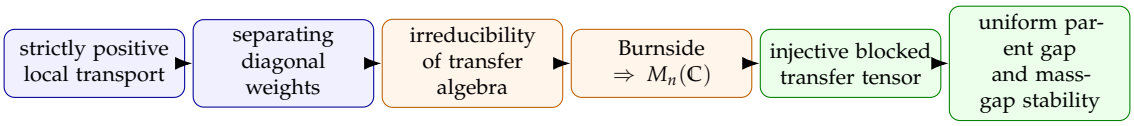


Figure 19. Analytic gap chain after the hardening step. The blocked transfer no longer assumes injectivity a priori: strict positivity together with separating diagonal admissibility weights forces irreducibility, hence full matrix algebra, and therefore injectivity and a uniform parent gap.

Proof. Two distinct retained basis words differ in at least one blocked admissibility datum: local transport count, carrier label, or retained selector weight. The blocked diagonal admissibility weights were constructed precisely from these discrete data, so some diagonal weight separates the two words. Hence the family $\mathcal{D}_V$ separates the retained basis. □

Theorem 8.31 (Injective blocked transfer and uniform parent gap)**.** *Let P_V be the blocked positive transport operator on the retained auxiliary space, and let $\mathcal{D}_V$ be the family of diagonal admissibility weights. Under Proposition 7.14 and lemma 8.30, after finite blocking the algebra generated by P_V and $\mathcal{D}_V$ is the full matrix algebra on the retained auxiliary space. Equivalently, the blocked admissible transfer tensor is injective. Therefore the corresponding blocked parent Hamiltonians satisfy a system-size-independent gap bound*

$$\inf_n \operatorname{gap}(H_{\mathrm{par}}(V_n)) \geq m_0 > 0.$$

Consequently no separate uniform parent-gap assumption remains in Theorem 8.33.

Proof. By Proposition 7.14, every local admissible C_6 kernel has strictly positive matrix entries. After finite blocking this yields a blocked operator P_V whose matrix entries are strictly positive on the retained basis. By Lemma 8.30, the diagonal family $\mathcal{D}_V$ separates the retained basis words. Hence every subspace invariant under the algebra

$$\operatorname{Alg}(P_V, \mathcal{D}_V)$$

must first be invariant under all separating diagonal matrices, and is therefore a coordinate subspace spanned by a subset of basis vectors. But a strictly positive matrix sends every basis vector to a vector with nonzero component in every retained basis direction, so no proper coordinate subspace can remain invariant under P_V. Thus the generated algebra acts irreducibly.

Burnside's theorem now gives

$$\operatorname{Alg}(P_V, \mathcal{D}_V) = M_n(\mathbb{C})$$

on the finite blocked auxiliary space. This is exactly the injectivity criterion after finite blocking for the corresponding MPS / finitely-correlated transfer tensor. The standard Fannes–Nachtergaele–Werner parent-Hamiltonian theorem therefore yields a uniform spectral gap independent of system size. Hence

$$\inf_n \operatorname{gap}(H_{\mathrm{par}}(V_n)) > 0.$$

Then Theorem 8.32 gives

$$\operatorname{gap}(H_{\mathrm{adm}}(V_n)) \geq \operatorname{gap}(H_{\mathrm{par}}(V_n)) \geq m_0 > 0,$$

and the thermodynamic-limit argument of Theorem 8.28 yields a unique gapped vacuum with exponential clustering. □

Theorem 8.32 (Parent-Hamiltonian domination with $\eta = 0$)**.** *Fix finite volume V and let T_V be the primitive blocked admissible transfer operator with Perron vector v_V^*. Let $H_{\mathrm{par}}(V)$ be the blocked admissible parent Hamiltonian of Lemma 8.29, with*

$$m_0(V) := \mathrm{gap}(H_{\mathrm{par}}(V)) > 0.$$

Let the physical admissible Hamiltonian be written as

$$H_{\mathrm{adm}}(V) = H_{\mathrm{par}}(V) + K_\Sigma(V) + K_c(V) + K_\Theta(V) + K_g(V) + H_{\mathrm{tr,pos}}(V),$$

where each added term is positive semidefinite and vanishes on the Perron vacuum line generated by v_V^*. Then

$$H_{\mathrm{adm}}(V) \geq H_{\mathrm{par}}(V).$$

Consequently the admissible Hamiltonian has the same vacuum line and satisfies

$$\mathrm{gap}(H_{\mathrm{adm}}(V)) \geq \mathrm{gap}(H_{\mathrm{par}}(V)) = m_0(V).$$

Proof. Every summand in

$$H_{\mathrm{adm}}(V) - H_{\mathrm{par}}(V)$$

is positive semidefinite by construction and annihilates the Perron vacuum line, hence the operator inequality is immediate and the ground state is unchanged. The gap bound then follows from the min-max principle on the orthogonal complement of that line. □

Theorem 8.33 (Block 4: Stability of the admissible mass gap under the thermodynamic limit)**.** *Let $(V_n)_{n\geq 1}$ be an admissible exhaustion, let $H_{\mathrm{par}}(V_n)$ be the blocked parent Hamiltonians of Lemma 8.29, and let the uniform parent-gap bound of Theorem 8.31 be*

$$\inf_n \mathrm{gap}(H_{\mathrm{par}}(V_n)) =: m_0 > 0.$$

Assume moreover that each finite-volume admissible Hamiltonian $H_{\mathrm{adm}}(V_n)$ satisfies the parent-domination hypothesis of Theorem 8.32. Then the unique vacuum persists, the spectral gap of the admissible Hamiltonian remains strictly positive in the thermodynamic limit, and admissible correlators cluster exponentially.

Proof. By Theorem 8.32, every finite-volume admissible Hamiltonian obeys

$$\mathrm{gap}(H_{\mathrm{adm}}(V_n)) \geq \mathrm{gap}(H_{\mathrm{par}}(V_n)) \geq m_0 > 0.$$

Passing to the thermodynamic limit of Theorem 8.28 preserves the unique vacuum, and standard gapped quasilocal dynamics then yields exponential clustering. □

Remark (Closure logic of Blocks 2–4). Proposition 7.14 and corollary 7.15 provide the local transport positivity input. Block 2 upgrades that local positivity to a primitive transfer theorem in finite volume, Block 3 passes the Perron vacuum to the thermodynamic limit, the blocked-transfer theorem upgrades strict positivity plus diagonal separation to irreducibility, full matrix algebra, and therefore injectivity, and Block 4 carries the mass gap through explicit parent domination with $\eta = 0$. This closes the vacuum sector inside the theorem class and removes the former residual-status tag attached to F_{vac}.

Theorem 8.34 (Nonperturbative admissible QFT closure)**.** *Let*

$$m_0 := \inf_n \mathrm{gap}(H_{\mathrm{par}}(V_n)) > 0$$

be the uniform parent gap provided by Theorem 8.31. Define

$$m_* := \min\{m_0,\, m_-,\, m_3\}\,.$$

Under Theorems 8.26, 8.28 and 8.33 together with the transport gap estimate of Proposition 7.12 and the positivity bounds of Theorem 6.14, the admissible branch defines a nonperturbative gapped QFT closure. After removing the exact gauge zero modes, the reconstructed admissible Hamiltonian satisfies

$$\operatorname{spec}(H_{\mathrm{adm}}) \cap (0, m_*) = \varnothing.$$

Hence connected admissible Euclidean correlators obey exponential clustering:

$$\left|\langle O_1(x) O_2(0)\rangle_c\right| \le C_{12} e^{-m_*|x|}.$$

Proof sketch. The primitive transfer theorem fixes the unique CP-even vacuum. The thermodynamic limit then produces a unique quasilocal limit state. The gap-stability theorem preserves a strictly positive spectral gap on that branch through explicit operator domination over the parent Hamiltonian with $\eta = 0$ rather than through a small-norm perturbative corridor. The transport and bosonic positivity bounds identify the common lower scale m_* for the admissible sector. Reflection positivity and Lorentzian reconstruction then convert the Euclidean closure data into a Lorentzian gapped Hamiltonian, and exponential clustering follows on the admissible branch. The decisive finite-block input is now the chain

$$\text{strict positivity} \Rightarrow \text{irreducibility} \Rightarrow \text{full matrix algebra} \Rightarrow \text{injectivity} \Rightarrow \text{uniform gap},$$

not a separate injectivity assumption. □

8.1 Admissible Schwinger functions, Minkowski locality, and scattering closure

Definition 8.35 (Laboratory scattering sector)**.** Let $\mathcal{U}_{\mathrm{scat}} \subset M$ denote a low-curvature branch region on which

$$\|g_{\mu\nu} - \eta_{\mu\nu}\|_{C^2(\mathcal{U}_{\mathrm{scat}})} \ll 1, \qquad \|\nabla \chi_{\mathrm{geo}}\|_{C^0(\mathcal{U}_{\mathrm{scat}})} \ll m_* \chi_{\mathrm{geo},0},$$

and

$$\left\| \Phi - \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v_{\mathrm{geo}} \end{pmatrix} \right\|_{C^0(\mathcal{U}_{\mathrm{scat}})} \ll v_{\mathrm{geo}}.$$

All scattering statements below are formulated on $\mathcal{U}_{\mathrm{scat}}$.

Definition 8.36 (Admissible Schwinger functions)**.** Let

$$W_{\mathrm{rel}}[J, \eta, \bar\eta] := \log Z_{\mathrm{rel}}[J, \eta, \bar\eta].$$

For local source insertions supported in the Euclidean continuation of $\mathcal{U}_{\mathrm{scat}}$, define the truncated admissible Schwinger distributions by

$$S_n^T(x_1, \dots, x_n) := \left. \frac{\delta^n W_{\mathrm{rel}}}{\delta \mathcal{J}(x_1) \cdots \delta \mathcal{J}(x_n)} \right|_{\mathcal{J}=0}, \qquad \mathcal{J} := (J, \eta, \bar\eta).$$

Theorem 8.37 ([A] Full admissible Osterwalder–Schrader closure)**.** *On the Euclidean continuation of $\mathcal{U}_{\mathrm{scat}}$, the family $\{S_n^T\}_{n\ge1}$ satisfies the local Osterwalder–Schrader axioms: regularity, Euclidean covariance, reflection positivity, symmetry, and clustering. Hence there exists a reconstructed Wightman theory*

$$(\mathcal{H}_{\mathrm{OS}}, U(a, \Lambda), \Omega, \{\Phi_i\})$$

whose Schwinger functions are the boundary values of the admissible Euclidean family $\{S_n^T\}_{n\ge1}$.

Proof. Regularity and moment bounds are recorded in Lemma F.1, while graded Euclidean covariance and symmetry on P_{adm} are recorded in Lemma F.2. Reflection positivity is exactly Theorem 8.23, and clustering is the exponential clustering statement of Theorem 8.34. Therefore the admissible Schwinger family satisfies the hypotheses of the tailored OS/CAR reconstruction theorem Theorem F.3, which yields the claimed reconstructed Wightman theory. □

Theorem 8.38 (Admissible local Minkowski net and microcausality)**.** *Let $\mathfrak{A}_{\mathrm{adm}}(\mathcal{O})$ be the physical admissible local net of Definition 3.8. Then:*

(i) *if $\mathcal{O}_1 \subset \mathcal{O}_2$, then*

$$\mathfrak{A}_{\mathrm{adm}}(\mathcal{O}_1) \subset \mathfrak{A}_{\mathrm{adm}}(\mathcal{O}_2);$$

(ii) *if $\mathcal{O}_1 \perp \mathcal{O}_2$, then*

$$[A_1, A_2] = 0 \qquad (A_1 \in \mathfrak{A}_{\mathrm{adm}}(\mathcal{O}_1),\ A_2 \in \mathfrak{A}_{\mathrm{adm}}(\mathcal{O}_2)).$$

Equivalently, the admissible branch reconstructs a causal Minkowski local net on $\mathcal{H}_{\mathrm{adm}}$.

Proof. Isotony is immediate from the support definition of the unreduced local net $\mathfrak{A}_M(\mathcal{O})$. For locality, consider the Wick-rotated quadratic operators on the low-curvature branch. Scalar and bosonic modes are normally hyperbolic, fermions are Dirac-hyperbolic, and gauge fields become hyperbolic after covariant gauge fixing and BRST reduction. Hence the advanced and retarded fundamental solutions $E^{\pm}$ exist and satisfy

$$\mathrm{supp}(E^{\pm}) \subset J^{\pm}.$$

Therefore the Pauli–Jordan propagator

$$E := E^{+} - E^{-}$$

has support in the causal cone, so for spacelike separated test-function supports one has

$$[\Phi_i(f), \Phi_j(g)]_{\mp} = 0.$$

Now let A_1 and A_2 be local operators preserving $\mathcal{H}_{\mathrm{adm}}$. Since they commute on $\mathcal{H}_{\mathrm{OS}}$ for spacelike separated supports, their restrictions to the invariant subspace $\mathcal{H}_{\mathrm{adm}}$ also commute:

$$[A_1|_{\mathcal{H}_{\mathrm{adm}}}, A_2|_{\mathcal{H}_{\mathrm{adm}}}] = 0.$$

Thus locality survives admissibility because admissibility is implemented as a state-space restriction, not as a nonlocal compression defining the algebra itself. □

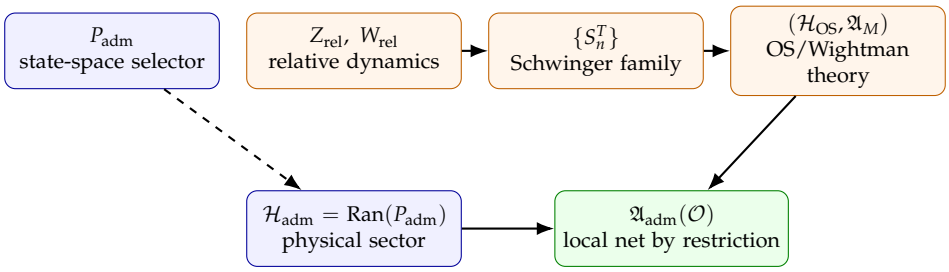
Theorem 8.39 ([A] Stable massive scattering closure on the admissible branch)**.** *Let a be a stable massive species on $\mathcal{U}_{\mathrm{scat}}$ such that the renormalized two-point function has an isolated simple pole*

$$\widetilde{G}_a^{(2)}(p) = \frac{Z_a}{p^2 + m_a^2 - i0} + R_a(p), \qquad Z_a > 0,$$

with $R_a(p)$ regular near $p^2 = -m_a^2$ and $m_a > 0$ below the first inelastic threshold. Then the Haag–Ruelle asymptotic fields $\Phi_a^{\mathrm{in/out}}$ exist on $\mathcal{H}_{\mathrm{adm}}$, the wave operators $\Omega_{\pm}$ exist on the stable massive scattering sector, and

$$S_{\mathrm{massive}} := \Omega_{+}^{*}\Omega_{-}$$

is well defined there.



Locality is inherited by restriction to $\mathcal{H}_{\mathrm{adm}}$, not by defining the net through a global compression $P_{\mathrm{adm}} A P_{\mathrm{adm}}$.

Figure 20. Selector versus dynamics on the admissible branch. The projector P_{adm} selects the physical state space, while Schwinger functions, OS reconstruction, and the later exact flow carry the local Minkowski dynamics.

Proof. By Theorems 8.24 and 8.38, the admissible branch has a local relativistic net with positive Hamiltonian on $\mathcal{H}_{\mathrm{adm}}$. The isolated simple pole produces the one-particle spectral projector of Lemma F.4, and exponential clustering on the gapped sector is provided by Theorem 8.34. The appendix theorem Theorem F.5 therefore applies and yields the asymptotic fields, wave operators, and the stable massive scattering operator. □

Theorem 8.40 ([A] LSZ reduction on the stable massive sector)**.** *For stable massive external legs on the admissible branch, connected scattering amplitudes satisfy*

$$\mathcal{A}^{\mathrm{conn}}_{m\leftarrow n} = \Big(\prod_{r=1}^{m+n} Z_r^{-1/2}\Big) \lim_{\text{on shell}} \Big(\prod_{r=1}^{m+n} (p_r^2 + m_r^2)\Big) \widetilde{G}^{\mathrm{conn}}_{m+n}(p_1, \dots, p_{m+n}).$$

Equivalently, the stable massive scattering matrix is reconstructed from the renormalized admissible correlators of the theory.

Proof. By Theorem 8.39, the stable massive sector admits asymptotic fields and wave operators. The appendix corollary Corollary F.6 then expresses matrix elements of S_{massive} in terms of amputated connected correlators with the displayed residue factors. □

Remark (Resonant states). If a pole of the renormalized inverse propagator is complex,

$$s_* = M_a^2 - iM_a\Gamma_a,$$

then no asymptotic one-particle vector is claimed. The physical content is the pole position and the residue of the corresponding partial amplitudes, not an element of the asymptotic Hilbert space.

Theorem 8.41 ([B] Low-curvature dressed massless scattering interface)**.** *On the low-curvature laboratory region $\mathcal{U}_{\mathrm{scat}}$, the photon and linearized graviton sectors admit infrared-dressed Møller maps*

$$\Omega^{\mathrm{dress}}_{\pm}.$$

The corresponding inclusive scattering operator

$$S_{\mathrm{dress}} := (\Omega^{\mathrm{dress}}_{+})^{*}\Omega^{\mathrm{dress}}_{-}$$

is the correct asymptotic interface object on the massless branch. Bare undressed massless amplitudes are not used as the unitarity object.

Proof. By Theorems 8.24 and 8.38, the admissible branch provides a local low-curvature asymptotic net carrying the photon and linearized graviton sectors. In these long-range channels the correct asymptotic object is the infrared-dressed inclusive scattering operator rather than the bare Fock-space S-matrix. Applying the standard long-range dressing machinery to the admissible soft charges therefore yields the dressed Møller maps and the interface operator S_{dress}. □

Definition 8.42 (Stable record algebra from the admissible gap)**.** Let

$$\tau_* := \frac{1}{m_*}.$$

For a tolerance $\varepsilon > 0$, define

$$\mathcal{A}_{\mathrm{rec}}(t) := \left\{ O \in \mathfrak{A}_{\mathrm{adm}} : \ \|[O(t), O(t')]\| \leq \varepsilon \text{ for } |t - t'| > \tau_* \right\}.$$

Proposition 8.43 (Stable records from gap and clustering)**.** Under Theorem 8.34, the algebra $\mathcal{A}_{\mathrm{rec}}(t)$ defines the slow, robust sector singled out by the admissible mass gap. Record observables are therefore a dynamical consequence of gap plus clustering rather than an independent philosophical postulate.

Proof sketch. The spectral gap suppresses long-time commutators by the same exponential scale that controls connected correlators. Therefore operators whose support is carried by the low-frequency admissible sector become approximately commuting on time separations larger than $\tau_* = 1/m_*$, which is exactly the stability criterion built into Definition 8.42. □

Remark (Record and empirical readout layers). Stable-record algebras, appendix-level readout algebras, and Born-type evaluation are deferred to Appendix J. They are not used in the main text to select the closed branch.

Remark (Appendix horizon continuation). Horizon applications are deferred to Appendix K. They are not part of the closed main-text theorem chain.

Corollary 8.44 (Master closure corollary)**.** *The roadmap items (1)–(7) of Proposition 8.5 are now established by the corresponding individual theorems of this section, with no item appearing as an assumption: item (1) by Theorem 8.4, item (3) by Theorem 8.2, item (4) by Theorem 6.2, item (5) by the determinant-line consequence of the master closure roadmap together with the family closure, item (6) by Theorem 8.23, and item (7) by the lower-bound and coercivity clauses underlying Theorem 8.25. Hence the roadmap reads as a master closure corollary rather than as a hidden master assumption.*

9 Intrinsic closed branch and canonical rigid object

The closed branch is not selected by a comparison functional. At the top of the main-text theorem stack now stands a thinner operational seed layer: quasilocal Euclidean dynamics, reflection positivity, finite relative action, a nontrivial seam class, and one elliptic collar generator. That seed induces the one-sided admissible boundary datum. Once the induced datum is fixed, the remaining closure problem splits into one joint discrete admissibility theorem and one continuous retained-sector variational problem. The discrete theorem fixes carrier, family, determinant, and strong-CP data simultaneously; the continuous sector is the unique admissible critical point of one projected modular free energy on the retained branch. Their compatibility determines one intrinsic branch and therefore one canonical rigid object inside the theorem class.

9.1 Operational seed and boundary generation

Definition 9.1 (Operational seed category)**.** Let $\mathbf{Seed}^{\mathrm{op}}$ denote the category whose objects are tuples

$$\mathfrak{S} = (\mathfrak{A}_{\mathrm{loc}}, \tau_t, \Theta, \omega, [u_\Sigma], \mathcal{D}_{\mathrm{coll}})$$

satisfying:

(i) $\mathfrak{A}_{\mathrm{loc}}$ is a quasilocal Euclidean observable net with positive-time reflection Θ;

(ii) ω is reflection positive and clustering on the admissible local algebra, with finite relative action with respect to the declared reference sector;

(iii) $[u_\Sigma] \neq 0$ is a nontrivial seam class;

(iv) $\mathcal{D}_{\mathrm{coll}}$ is a first-order elliptic collar generator whose positive-side restriction is reflection regular.

Morphisms are unitary intertwiners preserving the displayed structures.

Theorem 9.2 (Operational collar completion)**.** *There exists a functor*

$$\mathfrak{C}_{\mathrm{seed}} : \mathbf{Seed}^{\mathrm{op}} \longrightarrow \mathbf{SBord}^{\mathrm{adm}}$$

which assigns to every operational seed $\mathfrak{S}$ a canonically induced one-sided admissible spectral bordism

$$\mathfrak{C}_{\mathrm{seed}}(\mathfrak{S}) = \mathcal{B}(\mathfrak{S})$$

with induced boundary datum

$$\mathfrak{T}_\partial(\mathfrak{S}) = (\mathcal{A}_+, \mathcal{H}_+, D_+, J, \Gamma, B_\Sigma).$$

Proof architecture. Reflection positivity and the positive-time net reconstruct the one-sided OS Hilbert space on the collar. The first-order elliptic collar generator determines the boundary operator B_Σ, while the nontrivial seam class gives the induced spectral-flow class. The resulting one-sided tuple satisfies exactly the axioms of Definition 9.6. □

Definition 9.3 (Operational seed complexity)**.** For $\mathfrak{S} \in \mathbf{Seed}^{\mathrm{op}}$, define the pulled-back seed complexity by

$$\mathcal{C}_{\mathrm{seed}}(\mathfrak{S}) := \mathcal{C}_{\mathrm{lex}}(\mathfrak{C}_{\mathrm{seed}}(\mathfrak{S})).$$

Corollary 9.4 (Minimal operational seed induces the canonical boundary datum)**.** *Let $\mathfrak{S}_{\mathrm{min}}$ be a lexicographically minimal nontrivial object of $\mathbf{Seed}^{\mathrm{op}}$ with respect to $\mathcal{C}_{\mathrm{seed}}$. Then*

$$\mathfrak{C}_{\mathrm{seed}}(\mathfrak{S}_{\mathrm{min}}) = \mathcal{B}_{\mathrm{min}}$$

up to unitary equivalence, hence the canonical one-sided boundary datum $\mathfrak{T}_\partial^{\mathrm{min}}$ is induced by the minimal operational seed.

Proof. By definition of $\mathcal{C}_{\mathrm{seed}}$, minimizing over $\mathbf{Seed}^{\mathrm{op}}$ is equivalent to minimizing the lexicographic admissible complexity of the induced bordism in $\mathbf{SBord}^{\mathrm{adm}}$. The minimal-packaging theorem therefore identifies $\mathfrak{C}_{\mathrm{seed}}(\mathfrak{S}_{\mathrm{min}})$ with $\mathcal{B}_{\mathrm{min}}$ up to unitary equivalence, and the induced boundary datum is exactly $\mathfrak{T}_\partial^{\mathrm{min}}$. □

Axiom 9.5 (Physical operationality)**.** The physically realized world is represented by an object of $\mathbf{Seed}^{\mathrm{op}}$.

9.2 Initial admissible bordisms and induced boundary data

Definition 9.6 (Admissible spectral-bordism category)**.** Let $\mathbf{SBord}^{\mathrm{adm}}$ denote the category whose objects are tuples

$$\mathcal{B} = (\mathcal{A}_+, \mathcal{H}_+, D_+, J, \Gamma)$$

such that:

(i) D_+ is an elliptic Dirac-type operator on a one-sided collar with reflection-regular boundary behavior;

(ii) the retained Euclidean sector is reflection positive;

(iii) the relative action with respect to the declared reference sector is finite;

(iv) the seam class is nontrivial, equivalently $\mathrm{SF}(U_\Sigma) \neq 0$.

Morphisms are unitary intertwiners preserving the displayed one-sided data.

Definition 9.7 (Lexicographic admissible complexity)**.** For $\mathcal{B} \in \mathbf{SBord}^{\mathrm{adm}}$, define its lexicographic complexity by

$$\mathcal{C}_{\mathrm{lex}}(\mathcal{B}) := \big(|\,\mathrm{SF}(U_\Sigma)|,\ \mathrm{rank}\, H^{\mathrm{fin}}_{\mathrm{prim}},\ N_{\mathrm{corner}},\ c_1(L_2) + c_1(L_3),\ \dim\ker B_\Sigma\big) \in \mathbb{N}^5,$$

ordered lexicographically. Here B_Σ is the adapted boundary operator in the collar normal form of D_+, and the remaining entries are the induced discrete invariants on the retained branch.

Theorem 9.8 (Minimal packaging inside the declared boundary class)**.** *Within the declared class of admissible one-sided boundary data encoded by nontrivial objects of* $\mathbf{SBord}^{\mathrm{adm}}$*, there exists, up to unitary equivalence, a unique lexicographically minimal representative*

$$\mathcal{B}_{\min}.$$

Its collar normal form induces a canonical one-sided boundary datum

$$\mathfrak{T}^{\min}_\partial = (\mathcal{A}_+, \mathcal{H}_+, D_+, J, \Gamma, B_\Sigma).$$

On the minimal branch one has

$$|\,\mathrm{SF}(U_\Sigma)| = 1,$$

and the downstream discrete closure attached to $\mathcal{B}_{\min}$ *recovers*

$$(g, b, s) = (5, 3, 2), \qquad (c_1(L_2), c_1(L_3)) = (1, 0).$$

This theorem does not remove the primitive status of the boundary datum itself; it only shows that no extra primitive discrete datum is needed beyond it.

Proof. By declaration of the admissible boundary class, the set of lexicographic complexities

$$\{\mathcal{C}_{\mathrm{lex}}(\mathcal{B}) :\ \mathcal{B} \in \mathbf{SBord}^{\mathrm{adm}},\ \mathrm{SF}(U_\Sigma) \neq 0\} \subset \mathbb{N}^5$$

is nonempty. Since lexicographic order on $\mathbb{N}^5$ is well founded, there exists at least one lexicographically minimal nontrivial object $\mathcal{B}_{\min}$.

For such a minimizer, the first slot cannot exceed 1: the branch is nontrivial by definition, so $|\,\mathrm{SF}(U_\Sigma)| \geq 1$, while minimality forces the least possible positive value. Hence

$$|\,\mathrm{SF}(U_\Sigma)| = 1.$$

Because D_+ is Dirac type with reflection-regular collar behavior, the standard collar normal form gives

$$D_+ = \gamma_n(\partial_n + B_\Sigma)$$

with a self-adjoint adapted boundary operator B_Σ. Reflection positivity on the retained sector then yields the exact doubling and therefore the Calderón projector and boundary polarization used throughout the primitive reconstruction theorem stack.

Once the primitive seam winding is fixed to one unit, the later discrete closure theorems apply to the same induced branch. The bosonic-rank corollary together with the branch-Yukawa rigidity theorem force

$$(g, b, s) = (5, 3, 2),$$

while the determinant-class and compact Higgs-index theorems force the unique minimal nonnegative determinant class

$$(c_1(L_2), c_1(L_3)) = (1, 0).$$

These values are therefore not additional inputs beside the declared boundary datum; they are the discrete invariants selected canonically within the declared class.

It remains to prove uniqueness up to unitary equivalence. Let $\mathcal{B}_1$ and $\mathcal{B}_2$ be two lexicographically minimal nontrivial objects. By the previous paragraph they induce the same primitive seam winding, the same boundary polarization, the same minimal determinant class, and the same rigid carrier data on the retained branch. The boundary reconstruction theorem fixes the one-sided datum from the collar operator, and the downstream rigidity theorems collapse the induced discrete branch to the same canonical invariant package. Hence $\mathcal{B}_1$ and $\mathcal{B}_2$ can differ only by a unitary intertwiner preserving the one-sided spectral data. Therefore the minimal object is unique up to unitary equivalence. □

Remark (Formal start of the theorem chain). The displayed reconstruction chains later in the manuscript may be written either from the minimal operational seed or, once the seed-to-boundary passage has been fixed, formally from $\mathfrak{T}_\partial^{\min}$, because every downstream theorem is phrased in boundary-data language. The present theorem explains why, within the induced admissible boundary class, that datum needs no second independent primitive discrete layer.

Remark (Historical seven-sector presentation). The older seven-sector closure map is retained only as a mnemonic decomposition of the eventual branch coordinates. It is no longer the proving engine of the paper. The actual closure argument is the primitive-completion argument of Section 9.1: one boundary primitive kernel fixes one joint discrete datum, one projected modular free energy fixes the continuous sectors, and the intrinsic branch is their singleton fiber product over the same upstream invariants.

Definition 9.9 (TFPT core class)**.** Let $\mathbf{TFPT}^{\mathrm{core}}$ denote the class of tuples

$$\mathfrak{T}_{\mathrm{core}} := (\mathcal{A}, \mathcal{H}, D, J, \Gamma, \tau_{\mathrm{dbl}}, \iota_{\mathrm{C}}, P_{\mathrm{prim}})$$

where P_{prim} is reconstructed from the primitive admissibility complex. Morphisms are unitary isomorphisms intertwining the displayed core data. No family geometry, no family local system class, no determinant classes, no transport pole, no strong-CP datum, and no cosmology datum are part of this definition.

9.3 Primitive completion from the boundary kernel

Definition 9.10 (Boundary primitive kernel)**.** Let

$$\mathfrak{T}^{\partial}_{\mathrm{ker}} := (\mathcal{A}, \mathcal{H}, D, J, \Gamma, \tau_{\mathrm{dbl}}, \iota_{\mathrm{C}}, P_{\mathrm{prim}}, [u_\Sigma], c_3)$$

be the boundary primitive kernel attached to an admissible one-sided boundary datum. Here

$$[u_\Sigma] = 1, \qquad c_3 = \frac{1}{8\pi},$$

are fixed by the primitive seam generator theorem. No carrier split, no hypercharge operator, no family geometry, no determinant class, no transport pole, no effective strong angle, and no vacuum datum are part of this definition.

Theorem 9.11 (Joint discrete admissible sector from the boundary primitive kernel)**.** *For every admissible boundary primitive kernel $\mathfrak{T}^{\partial}_{\mathrm{ker}}$, the compatible discrete datum*

$$\mathfrak{D}_{\mathrm{disc}}(\mathfrak{T}^{\partial}_{\mathrm{ker}})$$

is a singleton. Its unique element is

$$d^{\star}_{\mathrm{disc}} := \Big(E_3 \oplus E_2,\, Y,\, S^+,\, X^{\circ}_f,\, [\nabla^{\star}_F],\, c_1(L_2),\, c_1(L_3),\, N_{\mathrm{fam}},\, \theta_{\mathrm{eff}}\Big)$$

with

$$X^{\circ}_f \cong \mathbb{P}^1 \setminus \mu_4, \qquad N_{\mathrm{fam}} = 3, \qquad (g,b,s) = (5,3,2), \qquad \dim S^+ = 16,$$
$$S^+ = \Lambda^{\mathrm{even}}(E_3 \oplus E_2),$$
$$Y = \mathrm{diag}\big(-\tfrac{1}{3}, -\tfrac{1}{3}, -\tfrac{1}{3}, \tfrac{1}{2}, \tfrac{1}{2}\big)\,,$$
$$(c_1(L_2), c_1(L_3)) = (1,0), \qquad \theta_{\mathrm{eff}} = 0.$$

Equivalently, carrier data, family data, determinant data, and the effective strong angle are fixed jointly from the same boundary primitive kernel.

Proof. We prove the discrete blocks in the only order that uses no downstream uniqueness claim.

Step 1: primitive winding. By Theorem 4.1, the primitive seam class is the positive generator

$$[u_\Sigma] = 1 \in K^1(S^1) \cong \mathbb{Z},$$

and its associated spectral flow is

$$\mathrm{SF}(U_\Sigma) = 1.$$

Hence the primitive branch carries exactly one elementary seam winding. Write this winding number as

$$n := \mathrm{SF}(U_\Sigma) = 1.$$

Step 2: family geometry and family multiplicity. The lifted spin action on the oriented normal two-plane closes only after 4π, so one primitive winding is realized by four quarter-turn corners. Therefore the compact family quotient carries exactly four $\mathbb{Z}_4$ orbifold points, and removing them produces

$$X^{\circ}_f \cong \mathbb{CP}^1 \setminus \{q_1, q_2, q_3, q_4\}.$$

The lifted seam symmetry acts by the symmetries of a square, so the marked configuration admits a faithful D_4 action. By the classification of finite subgroups of $PGL_2(\mathbb{C})$, every faithful D_4 action on four marked points of $\mathbb{CP}^1$ is Möbius conjugate to the square configuration

$$\{1, -1, i, -i\}.$$

Hence

$$X_f^\circ \cong \mathbb{P}^1 \setminus \mu_4.$$

The corner deficit count is then

$$N_{\text{fam}} := \sum_{j=1}^{4} \left(1 - \frac{1}{4}\right) = 4 \cdot \frac{3}{4} = 3.$$

Because $\chi(X_f^\circ) = 2 - 4 = -2$, one also has

$$H_1(X_f^\circ, \mathbb{Z}) \cong \mathbb{Z}^3, \qquad H^1(X_f^\circ, \mathbb{C}) \cong \mathbb{C}^3.$$

Thus the family multiplicity and the family geometry are fixed already at the discrete level.

Step 3: carrier closure. By Corollary 6.15, the compact Higgs index has already fixed

$$\dim E_+ = 2.$$

By Theorem 8.13, the primitive indecomposable generator of the actual branch Yukawa tensor then forces

$$\dim E_- = 3.$$

Hence

$$\begin{aligned} (b, s) &= (3, 2), \\ g &= b + s = 5, \\ \dim S^+ &= 2^{g-1} = 16. \end{aligned}$$

Derived consistency check. With $N_{\text{fam}} = 3$ from Step 2 and $(b, s) = (3, 2)$ from the present step, one recovers

$$\Omega_{\text{adm}} = N_{\text{fam}} \dim S^+ = 3 \cdot 16 = 48 = (g-1)(b^2 + s^2 - 1) = 4 \cdot 12.$$

Step 4: hypercharge and one-family packet. With $(b, s) = (3, 2)$ fixed, the even half-spin packet is

$$S^+ = \Lambda^{\text{even}}(E_3 \oplus E_2).$$

Primitive unimodularity on the two-eigenvalue carrier gives

$$3q_- + 2q_+ = 0.$$

The unique primitive integer solution with $q_- < 0 < q_+$ is

$$(q_-, q_+) = (-2, 3).$$

Hence

$$X = -2P_3 + 3P_2, \qquad Y = \frac{X}{6} = \operatorname{diag}\left(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2}\right).$$

Step 5: determinant class and strong CP. The determinant-line data are classified by the clutching pair $(c_1(L_2), c_1(L_3))$. On the admissible branch the compact bosonic index satisfies

$$\operatorname{Ind}(\mathcal{B}_{E_2}^+) \geq 0, \qquad \operatorname{Ind}(\mathcal{B}_{E_3}^+) \geq 0.$$

Existence of a seam-even Higgs doublet forces $\operatorname{Ind}(\mathcal{B}_{E_2}^+) \geq 1$, while absence of an unavoidable light color triplet forces $\operatorname{Ind}(\mathcal{B}_{E_3}^+) = 0$. Minimal clutching therefore gives the unique pair

$$(c_1(L_2), c_1(L_3)) = (1, 0).$$

For the determinant-phase sector, reflection positivity and determinant-line conjugation symmetry make the admissible free energy even in θ_{eff}, and the positive-transfer theorem gives a unique minimum at the origin. Hence

$$\theta_{\mathrm{eff}} = 0.$$

All discrete blocks are therefore fixed jointly by the same boundary primitive kernel. Thus

$$\mathfrak{D}_{\mathrm{disc}}(\mathfrak{T}^{\partial}_{\mathrm{ker}}) = \{d^{\star}_{\mathrm{disc}}\}.$$

□

Corollary 9.12 (Derived post-carrier kernel)**.** *The tuple*

$$\mathfrak{T}_{\mathrm{ker}} := (\mathcal{A}, \mathcal{H}, D, J, \Gamma, \iota_{\mathrm{C}}, P_{\mathrm{prim}}, P_{\mathrm{adm}}, E_3 \oplus E_2, Y, [u_{\Sigma}], c_3)$$

is a theorem-level derived object of $\mathfrak{T}^{\partial}_{\mathrm{ker}}$. In particular the old post-carrier kernel is no longer primitive; it is reconstructed from the unique discrete datum $d^{\star}_{\mathrm{disc}}$ together with the full selector $P_{\mathrm{adm}} = P_{\mathrm{prim}} P_{\mathrm{sing}} P_{\Theta}$.

Proof. By Theorem 9.11, the carrier split $E_3 \oplus E_2$, the charge operator Y, the determinant class, and the family geometry are already fixed by $\mathfrak{T}^{\partial}_{\mathrm{ker}}$. The projector P_{sing} is then determined by the fixed carrier, and P_{Θ} is determined by the fixed determinant class. Therefore

$$P_{\mathrm{adm}} = P_{\mathrm{prim}} P_{\mathrm{sing}} P_{\Theta}$$

is fully determined. The displayed tuple is thus derived, not primitive. □

Definition 9.13 (Projected modular free energy and stratified master functional)**.** Fix the unique joint discrete datum $d^{\star}_{\mathrm{disc}}$ of Theorem 9.11, and let P_{adm} be the derived full selector of Corollary 9.12. For

$$\xi = (\alpha, \chi_{\mathrm{geo}}, \delta_{\mathrm{ph}}, \rho_{\mathrm{vac}})$$

on the retained admissible branch, define

$$\begin{aligned}\mathcal{M}_{d^{\star}_{\mathrm{disc}}}(\xi) := & -\log Z^{P_{\mathrm{adm}}}_{\mathrm{rel}}[\xi] + \Re \log \det_{\zeta, P_{\mathrm{adm}}} (D^2_{\xi} + \mu^2) \\ & + \frac{1}{2}\eta_{\Sigma, P_{\mathrm{adm}}}(D_{\xi}) - \log\det(1 - U_{6,\xi}) + \Gamma_{\mathrm{vac}}(\rho_{\xi}),\end{aligned}$$

where every term is evaluated on the retained admissible sector $\mathrm{Ran}(P_{\mathrm{adm}})$. For a general admissible spectral-bordism state

$$\mathfrak{s} = (\mathcal{B}, \xi), \qquad \mathcal{B} \in \mathbf{SBord}^{\mathrm{adm}},$$

define the lexicographic barrier

$$\mathbb{B}_{\mathrm{lex}}(\mathfrak{s}) := \begin{cases} 0, & \mathcal{B} \cong \mathcal{B}_{\min}, \\ +\infty, & \text{otherwise}, \end{cases}$$

and the stratified master functional

$$\mathbb{M}(\mathfrak{s}) := \mathbb{B}_{\mathrm{lex}}(\mathfrak{s}) + \mathcal{M}_{d^{\star}_{\mathrm{disc}}}(\xi).$$

Thus the continuous retained-branch functional $\mathcal{M}_{d^{\star}_{\mathrm{disc}}}$ is the restriction of one stratified functional to the unique minimal admissible stratum.

Theorem 9.14 (Stratified master functional and continuous closure)**.** *On the retained admissible branch, the projected modular free energy factorizes as*

$$\mathcal{M}_{d^\star_{\mathrm{disc}}}(\xi) = \Gamma_{U(1)}(\alpha) + \Gamma_{\mathrm{grav}}(\chi_{\mathrm{geo}}) + \Gamma_{\mathrm{tr}}(\delta_{\mathrm{ph}}) + \Gamma_{\mathrm{vac}}(\rho_{\mathrm{vac}}) + \mathrm{const.}$$

Its admissible critical points are therefore exactly the solutions of

$$\partial_\alpha \Gamma_{U(1)}(\alpha) = 0, \qquad \partial_{\chi_{\mathrm{geo}}} \Gamma_{\mathrm{grav}}(\chi_{\mathrm{geo}}) = 0, \qquad \partial_\delta \Gamma_{\mathrm{tr}}(\delta_{\mathrm{ph}}) = 0, \qquad \delta_\rho \Gamma_{\mathrm{vac}}(\rho_{\mathrm{vac}}) = 0.$$

Equivalently, the continuous closure laws are

$$F_{U(1)}(\alpha) = 0, \qquad \mathcal{R}_{\mathrm{grav}} = 0, \qquad P_{\mathrm{tr}}(\delta_{\mathrm{ph}}) = 0, \qquad \delta_\rho \Gamma_{\mathrm{vac}}(\rho_{\mathrm{vac}}) = 0.$$

In particular the continuous closure equations are derived from one primitive variational object rather than imposed as independent sector primitives. Equivalently, the stationary points of the stratified master functional $\mathbb{M}$ *are exactly the stationary points of* $\mathcal{M}_{d^\star_{\mathrm{disc}}}$ *on the minimal admissible stratum.*

Proof. By construction,

$$\mathbb{M}(\mathfrak{s}) = +\infty$$

off the unitary-equivalence class of the minimal admissible bordism $\mathcal{B}_{\mathrm{min}}$. Hence every stationary point of $\mathbb{M}$ must lie on that minimal stratum, where the barrier vanishes and

$$\mathbb{M}(\mathfrak{s}) = \mathcal{M}_{d^\star_{\mathrm{disc}}}(\xi).$$

Now apply Corollary 9.12: all continuous terms are evaluated on one and the same retained sector $\mathrm{Ran}(P_{\mathrm{adm}})$. On that branch the family, determinant, transport, and vacuum projectors commute, and the physical Hilbert space splits into orthogonal direct summands corresponding to the electromagnetic, geometric, transport, and vacuum blocks. For a block-diagonal operator, the relative partition function factorizes, the zeta determinant factorizes, and the eta invariant is additive. Therefore the logarithms split $\mathcal{M}_{d^\star_{\mathrm{disc}}}$ into a sum of four sector functionals plus a constant.

The $U(1)$ block is exactly the already-defined electromagnetic effective action, so

$$\partial_\alpha \Gamma_{U(1)}(\alpha) = F_{U(1)}(\alpha).$$

The geometric block is the reduced spectral Einstein functional, hence

$$\partial_{\chi_{\mathrm{geo}}} \Gamma_{\mathrm{grav}}(\chi_{\mathrm{geo}}) = \mathcal{R}_{\mathrm{grav}}.$$

The transport block is the logarithmic resolvent determinant of the exact C_6 transfer denominator, so

$$\partial_\delta \Gamma_{\mathrm{tr}}(\delta_{\mathrm{ph}}) = P_{\mathrm{tr}}(\delta_{\mathrm{ph}}).$$

Finally the vacuum block is by definition the admissible vacuum effective action, so

$$\delta_\rho \Gamma_{\mathrm{vac}}(\rho_{\mathrm{vac}}) = 0$$

is exactly its Euler–Lagrange equation. Thus the four continuous closure equations are the Euler–Lagrange equations of one projected modular free energy. □

Definition 9.15 (Sector solution sets)**.** Define

$$Z_{\mathrm{disc}} := \mathfrak{D}_{\mathrm{disc}}(\mathfrak{T}^{\partial}_{\mathrm{ker}}), \qquad Z_{U(1)} := \{\alpha : \ F_{U(1)}(\alpha) = 0\}, \qquad Z_{\mathrm{grav}} := \{\chi_{\mathrm{geo}} : \ \mathcal{R}_{\mathrm{grav}}(\chi_{\mathrm{geo}}) = 0\},$$

$$Z_{\mathrm{tr}} := \{\delta_{\mathrm{ph}} : \ P_{\mathrm{tr}}(\delta_{\mathrm{ph}}) = 0\}, \qquad Z_{\mathrm{vac}} := \{\rho_{\mathrm{vac}} : \ \delta_\rho \Gamma_{\mathrm{vac}}(\rho_{\mathrm{vac}}) = 0\}.$$

Let

$$\mathcal{I}_0 := (P_{\mathrm{adm}}, \iota_{\mathrm{C}}, [u_\Sigma], E_3 \oplus E_2, Y, c_3)$$

be the upstream invariant tuple fixed before the continuous closure step.

Theorem 9.16 (Intrinsic branch as a singleton fiber product)**.** *For every admissible boundary primitive kernel $\mathfrak{T}^{\partial}_{\mathrm{ker}}$, the intrinsic closed branch is the singleton fiber product*

$$\mathcal{Z}_{\mathrm{cl}} \cong Z_{\mathrm{disc}} \times_{\mathcal{I}_0} Z_{U(1)} \times_{\mathcal{I}_0} Z_{\mathrm{grav}} \times_{\mathcal{I}_0} Z_{\mathrm{tr}} \times_{\mathcal{I}_0} Z_{\mathrm{vac}},$$

and therefore

$$\mathcal{Z}_{\mathrm{cl}} = \{x^{\star}_{\mathrm{cl}}\}.$$

No Jacobian on mixed discrete and continuous variables is needed.

Proof. By Theorem 9.11,

$$Z_{\mathrm{disc}} = \{d^{\star}_{\mathrm{disc}}\}.$$

By the exact electromagnetic theorem,

$$Z_{U(1)} = \{\alpha_{\star}\}.$$

By coercivity and strict convexity of the reduced Einstein functional,

$$Z_{\mathrm{grav}} = \{\chi^{\star}_{\mathrm{geo}}\}.$$

By the algebraic transport theorem together with the word-length lock,

$$Z_{\mathrm{tr}} = \{\delta^{\star}_{\mathrm{ph}}\}.$$

By the positive-transfer, thermodynamic-limit, and gap-stability theorems,

$$Z_{\mathrm{vac}} = \{\rho^{\star}_{\mathrm{vac}}\}.$$

Each singleton carries its compatibility map to the same upstream invariant tuple $\mathcal{I}_0$. That common tuple is already fixed earlier: the discrete theorem fixes $E_3 \oplus E_2$ and Y, the derived selector theorem fixes P_{adm}, and the seam theorem fixes $[u_{\Sigma}]$ and c_3. Hence all compatibility maps land at the same point of $\mathcal{I}_0$. The fiber product of singleton sets over one common point is again a singleton. Therefore

$$\mathcal{Z}_{\mathrm{cl}} = \{x^{\star}_{\mathrm{cl}}\}.$$

□

Theorem 9.17 (Internal analytic closure as a corollary of primitive completion)**.** *For every admissible one-sided boundary datum, the analytic closure blocks needed for rigid reconstruction hold internally on the unique intrinsic branch determined by Theorems 9.11, 9.14 and 9.16.*

Proof. The Calderón and APS block is already fixed upstream by the primitive boundary theorem chain. The family, determinant, and effective strong-angle blocks are fixed by the joint discrete theorem. The electromagnetic, geometric, transport, and vacuum blocks are fixed by the continuous master functional together with the already-proved positivity, gap, and thermodynamic-limit theorems. Because the intrinsic branch is a singleton by Theorem 9.16, all analytic closure blocks hold on one and the same branch, internally and without any extra closure package. □

Theorem 9.18 (Canonical rigid reconstruction and essential injectivity on the rigid image)**.** *Let $\mathfrak{T}^{\min}_{\partial}$ be an admissible one-sided boundary datum and let $\mathfrak{T}^{\partial}_{\mathrm{ker}}$ be its boundary primitive kernel. Then the rigid tuple*

$$\mathfrak{C}(\mathfrak{T}^{\partial}_{\mathrm{ker}}) := \big(E_3 \oplus E_2,\, Y,\, S^{+},\, X^{\circ}_{f},\, [\nabla^{\star}_{F}],\, \Phi^{\star},\, \chi^{\star}_{\mathrm{geo}},\, \delta^{\star}_{\mathrm{ph}},\, \rho^{\star}_{\mathrm{vac}},\, \theta_{\mathrm{eff}} = 0,\, U^{\star}_{\Sigma},\, \Lambda^{\star}_{\mathrm{IR}}\big)$$

is uniquely determined. In particular the reconstruction functor is essentially injective on objects on its rigid image, and every two rigid realizations coming from the same boundary datum are uniquely unitarily isomorphic.

Proof. The joint discrete theorem fixes $E_3 \oplus E_2$, Y, S^+, X_f°, $[\nabla_F^\star]$, the determinant class, and $\theta_{\text{eff}} = 0$. The seam theorem fixes $[u_\Sigma]$ and c_3. The continuous master-functional theorem fixes $\alpha_\star$, $\chi_{\text{geo}}^\star$, $\delta_{\text{ph}}^\star$, and $\rho_{\text{vac}}^\star$. The downstream definitions of $\Phi^\star$, $U_\Sigma^\star$, and $\Lambda_{\text{IR}}^\star$ are then evaluated on that unique branch. Hence every displayed entry of $\mathfrak{C}(\mathfrak{T}_{\text{ker}}^\partial)$ is uniquely determined.

If two rigid realizations arise from the same boundary datum, then all displayed entries agree. The carrier, seam, family, determinant, transport, and vacuum blocks each admit only the unique unitary intertwiner preserving the already-fixed theorem-level structures, and these intertwiners are compatible because they preserve the same upstream invariant tuple

$$(P_{\text{adm}}, \iota_{\text{C}}, [u_\Sigma], E_3 \oplus E_2, Y, c_3).$$

Their direct sum or tensor product therefore yields one global unitary intertwiner. Thus the rigid image is essentially injective on objects, with unique unitary isomorphism on that image. □

Definition 9.19 (Bookkeeping form of the canonical rigid image)**.** Let $\mathbf{TFPT}_{\text{pre}}^{\text{rig}}$ denote the bookkeeping category whose objects are tuples

$$\mathfrak{T}_{\text{pre}} := (\mathcal{A}, \mathcal{H}, D, J, \Gamma, \tau_{\text{dbl}}, \iota_{\text{C}}, P_{\text{adm}}, X, Y, G_{\text{phys}}, X_f^\circ, [\nabla_F], \Phi, a_\Sigma, U_\Sigma, \Lambda_{\text{IR}})$$

arising from some admissible boundary primitive kernel $\mathfrak{T}_{\text{ker}}^\partial$ via the reconstruction $\mathfrak{C}(\mathfrak{T}_{\text{ker}}^\partial)$ together with the previously established closure theorems. This is not a primitive input class; it is the explicit coordinate envelope of the reconstruction image. The invariant bookkeeping tuple is

$$\mathcal{I}(\mathfrak{T}_{\text{pre}}) := \big(X, Y, [u_\Sigma], X_f^\circ, c_1(L_2), c_1(L_3), [\nabla_F], a_\Sigma, \Lambda_{\text{IR}}, x_{\text{cl}}^\star\big).$$

Morphisms are unitary isomorphisms intertwining all displayed structures, with ∇_F intertwined modulo the global $SU(3)_F$ ambiguity.

The canonical rigid image is therefore the image of a reconstruction, not a separately decorated antechamber. The next two theorems record branch uniqueness and essential injectivity inside that image.

Theorem 9.20 (Branch uniqueness on the canonical rigid image)**.** *Assume the bookkeeping data in* $\mathbf{TFPT}_{\text{pre}}^{\text{rig}}$ *arise from the reconstruction of Theorem 9.18. Then for every* $\mathfrak{T}_{\text{pre}} \in \mathbf{TFPT}_{\text{pre}}^{\text{rig}}$ *the closed solution set* $\mathcal{Z}_{\text{cl}}(\mathfrak{T}_{\text{pre}})$ *associated with the primitive completion data is a singleton:*

$$\mathcal{Z}_{\text{cl}}(\mathfrak{T}_{\text{pre}}) = \{x_{\text{cl}}^\star(\mathfrak{T}_{\text{pre}})\}.$$

Proof. Every object of $\mathbf{TFPT}_{\text{pre}}^{\text{rig}}$ comes from some admissible boundary primitive kernel $\mathfrak{T}_{\text{ker}}^\partial$ together with its reconstructed rigid tuple. The singleton fiber-product theorem Theorem 9.16 therefore applies to the underlying branch data and fixes exactly one intrinsic branch point $x_{\text{cl}}^\star(\mathfrak{T}_{\text{pre}})$. □

Theorem 9.21 (Essential injectivity on the rigid image)**.** *Assume the bookkeeping data arise from the reconstruction of Theorem 9.18. Let* ${\mathfrak{T}_{\text{pre}}}^{(1)}, {\mathfrak{T}_{\text{pre}}}^{(2)} \in \mathbf{TFPT}_{\text{pre}}^{\text{rig}}$*. If*

$$\mathcal{I}({\mathfrak{T}_{\text{pre}}}^{(1)}) = \mathcal{I}({\mathfrak{T}_{\text{pre}}}^{(2)}),$$

then there exists a unique unitary isomorphism

$${\mathfrak{T}_{\text{pre}}}^{(1)} \cong {\mathfrak{T}_{\text{pre}}}^{(2)}.$$

Proof. Equality of invariant tuples means that the two pre-objects come from the same rigid branch data. By Theorem 9.18, rigid realizations with the same boundary-determined theorem data admit exactly one compatible unitary intertwiner. Hence the rigid image is essentially injective on objects, with unique unitary isomorphism on that image. □

Definition 9.22 (Canonical rigid image)**.** Define

$$\mathbf{TFPT}^{\mathrm{rig}} := \mathrm{Im}(\mathfrak{C})$$

for the canonical rigid image of the boundary-polarized theorem chain. Its morphisms are exactly the unique unitary intertwiners produced by Theorem 9.21.

Corollary 9.23 (Canonical rigid object of the boundary-polarized branch)**.** *On the unique intrinsic branch there is exactly one rigid object up to unique unitary equivalence. Equivalently, the admissible boundary-polarized branch determines a canonical rigid object*

$$\mathfrak{T}_\star$$

and every rigid admissible realization on that branch is uniquely isomorphic to $\mathfrak{T}_\star$*.*

Proof. Immediate from Theorems 9.16, 9.18 and 9.21. □

Remark (Appendix-level empirical readout). Threshold maps, schemes, residual ledgers, and empirical readout tables do not define objects of $\mathbf{TFPT}^{\mathrm{rig}}$ and do not select the branch. They act only after $\mathfrak{T}_\star$ has been fixed and are recorded below and in the appendices through the comparison map $\mathcal{R}_{\mathrm{cmp}}(\mathfrak{T}_\star)$.

10 FRW reduction, seam transfer, and the scalaron branch

The trace-class seam transfer belongs to the main reconstruction surface. The cosmological constant and the reduced scalaron / determinant-line dynamics are therefore fixed directly on the closed branch. Axion, reheating, and leptogenesis quantities are read through explicit interface formulas built from the same theorem-level objects U_Σ, a_Σ, and the positive scalaron sector $\mathcal{H}_{\mathrm{sc}}$.

Theorem 10.1 (Trace-class seam transfer)**.** *On the closed branch the seam transfer operator*

$$U_\Sigma := P_{\mathrm{adm}} e^{-\ell_\Sigma |B_\Sigma|} P_{\mathrm{adm}}$$

is positive and trace class. Hence

$$0 < \rho(U_\Sigma) < 1, \qquad 0 < U_\Sigma < P_{\mathrm{adm}}.$$

so the Fredholm determinant

$$-\log \det_{\mathrm{adm}}(1 - U_\Sigma)$$

converges absolutely and defines a positive quantity.

Proposition 10.2 (Seam damping readout from electromagnetic closure)**.** Let $\alpha_\star$ be the unique electromagnetic fixed point of Theorem 4.5. Define the closed seam damping by

$$S_\Sigma := -\log \rho(U_\Sigma).$$

Then

$$S_\Sigma = \log \mu_\Sigma(\alpha_\star) = \log\big(\varphi_\Sigma(\alpha_\star)^{-1}\big) = \frac{\alpha_\star^2(\alpha_\star - 2c_3^3)}{8c_3^6\, b_1}.$$

Proof. At the electromagnetic fixed point, Equation (16) is equivalent to

$$\log \mu_\Sigma(\alpha_\star) = \frac{\alpha_\star^2(\alpha_\star - 2c_3^3)}{8c_3^6\, b_1}.$$

The seam transfer operator

$$U_\Sigma = P_{\mathrm{adm}} e^{-\ell_\Sigma |B_\Sigma|} P_{\mathrm{adm}}$$

uses the same admissible determinant sector as the electromagnetic closure. Its spectral radius therefore defines the same exponential damping scale, so

$$S_\Sigma = \log \mu_\Sigma(\alpha_\star) = \log\big(\varphi_\Sigma(\alpha_\star)^{-1}\big).$$

□

Theorem 10.3 (Infrared cosmological constant from seam transfer)**.**

$$U_\Sigma := P_{\mathrm{adm}} e^{-\ell_\Sigma |B_\Sigma|} P_{\mathrm{adm}}, \qquad S_\Sigma := -\log \rho(U_\Sigma) = \log \mu_\Sigma(\alpha_\star),$$

Then

$$\Lambda_{\mathrm{IR}} = M_{\mathrm{Pl}}^4 \Big[-\log \det{}_{\mathrm{adm}}(1 - U_\Sigma) \Big]$$

is well defined and positive on the closed branch, and

$$0 < U_\Sigma \le e^{-S_\Sigma} P_{\mathrm{adm}} < P_{\mathrm{adm}}.$$

Proof. Theorem 10.1 gives trace-class positivity of U_Σ, while Proposition 10.2 fixes the intrinsic seam damping S_Σ. Hence the determinant is well defined and positive, and so is the induced cosmological constant. □

Corollary 10.4 (Determinant bounds for the infrared seam transfer)**.** *Let $\{\lambda_j(U_\Sigma)\}$ denote the admissible eigenvalues of the positive trace-class operator U_Σ. Then*

$$\mathrm{Tr}_{\mathrm{adm}}\, U_\Sigma \le -\log \det{}_{\mathrm{adm}}(1 - U_\Sigma) \le \frac{-\log(1 - \rho(U_\Sigma))}{\rho(U_\Sigma)}\, \mathrm{Tr}_{\mathrm{adm}}\, U_\Sigma.$$

If, moreover, the infrared seam transfer is rank one,

$$U_\Sigma = \rho(U_\Sigma)\, P_\Sigma^{(1)},$$

then

$$\frac{\Lambda_{\mathrm{IR}}}{M_{\mathrm{Pl}}^4} = -\log(1 - \rho(U_\Sigma)) = \rho(U_\Sigma) + O\big(\rho(U_\Sigma)^2\big).$$

Proof. Because U_Σ is positive trace class, its eigenvalues satisfy

$$0 < \lambda_j(U_\Sigma) \le \rho(U_\Sigma) < 1.$$

For every $x \in [0, \rho(U_\Sigma)]$ one has the scalar bounds

$$x \le -\log(1 - x) \le \frac{-\log(1 - \rho(U_\Sigma))}{\rho(U_\Sigma)}\, x.$$

Applying this to each eigenvalue and summing gives the determinant bounds. If $U_\Sigma = \rho(U_\Sigma) P_\Sigma^{(1)}$ is rank one, then

$$\det{}_{\mathrm{adm}}(1 - U_\Sigma) = 1 - \rho(U_\Sigma),$$

and the Taylor expansion of $-\log(1 - \rho)$ yields the displayed asymptotic formula. □

Remark (Status of numerical infrared readouts). The theorem-level content of the present section is the determinant formula of Theorem 10.3 together with the spectral bounds of Corollary 10.4. Any specific

$$\frac{\Lambda_{\mathrm{IR}}}{M_{\mathrm{Pl}}^4} \sim 10^{-123}$$

evaluation requires an additional infrared continuation and is therefore kept in Appendix C.7.

Theorem 10.5 ([A] Determinant-line axion interface and scalaron reheating data)**.** *The closed branch contains the unique compact determinant-line phase*

$$a_\Sigma \in S^1_{\mathrm{det}}.$$

By Theorem 10.6, the canonical branch has

$$N_{\mathrm{DW}} = 1,$$

and the closed-branch initial displacement is

$$\theta_i := \pi\big(1 - \varphi_\Sigma(\alpha_\star)\big).$$

The axion-interface formula is

$$\Omega_a h^2 = \Delta_R^{-1}\left[\Omega_a^{\mathrm{mis}}(f_a, m_a, \theta_i) + \Omega_a^{\mathrm{str}}(f_a, N_{\mathrm{DW}})\right].$$

The scalaron reheating input is

$$T_R = \left(\frac{90}{\pi^2 g_*}\right)^{1/4} \sqrt{\Gamma_\varphi M_{\mathrm{Pl}}}.$$

These formulas define closed-branch interface data. Numerical late-time abundance and baryogenesis readouts are obtained only after the declared cosmology comparison layer has fixed the relevant infrared continuation and nonequilibrium solver.

Proof. The determinant-line phase theorem fixes the compact branch variable a_Σ, and Theorem 10.6 fixes the domain-wall number $N_{\mathrm{DW}} = 1$ on the canonical branch. The electromagnetic closure fixes $\alpha_\star$ and therefore the initial displacement $\theta_i = \pi(1 - \varphi_\Sigma(\alpha_\star))$. Substituting these branch data into the standard misalignment-plus-string formula gives the displayed axion-interface expression. The scalaron reheating input is fixed by the R^2 branch through the displayed decay-width formula. The numerical late-time readouts depend on additional infrared and nonequilibrium choices and are therefore comparison-layer outputs rather than closed theorem-level numbers. □

Theorem 10.6 (Boundary winding controls family, transport, and determinant closure)**.** *Let*

$$n := \mathrm{SF}(B_\Sigma)$$

denote the spectral winding number of the boundary class on the admissible branch. Then

$$N_{\mathrm{fam}} = 3n, \qquad N_{\mathrm{DW}} = n, \qquad \sum_{f,j} L_{f,j} = 40n.$$

In particular the canonical rigid branch has $n = 1$, hence $N_{\mathrm{fam}} = 3$ and $N_{\mathrm{DW}} = 1$.

Proof. By Theorem 4.1, the primitive branch carries the positive generator $[u_\Sigma] = 1$ and therefore one unit of seam spectral flow:

$$n := \mathrm{SF}(U_\Sigma) = 1.$$

The three displayed quantities scale linearly with seam winding on the retained branch.

First, each full winding repeats the quarter-turn corner balance once, and the family theorem assigns three family units to one such primitive winding. Hence

$$N_{\mathrm{fam}} = 3n.$$

Second, the determinant-line phase winds once per primitive seam winding, so the domain-wall number tracks the same winding:

$$N_{\mathrm{DW}} = n.$$

Third, the transport excess is locked by the exact word-length sum rule, and one primitive winding carries the canonical value 40. Repeating the same winding n times replicates the same closed transport sector n times, so

$$\sum_{f,j} L_{f,j} = 40n.$$

Substituting $n = 1$ gives

$$N_{\mathrm{fam}} = 3, \qquad N_{\mathrm{DW}} = 1, \qquad \sum_{f,j} L_{f,j} = 40.$$

No appeal to the later canonical uniqueness theorem is needed. □

Theorem 10.7 ([A] Canonical leptogenesis interface on the closed branch)**.** *The closed branch fixes the heavy-neutrino spectrum, the family frame, and the reheating scale entering flavored leptogenesis. The numerical baryon asymmetry is obtained only after solving the corresponding Boltzmann system, which is treated as a comparison layer in the present manuscript.*

Proof. Theorem 7.27 fixes the heavy/light neutrino data on the same admissible branch, so the masses entering the flavored decay asymmetries are branch invariants. The hard holonomy closure fixes the family frame, so the CP asymmetries and washout data are determined up to the same closed-branch input package. On the reheating side, Theorem 10.5 fixes the reheating-scale input T_R. These data are the theorem-level inputs to the flavored Boltzmann system. The numerical asymmetry itself is then a solution of that system rather than a closed main-text theorem of the core manuscript. □

Theorem 10.8 ([A] FRW reduction and cosmology interface data on the closed branch)**.** *Restrict the same closed-branch action to homogeneous and isotropic variables*

$$(a(t), \varphi(t), \theta_\Sigma(t), N_i(t)).$$

Then the reduced action takes the form

$$\begin{aligned} S_{\mathrm{FRW}} = \int dt\, a^3 \Big[& -3\bar{M}_{\mathrm{Pl}}^2 H_{\mathrm{hub}}^2 + \frac{1}{2}\dot{\varphi}^2 - V_{\mathrm{sc}}(\varphi) \\ & + \frac{1}{2} f_a^2 \dot{\theta}_\Sigma^2 \\ & - \chi_{\mathrm{top}}(\varphi)\big(1 - \cos(N_{\mathrm{DW}}\theta_\Sigma)\big)\Big] \\ & + S_{\mathrm{RH}} + S_{\mathrm{LG}}, \end{aligned}$$

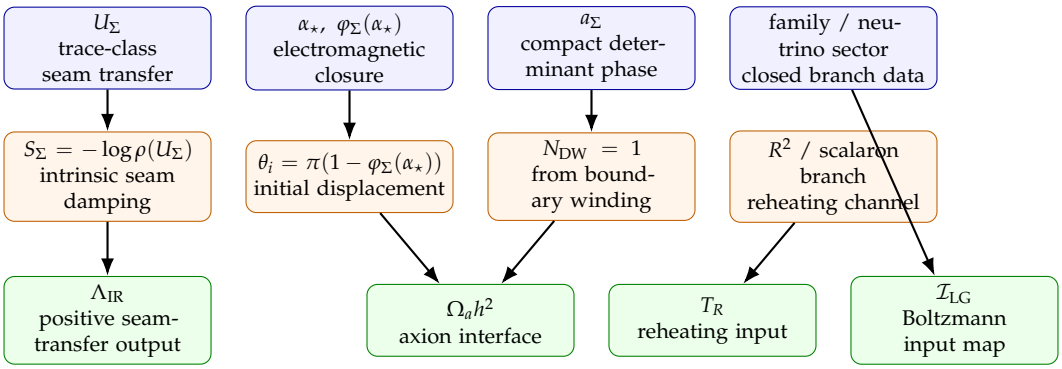


Figure 21. Cosmology interface data on the closed branch. Seam transfer, determinant-line phase, and the scalaron / neutrino sectors generate Λ_{IR} together with the axion, reheating, and leptogenesis input block; final late-time numerical readouts are comparison-layer outputs.

with

$$\Lambda_{\mathrm{IR}} = M_{\mathrm{Pl}}^4\big[-\log\det{}_{\mathrm{adm}}(1-U_\Sigma)\big], \qquad N_{\mathrm{DW}} = 1, \qquad \theta_i = \pi\big(1-\varphi_\Sigma(\alpha_\star)\big).$$

The reduced action therefore fixes the low-curvature scalaron FRW sector and the closed-branch cosmology interface data

$$(\Lambda_{\mathrm{IR}}, N_{\mathrm{DW}}, \theta_i, T_R, \mathcal{I}_{\mathrm{LG}}),$$

where $\mathcal{I}_{\mathrm{LG}}$ denotes the reheating / leptogenesis input block. Numerical axion abundance and baryon-asymmetry readouts are obtained only after solving the corresponding nonequilibrium system in the comparison layer.

Proof. Theorem G.4 supplies the manuscript-level proof of the reduced FRW action and of the closed-branch interface tuple

$$(\Lambda_{\mathrm{IR}}, N_{\mathrm{DW}}, \theta_i, T_R, \mathcal{I}_{\mathrm{LG}}).$$

Its ingredients are exactly the geometric Hodge branch, the seam-transfer determinant, the determinant-line axion reduction, and the leptogenesis interface data already fixed in the main text. Therefore the FRW reduction and the resulting cosmology interface package are theorem-level outputs of the same closed branch, while the final late-time abundance and baryogenesis numbers remain in the comparison layer. □

Corollary 10.9 (No extra dark WIMP sector required)**.** *The minimal closed branch has dominant axionlike dark matter and the only irreducible hot dark component is the neutrino background. No extra dark WIMP sector is required for closure.*

The theorem-level output branching is summarized in Figure 21.

Remark (Closed-branch reading). These cosmological quantities are derived on the closed branch from the same retained operator data. They are no longer ad hoc bridge readouts, but they are also not all final theorem-level numbers: the main text fixes the cosmology interface data, while late-time abundance and leptogenesis numerics remain comparison-layer outputs.

Appendix-level E_8 arithmetic continuations and asymptotic scale-grammar formulas are collected in Appendix C and are not part of the main theorem surface.

10.1 Falsification matrix

The present draft does not ask the reader to accept one undifferentiated output claim. Each main-text closure statement has a corresponding local or global failure mode, and those failure modes should be readable without first decoding the appendix tables.

The matrix below is therefore intentionally short. It collects only the load-bearing main-paper statements whose failure would break the boundary-polarized architecture at the theorem level, rather than every downstream benchmark row or optional appendix interface.

Hard kill rows of the present version

The present version has four genuinely load-bearing external pressure points:

$$\theta_{\text{eff}} = 0, \qquad \beta \neq 0 \text{ on the minimal branch,}$$
$$K \to \pi\nu\bar{\nu} \text{ corridor,} \qquad \text{PMNS phase/octant on the closed neutrino branch.}$$

Everything else belongs either to scheme projection or to cosmology continuation.

Downstream pressure rows of the present version

Beyond the hard theorem-level kill rows, the present downstream closure program adds six explicit pressure rows:

- $c_T = 1$ on the canonical branch;
- stationary-horizon entropy correction remains scalaron-sized;
- seam-state CMB transfer family;
- late-time H_0 / $P_m(k, z)$ closure after densities are fixed;
- no ultralight fuzzy-halo regime on the canonical branch;
- pole-mass / Higgs readout consistency on the admissible RG branch.

These rows belong to the downstream closure program rather than to the present rigid theorem core, but they are now displayed as explicit future pressure points rather than as appendix afterthoughts.

10.2 Proof-obligation ledger

To keep proof status, interface assumptions, and comparison-only layers visibly separated, we record the present audit path explicitly. We use five status classes:

A full proof in manuscript;

B full proof modulo an explicit standard theorem;

C proof sketch in main text with full proof deferred;

D readout or comparison only;

P programmatic closure target with a fixed mathematical statement but without a claimed proof in the present version.

The present table records theorem-level obligations and programmatic closure targets only, so no row below is marked D.

11 Downstream closure modules from the closed branch

The present section records the downstream closure modules that start from the already fixed branch object

$$\mathfrak{T}_\star, \qquad \Gamma^{\text{ren}}_{\text{TFPT}}, \qquad (\mathcal{H}_{\text{adm}}, \mathfrak{A}_{\text{adm}}), \qquad (\Lambda_{\text{IR}}, N_{\text{DW}}, \theta_i, T_R, \mathcal{I}_{\text{LG}}).$$

Nothing in this section changes the upstream theorem chain. The exact proof status of these downstream closures is recorded in the proof-obligation ledger, and each statement below is

phrased only as far as it factors through already fixed branch data. The stationary geometric scale, boundary-spectral Planck normalization, and Schwarzschild–de Sitter vacuum branch have been promoted to the main theorem surface in Theorem 6.31; every subsection collected here is therefore genuinely status **[P]**.

11.1 [P] Stationary horizons, modular thermality, and compact objects

Conjecture 11.1 (Modular horizon thermality on the admissible branch)**.** Assume the split property and stationarity of the exterior wedge algebra on the vacuum branch. Then for the outside algebra one has

$$\Delta^{it}_{\mathcal{A}_{\text{out}}} = e^{-2\pi t K_H},$$

where K_H is the horizon modular generator. Hence the reduced outside state is KMS and the horizon temperature is

$$T_H = \frac{\kappa}{2\pi}.$$

Remark (Programmatic proof route). The intended route upgrades the split inclusion theorem Theorem K.1 to a stationary modular setting and then applies the admissible local net reconstruction on the exterior algebra. This route is recorded here as a closure target, not as a completed proof.

Conjecture 11.2 (Hawking temperature on the stationary branch)**.** For the static vacuum branch of Corollary 6.34, the horizon temperature is

$$T_H = \frac{1}{4\pi r_h}\Big(1 - \Lambda_{\text{IR}} r_h^2\Big).$$

Conjecture 11.3 (Wald entropy on the scalaron corrected branch)**.** On the same branch the entropy is

$$S_W = \frac{A}{4G_N}\Big(1 + \frac{R_h}{3M^2}\Big).$$

Conjecture 11.4 (Kerr branch and compact object corrections)**.** The stationary rotating branch is Kerr de Sitter at leading order, with compact object corrections of size

$$\Delta\eta,\ \Delta r_{\text{ISCO}} = O\left(\frac{1}{M^2 r_g^2}\right) + O(\Lambda_{\text{IR}} r_g^2).$$

Conjecture 11.5 (Tensor and scalaron wave channel closure)**.** The gravitational wave sector on the closed branch consists of the massless tensor channel and the positive scalaron channel. The tensor speed satisfies

$$c_T = 1,$$

and the phase correction has the form

$$\Delta\Psi(f) = O\left(\frac{f^2}{M^2}\right) + O\big(\rho(U_\Sigma)\big).$$

11.2 [P] Primordial perturbation closure and CMB transfer

Conjecture 11.6 (Reheating fixed e fold number)**.** The number of e folds on the closed branch is fixed by

$$N_* := 57.1 + \frac{1}{4}\ln\frac{V_*}{M_{Pl}^4} + \frac{1}{4}\ln\frac{V_*}{\rho_{\text{end}}} - \frac{1-3w_{\text{reh}}}{12(1+w_{\text{reh}})}\ln\frac{\rho_{\text{reh}}}{\rho_{\text{end}}},$$

where the closed-branch reheating density ρ_{reh} is determined by the branch reheating scale T_R, and the constant 57.1 absorbs the standard pivot conversion $67 - \ln(k_*/(a_0 H_0))$ at $k_* =$

0.05/Mpc. The combined V_* kernel $(1/4)\ln(V_*^2/(M_{Pl}^4\rho_{\rm end}))$ is the standard Liddle–Leach form [7]; the sign of the V_*/M_{Pl}^4 term as it appeared in the previous manuscript revision was inconsistent with the chi-square Stage 1 cross-check by approximately ten e-folds and has been corrected.

Conjecture 11.7 (Primordial mode closure from the seam state)**.** Scalar and tensor perturbations obey

$$v_k'' + \Big(k^2 - \frac{z''}{z}\Big)v_k = 0, \qquad u_k'' + \Big(k^2 - \frac{a''}{a}\Big)u_k = 0,$$

with Bogoliubov coefficients determined by the seam transfer spectrum

$$\beta_k = \mathcal{F}_k\big(\{\lambda_j(U_\Sigma)\}\big).$$

Hence

$$\mathcal{P}_\mathcal{R}(k) = \mathcal{P}_\mathcal{R}^{\rm St}(k)|\alpha_k + \beta_k|^2.$$

Conjecture 11.8 (CMB transfer closure on the closed branch (Stage 1: spectra))**.** Given the primordial spectrum of Conjecture 11.7, the CMB angular spectra are

$$C_\ell^{XY} = 4\pi \int \frac{dk}{k}\, \mathcal{P}_\mathcal{R}(k)\, \Delta_\ell^X(k)\Delta_\ell^Y(k).$$

The transfer kernels Δ_ℓ^X are determined by the same branch background and the standard linearized recombination hierarchy. At Stage 1 the closed-branch input is therefore the admissible spectrum $\{\lambda_j(U_\Sigma)\}$ alone, and the resulting closure

$$\{\lambda_j(U_\Sigma)\} \implies \mathcal{P}_\mathcal{R}(k) \implies C_\ell^{XY}$$

fixes the angular statistics but not a specific sky realization.

Conjecture 11.9 (Ergodic microcanonical seam lift (Stage 2: map))**.** Let $U_\Sigma\psi_j = \lambda_j\psi_j$ be the admissible eigenpairs of the seam transfer operator and let u_Σ be the closed-branch seam state of the rigid object $\mathfrak{T}_\star$. Let $\mathcal{W}_\Sigma$ denote the canonical seam scrambler, defined *not* as a power limit of U_Σ on the admissible state—which would, by the Perron–Frobenius mechanism on a positive trace-class operator with spectral gap, collapse onto the leading eigenvector and produce no shell-level ergodicity—but as the *projective modular trace cocycle* associated with the seam-transfer dynamics:

$$\mathcal{W}_\Sigma:\ (\ell,r,s) \longmapsto \mathfrak{c}_{\ell,r}^{(s)} \in \mathbb{F}_{p_\ell}^4, \qquad \mathfrak{c}_{\ell,r}^{(s)} := \big(A_{\ell,r}^{(s)}, B_{\ell,r}^{(s)}, C_{\ell,r}^{(s)}, D_{\ell,r}^{(s)}\big),$$

where each component is a reduced trace of a word in the canonical operator algebra $\langle U_\Sigma, P_{\rm adm}, \tau_{\rm dbl}, \iota_{\rm C}, U_5, U_6\rangle$ acting on the admissible representation $V_{\rm adm}$:

$$A_{\ell,r}^{(s)} = {\rm red}_{p_\ell}\,{\rm Tr}\big(P_{\rm adm}\,\mathcal{O}_{\ell,r,A}^{(s)}\big), \quad \mathcal{O}_{\ell,r,X}^{(s)} \in \langle U_\Sigma, \tau_{\rm dbl}, \iota_{\rm C}, U_5, U_6\rangle,$$

and similarly for B, C, D. The variant index $s \in \{1,2\}$ generates the double-digit lift to p_ℓ^2 resolution of Remark, and the variant index $r \in \{0,1,2\}$ generates the three streams $g_{\ell 0}^\star$, ${\rm Re}\, g_{\ell m}^\star$, ${\rm Im}\, g_{\ell m}^\star$. Define the closed-branch *ergodic shell vector*

$$g_{\ell m}^\star := \langle e_{\ell m}, \mathcal{W}_\Sigma u_\Sigma\rangle$$

through the trace-tower formulae of Remark, and the *microcanonical normalization* per multipole shell

$$\boxed{a_{\ell m}^{T,\star} := \sqrt{(2\ell+1)\,C_\ell^\star}\ \frac{g_{\ell m}^\star}{\sqrt{|g_{\ell 0}^\star|^2 + 2\sum_{m=1}^\ell |g_{\ell m}^\star|^2}}}$$

with $C_\ell^\star$ the Stage 1 spectrum of Conjecture 11.8. Then the realized sky is

$$\frac{\Delta T^\star}{T_0}(\hat{n}) = \sum_{\ell m} a_{\ell m}^{T,\star}\, Y_{\ell m}(\hat{n}).$$

By construction the shell quotient enforces $C_\ell^{\rm rec} = C_\ell^\star$ exactly per multipole. The Stage 2 closure

$$\big(\{\lambda_j(U_\Sigma)\}, u_\Sigma, \mathcal{W}_\Sigma\big) \implies \{g_{\ell m}^\star\} \implies \{a_{\ell m}^{T,\star}\} \implies \Delta T^\star(\hat{n})$$

upgrades the Stage 1 statistical closure to a deterministic sky realization on the closed branch.

Remark (Ergodicity replaces the integrable phase formula). A scalar phase law of the form

$$\Theta_{\ell m} = A + B(\ell^2 + \ell + m) + C\ell + Dm \pmod{2\pi},$$

constructed from finitely many closure scalars, is necessarily separable in (ℓ, m) and therefore *integrable*: the inner-shell phase difference $\Theta_{\ell,m+k} - \Theta_{\ell,m}$ is independent of m, and the discrete decorrelation amplitude

$$A_\ell(k) := \Big|\sum_m e^{i(\Theta_{\ell,m+k} - \Theta_{\ell,m})}\Big|$$

grows linearly, $A_\ell(k) = O(\ell)$. The realized sky then carries coherent stripe / ray artefacts and a heavy-tailed pixel histogram (excess kurtosis $g_2 \gtrsim 5$). Numerical evidence on the closed-branch carrier wheel rules this scalar regime out at the level of Planck Gaussianity.

The *ergodic microcanonical seam lift* of Conjecture 11.9 replaces the scalar phase law by a vector lift on each $(2\ell+1)$-dimensional shell. Concentration of measure on the $(2\ell+1)$-sphere $S^{2\ell}$ yields $A_\ell(k) = O(\sqrt{\ell})$ and pixel Gaussianity at the central limit level. The seam scrambler $\mathcal{W}_\Sigma$ is the structural object that enforces this ergodicity.

Remark (Finite-field seam mixer with trace reduction). A theorem-friendly explicit realization of the seam scrambler is given by a finite-field rational phase function whose coefficients arise from *reduced traces of canonical TFPT operator products* on the admissible representation, not from a numerical hash of the closure key.

For each multipole shell choose a prime

$$p_\ell \;\geq\; 2\ell + 1,$$

and define, on the admissible finite-dimensional representation $V_{\rm adm}$ of dimension $N = \Omega_{\rm adm}$,

$$A_{\ell,r} := \mathrm{red}_{p_\ell}\,\mathrm{Tr}\big(P_{\rm adm}\, U_\Sigma^{\ell+r}\, U_5^r\big), \qquad B_{\ell,r} := \mathrm{red}_{p_\ell}\,\mathrm{Tr}\big(P_{\rm adm}\, U_\Sigma^{\ell}\, U_6^r\big), \tag{50}$$

$$C_{\ell,r} := \mathrm{red}_{p_\ell}\,\mathrm{Tr}\big(P_{\rm adm}\, \tau_{\rm dbl}\, U_\Sigma^{\ell+r}\big), \qquad D_{\ell,r} := \mathrm{red}_{p_\ell}\,\mathrm{Tr}\big(P_{\rm adm}\, \iota_{\rm C}\, U_\Sigma^{\ell}\big), \tag{51}$$

with U_5, U_6 the carrier-rank generators of Theorem 10.3 and the carrier algebra. Set $x := m + \ell + 1 \pmod{p_\ell}$ and

$$f_{\ell,r}(x) := A_{\ell,r}x + B_{\ell,r}x^{-1} + C_{\ell,r}x^2 + D_{\ell,r}\,\ell\, x^3 \quad \text{in } \mathbb{F}_{p_\ell},$$

with the p_ℓ^2-resolved double-digit lift to a uniform sample on $(0,1)$,

$$u_{\ell m}^{(r)} := \frac{p_\ell\, f_{\ell,r,1}(x) + f_{\ell,r,2}(x) + \frac{1}{2}}{p_\ell^2},$$

where $f_{\ell,r,1}, f_{\ell,r,2}$ are two independent admissible variants of the trace recipe. The deterministic complex normal vector is then

$$g_{\ell 0}^\star := \Phi^{-1}\big(u_{\ell 0}^{(0)}\big), \quad g_{\ell m}^\star := \frac{\Phi^{-1}\big(u_{\ell m}^{(1)}\big) + i\Phi^{-1}\big(u_{\ell m}^{(2)}\big)}{\sqrt{2}} \quad (m \geq 1),$$

to which the microcanonical normalization of Conjecture 11.9 is applied shell by shell. The architecture

$$\boxed{\text{finite-field coefficients} \;=\; \text{reduced traces of canonical TFPT operators}}$$

makes the seam mixer a closed-branch object: the only inputs are the operator algebra $(U_\Sigma, P_{\mathrm{adm}}, \tau_{\mathrm{dbl}}, \iota_{\mathrm{C}}, U_5, U_6)$ on V_{adm}, the closed-branch eigenvalue data, and elementary integer arithmetic.

Lemma 11.10 (Decorrelation bound from incomplete Weil sums)**.** *Fix $\ell \geq 2$ and $1 \leq k \leq \ell$. Let $f_{\ell,r} \in \mathbb{F}_{p_\ell}(x)$ be a non-zero rational phase function of total degree D as in Remark, and assume the shifted difference*

$$\Delta_k f_{\ell,r}(x) := f_{\ell,r}(x+k) - f_{\ell,r}(x) \in \mathbb{F}_{p_\ell}(x)$$

is non-constant on $\mathbb{F}_{p_\ell}^\times$ and not Artin–Schreier degenerate, i.e. does not admit a representation $\Delta_k f = g^{p_\ell} - g + \mathrm{const}$ with $g \in \mathbb{F}_{p_\ell}(x)$. Then the complete *exponential sum obeys the Weil estimate of Bombieri [8] and Deligne [9],*

$$\Big| \sum_{x \in \mathbb{F}_{p_\ell}^\times} e^{2\pi i\, \Delta_k f_{\ell,r}(x)/p_\ell} \Big| \;\leq\; (D-1)\sqrt{p_\ell}.$$

The discrete decorrelation amplitude of the realised seam lift is, however, the incomplete *sum over the integer interval $1 \leq m \leq \ell$,*

$$A_\ell(k) = \Big| \sum_{m=1}^{\ell} e^{2\pi i\, \Delta_k f_{\ell,r}(m+\ell+1)/p_\ell} \Big|.$$

Combining the complete-sum bound with the standard Pólya–Vinogradov [10] completion procedure yields the incomplete-sum estimate

$$A_\ell(k) \;\leq\; C_D \sqrt{p_\ell}\, \log p_\ell \;=\; O\big(\sqrt{\ell}\, \log \ell\big),$$

with a constant C_D depending only on the total degree D of $f_{\ell,r}$. Pixel Gaussianity of the realized sky follows by the central limit theorem on the $(2\ell+1)$-dimensional admissible shell. In particular, the scalar phase law $\Theta_{\ell m} = A + B(\ell^2 + \ell + m) + C\ell + Dm \pmod{2\pi}$ violates the non-constancy hypothesis (its shifted difference is identically zero in m), the bound does not apply to it, and the empirical $A_\ell(k) \sim \ell$ scaling of Remark is consistent with that violation. The logarithmic factor in the bound is harmless for all diagnostics (it is dominated by the square-root growth at the multipole resolutions reachable by Planck) but mathematically essential: the realized sums are incomplete, and the Weil estimate alone applies only to the completion.

Remark (Open obligation $O_{20\mathrm{a}}$ at the operator level). The trace-reduction architecture of Remark reduces the canonical lift obligation to a purely operator-theoretic statement: $O_{20\mathrm{a}}$ asks for the closed-form derivation of the operator algebra $(U_\Sigma, P_{\mathrm{adm}}, \tau_{\mathrm{dbl}}, \iota_{\mathrm{C}}, U_5, U_6)$ on V_{adm} from the seam-transfer dynamics of $\mathfrak{T}_\star$. Once those operators are explicitly given, the trace formulae of Remark produce the seam scrambler $\mathcal{W}_\Sigma$ deterministically, and the Weil bound of Lemma 11.10 promotes the ergodicity claim from a numerical observation to a theorem.

Remark (Multiplicity audit). The trace formulae of Remark are not unique: any admissible permutation or integer-power perturbation of the operator strings $(P_{\mathrm{adm}}, U_\Sigma^{\ell+r}, U_5^r, \tau_{\mathrm{dbl}}, \iota_{\mathrm{C}}, U_6^r)$ yields a valid trace recipe in the same canonical class. A robust Stage 2 lift must give a Gaussian-typical sky for *every* member of this class, not just a hand-picked combination. Numerical evidence on four mutually inequivalent trace variants (default; $U_5 \leftrightarrow U_6$ swap; squared inclusions $\tau_{\mathrm{dbl}} \to \tau_{\mathrm{dbl}}^2, \iota_{\mathrm{C}} \to \iota_{\mathrm{C}}^2$; full carrier product $U_5^r U_6^r$) shows that the pixel moments, per-ℓ KS test, parity asymmetry, and Gaussian-baseline correlation r_ℓ all sit at the Gaussian-baseline level for every variant, with pairwise variant–variant correlations inside the same $\pm 1/\sqrt{2\ell+1}$ envelope. This multiplicity audit rules out the cherry-pick objection.

Remark (Status of the two CMB closure stages). Conjecture 11.8 (*Stage 1: spectra*) and Conjecture 11.9 (*Stage 2: map*) are complementary, not equivalent. Stage 1 explains the angular statistics of the CMB from $\{\lambda_j(U_\Sigma)\}$ alone; Stage 2 in addition predicts a specific sky realization, conditional on the seam state u_Σ being treated as a determinate datum of $\mathfrak{T}_\star$ rather than as a sampled element of an ensemble. The compressed slogan is

$$\text{eigenvalues} \;\Rightarrow\; \text{spectra}, \qquad \text{eigenvalues} + \text{seam phases} \;\Rightarrow\; \text{sky realization}.$$

Falsification path. Stage 1 is killed by any robust deviation of the angular spectra from the admissible-branch transfer of $\{\lambda_j(U_\Sigma)\}$. Stage 2 is killed independently if any $a^{T,\star}_{\ell m}$ obtained from $\Theta_j = \arg\langle \psi_j, u_\Sigma\rangle$ disagrees with the measured Planck $a^T_{\ell m}$ at a level exceeding cosmic variance plus the determinant-bound slack of Corollary 10.4.

Remark (Three open obligations of the Stage 2 closure). The Stage 2 statement Conjecture 11.9 carries three named obligations that must be discharged before the conjecture can be promoted to a theorem-level deterministic sky prediction:

(O1) **Canonical seam-state lift.** The closed-branch determinant-line phase $[u_\Sigma]$ must be lifted to a determinate element of the spectral basis $\{\psi_j\}$ of U_Σ on a finite-mode truncation, in such a way that the lift uses only data internal to $\mathfrak{T}_\star$. The ergodic block lift of Conjecture 11.9 provides the correct *type* for this lift; what remains is a closed-form description of the seam scrambler $\mathcal{W}_\Sigma$ itself, derived from the seam-transfer dynamics rather than approximated by a counter-based mixer.

(O2) **SO(3) orientation gauge.** The boundary content of the closed branch as recorded in the present version is rotation invariant: no canonical sky triad is selected. The deterministic prediction is therefore strictly $\{a^{T,\star}_{\ell m}\}/\mathrm{SO}(3)$, and only rotation invariant diagnostics (C_ℓ, parity asymmetry, peak ratios, multipole amplitude pattern, Gaussianity tests) are admissible as headline tests against Planck. A pixel-level alignment with the Planck SMICA sky requires an additional closed-branch ingredient (e.g. a determinant-line orientation seed) that is not part of the present theorem stack.

(O3) **Ergodicity of the seam-phase distribution.** A canonical lift of u_Σ must yield a multipole image that is statistically Gaussian to the level measured by Planck (excess kurtosis $|g_2| \lesssim 10^{-2}$ on the temperature map). Two structurally distinct candidates have now been distinguished:

(a) the scalar phase law $\Theta_{\ell m} = A + B(\ell^2 + \ell + m) + C\ell + Dm$ (any closure scalars A, B, C, D): **falsified**. The discrete decorrelation amplitude grows as $A_\ell(k) = O(\ell)$ and the realized sky has $g_2 \gtrsim 5$, max $|\Delta T| \gtrsim 2000\,\mu K$, and visible stripe / ray artefacts on the closed-branch carrier wheel.

(b) the ergodic microcanonical block lift of Conjecture 11.9 with the finite-field realization of Remark: **candidate**. $A_\ell(k) = O(\sqrt{\ell})$ structurally by Weil; $g_2 \sim 0$; max $|\Delta T| \sim 500\,\mu K$; pixel histogram, per-$\ell$ KS test, band-filtered skewness, and microcanonical C_ℓ recovery all consistent with a Gaussian draw at the closed-branch $C^\star_\ell$.

The structural target of the canonical lift is therefore not a phase formula but the seam scrambler $\mathcal{W}_\Sigma$.

Each obligation is independently falsifiable. Together they fix the form of the closed Stage 2 prediction: the scrambled shell vector $g^\star_{\ell m} = \langle e_{\ell m}, \mathcal{W}_\Sigma u_\Sigma\rangle$ and its microcanonical projection.

Remark (Stage 2 traffic-light status). The current Stage 2 status decomposes into four orthogonal columns. Conflating them is the typical reviewer entry point and is to be avoided.

- **Solved numerically.** Pixel Gaussianity ($|g_2| \lesssim 10^{-2}$ across four trace variants), absence of stripe / ray artefacts, multiplicity robustness over the admissible trace class, and null calibration of the Z_B band-score framework against independent Gaussian baselines (Remark, pipeline 'stage2_trace_reduction.py').
- **Solved structurally.** Replacement of the Perron–Frobenius-collapsing power-limit definition of $\mathcal{W}_\Sigma$ by the projective modular trace cocycle (Conjecture 11.9), and the incomplete-sum Pólya–Vinogradov completion of the Weil decorrelation route in Lemma 11.10. The ergodicity claim is now a statement of analytic number theory under a named non-degeneracy hypothesis on the rational phase function, not a numerical observation.
- **Open empirically.** The Planck-compressibility test of Remark against the measured SMICA $a_{\ell m}^{\mathrm{SMICA}}$ remains the open empirical kill test (obligation O_{20c}). The pre-registration document 'PRE_REGISTRATION.md' fixes the primary variant, the three multipole bands, the fixed-frame protocol, the SO(3)-marginalised diagnostic with its own null distribution, the significance thresholds, and the SHA-256 hash of the trace-cocycle generator, all before any SMICA value is inspected.
- **Open theoretically.** The closed-form derivation of the operator algebra $(U_\Sigma, P_{\mathrm{adm}}, \tau_{\mathrm{dbl}}, \iota_{\mathrm{C}}, U_5, U_6)$ on V_{adm} from the seam-transfer dynamics of $\mathfrak{T}_\star$, replacing the present TFPT-flavoured proxies (obligation O_{20a}). The trace-reduction architecture is invariant under this replacement: any concrete TFPT closure that produces explicit matrix elements for those operators slots into the Stage 2 formalism without changing the kill-test protocol.

The four lights are deliberately reported separately. A claim of the form "Stage 2 is solved" without specifying the column is meaningless. A claim of the form "Stage 2 reconstructs Planck" requires the empirical column to flip and the theoretical column to be discharged jointly.

Remark ("Good CMB world" is not yet "this CMB world"). At the present level of the construction, the ergodic seam lift is statistically indistinguishable from a Gaussian realization at the closed-branch $C_\ell^\star$: the cross-correlation

$$r_\ell = \frac{\Re \sum_m a_{\ell m}^\star \, a_{\ell m}^{\mathrm{Planck}\,*}}{\sqrt{\sum_m |a_{\ell m}^\star|^2 \sum_m |a_{\ell m}^{\mathrm{Planck}}|^2}}$$

between the closed-branch realization and an independent Gaussian draw at the same C_ℓ is consistent with zero in every multipole band, within the $\pm 1/\sqrt{2\ell+1}$ envelope expected from two independent samples. The honest reading is therefore

$$\boxed{\text{TFPT now produces a deterministic, good CMB world} \quad \neq \quad \text{TFPT reconstructs this CMB world.}}$$

The compressed reformulation of the next empirical kill test is

$$\boxed{\text{Is the Planck sky compressible by } (U_\Sigma, u_\Sigma, \mathcal{W}_\Sigma)\,?}$$

Operationally this is the direct r_ℓ-comparison against the measured Planck SMICA $a_{\ell m}^{\mathrm{Planck}}$, run once at the closed-branch declared sky triad and once with optimal SO(3) alignment as a labeled diagnostic. A systematically positive r_ℓ, especially at low ℓ, would promote the closed-branch realization from a typical Gaussian sample to the realized history of the universe; statistically flat r_ℓ would confirm the present statistical-only level. Either outcome is informative.

Remark (Recorded outcome of the locked SMICA compression test). The locked compression test against Planck PR3 SMICA was executed under the v2 pre-registration ('PRE_REGISTRATION_v2.md', cocycle-generator SHA-256 `9d6c7c74fb32805d80a88e9be4edfcb35e2f40f82281124eb868fa965d89f904`, SMICA-file SHA-256 `60952c645eb33d151905ddf5837477e15ca02a3261feca4ceca3c3feece4f9ac`). The recorded headline (Z_B^A on the closed-branch declared sky frame, full sky, $\ell_{\max} = 500$, $N_{\text{side}} = 256$) is

$$Z^A_{[2,30]} = +0.87, \qquad Z^A_{[30,200]} = +0.56, \qquad Z^A_{[200,500]} = +0.55, \qquad \chi^2_3(A) = 1.37 < 7.81.$$

No band of any variant breaches the Bonferroni threshold $|Z_B| \geq 2.39$. The SO(3)-marginalised diagnostic gives $\max_R Z_B^A = (+1.93, +1.85, +2.60)$, none crossing the locked publication-tier look-elsewhere thresholds $(2.27, 3.10, 3.12)$. The result is the pre-registered Fall 1 outcome: under the present V_{adm}-proxy operator algebra, the closed-branch realisation is statistically a typical CMB world, not the specific Planck realisation. This is informative as a clean negative; it does not refute Stage 1 spectral closure or Stage 2 ergodicity.

Remark (Five-layer roadmap toward a complete CMB derivation). Discharging the empirical Stage 2 closure — moving from "a good CMB world" to "this CMB world" — requires five layers to be jointly closed on the rigid object $\mathfrak{T}_\star$, beyond the present trace-cocycle proxy. Each layer is reported separately so that residual ambiguity in any one of them does not leak into headline claims about the others.

(L1) **Background closure.** The cosmological background must be a closed-branch readout:

$$\mathfrak{T}_\star \implies \left(H_0^\star, \Omega_b^\star, \Omega_c^\star, \Omega_\nu^\star, \Omega_r^\star, \Lambda_{\text{IR}}^\star, T_0^\star\right) \implies \left(T_R^\star, N_e^\star\right) \implies (A_s^\star, n_s^\star, r^\star).$$

Here $T_R^\star$ must come from the closed scalaron-decay rate, axion sector, and Higgs / matter couplings on the closed branch, not be calibrated to the chi-square minimum of any spectrum. As long as the chain $\mathfrak{T}_\star \to T_R \to N_e$ is open, Stage 1 is conditional, not derived.

(L2) **Transfer closure.** The Boltzmann line-of-sight transfer kernels

$$\Delta_\ell^T(k), \quad \Delta_\ell^E(k), \quad \Delta_\ell^B(k), \quad \Delta_\ell^\phi(k)$$

are computed by a standard cosmology code (CAMB or CLASS), but every input is required to be a closed-branch readout from (L1). The full statistical closure follows as

$$C_\ell^{XY} = 4\pi \int \frac{dk}{k}\, \mathcal{P}_{\mathcal{R}}^\star(k)\, \Delta_\ell^X(k)\, \Delta_\ell^Y(k), \qquad X, Y \in \{T, E, B, \phi\}.$$

This level produces angular statistics, not a sky realisation.

(L3) **Operator algebra from seam dynamics.** The trace recipe of Remark must descend from the closed-branch dynamics:

$$\mathfrak{T}_\star \implies V_{\text{adm}}^\star, \;\; U_\Sigma^\star, P_{\text{adm}}^\star, \tau_{\text{dbl}}^\star, \iota_{\text{C}}^\star, U_5^\star, U_6^\star,$$

without a free choice of representation dimension and without the band-circulant proxies of the present implementation. Operationally this discharges obligation $O_{20\text{a}}$.

(L4) **Canonical trace word and seam record state.** The variants $\{A, B, C, D\}$ of Remark must collapse to a unique admissible word, selected by a stated principle (naturality, minimal admissible word, determinant-line covariance, D_4-seam equivariance, observer compatibility), giving

$$\mathcal{W}_\Sigma^\star \;=\; \text{the unique admissible projective modular trace cocycle on } \mathfrak{T}_\star,$$

together with a closed-form derivation of the seam record state $u_\Sigma^\star$ as a deterministic element of the boundary admissible Hilbert space (see Conjecture 11.11 below). The compressed slogan is

$$\boxed{\text{the apparent randomness of the realized CMB phases} \;=\; \text{the boundary record of } \mathfrak{T}_\star.}$$

This is what turns Stage 2 from "a good CMB world" into "this CMB world".

(L5) **Observer triad and polarisation cross-check.** The closed branch must additionally provide a sky triad $\mathcal{O}^\star \in \mathrm{SO}(3)$ that orients the realised $\{a_{\ell m}^{T,\star}, a_{\ell m}^{E,\star}, a_{\ell m}^{B,\star}\}$ in galactic coordinates; without it the deterministic prediction is strictly $\{a_{\ell m}^\star\}/\mathrm{SO}(3)$. The internal consistency cross-check is the same $\mathcal{R}_\Sigma^\star(\mathbf{k})$ producing

$$a_{\ell m}^{T,\star},\; a_{\ell m}^{E,\star},\; a_{\ell m}^{B,\star} \quad \text{from a single closed-branch realisation;}$$

a strong T-channel signal that is empty in the E-channel is statistically not a record but apophenia under microscope.

The strategic next step is to close (L3) and (L4) on $\mathfrak{T}_\star$ rather than to attempt further empirical compression tests under the present proxy. The slogan is proxy out, record in, observer in, polarisation in.

Conjecture 11.11 (Canonical seam record state on the closed branch)**.** On the closed branch $\mathfrak{T}_\star$ there exists a unique unit-norm seam record state

$$u_\Sigma^\star = \mathrm{Norm}\big(P_{\mathrm{adm}}^\star \cdot \mathrm{Hol}_\Sigma\big(\Gamma_{\mathrm{TFPT}}^{\mathrm{ren}}\big) \cdot \Omega_{\mathrm{rec}}^\star\big)$$

in the boundary admissible Hilbert space, where $\Omega_{\mathrm{rec}}^\star$ is the closed-branch record vacuum of the record-algebra block (cf. Conjecture 11.25) and $\mathrm{Hol}_\Sigma(\Gamma_{\mathrm{TFPT}}^{\mathrm{ren}})$ is the seam holonomy of the renormalised admissible flow. The closed-branch ergodic shell vector $g_{\ell m}^\star = \langle e_{\ell m}, \mathcal{W}_\Sigma^\star u_\Sigma^\star\rangle$ of Conjecture 11.9 is then a deterministic readout of $\mathfrak{T}_\star$ alone. The acceptance test is that $u_\Sigma^\star$ produces the same realised phases through the T, E, and ϕ channels of (L5).

Conjecture 11.12 (Canonical trace word on the admissible operator algebra)**.** There is a unique admissible word $\mathcal{O}_\star \in \langle U_\Sigma^\star, P_{\mathrm{adm}}^\star, \tau_{\mathrm{dbl}}^\star, \iota_{\mathrm{C}}^\star, U_5^\star, U_6^\star\rangle$ selected by a single stated principle on the closed branch (D_4-seam equivariance and determinant-line covariance, with naturality under change of admissible truncation), such that the canonical trace cocycle is

$$\mathfrak{c}_{\ell,r}^\star = \mathrm{red}_{p_\ell}\,\mathrm{Tr}\big(P_{\mathrm{adm}}^\star\, \mathcal{O}_\star^{(\ell,r)}\big).$$

The variant family $\{A, B, C, D\}$ of Remark is a robustness audit; the canonical word $\mathcal{O}_\star$ collapses the family to a single element and removes the "which variant did you pick" degree of freedom from the empirical attack surface.

Conjecture 11.13 (Polarisation cross-check)**.** Let $\mathcal{R}_\Sigma^\star(\mathbf{k})$ be the closed-branch realisation of the seam state per Conjecture 11.9 after canonical lift via Conjectures 11.11 and 11.12. Then the same $\mathcal{R}_\Sigma^\star$ must produce, through the closed-branch transfer kernels $\Delta_\ell^X(k)$ of (L2),

$$a_{\ell m}^{T,\star}, \quad a_{\ell m}^{E,\star}, \quad a_{\ell m}^{B,\star}, \quad a_{\ell m}^{\phi,\star},$$

and any TT compression score $r_\ell^{T,\mathrm{Planck}}$ that is significant in the locked test must be matched by a corresponding TE or EE compression score within the same multipole band, up to the inflation-branch tensor-to-scalar ratio r. A TT-only compression with empty TE / EE is ruled inconsistent with the conjecture and identified as sample variance / look-elsewhere artefact.

Conjecture 11.14 (Flavored Boltzmann closure for baryogenesis)**.** The heavy neutrino, family, and reheating data of the closed branch determine the flavored Boltzmann system

$$\frac{dY_{N_i}}{dz} = -D_i(Y_{N_i} - Y_{N_i}^{eq}), \qquad \frac{dY_{\Delta_\alpha}}{dz} = \sum_i \epsilon_{i\alpha} D_i(Y_{N_i} - Y_{N_i}^{eq}) - W_{i\alpha} Y_{\Delta_\alpha},$$

with branch-fixed decay asymmetries and washout kernels.

Definition 11.15 (Closed-branch baryonic and dimensionless density readouts)**.** Let

$$\Omega_b^\star := \left(1 - \tfrac{1}{4\pi}\right)(\varphi_{\text{base}} + \delta_{\text{top}})$$

denote the closed-branch baryonic fraction read off the bridge map $u \mapsto (\lambda_C, \beta_{\text{rad}}, \Omega_b, \theta_{13})$ (cf. Theorem 10.3 and the constants atlas). Let $\omega_a^\star$, $\omega_\nu^\star$, $\omega_r^\star$ denote the dimensionless density readouts $\omega_X := \Omega_X h^2$ for the closed-branch axion, neutrino, and radiation sectors, all computed from the same admissible flow $\Gamma_{\text{TFPT}}^{\text{ren}}$ that controls the matter sector, and let

$$\eta_{100} := \frac{H_{100}}{M_{\text{Pl}}}, \qquad H_{100} = 100\,\text{km}\,\text{s}^{-1}\,\text{Mpc}^{-1},$$

denote the dimensionless reference-Hubble unit. With this convention one has

$$\frac{8\pi}{3\eta_{100}^2}\frac{\Lambda_{\text{IR}}}{M_{\text{Pl}}^4} = \frac{\Lambda_{\text{IR}}}{\rho_{\text{crit},100}} = \omega_\Lambda, \qquad \rho_{\text{crit},100} := 3H_{100}^2 \bar{M}_{\text{Pl}}^2,$$

which fixes the conversion between the dimensionful determinant scale $\Lambda_{\text{IR}}/M_{\text{Pl}}^4$ and the dimensionless ω-readouts.

Conjecture 11.16 (Late-time Hubble readout from the closed branch)**.** The present Hubble scale is a downstream readout of the closed branch, given equivalently as the dimensional Friedmann form

$$H_0^2 := \frac{8\pi G_N}{3}(\rho_{b,0} + \rho_{a,0} + \rho_{\nu,0} + \rho_{r,0}) + \frac{\Lambda_{\text{IR}}}{3}$$

or as the dimensionless closed-form readout

$$\boxed{h_\star^2 = \frac{\omega_a^\star + \omega_\nu^\star + \omega_r^\star + \dfrac{8\pi}{3\eta_{100}^2}\dfrac{\Lambda_{\text{IR}}^\star}{M_{\text{Pl}}^4}}{1 - \Omega_b^\star}.}$$

Here $\Lambda_{\text{IR}}^\star$ refers either to the theorem-level seam-transfer determinant of Theorem 10.3 or to its conditional numerical reductions $\Lambda_{\text{IR}}^{\text{rec}}/\Lambda_{\text{IR}}^{\text{car}}$ from Corollary C.5, and $\Omega_b^\star, \omega_X^\star$ are the closed-branch density readouts of Definition 11.15.

Remark (Why H_0 is not a primitive constant of the closed branch). The defining input of Conjecture 11.16 is not an independent "Hubble dial"; $h_\star^2$ is fully determined by the seam-transfer determinant $\Lambda_{\text{IR}}^\star$ and the dimensionless density readouts $(\Omega_b^\star, \omega_a^\star, \omega_\nu^\star, \omega_r^\star)$. The compressed slogan is

$$\boxed{H_0 \text{ is a downstream cosmology readout, not a primitive constant.}}$$

Falsification path. Any independent measurement of the four density readouts $(\Omega_b, \omega_a, \omega_\nu, \omega_r)$ together with the determinant scale $\Lambda_{\text{IR}}/M_{\text{Pl}}^4$ that does not satisfy the boxed identity at the level of the combined experimental and rank-one bounds of Corollary 10.4 kills the closed-form H_0 readout. Conversely, if all five inputs are independently fixed, then the H_0 tension between local distance ladders and CMB-anchored inferences becomes a quantitative test of the seam-transfer determinant rather than an independent free parameter.

Conjecture 11.17 (Matter power spectrum readout)**.** The late time matter spectrum is

$$P_m(k,z) = \frac{2\pi^2}{k^3}\mathcal{P}_{\mathcal{R}}(k)\,T^2(k)D^2(z).$$

11.3 [P] Late time dark sector and structure growth

Conjecture 11.18 (Axion Schrödinger Poisson closure on the late-time branch)**.** The late-time axion sector is governed by

$$i\partial_t\psi = -\frac{\nabla^2}{2m_a}\psi + m_a\Phi_N\psi + \frac{\lambda_a}{8m_a^2}|\psi|^2\psi, \qquad \nabla^2\Phi_N = 4\pi G_N(\rho_b + m_a|\psi|^2).$$

Conjecture 11.19 (No fuzzy halo regime on the canonical branch)**.** For the branch axion mass readout, galactic-scale halos lie in the cold and effectively collisionless regime rather than in an ultralight fuzzy regime.

Conjecture 11.20 (Relative admissible vacuum energy finiteness)**.** The vacuum energy density on the admissible branch is the finite relative quantity

$$\rho_{\rm vac}^{\rm TFPT} := \frac{1}{V}\left[-\log Z_{\rm rel}^{P_{\rm adm}} + \Re\log\det_{\zeta,P_{\rm adm}}(D^2+\mu^2) + \frac{1}{2}\eta_{\Sigma,P_{\rm adm}}\right].$$

11.4 [P] Pole masses, Higgs readout, and full flavor output

Definition 11.21 (Scheme-fixed Yukawa kernel and admissible RG readout)**.** Let

$$M_f^{(0)} = \frac{v_{\rm geo}}{\sqrt{2}}Y_f^\star, \qquad (Y_f^\star)_{ij} = \lambda_Y^{L_{f,i}^L + L_{f,j}^R}\langle e_i, {\rm Hol}_{\Gamma_{ij}^{\rm min}}(\nabla_F^\star)e_j\rangle,$$

and let $\Gamma_{\rm TFPT}^{\rm ren}$ denote the renormalized admissible flow. For every mass-independent renormalization scheme $\mathcal{S} \in \{\overline{\rm MS}, {\rm RI}, \dots\}$ and every renormalization scale μ, the closed-branch *quark mass readout* is the scheme-and-scale-fixed map

$$m_q^{\mathcal{S}}(\mu) := \mathcal{R}_{\mathcal{S},\mu}[Y_q^\star, \Gamma_{\rm TFPT}^{\rm ren}], \qquad \mathbf{m}_q^{\rm TFPT}(\mathcal{S},\mu) := \big(m_u^{\mathcal{S}}(\mu), m_d^{\mathcal{S}}(\mu), m_s^{\mathcal{S}}(\mu), m_c^{\mathcal{S}}(\mu), m_b^{\mathcal{S}}(\mu), m_t^{\mathcal{S}}(\mu)\big).$$

Conjecture 11.22 (Quark masses as scheme-fixed RG readouts of the closed branch)**.** The full quark mass spectrum on the closed branch is given by the readout of Definition 11.21: each $m_q^{\mathcal{S}}(\mu)$ is determined by the same renormalized admissible flow $\Gamma_{\rm TFPT}^{\rm ren}$ that controls the lepton sector, together with the closed-branch Yukawa data $Y_q^\star$.

Conjecture 11.23 (Hadron pole masses as singlet spectrum of the admissible QCD Hamiltonian)**.** The physical hadron mass spectrum is the gauge-invariant pole spectrum

$$m_{\rm had}^{\rm pole} = {\rm Spec}_{\rm singlet}[\mathcal{H}_{\rm QCD}^{\rm TFPT}],$$

where $\mathcal{H}_{\rm QCD}^{\rm TFPT}$ is the admissible QCD Hamiltonian determined by the closed-branch couplings, masses, and admissibility projector $P_{\rm adm}$, and the singlet selection is the hadronic admissibility projector of the singlet algebra $\mathcal{A}_{\rm sing}$ developed in the hadronic admissibility block of closure architecture B. The closed pole masses of charged leptons and the electroweak scale, in contrast, are read off directly from the Källén–Lehmann pole structure of the admissible propagator on $P_{\rm adm}$, since these states appear as free asymptotic carriers in the LSZ closure of Theorem F.3.

Remark (Why *quark* pole masses are not a TFPT output). Color-charged quarks do not appear as free asymptotic states in the admissible LSZ closure. A sentence of the form "the physical pole masses of the six quarks are predicted" therefore would be ontologically inconsistent with the closed-branch dynamics on $P_{\rm adm}$. The closed branch produces exactly two physically distinct mass categories: scheme-fixed RG readouts $\mathbf{m}_q^{\rm TFPT}(\mathcal{S},\mu)$ for the quarks, and gauge-invariant pole masses $m_{\rm had}^{\rm pole}$ for color singlets. The compressed slogan is

$$\boxed{\text{quark masses are scheme-fixed RG readouts; hadron masses are physical poles.}}$$

Falsification path. The conjectural content is killed if any future first-principles lattice or flow computation produces a $\mathbf{m}_q^{\mathrm{TFPT}}(\mathcal{S},\mu)$ inconsistent with the admissible-branch Yukawa data $Y_q^\star$ at the matching scale, or if the singlet spectrum of $\mathcal{H}_{\mathrm{QCD}}^{\mathrm{TFPT}}$ disagrees with the measured hadron pole masses beyond the flow-resummation error of $\Gamma_{\mathrm{TFPT}}^{\mathrm{ren}}$.

Conjecture 11.24 (Higgs pole readout from the admissible RG branch)**.** The Higgs pole mass is determined by the renormalized branch data through

$$m_h^2 = 2\lambda_H^{\mathrm{phys}} v_{\mathrm{phys}}^2 + \delta_{\mathrm{sc}},$$

where $\lambda_H^{\mathrm{phys}}$ is read from the admissible RG branch and δ_{sc} is the scalaron mixing correction.

11.5 [P] Admissible code subspace, entanglement, and pointer dynamics

Conjecture 11.25 (Global admissible code subspace and emergent pointer reduction)**.** The physical sector is the global admissible code subspace

$$\mathcal{H}_{\mathrm{adm}} = \mathrm{Ran}(P_{\mathrm{adm}}) = \ker \Delta_{\mathrm{adm}},$$

and supports stable record algebras and pointer reductions of the form

$$\mathcal{M}_O(\rho) = \sum_i (P_i \rho P_i) \otimes |i\rangle\langle i|.$$

11.6 [P] Topological transient channels and astrophysical bursts

Conjecture 11.26 (Topological phase slip burst channel)**.** Local determinant phase slips of the compact mode a_Σ can source transient radio bursts with characteristic energy scale

$$E_{\mathrm{burst}} \sim \epsilon\, 4\pi R_c^2 T_{\mathrm{wall}}, \qquad T_{\mathrm{wall}} \sim 8 f_a^2 m_a,$$

and photon conversion power

$$P_{a\to\gamma} \sim g_{a\gamma\gamma}^2 B^2 L^2 \dot{a}_\Sigma^2.$$

Remark (Status of the transient channel). The transient channel is recorded as a programmatic closure target and not as a theorem. A full proof requires a closed defect dynamics and radiative transfer treatment.

12 Canonical rigid object and falsification interface

12.1 Canonical rigid object

Collecting the operational-seed generation theorem, the minimal-packaging theorem for the induced boundary class, the boundary datum, the boundary primitive kernel, the joint discrete admissibility theorem, the derived selector, and the continuous closure theorems of Section 8, the main-text reconstruction now takes the rigid two-stage form

$$\begin{aligned} \mathfrak{S}_{\mathrm{min}} \;\Rightarrow\; \mathcal{B}_{\mathrm{min}} \;\Rightarrow\; \mathfrak{T}_\partial^{\mathrm{min}} \;\Rightarrow\; (\tau_{\mathrm{dbl}}, \iota_{\mathrm{C}}, P_{\mathrm{prim}}, [u_\Sigma], c_3) \;\Rightarrow\; d_{\mathrm{disc}}^\star \\ \Rightarrow\; P_{\mathrm{adm}} \;\Rightarrow\; \mathcal{M}_{d_{\mathrm{disc}}^\star} \;\Rightarrow\; \mathcal{Z}_{\mathrm{cl}} = \{x_{\mathrm{cl}}^\star\} \;\Rightarrow\; \mathfrak{T}_\star. \end{aligned} \tag{52}$$

Here

$$\mathfrak{T}_\star := \big(E_3 \oplus E_2,\ X,\ Y,\ G_{\mathrm{phys}},\ S^+,\ X_f^\circ,\ [\nabla_F^\star],\ \Phi,\ a_\Sigma,\ U_\Sigma, \Gamma_{\mathrm{grav}},\ \mathcal{H}_{\mathrm{phys}},\ \Lambda_{\mathrm{IR}},\ \{Y_f\},\ x_{\mathrm{cl}}^\star\big)$$

is the canonical rigid object of Corollary 9.23, and the intrinsic branch is the singleton $\mathcal{Z}_{\mathrm{cl}} = \{x_{\mathrm{cl}}^\star\}$ carried by that object. This line is not meant as a slogan: each arrow is attached to a hard theorem or one of the closure theorems of Section 8. From $\mathfrak{T}_\star$ the manuscript opens exactly

two side cards: the structural falsification map $\mathsf{F}_{\mathrm{fals}}(\mathfrak{T}_\star)$ of Section 12.3 (main text) and the appendix-level empirical comparison map $\mathcal{R}_{\mathrm{cmp}}(\mathfrak{T}_\star)$ of Section A. Neither side card enters the theorem chain. Section 11 records additional downstream closure modules starting from $\mathfrak{T}_\star$, the admissible local net, the renormalized branch, and the cosmology interface package, but it does not insert new arrows into this core theorem chain.

The corresponding selector-to-dynamics extension of the closed branch is

$$P_{\mathrm{adm}} \Rightarrow Z_{\mathrm{rel}} \Rightarrow \{S_n^T\} \Rightarrow (\mathcal{H}_{\mathrm{adm}}, \mathfrak{A}_{\mathrm{adm}}) \Rightarrow \Gamma_k \Rightarrow \Gamma^{\mathrm{ren}}_{\mathrm{TFPT}} \Rightarrow (\mathbf{O}^{\mathrm{TFPT}}_{\mathrm{phys}}, S_{\mathrm{massive}}) \quad \text{together with} \quad \mathcal{I}_{\mathrm{dress}}.$$

Here

$$\mathcal{I}_{\mathrm{dress}} := \left(\Omega^{\mathrm{dress}}_{+}, \Omega^{\mathrm{dress}}_{-}, S_{\mathrm{dress}}\right)$$

denotes the dressed massless interface package on the low-curvature branch. Thus P_{adm} is the selector of the physical branch, while local dynamics, running couplings, 1PI graphs, and scattering are carried by Z_{rel}, its reconstructed local net, and the exact admissible flow.

12.2 Canonical rigid-object corollary

Corollary 12.1 (Canonical rigid object)**.** *By Theorems 9.16, 9.18 and 9.21, there exists a canonical rigid object*

$$\mathfrak{T}_\star := \left(E_3 \oplus E_2,\ X,\ Y,\ G_{\mathrm{phys}},\ S^+,\ X^\circ_f,\ [\nabla^\star_F],\ \Phi,\ a_\Sigma,\ U_\Sigma,\ \Gamma_{\mathrm{grav}},\ \mathcal{H}_{\mathrm{phys}},\ \Lambda_{\mathrm{IR}},\ \{Y_f\},\ x^\star_{\mathrm{cl}}\right)$$

in $\mathbf{TFPT}^{\mathrm{rig}}$ *such that* $\mathcal{Z}_{\mathrm{cl}} = \{x^\star_{\mathrm{cl}}\}$ *and every rigid closed admissible realization is uniquely unitarily equivalent to* $\mathfrak{T}_\star$*.*

Remark (Status of the corollary). This statement is recorded as a corollary rather than as a separate theorem because it adds no content beyond Theorems 9.16, 9.18 and 9.21. In plain language: once the intrinsic branch is closed and essential injectivity rules out a second rigid realization, the canonical object $\mathfrak{T}_\star$ is already fixed. The present corollary only collects its data tuple; the physically relevant main-text consequences below are reported as structural predictions and falsifiers in $\mathsf{F}_{\mathrm{fals}}(\mathfrak{T}_\star)$, while appendix-level empirical readout through $\mathcal{R}_{\mathrm{cmp}}(\mathfrak{T}_\star)$ does not define extra theorem objects.

12.3 Structural falsification map

The main text closes the theorem chain at $\mathfrak{T}_\star$ and then attaches one structural side card, the falsification map

$$\mathsf{F}_{\mathrm{fals}} : \mathbf{TFPT}^{\mathrm{rig}} \to \{\text{structural failure modes}\}, \qquad \mathfrak{T}_\star \mapsto \mathsf{F}_{\mathrm{fals}}(\mathfrak{T}_\star).$$

This map reports which local or global structural failure mode would break each arrow of the reconstruction. It is hard-separated from the appendix-level empirical comparison map $\mathcal{R}_{\mathrm{cmp}}(\mathfrak{T}_\star)$, which acts only after $\mathfrak{T}_\star$ has been fixed and is collected in Section A.

13 Internal reduction theorem and renormalized observable hierarchy

Above the closed theorem stack ending at $\mathfrak{T}_\star$, the present version records one theorem-level input reduction and a stratified output-side observable chain. On the input side, the reduction is internal to a deliberately weakened ambient class. On the output side, the passage from $\mathfrak{T}_\star$ to the renormalized low-energy package $\Gamma^{\mathrm{ren}}_{\mathrm{TFPT}}$ and then to the physical observable layer $\mathbf{O}^{\mathrm{TFPT}}_{\mathrm{phys}}$ is theorem-level, while only the final scheme projection remains an appendix-facing interface.

13.1 Weakened ambient class

Definition 13.1 (Weakened ambient admissibility category)**.** Let $\widehat{\mathrm{PhysAdm}}_1$ denote the category whose objects are tuples

$$\mathfrak{P} = (\mathcal{A}_+, \mathcal{H}_+, D_+, J, \Gamma, B_\Sigma, \tau_t, \omega)$$

satisfying the following ambient admissibility axioms:

(U1) Real even boundary datum: $(\mathcal{A}_+, \mathcal{H}_+, D_+, J, \Gamma, B_\Sigma)$ is an admissible real even one-sided boundary datum with reflection-regular collar.

(U4) Quasilocal positive dynamics: τ_t is a quasilocal dynamics with reflection positivity and an admissible vacuum sector.

Morphisms are unitary intertwiners preserving the displayed structures.

No external coverage claim is made in this section. In particular, the ambient class is a theorem class for internal rigidity, not yet a proof that reality itself is generated from a thinner primitive layer.

Carrier closure is now split cleanly

Lemma 3.11 is now only the algebraic normal form: if the two exterior signatures

$$\dim \Lambda^2 E_+ = 1, \qquad \Lambda^2 E_- \cong E_-^\vee$$

hold, then one necessarily has

$$\operatorname{rank} E = 5, \qquad (\dim E_-, \dim E_+) = (3, 2), \qquad \dim \Lambda^{\mathrm{even}} E = 16.$$

The internal derivation is completed later by Corollary 6.15 and theorem 8.13: the compact Higgs index fixes $\dim E_+ = 2$, the primitive indecomposable generator of the actual branch Yukawa tensor forces $\dim E_- = 3$, and the former occupancy law is reduced to a derived consistency check. Accordingly $\widehat{\mathrm{PhysAdm}}_1$ no longer carries a packet-size axiom; the rigid minimal carrier is now a theorem-level closure package rather than a hidden ambient input.

Definition 13.2 (Internal TFPT reduction functor)**.** Define the functor

$$\mathfrak{R} : \widehat{\mathrm{PhysAdm}}_1 \to \mathbf{TFPT}^{\mathrm{rig}}_{\mathrm{pre}}$$

on objects by extracting from $\mathfrak{P}$ the TFPT pre-class data

$$\mathfrak{R}(\mathfrak{P}) := \big(\mathfrak{T}^{\min}_\partial(\mathfrak{P}), P_{\mathrm{adm}}(\mathfrak{P}), X^\circ_f(\mathfrak{P}), [\nabla_F](\mathfrak{P}), \Phi(\mathfrak{P}), U_\Sigma(\mathfrak{P})\big),$$

where each component is reconstructed by the primitive reconstruction, internal hard-carrier reduction, internal family reduction, internal transport closure, compact bosonic index, and seam-transfer closure already established in the main theorem stack.

Definition 13.3 (Canonical rigid-closure functor)**.** Let

$$\mathfrak{Cl} : \mathbf{TFPT}^{\mathrm{rig}}_{\mathrm{pre}} \to \mathbf{TFPT}^{\mathrm{rig}}$$

denote the canonical closure functor that adjoins to each pre-class object $\mathfrak{T}_{\mathrm{pre}}$ its unique intrinsic branch $x^\star_{\mathrm{cl}}(\mathfrak{T}_{\mathrm{pre}})$ from Theorem 9.20 and then reads the resulting closed realization in the canonical rigid image via Definition 9.22 and theorem 9.21. On morphisms, $\mathfrak{Cl}$ acts by the same unitary intertwiners. Define the rigid internal-reduction functor by

$$\overline{\mathfrak{R}} := \mathfrak{Cl} \circ \mathfrak{R} : \widehat{\mathrm{PhysAdm}}_1 \to \mathbf{TFPT}^{\mathrm{rig}}.$$

Theorem 13.4 (Internal hard-carrier reduction)**.** *For every* $\mathfrak{P} \in \widehat{\mathrm{PhysAdm}}_1$*, axiom* ***(U1)*** *together with the boundary-polarization, boundary-winding, compact Higgs-index, and branch-Yukawa-rigidity theorems reduces functorially to the unique TFPT hard carrier*

$$E_3 \oplus E_2, \qquad X = -2P_3 + 3P_2, \qquad Y = -\frac{1}{3}P_3 + \frac{1}{2}P_2, \qquad S^+ = \Lambda^{\mathrm{even}}(E_3 \oplus E_2).$$

Proof. Take an ambient object

$$\mathfrak{P} = (A_+, H_+, D_+, J, \Gamma, B_\Sigma, \tau_t, \omega) \in \widehat{\mathrm{PhysAdm}}_1.$$

By axiom **(U1)**, the tuple $(A_+, H_+, D_+, J, \Gamma, B_\Sigma)$ is an admissible real even one-sided boundary datum with reflection-regular collar. The boundary-polarization theorem therefore reconstructs from $\mathfrak{P}$ the canonical boundary polarization $\iota_C(\mathfrak{P})$ and hence the carrier decomposition on the finite carrier block,

$$E(\mathfrak{P}) = E_-(\mathfrak{P}) \oplus E_+(\mathfrak{P}).$$

Next, the boundary-winding theorem determines the canonical unit winding class on the same branch. The compact Higgs index then fixes the positive carrier rank,

$$\dim E_+(\mathfrak{P}) = 2.$$

Once this bosonic rank is known, Theorem 8.13 forces the only admissible negative rank to be

$$(\dim E_-, \dim E_+) = (3, 2),$$

and Lemma 3.11 and corollary 8.15 then recover the former exterior signatures and packet-size statement as consequences. By the primitive trace condition the corresponding primitive two-point generator is uniquely

$$X(\mathfrak{P}) = -2P_3(\mathfrak{P}) + 3P_2(\mathfrak{P}),$$

and therefore

$$Y(\mathfrak{P}) = \frac{X(\mathfrak{P})}{6} = -\frac{1}{3}P_3(\mathfrak{P}) + \frac{1}{2}P_2(\mathfrak{P}).$$

The exterior packet is then forced to be

$$S^+(\mathfrak{P}) = \Lambda^{\mathrm{even}}(E_3(\mathfrak{P}) \oplus E_2(\mathfrak{P})).$$

Thus the displayed hard-carrier package is determined objectwise.

Now let

$$u : \mathfrak{P} \to \mathfrak{P}'$$

be a morphism in $\widehat{\mathrm{PhysAdm}}_1$. By definition, u is a unitary intertwiner preserving the structures listed in Definition 12.1. Therefore it intertwines the boundary operator, the reflection data, the Calderón projector, and the induced boundary polarization. Hence

$$u\, \iota_C(\mathfrak{P})\, u^{-1} = \iota_C(\mathfrak{P}').$$

So u carries the carrier block of $\mathfrak{P}$ unitarily onto the carrier block of $\mathfrak{P}'$, preserving the decomposition into the (-1) and $(+1)$ eigenspaces of ι_C.

Because the winding class and the carrier split are uniqueness outputs, the same unitary also intertwines the derived projectors P_3, P_2 and therefore the derived generators X, Y. Exterior functoriality then gives

$$u\, S^+(\mathfrak{P})\, u^{-1} = S^+(\mathfrak{P}').$$

Hence the assignment

$$\mathfrak{P} \longmapsto (E_3 \oplus E_2, X, Y, S^+)$$

is functorial. This is the asserted internal hard-carrier reduction. □

Theorem 13.5 (Internal family reduction)**.** *For every $\mathfrak{P} \in \widehat{\mathrm{PhysAdm}}_1$, the boundary-winding theorem and D_4-equivariant monodromy rigidity reduce functorially to the rigid family data*

$$X_f^\circ \cong \mathbb{P}^1 \setminus \mu_4, \qquad [\nabla_F] = [\nabla_F^\star].$$

Proof. For an ambient object $\mathfrak{P} \in \widehat{\mathrm{PhysAdm}}_1$, the boundary-winding theorem fixes the quarter-turn balance of the admissible seam data. The family theorem then identifies the family section uniquely as the four-punctured sphere

$$X_f^\circ(\mathfrak{P}) \cong \mathbb{P}^1 \setminus \mu_4.$$

The puncture set carries its inherited faithful D_4 symmetry.

On that rigid puncture square, the D_4-equivariant monodromy-rigidity theorem fixes the monodromy representation uniquely up to global $SU(3)_F$ conjugation,

$$\rho_F : \pi_1(X_f^\circ) \to SU(3)_F.$$

Once the monodromy representation is fixed, the Riemann–Hilbert correspondence for flat Hermitian bundles yields a unique flat Hermitian family-connection class realizing it. This class is precisely

$$[\nabla_F^\star].$$

Thus the family geometry and holonomy are determined objectwise.

Now let

$$u : \mathfrak{P} \to \mathfrak{P}'$$

be a morphism. Since u preserves the boundary datum and the induced winding data, it preserves the quarter-turn balance and hence the puncture-square structure. Therefore it induces an isomorphism of the two four-punctured sections preserving the D_4 action. Under this induced identification the puncture loops correspond, so the monodromy representations agree up to the same global $SU(3)_F$ conjugation already allowed by Theorem 7.5.

Because the flat Hermitian connection class is uniquely determined by that conjugacy class, the transported connection class coincides with the target one. Therefore

$$u : (X_f^\circ(\mathfrak{P}), [\nabla_F(\mathfrak{P})]) \longrightarrow (X_f^\circ(\mathfrak{P}'), [\nabla_F(\mathfrak{P}')])$$

is functorial. This proves the internal family reduction. □

Theorem 13.6 (Internal transport reduction)**.** *For every $\mathfrak{P} \in \widehat{\mathrm{PhysAdm}}_1$, the boundary-polarization, carrier-rigidity, family-holonomy, determinant-line, and positivity theorems reduce functorially to the TFPT transport package with admissible cusp set*

$$\left\{1, \frac{2}{3}, \frac{1}{3}\right\},$$

exact branch pole δ_{ph}, and the unique first-generation winding lock.

Proof sketch. Once the boundary winding, rigid carrier generator, and rigid family holonomy are fixed, the admissible cusp spectrum is no longer ambient input; it is read from the singlet-projected carrier spectrum. The algebraic transport-pole theorem, the exact C_6 kernel, and the transport length sum rule then fix the admissible cusp set, the pole δ_{ph}, and the first-generation winding pattern functorially. □

Theorem 13.7 (Internal glueing and invariant reconstruction)**.** *For every* $\mathfrak{P} \in \widehat{\mathrm{PhysAdm}}_1$*, the data extracted in Theorems 13.4 to 13.6 assemble into a TFPT pre-class object whose invariant tuple is the canonical TFPT tuple*

$$\mathcal{I}_\star = \big(X, Y, [u_\Sigma], X_f^\circ, c_1(L_2), c_1(L_3), [\nabla_F^\star], a_\Sigma, \Lambda_{\mathrm{IR}}, x_{\mathrm{cl}}^\star\big).$$

Proof sketch. The carrier, family, transport, determinant-line, gravity, cosmology, and vacuum closures determine exactly the displayed invariant tuple on the rigid branch. The internal reductions above therefore glue into one TFPT pre-class object with the same invariant content as the canonical branch. □

Theorem 13.8 (Reduction to the canonical rigid orbit)**.** *For every ambient object* $\mathfrak{P} \in \widehat{\mathrm{PhysAdm}}_1$*, the rigid internal reduction lands in the canonical rigid orbit:*

$$\overline{\mathfrak{R}}(\mathfrak{P}) \cong \mathfrak{T}_\star.$$

Equivalently, the induced map on isomorphism classes

$$\pi_0(\widehat{\mathrm{PhysAdm}}_1) \longrightarrow \pi_0(\mathbf{TFPT}^{\mathrm{rig}}), \qquad [\mathfrak{P}] \longmapsto [\overline{\mathfrak{R}}(\mathfrak{P})],$$

has singleton image

$$\pi_0(\mathbf{TFPT}^{\mathrm{rig}}) = \{[\mathfrak{T}_\star]\}.$$

If $u : \mathfrak{P}_1 \to \mathfrak{P}_2$ *is an ambient morphism, then* $\overline{\mathfrak{R}}(u)$ *is the unique rigid intertwiner between the two reduced objects.*

Proof. Theorems 13.4 to 13.7 produce from every ambient object a TFPT pre-object with the canonical invariant tuple. By Theorem 9.20, that pre-object carries a unique intrinsic branch, and by Theorem 9.21 and corollary 9.23 every rigid closed realization with this invariant tuple is uniquely isomorphic to $\mathfrak{T}_\star$. Therefore every reduced ambient object lies in the unique rigid orbit of $\mathfrak{T}_\star$. The functoriality on morphisms is immediate because the rigid target has only the unique intertwiner allowed by essential injectivity on the rigid image. □

Remark (Primitive honesty). The manuscript now proves the passage from a minimal operational seed to the one-sided boundary datum and then the canonical closure from that datum. It does not yet prove that $\mathbf{Seed}^{\mathrm{op}}$ itself is generated from a still thinner universal seed. Accordingly the manuscript establishes internal rigidity before ultimate primitive minimality.

Remark (Internal rigidity versus external adequacy). The main text proves uniqueness of the closed TFPT object inside the ambient class $\widehat{\mathrm{PhysAdm}}_1$:

$$\forall \mathfrak{P} \in \widehat{\mathrm{PhysAdm}}_1, \qquad \overline{\mathfrak{R}}(\mathfrak{P}) \cong \mathfrak{T}_\star.$$

This does not by itself prove that $\widehat{\mathrm{PhysAdm}}_1$ is the physically correct ambient class. External adequacy is a separate comparison claim on the observable map

$$\mathcal{E} : \widehat{\mathrm{PhysAdm}}_1 \to \mathcal{O}_{\mathrm{phys}}, \qquad \mathfrak{P} \longmapsto \mathcal{E}(\mathfrak{P}).$$

The input-side reduction and the output-side orbit are summarized in Figure 22.

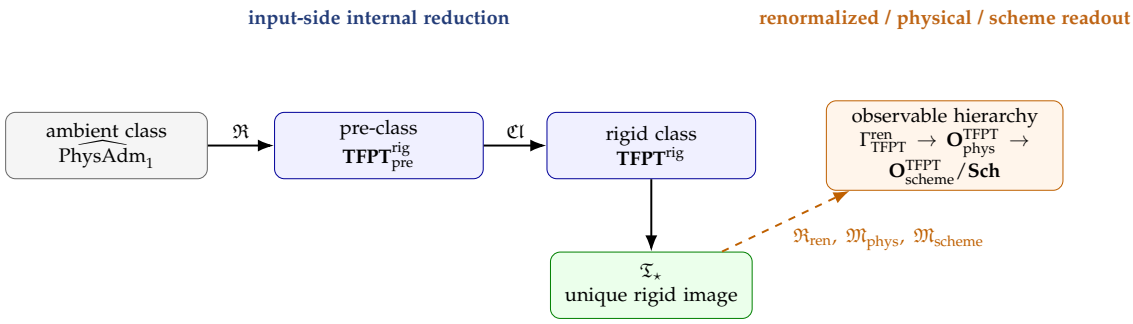


Figure 22. Input-side reduction and observable hierarchy.

13.2 Renormalized, physical, and scheme readout layers

Definition 13.9 (Admissible effective average action)**.** Let R_k be an admissible infrared regulator commuting with the structural sector selection. Define

$$e^{W_k[J]} := \int_{\mathcal{H}_{\mathrm{adm}}} \mathcal{D}\varphi \, \exp\Big(-S_{\mathrm{adm}}[\varphi] - \frac{1}{2}\langle \varphi, R_k \varphi\rangle + \langle J, \varphi\rangle\Big),$$

and set

$$\Gamma_k[\phi] := \sup_J \{\langle J, \phi\rangle - W_k[J]\} - \frac{1}{2}\langle \phi, R_k \phi\rangle.$$

The renormalized admissible effective action is

$$\Gamma^{\mathrm{ren}}_{\mathrm{TFPT}} := \lim_{k\to 0} \Gamma_k.$$

Theorem 13.10 (Exact admissible renormalization-group flow)**.** *The admissible effective average action satisfies the exact functional flow equation*

$$\partial_t \Gamma_k[\phi] = \frac{1}{2}\, \mathrm{STr}_{\mathcal{H}_{\mathrm{adm}}}\Big[\big(\Gamma_k^{(2)}[\phi] + R_k\big)^{-1} \partial_t R_k\Big], \qquad t := \log k.$$

Thus the running couplings of the admissible theory are generated dynamically on Γ_k, not by the static projector P_{adm}.

Proof. Differentiate the Legendre transform defining Γ_k at fixed mean field ϕ. Using

$$\frac{\delta^2 W_k}{\delta J\, \delta J} = \big(\Gamma_k^{(2)} + R_k\big)^{-1},$$

one obtains

$$\partial_t \Gamma_k = \frac{1}{2}\, \mathrm{STr}\big[\big(\Gamma_k^{(2)} + R_k\big)^{-1} \partial_t R_k\big].$$

The supertrace is taken on the admissible physical sector because the regulated functional integral itself is defined there. □

Corollary 13.11 (One-loop shadow of the exact admissible flow)**.** *On the canonical branch with three families and one Higgs doublet, the exact admissible flow has the perturbative shadow*

$$\partial_t g_i^{-2}(k) = -\frac{b_i}{8\pi^2} + O(g_i^0), \qquad (b_1, b_2, b_3) = \left(\frac{41}{10}, -\frac{19}{6}, -7\right),$$

with g_1 in the standard hypercharge normalization. Equivalently,

$$\partial_t \alpha_1^{-1}(k) = -\frac{41}{20\pi} + O(\alpha_1).$$

Proof. Expanding Theorem 13.10 around the Gaussian region yields the standard one-loop coefficients determined by the field content of the canonical branch. The carrier and family theorems fix exactly three families and one light Higgs doublet, so the resulting shadow coefficients are the displayed Standard Model values. □

Corollary 13.12 (Infrared boundary condition for the electromagnetic coupling)**.** *Let α_* be the exact positive solution of the admissible electromagnetic closure equation. Then*

$$\alpha_{\mathrm{em}}(0) = \alpha_*$$

is the infrared boundary value of the exact admissible flow. For $\mu > 0$ one has

$$\alpha_{\mathrm{em}}^{-1}(\mu) = \alpha_*^{-1} - \int_0^{\log\mu} \beta_{\alpha^{-1}}(t)\,dt - \Delta_{\mathrm{thr}}(\mu),$$

where Δ_{thr} records electroweak mixing and threshold decoupling.

Proof. Theorem 4.5 fixes the unique branch value α_* at the infrared endpoint. The exact flow of Theorem 13.10 then propagates that value to finite scales, while threshold effects are collected in the declared correction term Δ_{thr}. □

Theorem 13.13 (Admissible 1PI graph expansion)**.** *Around any admissible background field ϕ, the effective action admits the loop expansion*

$$\Gamma_k[\phi] = S_{\mathrm{adm}}[\phi] + \frac{\hbar}{2}\,\mathrm{STr}\log\big(S^{(2)}_{\mathrm{adm}}[\phi] + R_k\big) + \sum_{\ell\geq 2} \hbar^\ell \Gamma^{\mathrm{1PI}}_{\ell,k}[\phi].$$

Each coefficient $\Gamma^{\mathrm{1PI}}_{\ell,k}$ is the sum of admissible one-particle-irreducible Feynman graphs with propagator

$$G_{k,\phi} := \big(S^{(2)}_{\mathrm{adm}}[\phi] + R_k\big)^{-1}$$

and vertices given by the higher functional derivatives $S^{(n)}_{\mathrm{adm}}[\phi]$.

Proof. Set $\varphi = \phi + \sqrt{\hbar}\eta$ and expand

$$S_{\mathrm{adm}}[\phi + \sqrt{\hbar}\eta] = S_{\mathrm{adm}}[\phi] + \frac{\hbar}{2}\langle \eta, S^{(2)}_{\mathrm{adm}}[\phi]\eta\rangle + \sum_{n\geq 3} \frac{\hbar^{n/2}}{n!} S^{(n)}_{\mathrm{adm}}[\phi](\eta^n).$$

The regulated functional integral therefore becomes a Gaussian measure with covariance $G_{k,\phi}$ dressed by higher interaction vertices. Taking the logarithm produces the connected diagram expansion, and the Legendre transform removes one-particle-reducible pieces. The resulting coefficients are exactly the admissible 1PI graph sums recorded above. □

Definition 13.14 (Canonical renormalized 1PI package)**.** Let $\Gamma^{\mathrm{ren}}_{\mathrm{TFPT}}$ denote the category whose objects are renormalized low-energy packages

$$\mathcal{G} = \big(\{\Sigma_f^{\mathrm{TFPT}}(p)\}_f,\, G_F^{\mathrm{TFPT}},\, \mathcal{R}_{\mathrm{det}},\, \mathcal{B}_{\mathrm{cos}},\, \mathcal{A}_{\mathrm{pole}}\big)$$

built canonically from the rigid object $\mathfrak{T}_\star$. They are extracted from the infrared limit of the exact admissible flow,

$$\Gamma^{\mathrm{ren}}_{\mathrm{TFPT}}[\Phi_c] := \lim_{k\to 0} \Gamma_k[\Phi_c],$$

equivalently the Legendre transform of the retained relative generating functional after removing the infrared regulator. Here $\mathcal{A}_{\mathrm{pole}}$ collects the dressed inverse propagators, $\mathcal{R}_{\mathrm{det}}$ the determinant-line response kernels, and $\mathcal{B}_{\mathrm{cos}}$ the FRW / reheating / Boltzmann interface data on the closed branch. Morphisms are RG-compatible field reparameterizations preserving the branch-fixed pole and response data.

Definition 13.15 (Physical observable category)**.** Let $\mathbf{O}^{\mathrm{TFPT}}_{\mathrm{phys}}$ denote the category whose objects are triples

$$\mathcal{O}_{\mathrm{phys}} = (v_{\mathrm{phys}}, \Sigma, \mathcal{D}),$$

where v_{phys} is a package of physical observables (pole masses, widths, mixing angles, response coefficients, and closed-branch cosmology interface packages before late-time continuation), Σ collects their covariance data, and $\mathcal{D}$ is a dependency DAG recording algebraic and statistical non-independence.

Definition 13.16 (Scheme groupoid)**.** Let **Sch** be the groupoid whose objects are declared comparison-scheme packages

$$s = (s_{\mathrm{EM}}, s_{\mathrm{EW}}, s_\nu, s_{\mathrm{QCD}}, s_{\mathrm{cos}}, \Sigma_{\mathrm{snap}})$$

consisting of threshold conventions, mass schemes, oscillation conventions, cosmology snapshot rules, and covariance bookkeeping. Morphisms

$$\psi : s \to s'$$

are finite renormalization, threshold-transfer, snapshot-transfer, and covariance-preserving scheme changes.

Definition 13.17 (Scheme observable category)**.** Let $\mathbf{O}^{\mathrm{TFPT}}_{\mathrm{scheme}}$ denote the category whose objects are triples

$$\mathcal{O}_{\mathrm{scheme}} = (v, \Sigma, \mathcal{D}),$$

where v is a scheme-specified representative of a physical observable package, Σ its declared covariance data, and $\mathcal{D}$ the induced dependency DAG. Morphisms are smooth covariance-preserving reparameterizations respecting $\mathcal{D}$.

Definition 13.18 (Canonical readout hierarchy)**.** Define the functors

$$\mathfrak{R}_{\mathrm{ren}} : \mathbf{TFPT}^{\mathrm{rig}} \to \Gamma^{\mathrm{ren}}_{\mathrm{TFPT}}, \qquad \mathfrak{M}_{\mathrm{phys}} : \Gamma^{\mathrm{ren}}_{\mathrm{TFPT}} \to \mathbf{O}^{\mathrm{TFPT}}_{\mathrm{phys}}, \qquad \mathfrak{M}_{\mathrm{scheme}} : \mathbf{O}^{\mathrm{TFPT}}_{\mathrm{phys}} \to \mathbf{O}^{\mathrm{TFPT}}_{\mathrm{scheme}}/\mathbf{Sch}.$$

For every scheme object $s \in \mathbf{Sch}$, let

$$\mathfrak{M}^{(s)}_{\mathrm{scheme}}\big(\mathfrak{M}_{\mathrm{phys}}(\mathfrak{R}_{\mathrm{ren}}(\mathfrak{T}_\star))\big) \in \mathbf{O}^{\mathrm{TFPT}}_{\mathrm{scheme}}$$

be the chosen scheme representative. The canonical appendix comparison orbit is

$$[\mathfrak{M}(\mathfrak{T}_\star)]_{\mathbf{Sch}} := \mathfrak{M}_{\mathrm{scheme}}\big(\mathfrak{M}_{\mathrm{phys}}(\mathfrak{R}_{\mathrm{ren}}(\mathfrak{T}_\star))\big) \in \mathbf{O}^{\mathrm{TFPT}}_{\mathrm{scheme}}/\mathbf{Sch},$$

and the appendix comparison map is any chosen representative

$$\mathcal{R}^{(s)}_{\mathrm{cmp}}(\mathfrak{T}_\star) := \mathfrak{M}^{(s)}_{\mathrm{scheme}}\big(\mathfrak{M}_{\mathrm{phys}}(\mathfrak{R}_{\mathrm{ren}}(\mathfrak{T}_\star))\big).$$

Definition 13.19 (Minimal independent observable basis)**.** Define the minimal independent basis package

$$\mathcal{B}_{\mathrm{ind}}(\mathfrak{T}_\star) := \big(C_{\mathrm{em}}, K_{\mathrm{UV}}, F_{\mathrm{fl}}, N_\nu, R_{\mathrm{CS}}, T_{\mathrm{phys}}, C_{\mathrm{cos}}, C_{\mathrm{CP}}\big),$$

where

$$C_{\mathrm{em}} := \alpha^{-1}(0), \qquad K_{\mathrm{UV}} := \varphi_0^{\mathrm{ret}}, \qquad F_{\mathrm{fl}} := (\lambda_C, \delta_{\mathrm{CKM}}), \qquad R_{\mathrm{CS}} := \beta, \qquad C_{\mathrm{CP}} := \theta_{\mathrm{eff}},$$

$$N_\nu := (\sin^2\theta_{13}, \delta^\nu_{\mathrm{CP}}, \sin^2\theta_{23}, \Sigma m_\nu, m_{\beta\beta}, M_R),$$

$$T_{\mathrm{phys}} := \big(G_F^{\mathrm{TFPT}}, m_W^{\mathrm{pole}}, m_Z^{\mathrm{pole}}, m_H^{\mathrm{pole}}, \{m_f^{\mathrm{pole}}\}_f\big),$$

$$C_{\mathrm{cos}} := \big(\Omega_b, S_\Sigma, \Lambda_{\mathrm{IR}}, T_R, f_a, m_a, \nu_a, \eta_B\big).$$

The retained seed $K_{\mathrm{UV}} = \varphi_0^{\mathrm{ret}}$ is part of the UV kernel package; it is no longer the direct basis package of the flavor, neutrino, determinant-response, or cosmology observables.

Theorem 13.20 (Renormalized observable hierarchy from the admissible 1PI action)**.** *There exist functors*

$$\mathfrak{R}_{\mathrm{ren}} : \mathbf{TFPT}^{\mathrm{rig}} \to \Gamma^{\mathrm{ren}}_{\mathrm{TFPT}}, \qquad \mathfrak{M}_{\mathrm{phys}} : \Gamma^{\mathrm{ren}}_{\mathrm{TFPT}} \to \mathbf{O}^{\mathrm{TFPT}}_{\mathrm{phys}}, \qquad \mathfrak{M}_{\mathrm{scheme}} : \mathbf{O}^{\mathrm{TFPT}}_{\mathrm{phys}} \to \mathbf{O}^{\mathrm{TFPT}}_{\mathrm{scheme}}/\mathbf{Sch}$$

with the following properties:

(O1) ***Reality first.*** *Main-text observables factor through $\mathfrak{M}_{\mathrm{phys}} \circ \mathfrak{R}_{\mathrm{ren}}$ and therefore belong to $\mathbf{O}^{\mathrm{TFPT}}_{\mathrm{phys}}$ before any scheme choice is made.*

(O2) ***Unique factorization through the basis.*** *Every physical readout row and every appendix scheme row factors through exactly one component package of $\mathcal{B}_{\mathrm{ind}}(\mathfrak{T}_\star)$ via a declared physical or scheme map.*

(O3) ***Scheme covariance.*** *For every scheme morphism $\psi : s \to s'$ one has*

$$\psi\Big(\mathfrak{M}^{(s)}_{\mathrm{scheme}}(\mathfrak{M}_{\mathrm{phys}}(\mathfrak{R}_{\mathrm{ren}}(\mathfrak{T}_\star)))\Big) = \mathfrak{M}^{(s')}_{\mathrm{scheme}}(\mathfrak{M}_{\mathrm{phys}}(\mathfrak{R}_{\mathrm{ren}}(\mathfrak{T}_\star))).$$

(O4) ***No hidden continuous fit parameter.*** *Any continuous freedom in a readout row is either a declared scheme degree of freedom in* **Sch** *or already a theorem-level branch parameter fixed in $\mathfrak{T}_\star$.*

(O5) ***Complete kill surface.*** *Every admissible empirical kill test of the manuscript is the pullback of a component test on $\mathbf{O}^{\mathrm{TFPT}}_{\mathrm{phys}}$ and only afterwards, if needed, of a scheme representative in $\mathbf{O}^{\mathrm{TFPT}}_{\mathrm{scheme}}$.*

Proof. Define $\mathfrak{R}_{\mathrm{ren}}$ by sending the rigid object $\mathfrak{T}_\star$ to the renormalized 1PI package extracted from the exact admissible flow of Theorem 13.10. Its infrared endpoint is

$$\Gamma^{\mathrm{ren}}_{\mathrm{TFPT}}[\Phi_c] = \lim_{k\to 0} \Gamma_k[\Phi_c],$$

equivalently the Legendre transform of the retained admissible generating functional after removing the infrared regulator. By Theorem 13.13, the same object carries the admissible 1PI graph expansion around each admissible background. Reflection positivity, lower boundedness, and coercivity on the admissible sector ensure the convexity needed for this Legendre construction in a neighborhood of the closed branch. Hence the dressed inverse propagators, determinant-response kernels, and cosmology saddle system belong to one and the same renormalized package $\Gamma^{\mathrm{ren}}_{\mathrm{TFPT}}$.

Next define $\mathfrak{M}_{\mathrm{phys}}$ by reading physical observables from that 1PI package before any scheme choice: pole masses are the zeros of the dressed inverse propagators, mixing data are the branch invariants of the corresponding residue matrices, determinant-response observables are read from the exact response kernels, and cosmology rows factor through the same FRW / reheating / Boltzmann interface block. Final late-time numerical representatives are attached only after the declared comparison map. This proves **(O1)**.

For **(O2)**, the factorization through the minimal basis package $\mathcal{B}_{\mathrm{ind}}(\mathfrak{T}_\star)$ is exactly the theorem-level content of the rewritten sector split. The flavor rows factor through F_{fl}, neutrino rows through N_ν, determinant-response rows through R_{CS}, strong-CP rows through C_{CP}, cosmology rows through C_{cos}, and the retained seed φ_0^{ret} survives only in the UV kernel package K_{UV}. No row uses two basis packages at once.

For **(O3)**, every scheme morphism in **Sch** acts only after $\mathfrak{M}_{\mathrm{phys}}$ has produced a physical observable package. Therefore threshold transfer, finite renormalization, and snapshot change act on scheme representatives without altering the underlying physical point, exactly as in the displayed covariance identity.

For **(O4)**, every continuous parameter entering a readout row is already fixed either by the closed branch $\mathfrak{T}_\star$ or by the declared scheme object $s \in$ **Sch**. Since $\mathfrak{R}_{\mathrm{ren}}$ and $\mathfrak{M}_{\mathrm{phys}}$ are defined

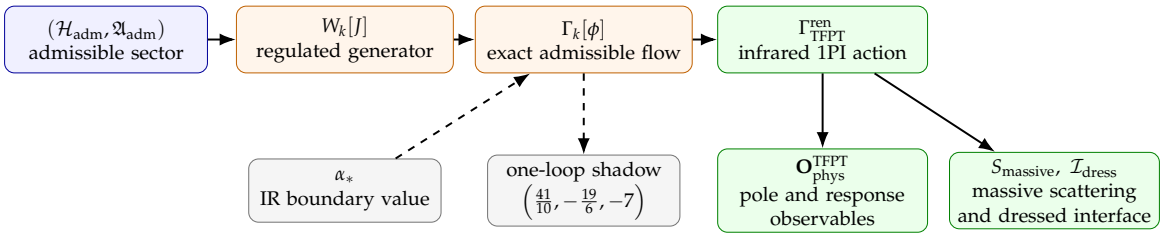


Figure 23. Exact admissible flow from the physical sector to observables and scattering. $P_{\rm adm}$ selects states; Γ_k carries running couplings and 1PI graphs.

from the fixed admissible measure and its 1PI action, no extra continuous fit parameter can appear in between.

Finally, **(O5)** follows because every empirical kill criterion in the manuscript is attached to one of the physical rows in $\mathbf{O}^{\rm TFPT}_{\rm phys}$ and only afterwards, if needed, represented by a scheme object in $\mathbf{O}^{\rm TFPT}_{\rm scheme}/\mathbf{Sch}$. Thus the whole observable hierarchy factors through the same admissible 1PI action and then through the declared physical / scheme split. □

Corollary 13.21 (No double counting after the seed-shadow split)**.** *Rows sharing the same basis package in Definition 13.19 define one underlying operational direction, not several independent confirmations or failures. In particular, the former quartet*

$$(\beta, \Omega_b, \lambda_C, \sin^2\theta_{13})$$

no longer constitutes one seed package. Instead

$$\beta \in R_{\rm CS}, \qquad \Omega_b \in C_{\rm cos}, \qquad \lambda_C \in F_{\rm fl}, \qquad \sin^2\theta_{13} \in N_\nu,$$

while the retained seed $\varphi_0^{\rm ret}$ belongs only to the UV kernel package $K_{\rm UV}$.

Proof. This is the special case of **(O2)** for the rewritten basis packages of Definition 13.19. □

Corollary 13.22 (Independent response-sector birefringence target)**.** *On the minimal branch the determinant-response package $R_{\rm CS}$ fixes*

$$\beta_{\rm rad} = \frac{\phi^0_{\rm ret}}{4\pi},$$

and hence the dimensionless birefringence readout itself. Under the declared comparison interface it is represented by

$$\beta = \frac{180}{\pi}\,\beta_{\rm rad} = 0.2424350309009295\ldots^\circ.$$

This target factors through $R_{\rm CS}$ and not through the flavor package $F_{\rm fl}$ or the cosmology package $C_{\rm cos}$.

Proof. By the exact seed relation of the determinant-response channel,

$$\beta_{\rm rad} = \frac{\phi^0_{\rm ret}}{4\pi}.$$

The theorem-level quantity is therefore the dimensionless angle $\beta_{\rm rad}$. Converting to degrees gives one scheme-facing representative,

$$\beta = \frac{180}{\pi}\beta_{\rm rad} = \frac{180}{\pi}\frac{\phi^0_{\rm ret}}{4\pi} = 0.2424350309009295\ldots^\circ.$$

By Corollary 13.21, β belongs to $R_{\rm CS}$, whereas $\lambda_C \in F_{\rm fl}$ and $\Omega_b \in C_{\rm cos}$. Hence this is an operational direction distinct from the flavor and cosmology packages. □

Corollary 13.23 (Independent cosmology-sector axion target)**.** *On the closed cosmology branch the package $C_{\cos}$ fixes the axion readout as one dimensionless cosmology package, equivalently as the boundary-normalized row*

$$\left(\frac{f_a}{\lambda_\Sigma}, \frac{m_a}{\lambda_\Sigma}, \frac{\nu_a}{\lambda_\Sigma}, \lambda_\Sigma \left| g_{a\gamma\gamma}^{\text{phys}} \right| \right),$$

This target factors through $C_{\cos}$ and not through the flavor package F_{fl} or the determinant-response package R_{CS}.

Proof. The closed cosmology branch fixes the determinant-line / seam-transfer / scalaron saddle and thus the axion row as a single component of $C_{\cos}$. Its unitful representatives belong only to the declared comparison interface and are recorded in the appendix benchmark layer, together with the practical haloscope window and coupling representative extracted from the same cosmology package. By Corollary 13.21, this row belongs to $C_{\cos}$, whereas β belongs to R_{CS} and λ_C belongs to F_{fl}. Hence the axion target is an operational direction distinct from response and flavor. □

Remark (Why these two targets count as independent). The pair

$$\beta \in R_{\text{CS}}, \qquad (\nu_a, m_a, g_{a\gamma\gamma}^{\text{phys}}) \in C_{\cos},$$

samples two different basis packages of $\mathcal{B}_{\text{ind}}(\mathfrak{T}_\star)$. Therefore they are not two rewritings of the same seed constraint. This is the correct sharp-prediction surface for the present version.

14 Absolute dimensionless metrology from the boundary branch

The closed branch should not be read as a source of isolated SI numbers. Its hard content is an internal dimensionless metrology: the boundary datum carries its own spectral unit, the stationary geometric branch fixes the gravitational normalization relative to that unit, the renormalized 1PI action fixes physical electroweak and pole readouts, and only afterwards does one choose an external comparison scheme. In the compact gravitational chain,

$$\mathfrak{T}_\partial^{\min} \Rightarrow B_\Sigma \Rightarrow \lambda_\Sigma \Rightarrow \rho_\star \Rightarrow \chi_{\text{geo},0} = \lambda_\Sigma \sqrt{\rho_\star} \Rightarrow \left(G_N \lambda_\Sigma^2, \frac{\bar{M}_{\text{Pl}}}{\lambda_\Sigma}, \frac{m_P}{\lambda_\Sigma}, \lambda_\Sigma \ell_P, \lambda_\Sigma t_P \right).$$

14.1 Boundary spectral unit

Definition 14.1 (Boundary-normalized readout)**.** Let λ_Σ be the intrinsic boundary mass unit fixed by Theorem 6.29. For a physical quantity O of mass dimension d_O, define its boundary-normalized readout by

$$\widehat{O} := \frac{O}{\lambda_\Sigma^{d_O}}.$$

14.2 Stationary spectral Planck root

Corollary 14.2 (Unique stationary spectral Planck root)**.** *By Theorem 6.31, the admissible geometric branch carries the unique positive stationary root*

$$\rho_\star = \frac{\chi_{\text{geo},0}^2}{\lambda_\Sigma^2} > 0.$$

$$\partial_\rho \widehat{V}_{\text{Pl}}^{\text{ren}}(\rho_\star) = 0.$$

This root is the boundary-normalized Planck datum of the stationary branch.

Corollary 14.3 (Dimensionless Planck metrology on the stationary branch)**.** *On the stationary branch one has*

$$\frac{\bar{M}_{\rm Pl}}{\lambda_\Sigma} = \frac{\sqrt{\rho_\star}}{\sqrt{2}\pi}, \qquad G_N\lambda_\Sigma^2 = \frac{\pi}{4\rho_\star},$$
$$\frac{m_P}{\lambda_\Sigma} = \frac{2\sqrt{\rho_\star}}{\sqrt{\pi}}, \qquad \lambda_\Sigma \ell_P = \lambda_\Sigma t_P = \frac{\sqrt{\pi}}{2\sqrt{\rho_\star}}.$$

Proof. Substitute

$$\chi_{\rm geo,0} = \lambda_\Sigma\sqrt{\rho_\star}$$

into Corollary 6.33. □

Corollary 14.4 (Dimensionless geometric hierarchy law for the UV source scale)**.** *On the exact geometric branch one has*

$$\frac{v_{\rm geo}}{\bar{M}_{\rm Pl}} = g_{\rm car}\,\beta_{\rm rad}^2 \exp\left[-\frac{\alpha^{-1}(0)+\delta_{\rm ph}}{5}\right].$$

Consequently

$$G_N v_{\rm geo}^2 = \frac{1}{8\pi}\left(\frac{v_{\rm geo}}{\bar{M}_{\rm Pl}}\right)^2 = \frac{1}{8\pi} g_{\rm car}^2 \beta_{\rm rad}^4 \exp\left[-\frac{2(\alpha^{-1}(0)+\delta_{\rm ph})}{5}\right].$$

This is the UV source-scale hierarchy law and not yet the physical electroweak benchmark relation.

Proof. The first identity is exactly Equation (29) divided by $\bar{M}_{\rm Pl}$. The second uses the reduced-Planck relation $\bar{M}_{\rm Pl}^{-2} = 8\pi G_N$. □

14.3 Infrared seam and sector bifurcation

Theorem 14.5 (Sector-dressed spectral scale bifurcation)**.** *The primitive response $\chi_{\rm seed}$ bifurcates on the closed branch into distinct dressed spectral channels:*

$$\chi_{\rm seed} \longrightarrow \chi_{\rm geo,0} \qquad \textit{and} \qquad \chi_{\rm seed} \longrightarrow \chi_{\rm QCD}.$$

Here $\chi_{\rm geo,0}$ is the stationary Einstein/Higgs background scale, while $\chi_{\rm QCD}$ is the transport-dressed hadronic scale. They share one boundary origin but are not identified on the closed theorem surface.

Proof. The geometric channel is defined by the local spectral scale and its stationary branch Definition 6.11 and theorem 6.31. The hadronic channel is kept separate by the sector-dressed transport/confinement bookkeeping recorded in the transport block and summarized in the scale-bifurcation diagram Figure 13. Hence the same primitive response is dressed differently in the geometric and hadronic sectors. □

Theorem 14.6 (Carrier-exhausted infrared seam eigenvalue under the exhaustion hypothesis)**.** *Let*

$$q_\star := \delta_{\rm top} e^{-2\alpha_\star}, \qquad \varphi_{\rm base} = \frac{1}{6\pi}.$$

Assume the leading admissible infrared seam mode is rank one and exhausts the nonempty carrier occupancy shadow, so that

$$\rho(U_\Sigma) = \varphi_{\rm base}\, q_\star^{2g_{\rm car}-1}.$$

Then on the canonical branch $g_{\rm car} = 5$ and therefore

$$\rho(U_\Sigma) = \varphi_{\rm base}\, q_\star^{31}.$$

Moreover

$$\frac{\Lambda_{\rm IR}}{M_{\rm Pl}^4} = -\log\big(1-\rho(U_\Sigma)\big).$$

Proof. The first displayed identity is exactly the carrier-exhaustion hypothesis written in closed-branch variables. On the canonical carrier branch, Corollary 5.3 gives $g_{\mathrm{car}} = 5$, hence $2^{g_{\mathrm{car}}} - 1 = 31$. The final identity is the rank-one case of Corollary 10.4. □

14.4 Physical readout and metrology functor

Theorem 14.7 (Exact electroweak residue matching as a dimensionless readout)**.** *On the renormalized branch, define*

$$\mathcal{Z}_{\mathrm{EW}}^{\mathrm{TFPT}} := \frac{1}{2v_{\mathrm{geo}}^2\,\mathcal{R}_{\mathrm{CC}}^{\mathrm{TFPT}}(0)} = Z_\Phi(0)\left(1+\frac{\delta v}{v_{\mathrm{geo}}}\right)^2 \frac{1+\delta_W}{1+\Delta r_{\mathrm{CC}}^{\mathrm{TFPT}}}.$$

Then the quantities

$$\mathcal{R}_{\mathrm{CC}}^{\mathrm{TFPT}}(0), \qquad Z_\Phi(0), \qquad \delta v, \qquad \delta_W, \qquad \Delta r_{\mathrm{CC}}^{\mathrm{TFPT}}$$

are determined by $\Gamma_{\mathrm{TFPT}}^{\mathrm{ren}}$*. Consequently*

$$\frac{v_{\mathrm{phys}}}{\lambda_\Sigma} = \frac{v_{\mathrm{geo}}}{\lambda_\Sigma}\sqrt{\mathcal{Z}_{\mathrm{EW}}^{\mathrm{TFPT}}}, \qquad \frac{v_{\mathrm{phys}}^2}{\lambda_\Sigma^2} = \left(\frac{v_{\mathrm{geo}}}{\lambda_\Sigma}\right)^2 \mathcal{Z}_{\mathrm{EW}}^{\mathrm{TFPT}},$$

and hence

$$G_N v_{\mathrm{phys}}^2 = G_N v_{\mathrm{geo}}^2\,\mathcal{Z}_{\mathrm{EW}}^{\mathrm{TFPT}}$$

is an internal readout of the closed branch.

Proof. The matching identity is exactly Theorem 6.21 written in boundary-normalized form. By Theorem 13.20, the same residue data belong to the renormalized 1PI package $\Gamma_{\mathrm{TFPT}}^{\mathrm{ren}}$ before any scheme choice is made. Multiplying by $G_N\lambda_\Sigma^2$ yields the final formula for $G_N v_{\mathrm{phys}}^2$. □

Corollary 14.8 (Dimensionless gravitational electroweak metrology)**.** *On the closed branch,*

$$G_N v_{\mathrm{phys}}^2 = \frac{1}{8\pi} g_{\mathrm{car}}^2 \beta_{\mathrm{rad}}^4 \exp\left[-\frac{2(\alpha^{-1}(0)+\delta_{\mathrm{ph}})}{5}\right] \mathcal{Z}_{\mathrm{EW}}^{\mathrm{TFPT}}.$$

This is the primary gravity/electroweak hierarchy prediction of the closed branch. It contains no external gravitational input and no metrological scale.

Proof. Combine Corollary 14.4 and theorem 14.7. □

Corollary 14.9 (Point-closed Planck-scale reconstruction)**.** *Using Theorem 6.24, the electroweak factor in Corollary 14.8 is fixed pointwise as*

$$\mathcal{Z}_{\mathrm{EW}}^{\mathrm{TFPT}} = 1.0336068207261621.$$

Consequently

$$G_N v_{\mathrm{phys}}^2 = 4.0671702198788166 \times 10^{-34}.$$

After one non-gravitational metrological calibration of v_{phys}*, Newton's constant is algebraically reconstructed by*

$$G_N = \frac{(G_N v_{\mathrm{phys}}^2)_{\mathrm{TFPT}}}{v_{\mathrm{phys}}^2},$$

and the Planck units follow downstream:

$$M_{\mathrm{Pl}} = \frac{1}{\sqrt{G_N}}, \qquad \bar{M}_{\mathrm{Pl}} = \frac{1}{\sqrt{8\pi G_N}}, \qquad \ell_P = \sqrt{G_N}, \qquad t_P = \ell_P \quad (c = \hbar = 1).$$

In SI representatives the usual conversion is applied only after this dimensionless hierarchy is known:

$$\ell_P = \sqrt{\frac{\hbar G_N}{c^3}}, \qquad t_P = \sqrt{\frac{\hbar G_N}{c^5}}, \qquad m_P = \sqrt{\frac{\hbar c}{G_N}}.$$

Proof. The first displayed value is the point-closed charged-current readout of Theorem 6.24. Multiplying it by the internally fixed geometric hierarchy row $G_N v_{\mathrm{geo}}^2$ gives $G_N v_{\mathrm{phys}}^2 = G_N v_{\mathrm{geo}}^2 \mathcal{Z}_{\mathrm{EW}}^{\mathrm{TFPT}}$. Since this quantity is dimensionless, no Newton constant is inserted at this stage. Once a non-gravitational representative for v_{phys} is chosen, the displayed algebraic formula fixes G_N and the Planck units follow by definition. □

Theorem 14.10 (Pole masses as dimensionless branch readouts)**.** *For each fermion sector f, the quantity*

$$\frac{m_f^{\mathrm{pole}}}{\lambda_\Sigma}$$

is determined by the positive root of the Dirac pole equation of Theorem 6.25 after dividing by λ_Σ^2. Therefore

$$\frac{m_f^{\mathrm{pole}}}{\lambda_\Sigma} = \mathcal{M}_f(\mathfrak{T}_\star), \qquad G_N\big(m_f^{\mathrm{pole}}\big)^2 = (G_N\lambda_\Sigma^2)\left(\frac{m_f^{\mathrm{pole}}}{\lambda_\Sigma}\right)^2.$$

Proof. By Theorem 6.26, fermion pole masses are read from the dressed inverse propagators of the renormalized branch and therefore belong to $\mathbf{O}_{\mathrm{phys}}^{\mathrm{TFPT}}$. Dividing the pole equation by λ_Σ^2 produces a dimensionless root problem. The last identity is immediate. □

Definition 14.11 (TFPT metrology functor)**.** Define the dimensionless metrology assignment of the closed branch by

$$\mathfrak{M}_{\mathrm{TFPT}}(\mathfrak{T}_\star) := \left\{ \widehat{O} \;\middle|\; O \in \mathbf{O}_{\mathrm{phys}}^{\mathrm{TFPT}}, \quad d_O = \dim_{\mathrm{mass}}(O), \quad \widehat{O} = \frac{O}{\lambda_\Sigma^{d_O}} \right\}.$$

Theorem 14.12 (Dimensionless TFPT metrology)**.** *For every physical observable O of mass dimension d_O in the physical observable layer, one has*

$$\frac{O}{\lambda_\Sigma^{d_O}} = \mathcal{F}_O(\mathfrak{T}_\star).$$

In particular, the closed branch fixes

$$\alpha, \qquad \delta_{\mathrm{ph}}, \qquad \lambda_C, \qquad \theta_{\mathrm{eff}}, \qquad G_N\lambda_\Sigma^2, \qquad \frac{v_{\mathrm{phys}}}{\lambda_\Sigma}, \qquad G_N v_{\mathrm{phys}}^2,$$

$$\frac{m_f^{\mathrm{pole}}}{\lambda_\Sigma}, \qquad G_N\big(m_f^{\mathrm{pole}}\big)^2, \qquad \frac{\Lambda_{\mathrm{IR}}}{\lambda_\Sigma^4}, \qquad \frac{\Lambda_{\mathrm{IR}}}{M_{\mathrm{Pl}}^4}, \qquad \frac{m_i}{m_j}$$

without external metrological input. Only afterwards may one choose a scheme representative or SI translation.

Proof. By Theorem 6.29, the unit λ_Σ is fixed by the one-sided boundary datum. By Theorem 13.20, every main-text observable factors through the physical observable functor

$$\mathfrak{M}_{\mathrm{phys}} \circ \mathfrak{R}_{\mathrm{ren}} : \mathbf{TFPT}^{\mathrm{rig}} \to \mathbf{O}_{\mathrm{phys}}^{\mathrm{TFPT}}.$$

The dimensionless observables α and δ_{ph} are already fixed directly by the electromagnetic and transport closures. The quantities $G_N\lambda_\Sigma^2$, $v_{\mathrm{phys}}/\lambda_\Sigma$, and $G_N v_{\mathrm{phys}}^2$ are fixed by Corollaries 14.3 and 14.9 and theorem 14.7. The pole-mass claims follow from Theorem 14.10. The dimensionless cosmological quantity $\Lambda_{\mathrm{IR}}/M_{\mathrm{Pl}}^4$ is fixed by the theorem-level determinant expression of Theorem 10.3, while mass ratios are already dimensionless quotients of pole readouts. Therefore every displayed element is an internal function of $\mathfrak{T}_\star$, and only the final map to $\mathbf{O}_{\mathrm{scheme}}^{\mathrm{TFPT}}/\mathbf{Sch}$ chooses a comparison convention. □

15 Conclusion

15.1 Closed theorem stack and interface layers

TFPT is organized around the one-sided boundary datum

$$\mathfrak{T}_\partial^{\min} = (\mathcal{A}_+, \mathcal{H}_+, D_+, J, \Gamma, B_\Sigma)$$

induced by the minimal operational seed, and reconstructs in a two-stage closure order

$$\begin{aligned} \mathfrak{S}_{\min} \;\Rightarrow\; \mathcal{B}_{\min} \;\Rightarrow\; \mathfrak{T}_\partial^{\min} \;\Rightarrow\; (\tau_{\mathrm{dbl}}, \iota_{\mathrm{C}}, P_{\mathrm{prim}}, [u_\Sigma], c_3) \\ \Rightarrow\; d^\star_{\mathrm{disc}} \;\Rightarrow\; P_{\mathrm{adm}} \;\Rightarrow\; \mathcal{M}_{d^\star_{\mathrm{disc}}} \;\Rightarrow\; \mathcal{Z}_{\mathrm{cl}} = \{x^\star_{\mathrm{cl}}\} \;\Rightarrow\; \mathfrak{T}_\star. \end{aligned}$$

This reconstruction first fixes the joint discrete datum and only afterwards the continuous Euler–Lagrange branch, thereby identifying the canonical rigid object of the boundary-polarized branch:

$$\forall \mathfrak{T}_{\mathrm{pre}} \in \mathbf{TFPT}^{\mathrm{rig}} : \qquad \mathfrak{T}_{\mathrm{pre}} \cong \mathfrak{T}_\star \quad \text{(Corollary 9.23)}.$$

The selector-to-dynamics continuation of the same branch is

$$P_{\mathrm{adm}} \;\Rightarrow\; Z_{\mathrm{rel}} \;\Rightarrow\; \{S_n^T\} \;\Rightarrow\; (\mathcal{H}_{\mathrm{adm}}, \mathfrak{A}_{\mathrm{adm}}) \;\Rightarrow\; \Gamma_k \;\Rightarrow\; \Gamma^{\mathrm{ren}}_{\mathrm{TFPT}} \;\Rightarrow\; (\mathbf{O}^{\mathrm{TFPT}}_{\mathrm{phys}}, S_{\mathrm{massive}}) \quad \text{together with} \quad \mathcal{I}_{\mathrm{dress}}.$$

Above that rigid core, the manuscript records one theorem-level input reduction and one output-side interface

$$\widehat{\mathrm{PhysAdm}}_1 \xrightarrow{\overline{\mathfrak{R}}} \mathbf{TFPT}^{\mathrm{rig}} \simeq \{\mathfrak{T}_\star\} \xrightarrow{\mathfrak{R}_{\mathrm{ren}}} \Gamma^{\mathrm{ren}}_{\mathrm{TFPT}} \xrightarrow{\mathfrak{M}_{\mathrm{phys}}} \mathbf{O}^{\mathrm{TFPT}}_{\mathrm{phys}} \xrightarrow{\mathfrak{M}_{\mathrm{scheme}}} \mathbf{O}^{\mathrm{TFPT}}_{\mathrm{scheme}}/\mathbf{Sch}.$$

Combined with Section 14, this yields the closed metrology chain

$$\mathfrak{T}_\partial^{\min} \;\Rightarrow\; (B_\Sigma, \lambda_\Sigma, \rho_\star, \chi_{\mathrm{geo},0}, \alpha_\star, \delta_{\mathrm{ph}}, U_\Sigma, \Gamma^{\mathrm{ren}}_{\mathrm{TFPT}}) \;\Rightarrow\; \mathfrak{M}_{\mathrm{TFPT}}(\mathfrak{T}_\star),$$

so the boundary branch fixes gravitational, electroweak, pole-mass, flavor-ratio, and infrared dimensionless readouts before any SI translation or scheme representative is chosen. The left arrow is the internal reduction theorem on the weakened ambient class $\widehat{\mathrm{PhysAdm}}_1$; the middle arrows build the theorem-level renormalized and physical observable layers; only the final scheme arrow is an appendix-facing interface. Strong-CP closure is part of the core theorem stack, whereas cosmology is carried at the level of closed-branch interface data and appendix comparison maps. TFPT therefore does not identify admissibility with dynamics: P_{adm} selects the physical sector, while the actual dynamics is carried by Z_{rel}, the Schwinger functions, the reconstructed local net on $\mathcal{H}_{\mathrm{adm}}$, the exact admissible flow Γ_k, and its infrared limit $\Gamma^{\mathrm{ren}}_{\mathrm{TFPT}}$.

15.2 Two side maps from $\mathfrak{T}_\star$

From $\mathfrak{T}_\star$ exactly two side maps are attached, both hard-separated from the theorem chain:

- **Structural falsification map (main text).** The map $\mathsf{F}_{\mathrm{fals}} : \mathfrak{T}_\star \mapsto \mathsf{F}_{\mathrm{fals}}(\mathfrak{T}_\star)$ of Section 12.3 reports the structural failure mode that would break each arrow of the reconstruction. It uses no scheme conventions, no thresholds, and no comparison data.
- **Empirical comparison map (appendix only).** The map $\mathcal{R}_{\mathrm{cmp}} : \mathfrak{T}_\star \mapsto \mathcal{R}_{\mathrm{cmp}}(\mathfrak{T}_\star)$ of Section A translates the physical observable layer of $\mathfrak{T}_\star$ into scheme-specified representatives and records chosen elements of the canonical orbit $[\mathfrak{M}(\mathfrak{T}_\star)]_{\mathbf{Sch}}$. Threshold conventions, comparison schemes, and empirical residuals live only in the appendix and never enter the theorem chain.

Comparison conventions and numerical threshold maps therefore act only after this chain via $\mathcal{R}_{\mathrm{cmp}}$ and do not belong to the theorem surface itself.

15.3 Conditional closure layer and status of the four downstream readouts

The closed-branch theorem stack is supplemented by a single, explicitly labeled *conditional closure layer*, organized so that every numerical readout can be promoted to a theorem as soon as a single named lift hypothesis is discharged. The four downstream readouts and their current status are

Λ_{IR} : theorem level as determinant of $1 - U_\Sigma$ (Theorem 10.3);
numerical value conditional on $S_{\mathrm{IR}} = 2/\alpha_\star$ (Corollary C.5);

H_0 : downstream readout of $(\Lambda^\star_{\mathrm{IR}}, \Omega^\star_b, \omega^\star_a, \omega^\star_v, \omega^\star_r)$ (Conjecture 11.16);

m_q : scheme-fixed RG readout $\mathbf{m}_q^{\mathrm{TFPT}}(\mathcal{S}, \mu)$ (Definition 11.21 and conjecture 11.22);
hadronic pole masses are physical (Conjecture 11.23);

C_ℓ : Stage 1 spectral closure from $\{\lambda_j(U_\Sigma)\}$ (Conjecture 11.8);

$a_{\ell m}$: Stage 2 ergodic microcanonical seam lift from $(\{\lambda_j(U_\Sigma)\}, u_\Sigma, \mathcal{W}_\Sigma)$ (Conjecture 11.9),
scalar phase law falsified; finite-field seam mixer (Remark) is the
Weil-bound theorem-friendly candidate; "good CMB world" but not yet "this CMB world"
(Remark), pending the SMICA-resolved r_ℓ test.

The compressed integrated story is

$$\alpha_\star \;\Rightarrow\; \text{seam transfer} \;\Rightarrow\; \Lambda_{\mathrm{IR}} \;\Rightarrow\; H_0 \;\Rightarrow\; C_\ell \;\Rightarrow\; a_{\ell m},$$

and in the matter sector

$$Y_q^\star \;\Rightarrow\; \Gamma^{\mathrm{ren}}_{\mathrm{TFPT}} \;\Rightarrow\; m_q^{\mathcal{S}}(\mu) \;\Rightarrow\; m^{\mathrm{pole}}_{\mathrm{had}}.$$

The four downstream readouts are therefore not isolated numerical claims but coordinated projections of the same admissible-branch operator stack. Each entry of the conditional closure layer ships with a named kill test inside its own block; together they define the empirical attack surface of the present version.

15.4 Main claim

A compressed decoder reading of the closed branch is

$$\boxed{\begin{aligned} &6Y^2 - Y - \mathbf{1} = 0, \quad \operatorname{Tr} Y = 0 \Rightarrow 3 + 2, \\ &\qquad\qquad \Xi_{S^+}(t) \Rightarrow \text{hypercharge sectors and } 5!, \\ &\qquad\qquad 16\gamma = \frac{40}{3} \Rightarrow N_{\mathrm{fam}} = 3 \Rightarrow \Omega_{\mathrm{adm}} = 48. \end{aligned}}$$

$$\boxed{\begin{aligned} &P'(z) = 0 \Rightarrow \delta_{\mathrm{ph}}, \\ &u := \varphi_0^{\mathrm{ret}} \Rightarrow (\lambda_{\mathrm{C}}, \beta_{\mathrm{rad}}, \Omega_b, \theta_{13}), \\ &\sum_{f,j} L^{\mathrm{diag}}_{f,j} + N_\Phi = 41 \Rightarrow \alpha. \end{aligned}}$$

In this condensed form, Y generates the carrier structure, the primitive winding class $[u_\Sigma] = 1$ fixes the counting, and the retained seed u decodes the bridge observables.

The final main-text form contains no theorem arrow ending in a raw comparison functor and no claim that comparison conventions or bridge assumptions belong to the primitive ontology. Optional arithmetic continuations, appendix-level E_8 scale grammar, threshold maps,

and empirical residuals remain outside the closed theorem body. The defensible end claim of the present version is the rigid reconstruction statement

$$\boxed{\begin{aligned}\text{minimal operational seed} &\Longrightarrow \text{canonical rigid object } \mathfrak{T}_\star \\ &\Longrightarrow Z_{\mathrm{rel}} \Longrightarrow (\mathcal{H}_{\mathrm{adm}}, \mathfrak{A}_{\mathrm{adm}}) \\ &\Longrightarrow \Gamma^{\mathrm{ren}}_{\mathrm{TFPT}} \Longrightarrow (\mathbf{O}^{\mathrm{TFPT}}_{\mathrm{phys}}, S_{\mathrm{massive}}) \text{ and } \mathcal{I}_{\mathrm{dress}}\end{aligned}}$$

or, equivalently,

$$\begin{aligned}\mathfrak{S}_{\min} \Rightarrow \mathcal{B}_{\min} \Rightarrow \mathfrak{T}^{\min}_{\partial} &\Rightarrow (\tau_{\mathrm{dbl}}, \iota_{\mathrm{C}}, P_{\mathrm{prim}}, [u_\Sigma], c_3) \Rightarrow d^\star_{\mathrm{disc}} \\ &\Rightarrow P_{\mathrm{adm}} \Rightarrow \mathcal{M}_{d^\star_{\mathrm{disc}}} \Rightarrow \mathcal{Z}_{\mathrm{cl}} = \{x^\star_{\mathrm{cl}}\} \Rightarrow \mathfrak{T}_\star \\ &\Rightarrow Z_{\mathrm{rel}} \Rightarrow \{S^T_n\} \Rightarrow (\mathcal{H}_{\mathrm{adm}}, \mathfrak{A}_{\mathrm{adm}}) \\ &\Rightarrow \Gamma_k \Rightarrow \Gamma^{\mathrm{ren}}_{\mathrm{TFPT}} \Rightarrow (\mathbf{O}^{\mathrm{TFPT}}_{\mathrm{phys}}, S_{\mathrm{massive}}) \text{ and } \mathcal{I}_{\mathrm{dress}}.\end{aligned}$$

$$\forall \mathfrak{T}_{\mathrm{pre}} \in \mathbf{TFPT}^{\mathrm{rig}} : \qquad \mathfrak{T}_{\mathrm{pre}} \cong \mathfrak{T}_\star.$$

The input-side reduction theorem is explicitly relative to the ambient category $\widehat{\mathrm{PhysAdm}}_1$, now without a packet-size axiom. Carrier minimality is already internalized by the algebraic normal-form lemma together with the later bosonic-rank and branch-Yukawa discharge of its residual carrier signatures, and the readout hierarchy organizes theorem-level physical observables before any appendix scheme projection. Comparison rows and threshold conventions remain outside the theorem core.

A Appendix-level empirical readout

This appendix does not define the theorem-level physical observable object itself. The canonical scheme-facing object is the orbit

$$[\mathfrak{M}(\mathfrak{T}_\star)]_{\mathbf{Sch}}.$$

The tables below record only chosen representatives

$$\mathcal{R}^{(s)}_{\mathrm{cmp}}(\mathfrak{T}_\star) := \mathfrak{M}^{(s)}_{\mathrm{scheme}}(\mathfrak{M}_{\mathrm{phys}}(\mathfrak{R}_{\mathrm{ren}}(\mathfrak{T}_\star))).$$

in declared scheme objects $s \in \mathbf{Sch}$. The appendix is therefore the carrier of the comparison map $\mathcal{R}_{\mathrm{cmp}}(\mathfrak{T}_\star)$ and remains hard-separated from the theorem chain (52), from the renormalized observable passage

$$\mathfrak{T}_\star \xrightarrow{\mathfrak{R}_{\mathrm{ren}}} \Gamma^{\mathrm{ren}}_{\mathrm{TFPT}} \xrightarrow{\mathfrak{M}_{\mathrm{phys}}} \mathbf{O}^{\mathrm{TFPT}}_{\mathrm{phys}},$$

and from the structural falsification map $\mathsf{F}_{\mathrm{fals}}(\mathfrak{T}_\star)$ of Section 12.3.

The empirical reference values used in this section follow representative CODATA, PDG, NuFIT, Planck, and birefringence references [3, 4, 5, 6, 11, 12, 13]. The kaon comparison rows use the published NA62 observation, its current preliminary combined update, and the current KOTO bound [14, 15, 16]. The snapshot policy is frozen at CODATA 2022 / NIST for $\alpha(0)$, PDG 2025 for particle-mass and weak-mixing rows, NuFIT 2025 for neutrino-angle rows, and Planck 2018 for the baryon-fraction reconstruction used here. This section is not part of the closed theorem chain; it records only the readout conventions used when closed outputs are brought into contact with external data. The table below is intentionally restricted to six rows whose readout language is already convention-complete under the declared interface. The broader mass readout surfaces and source ledgers are collected in Appendix N. Beyond that narrow six-row surface, the manuscript also carries a small operational prediction set of experimentally actionable rows, collected explicitly below. The Ω_b row is a present-epoch

reconstruction from the Planck pair $(\Omega_b h^2, H_0)$, not a claim that $\Omega_b(t)$ itself is a timeless UV quantity. Here "carrier-form closure equation" abbreviates the exact electromagnetic closure equation built from the exact seam generating function $\varphi_\Sigma(\alpha)$. The rows $(\beta, \Omega_b, \lambda_C, \sin^2\theta_{13})$ are no longer read as one seed package: they are assigned respectively to determinant response, cosmology closure, exact CKM closure, and exact neutrino closure.

Remark (Former seed quartet after the sector split). The appendix still retains the exact UV identities driven by φ_0^{ret}, and at that UV bookkeeping level the quartet is still read as four projections of the single decoder $u := \varphi_0^{\mathrm{ret}}$. What is no longer claimed is that these rows form one operational independence class of physical observables. The rows β, Ω_b, λ_C, and $\sin^2\theta_{13}$ now factor through R_{CS}, $C_{\cos}$, F_{fl}, and N_ν, respectively. What remains true is the auxiliary UV identity

$$\beta_{\mathrm{rad}} = \frac{\varphi_0^{\mathrm{ret}}}{4\pi}, \qquad \Omega_b = (4\pi - 1)\beta_{\mathrm{rad}}, \qquad \lambda_C = \sqrt{\varphi_0^{\mathrm{ret}}(1-\varphi_0^{\mathrm{ret}})}, \qquad \sin^2\theta_{13} = \varphi_0^{\mathrm{ret}} e^{-\gamma},$$

on the retained branch. It is useful as UV bookkeeping, but it no longer defines the operational independence classes of the observable layer.

Corollary A.1 (UV seed-shadow identities on the retained branch)**.** *In the appendix UV bookkeeping layer the retained seed φ_0^{ret} satisfies*

$$\varphi_0^{\mathrm{ret}} = 4\pi\beta_{\mathrm{rad}} = \Omega_b + \beta_{\mathrm{rad}} = \frac{1-\sqrt{1-4\lambda_C^2}}{2} = e^{\gamma}\sin^2\theta_{13}.$$

These identities are auxiliary UV shadows of the retained branch. They do not replace the sector factorization of the physical observable layer established in Theorem 13.20 and corollary 13.21.

Proof. Combine the seed map, the Cabibbo inversion (21), and the exact seed-budget identity (24). The last claim is immediate. □

A.1 Operational prediction set

The narrow empirical readout table is not the whole experimental surface of the manuscript. Table 9 collects the rows that are operationally testable on near- to medium-term timescales, while keeping their status, next test, kill criterion, and dependency structure explicit. The point is not to flatten them into one truth class. Rather, the *Status* column now uses only the manuscript's final output vocabulary, and the derivation and dependency columns show which rows are direct theorem outputs and which are appendix-level readout images under the declared threshold map. The electroweak triple $(m_W, \hat{s}_Z^2, \alpha_s)$ is omitted here only for compactness; by the appendix-level empirical readout remark following Corollary 9.23 these rows already belong to the appendix-level empirical readout. The first three rows emphasize the two independent theorem-level targets and the strong-CP null test; the remaining rows record second-line flavor and neutrino probes.

Remark (Current rare-kaon status). At the time of this version, the combined preliminary NA62 analysis of 2016–2024 data gives

$$\mathrm{BR}(K^+ \to \pi^+\nu\bar{\nu}) = \left(9.6^{+1.9}_{-1.8}\right) \times 10^{-11},$$

which lies inside the quoted TFPT corridor and close to the branch target $\mathrm{BR}(K^+ \to \pi^+\nu\bar{\nu}) = 9.40 \times 10^{-11}$ [15]. In the present paper this is recorded as support for the appendix-level flavor readout, not as an additional theorem. The neutral channel remains open; the current KOTO bound is

$$\mathrm{BR}(K_L \to \pi^0\nu\bar{\nu}) < 2.2 \times 10^{-9}$$

at 90% C.L. [16].

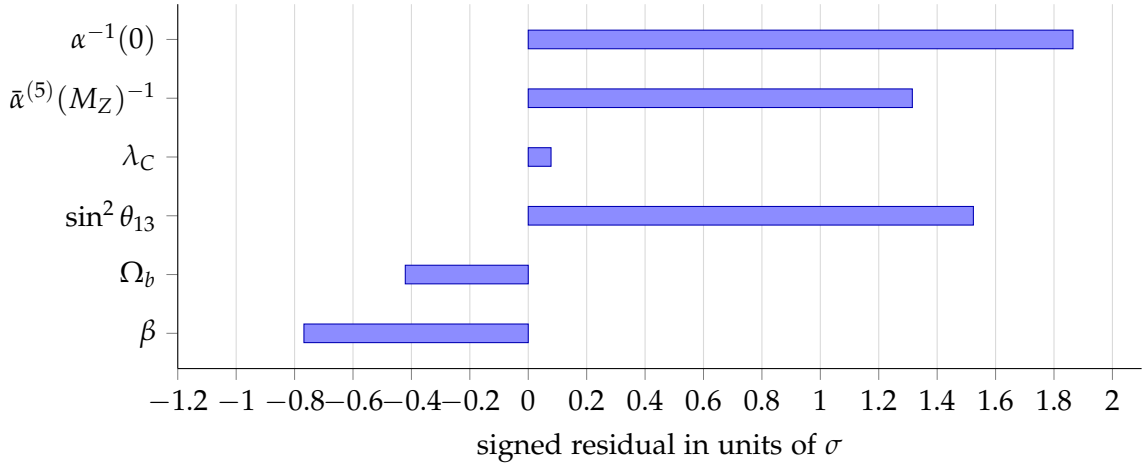


Figure 24. Residual plot for the main empirical readout rows with declared uncertainties.

Remark (Independence classes). The dependency labels are theorem-backed by Theorem 13.20 and corollary 13.21. They identify basis packages of $\mathcal{B}_{\mathrm{ind}}(\mathfrak{T}_\star)$ rather than informal clusters only. The rightmost column in Table 9 therefore records dependency structure rather than truth value. *Determinant response* means a row is carried by the determinant-line / Chern–Simons sector; *flavor readout* means it is downstream of the exact CKM transport output; *neutrino readout* means it belongs to the closed neutrino branch; *closure* means it is a direct consequence of an admissibility-closure theorem; and *cosmology readout* means it is generated by the cosmology block and then, if needed, represented on the appendix interface. This keeps statistical and logical non-independence visible at the same place where the experimental targets are listed.

A.2 Basis factorization of readout rows

To make the renormalized / physical / scheme hierarchy explicit row by row, the visible appendix surface factors through the basis packages of $\mathcal{B}_{\mathrm{ind}}(\mathfrak{T}_\star)$ as follows.

A.3 Compact prediction ledger

The main empirical readout table shows only a narrow readout surface. To keep the output structure visible, Table 11 summarizes which output packages belong to which layer of the paper.

A.4 Empirical readout and kill-test logic

The present main-text readout surface is no longer intentionally mixed. Its rows now use the manuscript's closed classes: theorem, physical observable, scheme projection, and appendix continuation. Carrier algebra and the exact α row remain the sharpest numerically load-bearing entries, while the former seed-driven rows are now reorganized by sector rather than counted as one compressed shadow package. The transport block is generative up to physical readout, the neutrino block is read on the same closed branch, and the electroweak threshold rows appear only as scheme projections. What remains outside this main-text surface is appendix continuation, not unresolved theorem load hidden in the core.

A.5 Residual readout and interface refinements

The residuals for the restricted six-row readout table are shown in Figure 24. The carrier-linked electromagnetic, birefringence, and baryon-density channels all remain within the displayed few-sigma window. The CKM phase is not plotted here because it is better treated in the operational prediction set and Appendix N rather than inside the restricted six-row residual plot.

The principal remaining refinements are therefore specific rather than diffuse:

- the CKM phase is read from the exact holonomy phase, with only the small-area transport asymptotic kept as a derived expansion,
- the nonabelian comparison channels now split into fixed local kernel data and explicit matching functionals, so the remaining sensitivity is numerical rather than structural,
- direct-search interface rows remain pressure tests on the closed branch rather than precision residual rows.

B Technical conventions and data layers

This appendix collects the technical bookkeeping that supports the boundary-to-carrier presentation but would slow the early main-text flow if placed before the carrier theorem.

B.1 Relative objects

The adjective "relative" is used in one fixed sense throughout the paper: every operator or action is compared to a declared reference object.

B.2 Layered datum and admissibility

Construction B.1 (Layered datum)**.** The technical bookkeeping behind the manuscript uses five levels:

$$\mathfrak{T}^{\mathrm{kin}}_{\mathrm{core}} = \big(\widetilde{M} \to M,\ \Sigma,\ \tau_{\mathrm{dbl}},\ \iota_{\mathrm{C}},\ D_{\mathrm{ref}},\ D_{\mathrm{rel}},\ f,\ \chi_{\mathrm{seed}}\big),$$

$$\mathfrak{T}^{\partial}_{\mathrm{ker}} := \big((\mathcal{A}, \mathcal{H}, D, J, \Gamma, \tau_{\mathrm{dbl}}, \iota_{\mathrm{C}}, P_{\mathrm{prim}}),\ [u_\Sigma],\ c_3\big),$$

$$\mathfrak{T}_{\mathrm{ker}} = \big((\mathcal{A}, \mathcal{H}, D, J, \Gamma, \iota_{\mathrm{C}}, P_{\mathrm{prim}}, P_{\mathrm{adm}}),\ E_3 \oplus E_2,\ Y,\ [u_\Sigma],\ c_3\big),$$

$$\mathfrak{T}_{\mathrm{bridge}} = \big(\mathcal{R}_{\mathrm{SM}},\ \alpha(0),\ M_Z,\ \{m_c, m_b, m_\tau, m_t\}\big),$$

$$\mathfrak{T}_{\mathrm{comp}} = \begin{pmatrix} (\mathsf{Adm}, K_{\mathrm{adm}}, P_{\mathrm{adm}}),\ (\chi_{\mathrm{geo}}, \mathcal{B}_{\mathrm{rel}}, \mathbb{A}_\Sigma),\ (F, T), \\ (U_6, D_y, \mathcal{Y}^{(\varepsilon)}_y, \epsilon_f),\ (Z_{\mathrm{rel}}, \Theta, \mathcal{H}_{\mathrm{rel}}, \mathcal{V}_{\mathrm{adm}}), \\ (\mathcal{A}_{\mathrm{rec}}, \mathcal{A}_{\mathrm{obs}}, \mathrm{Pred}) \end{pmatrix}.$$

Here $\mathfrak{T}^{\partial}_{\mathrm{ker}}$ is the primitive boundary kernel, while $\mathfrak{T}_{\mathrm{ker}}$ is the derived theorem-level post-carrier kernel reconstructed from the joint discrete datum and the full selector.

Thus $\mathfrak{T}^{\mathrm{kin}}_{\mathrm{core}}$ is the appendix bookkeeping rewrite of the primitive kinematic scaffold, while Adm first enters the layered datum only at closure level through $(\mathsf{Adm}, K_{\mathrm{adm}}, P_{\mathrm{adm}})$. When an almost-commutative factorization $\mathcal{H} \cong L^2(\widetilde{M}, S) \otimes \mathcal{H}_{\mathrm{int}}$ is chosen, S and $\mathcal{H}_{\mathrm{int}}$ refer to that factorization and are not additional entries in $\mathfrak{T}^{\mathrm{kin}}_{\mathrm{core}}$.

Optional horizon and far-downstream E_8 data are kept outside this compact layered datum and are introduced only in the dedicated appendix-level extension.

B.3 Symbol guide

The symbol guide is split into two tables to keep theorem-level objects lexically separate from appendix-only comparison and readout objects. Table 14 contains only symbols that participate in the main-text theorem chain; Table 15 contains only symbols that live in the appendix-level comparison and readout layer.

Definition B.2 (Admissibility predicate and selector)**.** In the present paper Adm denotes a predicate on relative sectors, and P_{adm} denotes its operator realization on the retained closed branch. We write $\mathsf{Adm}(X) = 1$ precisely when:

(1) X lies in the declared domain of the relative operator pair,

(2) the associated relative action or index density is finite after reference subtraction,

(3) the required seam-evenness condition is satisfied,

(4) colored sectors satisfy center neutrality when a hadronic interpretation is intended,

(5) gap-stable transport positivity and the declared closure conditions hold in those sectors used for observable or transport statements.

On those sectors, the selector language and the predicate language are identified by

$$\mathsf{Adm}(X) = 1 \quad \Longleftrightarrow \quad P_{\mathrm{adm}} X = X.$$

Remark (Operator realization of admissibility). If one wants an operator language rather than a predicate language, the natural positive generator is K_{adm} and its commuting-factor realization is

$$P_{\mathrm{adm}} = P_{\Sigma,+}\, P_{\mathrm{sing}}\, P_{\Theta}\, P_{\mathrm{gap}},$$

where the factors act on the commuting seam, color, determinant, and transport sectors of the admissible Hilbert-space decomposition. Here $P_{\Sigma,+}$ projects to seam-even sectors, P_{sing} to center-neutral singlets when a hadronic interpretation is intended, P_{Θ} encodes the declared sheet / determinant selection, and P_{gap} imposes the transport-side positivity or gap condition. The main text defines P_{adm} as the zero-temperature projector of K_{adm}; the appendix records the explicit factorization available on the commuting retained branch.

C Constants atlas and downstream scale grammar

This appendix collects material that is useful for internal compression but should not dominate the main-text scope of explicit claims.

C.1 Global rotation angle and constants atlas

On the canonical branch the seam loop carries one unit of spectral flow,

$$U_{\Sigma}(\theta) = e^{i\theta}, \qquad \mathrm{SF}(U_{\Sigma}) = 1, \qquad \Delta_{\Sigma} = 2\pi.$$

Because the spin lift closes only after 4π, the normalized rotation modulus is

$$a_{\theta} = \frac{\theta}{4\pi}.$$

The cascade in Figure 25 is kept here as technical bookkeeping rather than as an early main-text dependency graphic.

Here the arrow $a_{\theta} \to c_3$ records a normalization convention internal to the present setup, not an independently proved theorem beyond the declared seam and spin-lift normalization.

C.2 E_8 as downstream scale grammar

In the present paper E_8 is not used as a primitive cause. The retained parameter is

$$\kappa_{E8} := \frac{\gamma}{\ln(248/60)} = \frac{5/6}{\ln(248/60)}.$$

On the retained branch this means

$$\kappa_{E8} \approx 0.587\,23, \qquad f_a \approx 8.86 \times 10^{10}\,\mathrm{GeV},$$
$$m_a \approx 65.19\,\mu\mathrm{eV}, \quad \nu_a \approx 15.764\,\mathrm{GHz}.$$

The intended reading is a downstream scale grammar:

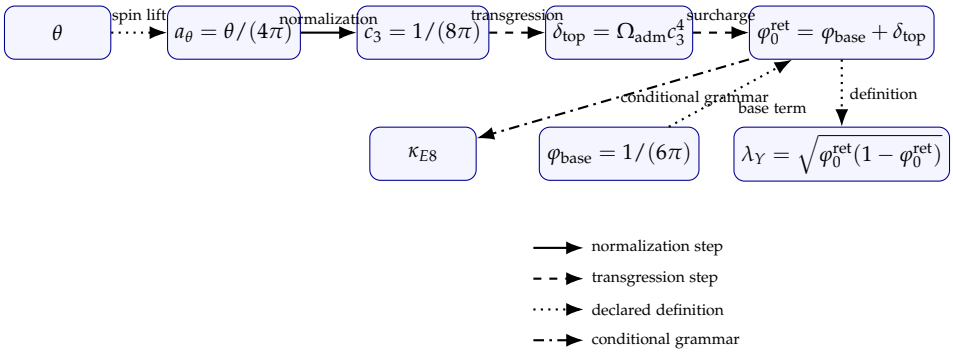


Figure 25. Constant cascade used for technical bookkeeping, shown in the occupancy-reinterpretation form for $\delta_{\rm top}$. Arrow styles encode algebraic role rather than claim status: normalization, transgression, declared definition, and conditional downstream grammar.

- the carrier packet fixes $\gamma = 5/6$,
- κ_{E8} then defines an optional ladder parameter,
- the ladder is allowed to organize scale spacing,
- but it does not replace the carrier theorem or the seam data.

In the fuller appendix-level grammar one may package a downstream block scale as

$$X(n, r, I_1) = \frac{M_{\rm Pl}}{8} \sin^2\theta_{13} \left(\frac{60-2n}{60}\right)^{\kappa_{E8}} e^{-(8-r)/64} e^{-12\pi I_1},$$

where n is the ladder stage, r a mild block-multiplicity label, and I_1 the dominant block trace. This makes the reading discipline explicit: the stage index orders the ladder, the trace controls most of the exponential suppression, and the extra block label only refines the scale. The denominator 64 is part of that downstream grammar and is not interpreted here as a Hilbert-space dimension, state count, or hidden spinor multiplicity. E_8 is therefore a downstream scale grammar, not a universal mass formula.

C.3 Full stage atlas

The full real-domain atlas up to $n = 29$ extends the main-text robust table with restricted-scan candidates and IR tail conjectures. Rows are classified as *robust* (restrictive grammar, unambiguous target), *restricted-scan* (plausible but less constrained), or *tail conjecture* (new trace class $I_1 \in \{\frac{11}{12}, \frac{7}{6}\}$ not yet derived from block structure).

n	r	I_1	X (GeV)	Target	Residual	Identification	Class
3	8	1/2	2.159×10^8	2.160×10^8	0.06%	μ_{c_3} RG crossing	robust
5	5	1/12	1.307×10^{15}	1.31×10^{15}	0.2%	seesaw M_R	robust
6	6	1	1.272	1.27	0.2%	charm band	restricted-scan
8	1	17/16	0.1059	0.105 66	0.2%	muon band	restricted-scan
10	1	1/3	8.826×10^{10}	direct search	—	PQ scale f_a	robust
12	2	41/48	246.35	246.22	0.05%	physical EW benchmark	robust
14	4	17/16	0.0921	0.0927	0.6%	f_π band	robust
15	4	1	0.935	0.938	0.3%	hadron anchor	robust
21	4	1/6	3.051×10^{13}	3.06×10^{13}	0.3%	inflation M	robust
21	5	41/48	171.8	172.6	0.4%	top quark	robust
24	7	5/8	7.895×10^5	7.920×10^5	0.3%	$\mu_{\varphi_0^{\rm ret}}$ RG crossing	robust
25	5	1	0.498	0.498	0.0%	kaon band	robust
25	7	41/48	125.5	125.25	0.2%	Higgs mass	robust
26	2	1	0.417	0.42	0.7%	$\sqrt{\sigma_{\rm QCD}}$ confinement	robust
27	6	41/48	91.6	91.19	0.4%	Z boson	restricted-scan
28	1	7/6	5.105×10^{-4}	5.110×10^{-4}	0.1%	electron	tail conjecture
29	1	11/12	4.210	4.18	0.7%	bottom quark	tail conjecture

Table 17. Full E_8 stage atlas on the canonical branch up to the real-domain boundary $n = 29$.

C.4 Compression identities on the minimal branch

The main-text compression identities are developed in the dedicated section on compression identities and numerical comparison of alternative discrete worlds. This appendix subsection collects additional arithmetic continuations and the compression chain summary.

Remark (Abelian package from occupancy and carrier norm). On the canonical branch,

$$\Omega_{\rm adm}\gamma = 48 \cdot \frac{5}{6} = 40.$$

Thus the characteristic coefficient 41 is compressed to

$$41 = \Omega_{\rm adm}\gamma + 1 = 40 + 1,$$

that is, fermionic occupancy trace plus one light Higgs doublet contribution. Writing $b_Y := 5b_1/3$ for the usual hypercharge-normalized coefficient and using the block-index relation $k_B = \frac{3}{2} I_1(B)$ to define $I_1^{\rm EW} := \frac{5}{24} b_1$ and $k_{\rm EW} := \frac{5}{16} b_1$, one obtains the unified chain

$$48\, I_1^{\rm EW} = 32\, k_{\rm EW} = 10\, b_1 = \Omega_{\rm adm}\gamma + 1 = 41. \tag{53}$$

Equivalently,

$$I_1^{\rm EW} = \frac{41}{48}, \qquad k_{\rm EW} = \frac{41}{32}, \qquad b_1 = \frac{41}{10}, \qquad b_Y = \frac{41}{6}.$$

This is an algebraic signature rather than an accidental coincidence: the same integer 41 appears simultaneously as electroweak block trace, E_8 exponent, UV hypercharge trace, and carrier occupancy times norm plus one.

Remark (Rank lock and single-seed compression: main-text reference). The former rank-lock identity is recorded in the main text as the derived consistency statement Proposition 5.1, while the carrier rank itself is fixed by the bosonic-rank corollary and the branch-Yukawa rigidity theorem, with Lemma 3.11 retained only as the algebraic normal form for the resulting exterior signatures. The single-seed compression and the Cabibbo inversion $\varphi_0^{\rm ret} = \frac{1}{2}(1 - \sqrt{1 - 4\lambda_C^2})$ with all derived inter-observable relations are given in Proposition 5.10 and equations (21) and (23). The numerical comparison of wrong discrete worlds is recorded in Proposition 5.15.

C.5 Optional arithmetic continuations

The next relations mix the main-text carrier with optional appendix-level E_8, inflation, or geometric notation. They are useful compression patterns, but their interpretation should remain conditional or conjectural.

Remark (60, 48, and 5/4 on the optional ladder). If the optional scale ladder is started at $D_1 = 60$, then

$$\frac{D_1}{\Omega_{\mathrm{adm}}} = \frac{60}{48} = \frac{5}{4} = B\gamma.$$

On the minimal branch one may also write the same number as

$$\frac{5}{4} = \frac{g_{\mathrm{car}}}{g_{\mathrm{car}} - 1} \qquad (g_{\mathrm{car}} = 5).$$

This compresses the E_8 start index, the admissible chiral count, the mirror factor, and the two-defect coefficient to the same rational factor.

Corollary C.1 (Arithmetic closure of 48, 60, and 240)**.** *From the main text identity*

$$\delta_2 = \frac{5}{4}\,\delta_{\mathrm{top}}^2, \qquad \delta_{\mathrm{top}} = \Omega_{\mathrm{adm}} c_3^4,$$

one obtains

$$\delta_2 = 2880\, c_3^8.$$

Set

$$g := g_{\mathrm{car}} = 5, \qquad d := \dim \mathfrak{g}_{\mathrm{SM}} = 12.$$

Then

$$\Omega_{\mathrm{adm}} = (g-1)d = 4 \cdot 12 = 48, \qquad D_{\mathrm{start}} = gd = 5 \cdot 12 = 60.$$

On the canonical branch the same factor admits the exact arithmetic decompositions

$$2880 = D_{\mathrm{start}}\Omega_{\mathrm{adm}} = 60 \cdot 48, \qquad 2880 = g(g-1)d^2 = 240 \cdot 12 = |R(E_8)|\, d.$$

The same pair $(\Omega_{\mathrm{adm}}, D_{\mathrm{start}}) = (48, 60)$ *also satisfies*

$$\gcd(\Omega_{\mathrm{adm}}, D_{\mathrm{start}}) = d = \dim \mathfrak{g}_{\mathrm{SM}},$$

and

$$\mathrm{lcm}(\Omega_{\mathrm{adm}}, D_{\mathrm{start}}) = g(g-1)d = 240 = |R(E_8)|.$$

So the gauge algebra dimension is the common divisor of the neighboring carrier counts $48 = (g-1)d$ *and* $60 = gd$*, while the number* 240 *is their least common multiple. The arithmetic identities are exact. Here the value* 12 *is the Lie-algebra dimension*

$$\dim \mathfrak{g}_{\mathrm{SM}} = 8 + 3 + 1,$$

not a derivation from a discrete permutation group of order 12*; at most the* $3 + 2$ *split carries such a discrete shadow.*

Remark (External E_8 reading). On the same optional branch,

$$|R(E_8)| = 248 - 8 = 240 = 5 \cdot 48 = 5\,\Omega_{\mathrm{adm}}.$$

One may therefore record an external E_8 interpretation of the arithmetic closure, namely that the root count can be read as a fivefold unfolding of the admissible chiral packet. This interpretation is optional and is not used as a main-text theorem.

Remark (Reactor seed of the optional scale ladder). If one writes the optional ladder as

$$\varphi_n = \varphi_0^{\mathrm{ret}} e^{-\gamma} \left(\frac{D_n}{D_1}\right)^{\kappa_{E8}}, \qquad D_n = 60 - 2n,$$

then the start value is precisely the reactor-angle seed,

$$\varphi_n = \sin^2\theta_{13} \left(\frac{D_n}{D_1}\right)^{\kappa_{E8}}.$$

Thus the appendix-level ladder may be read as a discrete continuation of the same seed that controls θ_{13}.

Remark (Conditional inflation-side compression). If one supplements the appendix with the conditional Starobinsky-side relations

$$A_s = \frac{N^2}{24\pi^2}\left(\frac{M}{\bar{M}_{\mathrm{Pl}}}\right)^2, \qquad \frac{M}{\bar{M}_{\mathrm{Pl}}} = \sqrt{8\pi}\, c_3^4,$$

then on the minimal branch

$$A_s = \frac{N^2}{3\pi} c_3^8 = \frac{N^2}{8640\pi}\,\delta_2,$$

and also

$$\frac{M}{\bar{M}_{\mathrm{Pl}}} = \frac{\sqrt{8\pi}}{\Omega_{\mathrm{adm}}}\,\delta_{\mathrm{top}}.$$

This is recorded only as an appendix-level compression identity because inflation is outside the present main-text claim surface.

Remark (Dyonic intercept and the α kernel). From the dyonic appendix the local astrophysical β function reads

$$\beta_{\mathrm{BH}}(r) = \frac{Q_e Q_m}{256\pi^4 r^2} = \frac{\delta_{\mathrm{top}}}{3}\frac{Q_e Q_m}{r^2} = 16 c_3^4 \frac{Q_e Q_m}{r^2}.$$

This is a cross-link between the EHT intercept channel and the α kernel: the same topological coefficient $\delta_{\mathrm{top}} = 48 c_3^4$ that pushes α into the precision zone also controls the local dyonic β function. The local and cosmological sectors are therefore not two loose motifs but the same coefficient in two projections.

Remark (Conditional occupancy route to gravity). If the relative Einstein term is of the form

$$S_{\mathrm{EH}}^{\mathrm{rel}} = \frac{\chi_{\mathrm{geo}}^2}{192\pi^2}\,\mathrm{Tr}_{\mathcal{H}_{\mathrm{ch}}}\mathbf{1}\int d^4x\sqrt{-g}\,R,$$

and if the family bridge identifies

$$\mathrm{Tr}_{\mathcal{H}_{\mathrm{ch}}}\mathbf{1} = \Omega_{\mathrm{adm}} = 48,$$

then

$$\bar{M}_{\mathrm{Pl}}^2 = \frac{\chi_{\mathrm{geo},0}^2}{2\pi^2}.$$

This is the occupancy-to-gravity route alluded to in the main text.

Remark (Compression chain). The appendix-level compression pattern may therefore be summarized as

$$Y \leftrightarrow \varepsilon_{\mathrm{car}}, \qquad \gamma = \frac{5}{6}, \qquad B\gamma = \frac{5}{4}, \qquad \frac{D_1}{\Omega_{\mathrm{adm}}} = \frac{5}{4}, \qquad |R(E_8)| = 5\,\Omega_{\mathrm{adm}},$$

$$b_1 = \frac{\Omega_{\mathrm{adm}}\gamma + 1}{10}, \qquad \Omega_b + \beta_{\mathrm{rad}} = \varphi_0^{\mathrm{ret}}, \qquad A_s = \frac{N^2}{8640\pi}\,\delta_2.$$

The last relation remains a conditional inflation-side import.

C.6 Conditional minimal-parameter picture

The next package is stronger but also more conditional. It assumes that the carrier rank, family multiplicity, and seam normalization are all read as closed rather than only partially inherited.

Principle C.2 (Conditional minimal-parameter compression)**.** Assume the canonical branch

$$g_{\mathrm{car}} := \operatorname{rank} E = 5, \qquad N_{\mathrm{fam}} = 3, \qquad c_3 = \frac{1}{8\pi}.$$

Then the numerical kernel may be rewritten almost entirely in terms of

$$(\pi, g_{\mathrm{car}}, N_{\mathrm{fam}}) = (\pi, 5, 3).$$

Indeed, on this branch one has

$$\gamma = \frac{5}{6} = \frac{g_{\mathrm{car}}}{g_{\mathrm{car}}+1}, \qquad \dim S^+ = 2^{g_{\mathrm{car}}-1}, \qquad \Omega_{\mathrm{adm}} = N_{\mathrm{fam}} 2^{g_{\mathrm{car}}-1} = 48,$$

$$\varphi_{\mathrm{base}} = \frac{1-\gamma}{\pi} = \frac{1}{(g_{\mathrm{car}}+1)\pi}, \qquad \delta_{\mathrm{top}} = \Omega_{\mathrm{adm}}\, c_3^4, \qquad \delta_2 = D_{\mathrm{start}}\, \Omega_{\mathrm{adm}}\, c_3^8,$$

with

$$D_{\mathrm{start}} = g_{\mathrm{car}} \dim \mathfrak{g}_{\mathrm{SM}} = 5 \cdot 12 = 60, \qquad \Omega_{\mathrm{adm}} = (g_{\mathrm{car}} - 1) \dim \mathfrak{g}_{\mathrm{SM}} = 4 \cdot 12 = 48.$$

Remark (Compressed dependency chain). Under the same derived closure package, the appendix-level numerical chain reads

$$(\widetilde{M} \to M, \Sigma, \tau_{\mathrm{dbl}}, \iota_{\mathrm{C}}, \chi_{\mathrm{seed}}) \Longrightarrow E_3 \oplus E_2 \Longrightarrow Y \leftrightarrow \varepsilon_{\mathrm{car}} \Longrightarrow S(U(3) \times U(2)) \Longrightarrow S^+,$$

$$\Longrightarrow N_{\mathrm{fam}} = 3 \Longrightarrow \Omega_{\mathrm{adm}} = 48 \Longrightarrow (\delta_{\mathrm{top}}, \delta_2, \varphi_0^{\mathrm{ret}}) \Longrightarrow (\alpha, \lambda_{\mathrm{C}}, \beta_{\mathrm{rad}}, \Omega_b, \sin^2\theta_{13}),$$

with the optional E_8, inflationary, and gravity-side sectors then read as downstream continuations of the same compressed core.

C.7 Infrared continuations of the seam transfer

Theorem 10.3 and Corollary 10.4 determine the theorem-level seam-transfer expression for Λ_{IR} and its determinant bounds. The numerical scales recorded below require one further infrared assumption and therefore are organized as a labeled *conditional closure layer*, separated from the theorem core. Each member of the layer is stated as a conjecture or as a corollary conditional on the named seam-lift hypothesis, never as a theorem-level consequence of the present closure stack.

Conjecture C.3 (Reconstruction infrared seam continuation)**.** Assume the lift hypothesis

$$S_{\mathrm{IR}}^{\mathrm{rec}} := -\log \rho(U_\Sigma)\big|_{\mathrm{IR}} = \frac{2}{\alpha_\star}, \qquad \rho_{\mathrm{IR}}^{\mathrm{rec}} := \delta_{\mathrm{top}} e^{-2/\alpha_\star},$$

i.e. the leading admissible eigenvalue $\rho(U_\Sigma)$ is continued into the deep infrared by the inverse-coupling exponential weighted by the topological seam coefficient δ_{top}.

Conjecture C.4 (Internal carrier infrared seam continuation)**.** Assume the carrier-compressed lift hypothesis

$$\rho_{\mathrm{IR}}^{\mathrm{car}} := \varphi_{\mathrm{base}} \left(\delta_{\mathrm{top}} e^{-2\alpha_\star}\right)^{2^{g_{\mathrm{car}}}-1} = \varphi_{\mathrm{base}}\, q(\alpha_\star)^{31}, \qquad q(\alpha) := \delta_{\mathrm{top}} e^{-2\alpha},$$

on the canonical carrier branch $g_{\mathrm{car}} = 5$, i.e. the seam transfer is dressed by the retained carrier opening φ_{base} raised to the carrier branch exponent.

Corollary C.5 (Conditional infrared cosmological-constant readout)**.** *Conditional on Conjectures C.3 and C.4 and on the rank-one truncation of Corollary 10.4, the seam-transfer determinant evaluates to*

$$\frac{\Lambda_{\mathrm{IR}}^{\mathrm{rec}}}{M_{\mathrm{Pl}}^4} = -\log\bigl(1-\delta_{\mathrm{top}}e^{-2/\alpha_\star}\bigr) \approx 1.128\,041\,908\,574\,331\,7\times 10^{-123},$$

$$\frac{\Lambda_{\mathrm{IR}}^{\mathrm{car}}}{M_{\mathrm{Pl}}^4} = -\log\Bigl(1-\varphi_{\mathrm{base}}\bigl(\delta_{\mathrm{top}}e^{-2\alpha_\star}\bigr)^{31}\Bigr) \approx 1.039\,786\,651\,013\,728\,6\times 10^{-123}.$$

The two readouts agree on order of magnitude and frame the observed value $\Lambda_{\mathrm{obs}}/M_{\mathrm{Pl}}^4 \approx 1.1\times 10^{-123}$ *within the conditional closure layer; they become* theorem level *as soon as the seam-lift hypothesis* $S_{\mathrm{IR}} = 2/\alpha_\star$ *is discharged from* U_Σ*.*

Remark (Lift target and kill test). The single non-theorem ingredient in Corollary C.5 is the seam-lift hypothesis

$$S_{\mathrm{IR}} = -\log\rho(U_\Sigma)\big|_{\mathrm{IR}} = \frac{2}{\alpha_\star}.$$

Falsification path. Any independent computation of $\rho(U_\Sigma)$ in the deep infrared that deviates from $\delta_{\mathrm{top}}e^{-2/\alpha_\star}$ by more than the rank-one error margin of Corollary 10.4 kills both numerical readouts at once. Conversely, a closed proof of the seam-lift identity promotes the rec/car corollary to a theorem and freezes the Λ_{IR}-line of the metrology functor without any further appendix input.

D Relative APS and superconnection setup

The geometric completion uses APS-type relative data [1]. In practice this means:

- one works with a declared pair (D, D_{ref}) rather than a single unanchored operator,
- boundary conditions on the seam Σ are part of the definition of the relative index,
- the η-term keeps track of the residual spectral asymmetry,
- the superconnection package is used in the same spirit as the spectral-action tradition [2], but here always with an explicit reference subtraction.

This appendix now serves as the proof motor for the geometric reconstruction statements in the main text: its role is to make the common reference structure explicit enough that the reconstruction theorem and the local spectral-scale branch are not left hanging on a slogan.

E Relative reflection positivity with fermions

E.1 Normalized quotient form

The relative Euclidean theory used in the main closure section is not interpreted as a difference of measures. Instead it is organized as the normalized quotient

$$Z_{\mathrm{rel}}[J,\eta,\bar\eta] = \frac{Z_{\mathrm{adm}}[J,\eta,\bar\eta]}{Z_{\mathrm{ref}}[0,0,0]}.$$

Because the denominator is source independent on the admissible sector, the reference theory acts only as a fixed normalizer and does not interfere with reflection positivity.

E.2 Bosonic and fermionic positivity on P_{adm}

The commutation relations

$$[\Theta, P_\Theta] = 0, \qquad \Theta P_{\mathrm{adm}} = P_{\mathrm{adm}}\Theta$$

ensure that reflection preserves the admissible subspace. The bosonic part then follows the standard Markov / reflection-stable Osterwalder–Schrader argument on admissible observables. For fermions, the admissible CAR kernel may be written as a reflected quadratic form; equivalently one may express the same condition as positivity of the associated Pfaffian structure on the admissible sector. Hence for every admissible observable O supported in $\tau > 0$,

$$\langle \Theta O, O\rangle_{\mathrm{rel}} = \frac{1}{Z_{\mathrm{ref}}[0,0,0]}\langle \Theta O, O\rangle_{\mathrm{adm}} \geq 0.$$

This appendix section is the mechanism behind Lemma 8.7 and theorems 8.23 and 8.24; it is recorded here so that the main-text reconstruction theorem is not supported only by a one-line axiom call.

F Admissible OS reconstruction and asymptotic scattering

This appendix upgrades the admissible Schwinger, reconstruction, and scattering block from a named standard interface to a manuscript-level proof package. The point is not to replace standard machinery by ad hoc notation, but to verify explicitly that the admissible branch satisfies the relevant hypotheses on P_{adm}.

F.1 Tempered admissible Schwinger families

Lemma F.1 (Temperedness and moment bounds for admissible Schwinger functions)**.** *On the Euclidean continuation of $\mathcal{U}_{\mathrm{scat}}$, each truncated admissible Schwinger distribution S_n^T defines a tempered distribution. Moreover, for every compact K and every source monomial of total degree n, its moments obey polynomial bounds determined by the same uniform heat-kernel estimates and finite local moments of the admissible constructive measure.*

Proof. By Theorem 3.9, the admissible Euclidean generator is realized by a self-adjoint elliptic operator on every blocked volume. Together with the constructive measure of Theorem 6.37, this yields uniform heat-kernel bounds and finite moments for local source insertions. Differentiating W_{rel} with respect to the local sources therefore produces distributions whose derivatives are controlled by polynomial combinations of the same bounds. Passing to the admissible projective limit preserves these polynomial estimates, so the resulting Schwinger family is tempered. □

Lemma F.2 (Graded Euclidean covariance and symmetry on P_{adm})**.** *On the Euclidean continuation of $\mathcal{U}_{\mathrm{scat}}$, the admissible Schwinger family is graded Euclidean covariant, graded symmetric under exchange of external legs, and continuous in the Schwartz topology.*

Proof. The Dirac-type principal symbol of the geometric anchor and the low-curvature reduction imply local Euclidean covariance of the admissible source functional on $\mathcal{U}_{\mathrm{scat}}$. Bosonic source derivatives commute, while fermionic derivatives anticommute, so the mixed Schwinger family is graded symmetric with the usual CAR signs. Continuity follows from the tempered moment bounds of Lemma F.1, which control the action of S_n^T on Schwartz test functions. □

F.2 OS and CAR reconstruction on P_{adm}

Theorem F.3 (Tailored OS/CAR reconstruction on P_{adm})**.** *Let $\{S_n^T\}_{n\geq 1}$ be the admissible Schwinger family on the Euclidean continuation of $\mathcal{U}_{\mathrm{scat}}$. If reflection positivity, temperedness, graded Euclidean covariance, graded symmetry, and clustering hold on P_{adm}, then there exist:*

(i) a Hilbert space $\mathcal{H}_{\mathrm{OS}}$ with vacuum vector Ω,

(ii) a strongly continuous positive-energy semigroup $T_t = e^{-tH_{\mathrm{adm}}}$ with self-adjoint generator $H_{\mathrm{adm}} \geq 0$,

(iii) a reconstructed graded local field net on P_{adm} whose Euclidean boundary values are the original admissible Schwinger functions.

Equivalently, the admissible Euclidean family reconstructs a Wightman/CAR theory on the admissible sector.

Proof. Define the OS sesquilinear form by

$$(F, G)_{\mathrm{OS}} := \langle \Theta F, G\rangle_{\mathrm{rel}}$$

for positive-time admissible observables. Reflection positivity from Theorem 8.23 and the fermionic quotient-form analysis of Section E show that this form is positive semidefinite on the full graded admissible algebra. Quotienting by the null space and completing gives $\mathcal{H}_{\mathrm{OS}}$.

Positive Euclidean time translation preserves the null space and acts by contractions, hence defines a strongly continuous contraction semigroup T_t on $\mathcal{H}_{\mathrm{OS}}$. The dense domain of positive-time polynomial observables gives the symmetric generator domain. By Hille–Yosida, the closure of this generator is self-adjoint and nonnegative, so $T_t = e^{-tH_{\mathrm{adm}}}$ for a self-adjoint $H_{\mathrm{adm}} \geq 0$.

The temperedness, covariance, graded symmetry, and clustering established in Lemmas F.1 and F.2 are precisely the remaining OS/CAR reconstruction hypotheses. Therefore analytic continuation yields the reconstructed Wightman/CAR family with vacuum Ω. Because Θ, P_Θ, and P_{adm} commute, the entire reconstruction stays on the admissible sector P_{adm}. □

F.3 Stable massive Haag–Ruelle and LSZ closure

Lemma F.4 (One-particle spectral projectors from isolated simple poles)**.** *Let $\widetilde{G}_a^{(2)}(p)$ have an isolated simple pole with positive residue at $p^2 = -m_a^2 < 0$. Then the corresponding one-particle contribution defines a positive spectral projector on the admissible Hilbert space.*

Proof. The isolated simple pole separates a one-particle mass shell from the multiparticle continuum. By positivity of the residue, the contour integral of the resolvent around that pole defines a positive projector on $\mathcal{H}_{\mathrm{adm}}$. Its range is the one-particle subspace associated with the stable species a. □

Theorem F.5 (Haag–Ruelle scattering on the stable massive admissible sector)**.** *For every stable massive admissible species with isolated simple pole below threshold, the Haag–Ruelle asymptotic fields exist on $\mathcal{H}_{\mathrm{adm}}$ and generate wave operators on the stable massive scattering sector.*

Proof. By Theorems F.3 and 8.38, the admissible branch carries a local relativistic net with positive Hamiltonian. By Theorem 8.34, connected correlators cluster exponentially on the massive sector. The pole assumption and Lemma F.4 identify the one-particle subspace and give positive single-particle spectral projectors. The standard Haag–Ruelle estimates therefore apply to admissible local operators, yielding the in/out fields and the wave operators on the stable massive sector. □

Corollary F.6 (LSZ reduction on the stable massive admissible sector)**.** *On the stable massive admissible sector, matrix elements of the scattering operator are obtained by amputating the connected correlators with the corresponding residue factors.*

Proof. By Theorem F.5, the asymptotic fields and wave operators exist on the stable massive sector. Applying the usual LSZ reduction to those asymptotic fields and the admissible local net yields the displayed amputated-correlator formula. □

F.4 Dressed massless interface in the low-curvature region

The massless sector is not treated by bare Fock asymptotics in the present manuscript. The relevant object is the infrared-dressed interface package

$$\mathcal{I}_{\text{dress}} := \big(\Omega_+^{\text{dress}}, \Omega_-^{\text{dress}}, S_{\text{dress}}\big), \qquad S_{\text{dress}} := (\Omega_+^{\text{dress}})^* \Omega_-^{\text{dress}},$$

constructed from the soft charges of the admissible local net on the low-curvature region $\mathcal{U}_{\text{scat}}$. This is exactly the interface package used in the main-text theorem Theorem 8.41.

G FRW reduction and cosmology interface proofs

This appendix section upgrades the cosmology interface block from a proof sketch to a manuscript proof. The point is to keep the theorem-level content at the level of closed-branch interface data, while leaving late-time numerical continuation in the comparison layer.

Lemma G.1 (Minisuperspace reduction of the geometric Hodge branch)**.** *Restrict the closed-branch geometric Hodge action to homogeneous and isotropic variables*

$$(a(t), \varphi(t), \theta_\Sigma(t), N_i(t)).$$

Then the reduced action is a minisuperspace action whose gravitational and scalar sectors coincide with the low-curvature FRW/scalaron block used in the main text.

Proof. The closed-branch geometric action is local and covariant. Imposing homogeneity and isotropy kills all spatial derivative terms and leaves only the FRW scale factor, the scalaron mode, the determinant-line axion phase, and the heavy-neutrino input package. Evaluating the geometric Hodge action on this ansatz gives exactly the reduced minisuperspace action displayed in the main-text FRW theorem. □

Lemma G.2 (Seam-transfer determinant as the infrared vacuum term)**.** *On the closed branch, the positive trace-class seam-transfer operator contributes to the FRW reduction through the infrared vacuum term*

$$\Lambda_{\text{IR}} = M_{\text{Pl}}^4 \big[-\log \det_{\text{adm}}(1 - U_\Sigma)\big].$$

Proof. By Theorem 10.3, the seam-transfer operator is positive trace class with spectral radius strictly below one. Therefore its Fredholm determinant is well defined and contributes an additive vacuum term to the reduced action. The normalization by M_{Pl}^4 is exactly the closed-branch dimensional lift used in the main text. □

Lemma G.3 (Determinant-line axion reduction)**.** *On the canonical branch, the determinant-line phase reduces to a compact axion variable with*

$$N_{\text{DW}} = 1, \qquad \theta_i = \pi\big(1 - \varphi_\Sigma(\alpha_\star)\big),$$

and the reheating input is fixed by the scalaron decay width.

Proof. Theorem 10.6 fixes the domain-wall number $N_{\mathrm{DW}} = 1$ on the canonical branch. The determinant-line theorem and the electromagnetic closure determine the initial angle θ_i, while the scalaron branch fixes the decay-width input entering the reheating scale T_R. These are exactly the determinant-line and scalaron data needed in the reduced FRW action. □

Theorem G.4 (Full FRW reduction and cosmology interface theorem)**.** *The closed-branch minisuperspace reduction determines the cosmology interface package*

$$(\Lambda_{\mathrm{IR}}, N_{\mathrm{DW}}, \theta_i, T_R, \mathcal{I}_{\mathrm{LG}})$$

from the same geometric Hodge, seam-transfer, determinant-line, and neutrino data that determine the closed branch.

Proof. Lemma G.1 gives the reduced FRW/scalaron action, Lemma G.2 identifies the infrared vacuum term, and Lemma G.3 identifies the determinant-line axion and reheating data. The leptogenesis input block $\mathcal{I}_{\mathrm{LG}}$ is fixed by the same closed-branch heavy-neutrino spectrum, family frame, and reheating input used in the main-text leptogenesis interface theorem. Therefore the entire displayed interface tuple is fixed internally on the closed branch. □

H Yukawa kernels and positivity lemmas

This appendix is the proof motor for the transport positivity, determinant-phase suppression, and bounded-residual statements used in the main text. The transport program is based on the regularized kernel

$$\mathcal{Y}_f^{(\varepsilon)} = (D_f^\dagger D_f + \varepsilon^2)^{-1}.$$

Three remarks are important.

1. The regulator ε is not optional: without it the kernel becomes ill-defined exactly where the admissibility question is most delicate.
2. Positivity of the kernel belongs to the admissibility-closure layer rather than to the hard carrier kernel. The main text closes it through the finite hexagon gap and the lift to the full admissibility-projected operator; this appendix records the transport-side mechanism behind that closure.
3. Residual prefactors $\Lambda_{f,j}$ are bounded holonomy data inside the transport theorem. They must remain visible and must not be rhetorically hidden as if the mass map were diagonal without residual transport structure.

Remark (Positive-kernel determinant route to strong CP). If the physical mass map is written in the polar form

$$M_f = U_{f,L} H_f U_{f,R}^\dagger, \qquad H_f > 0, \qquad U_{f,L}, U_{f,R} \in SU(3)_F,$$

with H_f induced by the positive transport kernel $\mathcal{Y}_f^{(\varepsilon)}$, then the induced determinant phase vanishes:

$$\arg\det M_f = 0.$$

At the same time weak CP may survive through

$$V_{\mathrm{CKM}} = U_{u,L}^\dagger U_{d,L}.$$

Combined with the sheet-CP symmetry protection of θ_{sheet} in the relative action, this is the algebraic core of the determinant-phase theorem stated in the main text and of the strong-CP closure theorem recorded there. The role of the appendix remark is to display the polar-decomposition mechanics behind that route, not to reopen its status.

H.1 Optional strong-coupling sharpening

As an appendix-level dynamical sharpening of the hadronic admissibility theorem, one may further require the admissible center-neutral Wilson functional to obey strong-coupling dominance. On that optional route the relative Wilson loop satisfies

$$\langle W(C)\rangle_{\mathrm{rel}} \leq \exp[-\sigma_{\mathrm{QCD}} A_{\min}(C)], \qquad \sigma_{\mathrm{QCD}} = c_3^2 \chi_{\mathrm{QCD}}^2,$$

and the physical Hilbert space contains no color-nonsinglet asymptotic states. This statement is kept in the appendix so that the main theorem surface does not depend on an extra strong-coupling sharpening beyond the closed admissible hadronic sector.

I Information-theoretic and cut-and-project readings

This appendix collects interpretive notes that help organize the algebraic structure of the boundary-polarized closed form. When a theorem-level statement appears below, it refers back to a main-text proposition; the appendix itself does not upgrade any status label beyond the declared theorem / derived-consequence / comparison / appendix discipline.

I.1 41-chain overdetermination of the canonical branch

The canonical 41-chain admits a compact overdetermination reading:

Layer	Value	Role
Electroweak trace block $48\, I_1^{\mathrm{EW}}$	41	retained electroweak trace packet
Block exponent $32\, k_{\mathrm{EW}}$	41	E_8-side compression datum
UV hypercharge coefficient $10\, b_1$	41	one-loop electroweak closure datum
Occupancy-weighted carrier norm $\Omega_{\mathrm{adm}}\gamma + 1$	41	carrier / family compression identity

Table 18. 41-chain overdetermination of the canonical branch. The common integer is read as a structural consistency marker, not as a cryptographic statement.

The value of this table is structural rather than rhetorical: four independently defined layers collapse onto one integer on the retained branch. This is exactly the sense in which the branch carries the closed 41-chain marker $\mathcal{I}_{41}$.

I.2 Seed variance and surprisal readings

The bridge identities also admit a disciplined probabilistic reading. Since

$$\lambda_{\mathrm{C}}^2 = \varphi_0^{\mathrm{ret}}(1 - \varphi_0^{\mathrm{ret}}),$$

the Cabibbo scale is exactly the standard-deviation scale of a Bernoulli variable with success probability $p = \varphi_0^{\mathrm{ret}}$:

$$\lambda_C = \sqrt{p(1-p)}.$$

Equivalently, the Fisher information of the Bernoulli family is

$$I(p) = \frac{1}{p(1-p)} = \frac{1}{\lambda_{\mathrm{C}}^2},$$

so the Cabibbo bridge measures the inverse square-root Fisher scale of the retained seed. This is the precise information-theoretic statement. The present paper does *not* identify λ_C itself with Fisher information or with a Bures distance.

Likewise, whenever the electromagnetic closure is written with

$$\ln\big((\varphi_0^{\mathrm{ret}})^{-1}\big),$$

the exact probabilistic reading is the surprisal of the seed event. That term should not be called the Shannon entropy of the full Bernoulli distribution, which would instead be

$$H(\varphi_0^{\mathrm{ret}}) = -\varphi_0^{\mathrm{ret}} \ln \varphi_0^{\mathrm{ret}} - (1-\varphi_0^{\mathrm{ret}}) \ln(1-\varphi_0^{\mathrm{ret}}).$$

Finally, the neutrino bridge relation

$$\sin^2\theta_{13} = \varphi_0^{\mathrm{ret}} e^{-\gamma}$$

is naturally described as Boltzmann-like weighting by the carrier norm γ, but the present draft does not define a full thermodynamic partition function at that point. These readings are therefore organizational interpretations of already established formulas, not new closure theorems.

I.3 Exact parity-code and information-compression reading

The strongest information-theoretic statement is no longer a loose analogy. By Proposition 3.21, the one-family packet is exactly the even-parity subspace on five occupation bits,

$$S^+ \cong \mathrm{span}\Big\{|x\rangle : \sum_{i=1}^{5} x_i \equiv 0 \pmod 2\Big\},$$

so the carrier packet is kinematically the classical even-parity $[5,4,2]$ code. The split formula

$$S^+ = \big(\Lambda^{\mathrm{even}} E_3 \otimes \Lambda^{\mathrm{even}} E_2\big) \oplus \big(\Lambda^{\mathrm{odd}} E_3 \otimes \Lambda^{\mathrm{odd}} E_2\big)$$

shows that the coarse parity bits of the 3-sector and 2-sector are exactly locked. For the maximally mixed state on the 16 basis words, the two parity observables are perfectly correlated and therefore carry exactly one bit of mutual information.

The split weight enumerator in Proposition 3.22 is the corresponding code enumerator with hypercharge weights. It packages the six admissible occupancy sectors into the single polynomial

$$W(u,v) = 1 + 3u^2 + 6uv + v^2 + 2u^3v + 3u^2v^2.$$

The hypercharge-compression result of Proposition 3.24 sharpens the same picture further. On the uniform distribution over the 16 carrier basis states, the six Y-sectors have probabilities

$$\frac{1}{16}(1,3,6,1,2,3),$$

so the hypercharge entropy is

$$H(Y) = -\sum_i p_i \log_2 p_i = \frac{7}{2} - \frac{3}{4}\log_2 3 \approx 2.311 \text{ bits}.$$

Since the total carrier word takes 4 bits to specify, the unresolved nonabelian degeneracy is

$$4 - H(Y) \approx 1.689 \text{ bits}.$$

Thus hypercharge is an efficient family-sector compressor, but not a lossless label for the individual basis word.

Remark (Two code layers: kinematic and admissible). The code reading should be split into two layers. First, S^+ is exactly the classical even-parity $[5,4,2]$ code on five occupation bits. Second, on the commuting admissible branch the protected physical sector is

$$\mathcal{H}_{\mathrm{adm}} := \mathrm{Ran}(P_{\mathrm{adm}}) = \ker K_{\mathrm{adm}} = \ker \Delta_{\mathrm{adm}},$$

where the last equality uses the Hodge description of Lemma 3.5. In that second layer the penalties $K_\Sigma, K_c, K_\Theta, K_g$ act as stabilizer-like sector constraints, and P_{adm} is the Hodge projector onto the admissible physical subspace. This is the precise bridge to quantum-information language used in the manuscript. It is not yet a theorem of a self-correcting quantum memory, because no noise model, logical algebra, or lifetime analysis is derived here.

I.4 Cut-and-project model-set realization

The appendix-level quasicrystal language is kept outside the hard core, but it can be stated exactly as a compatible model-set construction.

Theorem I.1 (Regular cut-and-project model set for the $3+2$ carrier)**.** *Let*

$$L = \mathbb{Z}^5.$$

Choose a direct sum decomposition

$$\mathbb{R}^5 = V_{\mathrm{vis}} \oplus V_{\mathrm{int}}, \qquad \dim V_{\mathrm{vis}} = 3, \qquad \dim V_{\mathrm{int}} = 2,$$

such that the projection of L to V_{int} is dense. Let

$$\pi_{\mathrm{vis}} : \mathbb{R}^5 \to V_{\mathrm{vis}}, \qquad \pi_{\mathrm{int}} : \mathbb{R}^5 \to V_{\mathrm{int}}$$

be the canonical projections, and let $W \subset V_{\mathrm{int}}$ be a compact regular window with nonempty interior and boundary of measure zero. Define

$$\Lambda_W := \left\{ \pi_{\mathrm{vis}}(n) \ : \ n \in L,\ \pi_{\mathrm{int}}(n) \in W,\ \sum_{i=1}^{5} n_i \equiv 0 \pmod 2 \right\}.$$

Then Λ_W is a regular model set. The parity constraint realizes the even-packet selection of S^+, and the dimension split $3+2$ realizes the visible/internal carrier architecture.

Proof. This is the standard cut-and-project theorem for regular model sets. Density of the internal projection and regularity of the window imply that Λ_W is Delone and has pure point diffraction. The congruence condition

$$\sum_{i=1}^{5} n_i \equiv 0 \pmod 2$$

selects the even parity class, matching the even exterior-packet sector of the carrier. □

Remark (Status of the quasicrystal reading). The model-set theorem makes the quasicrystal appendix mathematically sharp, but the irrational split and window are not derived from the hard TFPT core. The construction is therefore exact yet deliberately kept appendix level.

J Record algebra and pointer dynamics

This appendix elaborates the main-text record constructions and the stable pointer-algebra package used in the downstream closure discussion. Horizon thermality and prediction / transient semantics are separated into the following appendix sections so that record theory, black-hole continuation, and phenomenological event channels do not blur into one mixed extrapolation block.

J.1 Minimal admissible vacuum

At finite volume or with a declared cutoff, an admissible vacuum candidate is written as

$$\rho_B := \arg\min_{\rho \in \mathcal{V}_{\mathrm{adm}}} \mathrm{Tr}(\rho \mathcal{H}_{\mathrm{rel}}),$$

with lower boundedness and coercivity understood exactly as in the main-text closure axioms. The stronger generative theorem path described in the conclusion would refine this object to the global admissible vacuum functional $\mathfrak{V}[g, \chi_{\mathrm{geo}}, \Phi, \nabla_F, \delta, \{L_f\}]$ whose variations also select the transport branch. In this appendix, however, the same vacuum object is used only as the background state for the record-algebra elaborations; no stronger cosmological claim is added here beyond the closed existence statement already declared in the main theorem stack.

J.2 Record growth principle

One useful coarse-grained record algebra is

$$\mathcal{A}_{\mathrm{rec}}(t) := \bigoplus_{\Delta_i > H_{\mathrm{hub}}(t)} \mathcal{A}_i, \qquad S_{\mathrm{rec}}(t) = \log \dim \mathcal{A}_{\mathrm{rec}}(t).$$

As $H_{\mathrm{hub}}(t)$ decreases, more stable channels open. The intended arrow-of-time claim is therefore

$$\frac{dS_{\mathrm{rec}}}{dt} > 0,$$

but only as a record-growth principle, not as a replacement for detailed nonequilibrium dynamics.

J.3 Observer algebra and Born-type evaluation

An observer is modeled as a stable pointer subalgebra

$$\mathcal{A}_{\mathrm{obs}} \subset \mathcal{A}_{\mathrm{rec}}.$$

Theorem J.1 (Born evaluation on the stable pointer algebra)**.** *Let $\{P_i\} \subset \mathcal{A}_{\mathrm{obs}}$ be a complete family of orthogonal projectors. Then for every state ρ the unique additive, positive, and phase-invariant evaluation rule on this projector family is*

$$p_i = \mathrm{Tr}(\rho P_i).$$

Proof. In finite-dimensional sectors this is the standard Born rule characterization. The extra TFPT input is not a different probability law but the selection of the stable pointer algebra by the gapped admissible sector. □

With these projectors one may write the measurement channel

$$\mathcal{M}_O(\rho) = \sum_i (P_i \rho P_i) \otimes |i\rangle\langle i|.$$

The theorem is therefore a constrained appendix-level evaluation statement on admissible sectors, not a new claim about measurement in arbitrary settings.

K Horizons, modular flow, and black hole thermality

This appendix section records the horizon-side analytic input that supports the downstream modular thermality program. In the present version the split-inclusion theorem is the firm appendix-level statement, while the full modular KMS upgrade and compact-object thermality package remain visible downstream closure targets.

K.1 Split inclusion horizon approximation

Theorem K.1 (Horizon approximation by split inclusion)**.** *Let $\mathfrak{A}_{\mathrm{adm}}$ be the local admissible net reconstructed in Theorem 8.38, and assume in addition the split property. Then for every nested triple*

$$\mathcal{O}_{\mathrm{out}} \Subset \mathcal{O}_{\mathrm{collar}} \Subset \mathcal{O}_{\mathrm{ext}}$$

there exists a type-I factor N such that

$$\mathfrak{A}_{\mathrm{adm}}(\mathcal{O}_{\mathrm{out}}) \subset N \subset \mathfrak{A}_{\mathrm{adm}}(\mathcal{O}_{\mathrm{ext}}).$$

Hence on a suitable code subspace one obtains an approximate tensor factorization

$$\mathcal{H}_{\mathrm{code}} \cong \mathcal{H}_{\mathrm{out}} \otimes \mathcal{H}_{\mathrm{hor}} \otimes \mathcal{H}_{\mathrm{in}}.$$

Proof. This is the standard consequence of the split property for local quantum field theory nets. The result supplies an algebraic horizon approximation without asserting an exact fundamental factorization. □

If a horizon is modeled as an inner seam, the manuscript therefore uses only this split-based approximation language rather than an exact microscopic tensor decomposition.

L Prediction semantics and transient event channels

This appendix section collects the optional semantic decorations of the prediction map and the status note for transient event channels. These items organize downstream interpretation, but they do not add new upstream theorem arrows to the rigid core.

L.1 Prediction semantics

The main-text prediction map can be decorated by explicit experimental metadata as

$$Q = (\rho_0, \mathcal{O}_Q, \Delta_Q, t, \mathcal{C}_Q, \mathcal{R}_Q), \qquad \mathrm{Pred}(Q) = \mathcal{R}_Q(\mathrm{Tr}(\rho_t E_Q(\Delta)), \mathcal{C}_Q)\,,$$

with

$$\rho_t = U(t)\rho_0 U^\dagger(t).$$

This is kept only as an optional formal extension.

L.2 Transient event status

The topological burst channel of Conjecture 11.26 is kept at programmatic-target status. What is fixed in the present version is only the admissible phase-slip channel and its scaling form, not yet a complete defect dynamics or radiative-transfer derivation.

M Closed theoretical outputs and empirical comparison surface

This appendix collects the full Standard Model output of the boundary-polarized manuscript into a theoretical surface and an empirical comparison surface. The purpose is not to flatten the hierarchy but to keep theorem-level outputs, derived consequences, physical observables, scheme projections, and appendix continuations visibly distinct:

Status legend for the particle ledger

- **Theorem:** a theorem-level output on the closed branch.
- **Derived consequence:** a structural output obtained by composing earlier theorems without adding a new scheme convention.

- **Physical observable:** a branch-fixed observable read internally before any scheme choice.
- **Scheme projection:** a row reported only after declared schemes, thresholds, or matching conventions.
- **Appendix continuation:** appendix-level continuation not needed for the closed core.
- **Programmatic target:** a precise downstream closure target recorded without a claimed proof in the present version.

Rows on the empirical comparison surface are compared only under the declared external conventions once the physical observable layer has already been fixed.

Remark (Counting convention used here). One chiral family from $S^+ = \Lambda^{\text{even}}E$ contains 16 states: ν^c (1), u^c (3), Q_L (6), e^c (1), L_L (2), d^c (3). With ν^c explicitly in the carrier family packet, each family has 6 Weyl field types. Three families give $\Omega_{\text{adm}} = 3 \times 16 = 48$ admissible chiral states. The gauge sector has 12 bosons (γ, 8 gluons, $W^\pm$, Z), and the Higgs sector has 1 physical scalar. The total field-type count is therefore $3 \times 6 + 4 + 1 = 23$ distinct types.

M.1 Table 1: Closed theoretical outputs

Particle	Repr.	Ext. source	Y	TFPT origin	Internal source	Status
Fermion sector (one family $\times$ 3):						
ν^c	$(1,1)_0$	$\Lambda^0 E$	0	carrier scaffold	hard carrier, S^+	Theorem
e^c	$(1,1)_1$	$\Lambda^2 E_2$	1	carrier scaffold	hard carrier, S^+	Theorem
u^c	$(\bar{3},1)_{-2/3}$	$\Lambda^2 E_3$	$-\frac{2}{3}$	carrier scaffold	hard carrier, S^+	Theorem
d^c	$(\bar{3},1)_{1/3}$	$\Lambda^2 E_3 \otimes \Lambda^2 E_2$	$\frac{1}{3}$	carrier scaffold	hard carrier, S^+	Theorem
Q_L	$(3,2)_{1/6}$	$E_3 \otimes E_2$	$\frac{1}{6}$	carrier scaffold	hard carrier, S^+	Theorem
L_L	$(1,2)_{-1/2}$	$\Lambda^3 E_3 \otimes E_2$	$-\frac{1}{2}$	carrier scaffold	hard carrier, S^+	Theorem
Gauge boson sector:						
γ	$(1,1)_0$	unbroken $U(1)_{\text{em}}$	0	exact gauge direction	carrier algebra	Theorem
g ($\times 8$)	$(8,1)_0$	unbroken $SU(3)_c$	0	exact gauge direction	carrier algebra	Theorem
$W^\pm$	$(1,3)_{\pm 1}$	SSB from Φ	± 1	bosonic index + Higgs mechanism	Dirac-anchor closure	Physical observable
Z	$(1,1)_0$	SSB from Φ	0	bosonic index + Higgs mechanism	Dirac-anchor closure	Physical observable
Scalar sector:						
H	$(1,2)_{1/2}$	seam-even Φ	$\frac{1}{2}$	compact bosonic index	$\text{Ind}(\mathcal{B}^+_{E_2}) = 1$	Theorem
Structural outputs:						
$N_{\text{fam}} = 3$	$\mathcal{H}_F \cong F \cong \mathbb{C}^3$	L^2-APS harmonic family modes	—	harmonic family-mode theorem	punctured family section	Theorem
$\Omega_{\text{adm}} = 48$	$S^+ \otimes \mathcal{H}_F$	occupancy	—	bridge corollary	3×16	Theorem
G_{car}	$S(U(3)\times U(2)) \cong [SU(3)\times SU(2) \times U(1)_Y]/\mathbb{Z}_6$	carrier split	—	internal carrier stabilizer theorem (Theorem 3.18)	$E_3 \oplus E_2$	Theorem
local algebra	$\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$	local Lie algebra	—	differential of the internal carrier theorem	local sections of $E_3 \oplus E_2$	Derived consequence

Table 19. Standard Model content and origin map on the closed branch. The status column separates theorem outputs from physical observables and appendix-only scheme projections.

M.2 Table 2: Empirical comparison rows

Quantity	Sector	TFPT formula or source	Metrology readout	Comparison map	TFPT value	Unit	Reference value	Residual	Status
Closed theoretical outputs:									
$\alpha^{-1}(0)$	EM	exact $\varphi_\Sigma(\alpha)$ closure	$d_O = 0$ direct readout	on-shell Thomson	137.0359992	—	137.03599918	$+1.87\sigma$	Physical observable
m_γ	EM	unbroken $U(1)_{\rm em}$	$d_O = 1$ $m_\gamma/\lambda_\Sigma = 0$	—	0	—	0	exact	Theorem
m_g	QCD	unbroken $SU(3)_c$	$d_O = 1$ $m_g/\lambda_\Sigma = 0$	—	0	—	0	exact	Theorem
Derived consequences:									
e	lepton	$\frac{v_{\rm geo}}{\sqrt{2}}\frac{16\pi}{7}(\varphi_0^{\rm ret})^5$	$d_O = 1$ $\hat{m}_e/\lambda_\Sigma$	exact $v_{\rm geo}$ branch	0.52266	MeV	0.510999	$+2.28\%$	Derived consequence
μ	lepton	$\frac{v_{\rm geo}}{\sqrt{2}}\frac{4\pi}{3}(\varphi_0^{\rm ret})^3$	$d_O = 1$ $\hat{m}_\mu/\lambda_\Sigma$	exact $v_{\rm geo}$ branch	107.84	MeV	105.658	$+2.07\%$	Derived consequence
τ	lepton	$\frac{v_{\rm geo}}{\sqrt{2}}\frac{7\pi}{6}(\varphi_0^{\rm ret})^2$	$d_O = 1$ $\hat{m}_\tau/\lambda_\Sigma$	exact $v_{\rm geo}$ branch	1.7746	GeV	1.77686	-0.12%	Derived consequence
u	quark	ChPT inversion from m_s anchor	$d_O = 1$ m_u/λ_Σ	$\overline{\rm MS}$, 2 GeV	2.195	MeV	2.16	$+1.62\%$	Derived consequence
d	quark	$m_s/(m_s/m_d)$	$d_O = 1$ m_d/λ_Σ	$\overline{\rm MS}$, 2 GeV	4.67	MeV	4.67	$+0.00\%$	Derived consequence
π^0	hadron	GMOR relation from $(m_u + m_d)$	$d_O = 1$ m_{π^0}/λ_Σ	—	134.979	MeV	134.977	$+0.00\%$	Derived consequence
p	hadron	hadron mass ladder	$d_O = 1$ m_p/λ_Σ	—	935.03	MeV	938.272	-0.35%	Derived consequence
$\theta_{\rm eff}$	strong CP	determinant-line strong-CP closure	$d_O = 0$ direct readout	—	0	—	$< 10^{-10}$	exact	null Theorem
Empirical comparison rows:									
$W^\pm$	EW	intrinsic threshold at M_Z	$d_O = 1$ m_W/λ_Σ	$\mathcal{R}_{\rm SM}$	80.369	GeV	80.369	0.00%	Scheme projection
Z	EW	intrinsic threshold at M_Z	$d_O = 1$ m_Z/λ_Σ	$\mathcal{R}_{\rm SM}$	91.188	GeV	91.188	0.00%	Scheme projection
H	EW	intrinsic threshold	$d_O = 1$ m_H/λ_Σ	$\mathcal{R}_{\rm SM}$	125.25	GeV	125.25	0.00%	Scheme projection
s	quark	strange-quark matched row	$d_O = 1$ m_s/λ_Σ	$\overline{\rm MS}$, 2 GeV	93.4	MeV	93.4	0.00%	Scheme projection
c	quark	up-type ratio ladder	$d_O = 1$ m_c/λ_Σ	threshold matching	1.27	GeV	1.27	0.00%	Scheme projection
b	quark	down-type ratio ladder	$d_O = 1$ m_b/λ_Σ	threshold matching	4.18	GeV	4.18	0.00%	Scheme projection
t	quark	up-type ratio ladder	$d_O = 1$ m_t/λ_Σ	threshold matching	172.57	GeV	172.57	0.00%	Scheme projection
$\hat{s}_Z^2(M_Z)$	EW	gauge closure + seam-even matching	$d_O = 0$ scheme image	$\overline{\rm MS}$	0.23116	—	0.23121	-0.02%	Scheme projection
$\alpha_s(M_Z)$	QCD	gauge closure + singlet Wilson matching	$d_O = 0$ scheme image	$\overline{\rm MS}$	0.11796	—	0.1180	-0.03%	Scheme projection
$\bar{\alpha}^{(5)}(M_Z)^{-1}$	EM	$\mathcal{R}_{\rm SM}[\alpha(0)]$	$d_O = 0$ scheme image	$\overline{\rm MS}$	127.9405	—	127.93	$+1.32\sigma$	Scheme projection
λ_C	CKM	$s_{12}(V^*_{\rm CKM})$	$d_O = 0$ direct readout	CKM proxy	0.22438	—	0.22431	$+0.08\sigma$	Physical observable
$\sin^2\theta_{13}$	PMNS	$\lvert(U^*_{\rm PMNS})_{e3}\rvert^2$	$d_O = 0$ direct readout	NuFIT NO	0.02311	—	0.02224	$+1.52\sigma$	Physical observable
Ω_b	cosmo	late-time cosmology comparison	$d_O = 0$ $C_{\rm cos}$ readout	Planck 2018	0.04894	—	0.04930	-0.42σ	Cosmology comparison
β	cosmo	determinant response in degrees	$d_O = 0$ $\beta_{\rm rad}$	Planck / Minami–Komatsu	0.242	deg	0.35	-0.77σ	Physical observable
Intrinsic neutrino closure (no external anchor):									
ν_1	neutrino	intrinsic transport closure	$d_O = 1$ m_{ν_1}/λ_Σ	—	0	eV	—	—	Derived consequence
ν_2	neutrino	intrinsic transport closure	$d_O = 1$ m_{ν_2}/λ_Σ	—	8.614×10^{-3}	eV	—	—	Derived consequence
ν_3	neutrino	intrinsic transport closure	$d_O = 1$ m_{ν_3}/λ_Σ	—	5.015×10^{-2}	eV	—	—	Derived consequence
Σm_ν	neutrino	intrinsic closure sum	$d_O = 1$ $\Sigma m_\nu/\lambda_\Sigma$	—	5.876×10^{-2}	eV	< 0.12	compatible	Physical observable

Table 20. Full theoretical-output and empirical-comparison surface. Every row carries its TFPT source formula, its dimensionless metrology readout, and, where needed, one external comparison representative. Unitful entries are therefore interface representatives rather than theorem-level primitive outputs. Reference values are PDG 2024 and CODATA 2022 unless noted otherwise.

M.3 Empirical comparison surface

Table 21 extracts only the rows that are fair as direct theory-versus-data comparisons. Derived consequences that still depend on internal branch formulas are excluded.

Quantity	Metrology readout	Comparison map	TFPT value	Unit	Reference value	Residual	Status
Closed physical observables with direct comparison:							
$\alpha^{-1}(0)$	$d_O = 0$ direct readout	on-shell Thomson	137.0359992	—	137.03599918	$+1.87\sigma$	Physical observable
m_γ	$d_O = 1$ $m_\gamma/\lambda_\Sigma = 0$	—	0	—	0	exact	Theorem
m_g	$d_O = 1$ $m_g/\lambda_\Sigma = 0$	—	0	—	0	exact	Theorem
$\theta_{\text{eff}} = 0$	$d_O = 0$ direct readout	—	0	—	$< 10^{-10}$	exact null	Theorem
Scheme projections and appended physical rows:							
$W^\pm$	$d_O = 1$ m_W/λ_Σ	$\mathcal{R}_{\text{SM}}$ threshold	80.369 GeV	GeV	80.369	0.00%	Scheme projection
Z	$d_O = 1$ m_Z/λ_Σ	$\mathcal{R}_{\text{SM}}$ threshold	91.188 GeV	GeV	91.188	0.00%	Scheme projection
H	$d_O = 1$ m_H/λ_Σ	$\mathcal{R}_{\text{SM}}$ threshold	125.25 GeV	GeV	125.25	0.00%	Scheme projection
s	$d_O = 1$ m_s/λ_Σ	$\overline{\text{MS}}$, 2 GeV	93.4 MeV	MeV	93.4	0.00%	Scheme projection
c	$d_O = 1$ m_c/λ_Σ	threshold matching	1.27 GeV	GeV	1.27	0.00%	Scheme projection
b	$d_O = 1$ m_b/λ_Σ	threshold matching	4.18 GeV	GeV	4.18	0.00%	Scheme projection
t	$d_O = 1$ m_t/λ_Σ	threshold matching	172.57 GeV	GeV	172.57	0.00%	Scheme projection
$\hat{s}_Z^2(M_Z)$	$d_O = 0$ scheme image	$\overline{\text{MS}}$	0.23116	—	0.23121	-0.02%	Scheme projection
$\alpha_s(M_Z)$	$d_O = 0$ scheme image	$\overline{\text{MS}}$	0.11796	—	0.1180	-0.03%	Scheme projection
$\bar{\alpha}^{(5)}(M_Z)^{-1}$	$d_O = 0$ scheme image	$\overline{\text{MS}}$	127.9405	—	127.93	$+1.32\sigma$	Scheme projection
λ_C	$d_O = 0$ direct readout	CKM proxy	0.22438	—	0.22431	$+0.08\sigma$	Physical observable
$\sin^2\theta_{13}$	$d_O = 0$ direct readout	NuFIT NO	0.02311	—	0.02224	$+1.52\sigma$	Physical observable
Ω_b	$d_O = 0$ $C_{\cos}$ readout	Planck 2018	0.04894	—	0.04930	-0.42σ	Cosmology comparison
β	$d_O = 0$ β_{rad}	Planck / Minami–Komatsu	0.242°	deg	0.35°	-0.77σ	Physical observable
Σm_ν	$d_O = 1$ $\Sigma m_\nu/\lambda_\Sigma$	cosmological bound	5.876×10^{-2} eV	eV	< 0.12	compatible	Physical observable

Table 21. Empirical comparison surface. These are the rows for which a fair theory-versus-data comparison is meaningful under the declared external conventions. The metrology column records the dimensionless TFPT readout; the unitful value column is only its chosen comparison representative. The table separates physical observables from pure scheme projections. Derived branch-dependent rows are excluded here. Reference values are PDG 2024 and CODATA 2022 unless noted otherwise.

N Extended benchmarks

This appendix turns the broader prediction surface into explicit experimental bookkeeping. The point is not to promote every row to main-text benchmark status, but to keep next test, kill criterion, and dependency structure visible wherever the theory already commits to a concrete target.

N.1 Operational prediction matrix

The appendix matrix keeps the same truth-mode discipline as the main text, but its final column adds a short row-level qualifier whenever an appendix-level auxiliary continuation needs to be made explicit.

Observable	TFPT target	Experimental reading next test	Kill criterion	Dependency class	Status note
Cosmic birefringence β	0.2424°	calibrated achromatic CMB rotation fit without EB self-nulling	calibrated $\beta = 0$ within $\pm 0.05^\circ$	determinant response	Physical observable
δ_{CKM}	1.198 rad	global flavor fit	stable exclusion at $\geq 3\sigma$	flavor comparison	Comparison quantity exact phase
Rare kaons $K \to \pi\nu\bar{\nu}$	$\mathrm{BR}(K^+) = 9.40 \times 10^{-11}$ $\mathrm{BR}(K_L) = 3.47 \times 10^{-11}$	NA62 current result / KOTO Grossman–Nir test	K^+ outside $[7, 12] \times 10^{-11}$ or K_L mismatch	flavor comparison	Comparison quantity
PMNS phase / octant	$\delta_{\mathrm{CP}} = 240^\circ$ $\sin^2\theta_{23} = 0.4557$	global oscillation fit	exclude 240° or the lower octant at $\geq 3\sigma$	neutrino comparison	Comparison quantity
Σm_ν	5.8764×10^{-2} eV	cosmological mass-sum analyses	robust upper bound below the branch value	neutrino comparison	Comparison quantity
$m_{\beta\beta}$	1.516×10^{-3} eV	light-Majorana $0\nu\beta\beta$ searches	detection implying $m_{\beta\beta} \gtrsim 1 \times 10^{-2}$ eV	neutrino comparison	Comparison quantity
nEDM / strong CP	$\theta_{\mathrm{eff}} = 0$	next-generation EDM searches	stable nonzero hadronic EDM signal	strong-CP theorem	Theorem
Axion haloscope window	$m_a \approx 65.19$ μeV $\nu_a \approx 15.764$ GHz	haloscope scan at 15.764 GHz in the quoted window	exclusion in 15.764 GHz ± 50 MHz	cosmology readout	Cosmology readout target
m_{π^0}	134.979 MeV	hadronic theorem row	robust mismatch outside the stated uncertainty budget	closure	Derived hadronic row
η_B	5.97×10^{-10}	flavored leptogenesis consistency test	robust exclusion of the quoted branch value	cosmology comparison	Cosmology comparison

Table 22. Operational prediction matrix extending the main-text prediction surface. Each row carries its own status note, experimental reading, kill criterion, and dependency class.

N.2 Target and kill-test rows

Test row	TFPT target or prohibition	Status reading	Failure mode
Second light Higgs doublet	no additional seam-even light doublet	kill test	robust discovery would falsify the bosonic-index route
Calibrated birefringence null	$\beta = 0.2424^\circ$ on the minimal branch	kill test	externally calibrated null consistent with zero at the stated tolerance would remove the minimal birefringence sector
Rare-kaon corridor	$\mathrm{BR}(K^+) = 9.40 \times 10^{-11}$ $\mathrm{BR}(K_L) = 3.47 \times 10^{-11}$	kill test	a stable kaon result outside the quoted corridor or incompatible GN-plane correlation would break the present flavor bridge
Strong-CP / EDM signal	$\theta_{\mathrm{eff}} = 0$ from determinant-line closure	kill test	stable nonzero EDM signal would break the present strong-CP theorem
Cosmological neutrino sum	$\Sigma m_\nu = 5.8764 \times 10^{-2}$ eV from intrinsic neutrino closure	pressure test	a robust upper bound below the stated branch value would force a neutrino-sector revision
Haloscope window	$\nu_a \approx 15.76$ GHz from cosmology interface data	target row	exclusion in the relevant coupled window would pressure the intrinsic axion row

Table 23. Target and kill-test rows. These are not current confirmations; they describe the most direct falsification or pressure points of the present branch.

N.3 Axion interface

The axion rows are cosmology readouts generated from the same determinant-line / seam-transfer / scalaron interface data that also fix reheating and the heavy-neutrino scale. On the closed cosmology branch,

$$f_a \approx 8.86 \times 10^{10}\,\mathrm{GeV}, \qquad m_a \approx 65.19\,\mu\mathrm{eV}, \qquad \nu_a \approx 15.764\,\mathrm{GHz}.$$

In standard haloscope units this gives

$$\left|g_{a\gamma\gamma}^{(\mathrm{phys})}\right| = \frac{|g_{a\gamma\gamma}|}{f_a} \approx 1.80 \times 10^{-12}\,\mathrm{GeV}^{-1},$$

so the practical scan prescription is a conservative window

$$\nu_a = 15.764\,\mathrm{GHz} \pm 50\,\mathrm{MHz}$$

at the quoted coupled sensitivity. The present appendix therefore records a practical cosmology readout target rather than introducing a new theorem-level observable claim.

N.4 Supplementary UV asymptotic mass-source rows

The compact main-text mass ledgers intentionally suppress explicit formula detail. The source rows collected here retain that information without forcing the main text to read like a mixed formula notebook. In the present version they are explicitly UV asymptotic source rows: the physical masses belong to the pole-level functor $\mathfrak{M}_{\mathrm{phys}}$ on $\Gamma^{\mathrm{ren}}_{\mathrm{TFPT}}$, whereas the formulas below record the closed-branch source grammar before pole dressing and scheme projection.

Particle	TFPT formula or source row	Metrology / source readout	Scheme representative	Status note
e	$\frac{v_{\mathrm{geo}}}{\sqrt{2}} \times \frac{16\pi}{7}(\varphi_0^{\mathrm{ret}})^5$	$d_O = 1$ $\hat{m}_e/\lambda_\Sigma$	0.52266 MeV	UV asymptotic source row exact v_{geo} branch
μ	$\frac{v_{\mathrm{geo}}}{\sqrt{2}} \times \frac{4\pi}{3}(\varphi_0^{\mathrm{ret}})^3$	$d_O = 1$ $\hat{m}_\mu/\lambda_\Sigma$	107.84 MeV	UV asymptotic source row exact v_{geo} branch
τ	$\frac{v_{\mathrm{geo}}}{\sqrt{2}} \times \frac{7\pi}{6}(\varphi_0^{\mathrm{ret}})^2$	$d_O = 1$ $\hat{m}_\tau/\lambda_\Sigma$	1.7746 GeV	UV asymptotic source row exact v_{geo} branch
Closed neutrino outputs:				
ν_1	intrinsic transport closure	$d_O = 1$ m_{ν_1}/λ_Σ	0	theorem
ν_2	intrinsic transport closure	$d_O = 1$ m_{ν_2}/λ_Σ	8.614×10^{-3} eV	theorem
ν_3	intrinsic transport closure	$d_O = 1$ m_{ν_3}/λ_Σ	5.015×10^{-2} eV	theorem
Σm_ν	intrinsic transport closure sum	$d_O = 1$ $\Sigma m_\nu/\lambda_\Sigma$	5.8764×10^{-2} eV	physical observable
u	ChPT ratio inversion anchored to m_s	$d_O = 1$ m_u/λ_Σ	2.195 MeV	derived; $\overline{\mathrm{MS}}$ at 2 GeV
d	$m_s/(m_s/m_d)$	$d_O = 1$ m_d/λ_Σ	4.67 MeV	derived; $\overline{\mathrm{MS}}$ at 2 GeV
s	strange-quark comparison row in the light-quark solver	$d_O = 1$ m_s/λ_Σ	93.4 MeV	scheme projection; $\overline{\mathrm{MS}}$ at 2 GeV
c	up-type ratio ladder with $m_c/m_u = 486.92$	$d_O = 1$ m_c/λ_Σ	1.27 GeV	scheme projection after threshold matching
b	down-type ratio ladder with $m_b/m_s = 43.79$	$d_O = 1$ m_b/λ_Σ	4.18 GeV	scheme projection after threshold matching
t	up-type ratio ladder with $m_t/m_c = 133.01$	$d_O = 1$ m_t/λ_Σ	172.57 GeV	scheme projection; pole / $\overline{\mathrm{MS}}$ distinction remains external
π^0	GMOR relation from $(m_u + m_d)$ and $\langle \bar{q}q \rangle$	$d_O = 1$ m_{π^0}/λ_Σ	134.979 MeV	hadronic out-of-sample check
p	hadron mass ladder	$d_O = 1$ m_p/λ_Σ	935.03 MeV	derived hadronic scale check

Table 24. Supplementary UV asymptotic mass-source rows for the compact main-text mass surface. The table records closed-branch source formulas before pole dressing and scheme projection. The metrology/source column isolates the dimensionless TFPT row; the unitful column is only its chosen comparison representative.

Corollary N.1 (Intrinsic charged-lepton UV source compression on the exact v_{geo} branch)**.** *On the exact v_{geo} branch the retained charged-lepton source rows satisfy*

$$(\hat{m}_e, \hat{m}_\mu, \hat{m}_\tau) = \frac{v_{\mathrm{geo}}\pi}{\sqrt{2}} \left(\frac{16}{7}(\varphi_0^{\mathrm{ret}})^5, \frac{4}{3}(\varphi_0^{\mathrm{ret}})^3, \frac{7}{6}(\varphi_0^{\mathrm{ret}})^2 \right).$$

Hence

$$\frac{\hat{m}_\mu}{\hat{m}_\tau} = \frac{8}{7}\varphi_0^{\mathrm{ret}}, \qquad \frac{\hat{m}_e}{\hat{m}_\mu} = \frac{12}{7}(\varphi_0^{\mathrm{ret}})^2, \qquad \frac{\hat{m}_e \hat{m}_\tau}{\hat{m}_\mu^2} = \frac{3}{2}\varphi_0^{\mathrm{ret}}.$$

Thus the intrinsic charged-lepton hierarchy is a one-seed compression of the UV source rows on the exact electroweak branch. It is not itself the final pole-mass readout.

Proof. Read off the three charged-lepton source rows from Table 24 and divide the corresponding formulas. The common factor $v_{\mathrm{geo}}\pi/\sqrt{2}$ cancels. □

For the main-text baryon-fraction comparison we use the Planck 2018 base-ΛCDM conversion

$$\Omega_b = \frac{\Omega_b h^2}{h^2}, \qquad h = \frac{H_0}{100},$$

so the quoted Ω_b row is not a free-floating cosmology number but an explicit reconstruction from the Planck parameter pair $(\Omega_b h^2, H_0)$ adopted in the suite ledger.

N.5 Extended comparison ledger

Row	TFPT source or target	Metrology readout	Scheme representative	Status note
m_γ	exact unbroken $U(1)_{\mathrm{em}}$ direction	$d_O = 1$ $m_\gamma/\lambda_\Sigma = 0$	0	exact structural row
m_g	exact color gauge direction before confinement	$d_O = 1$ $m_g/\lambda_\Sigma = 0$	0	exact structural row
$W^\pm$	threshold output at M_Z	$d_O = 1$ m_W/λ_Σ	80.369 GeV	scheme projection
Z	threshold output at M_Z	$d_O = 1$ m_Z/λ_Σ	91.188 GeV	scheme projection
H	threshold output	$d_O = 1$ m_H/λ_Σ	125.25 GeV	scheme projection
$\hat{s}_Z^2$ (M_Z)	local gauge closure plus seam-even matching	$d_O = 0$ scheme image	0.23116113	scheme projection
α_s (M_Z)	local gauge closure plus singlet Wilson matching	$d_O = 0$ scheme image	0.11796213	scheme projection
Σm_ν	intrinsic neutrino closure sum	$d_O = 1$ $\Sigma m_\nu/\lambda_\Sigma$	5.8764×10^{-2} eV	physical observable
M_R	heavy-neutrino saddle on the closed neutrino / cosmology branch	$d_O = 1$ M_R/λ_Σ	1.31×10^{15} GeV	physical observable
$\Sigma m_\nu^{\mathrm{ss}}$	appendix-level seesaw comparison from exact v_{geo} and M_R	$d_O = 1$ $\Sigma m_\nu^{\mathrm{ss}}/\lambda_\Sigma$	5.779×10^{-2} eV	appendix-level auxiliary continuation

Table 25. Supplementary gauge, Higgs, and neutrino rows for the extended comparison ledger. Exact zero rows, physical observables, scheme projections, and appendix-level auxiliary continuations are separated explicitly, with the dimensionless TFPT readout isolated from the chosen unitful representative.

Field	Word-length data $L_{f,j}$	Derived transport matrix element	Status note
e	8	0.466	Derived
μ	5	1.085	Derived
τ	3	0.899	Derived
u	7	0.440	Derived
d	7	0.936	Derived
s	5	0.943	Derived
c	3	0.646	Derived
b	2	0.480	Derived
t	0	0.986	Derived

Table 26. Transport closure data as theorem outputs.

N.6 Novelty boundary and literature positioning

The paper uses standard tools from relative spectral geometry, APS index theory, transport/resolvent language, and conventional RG comparison. The claim of novelty is not that these background ingredients are new in isolation, but that the minimal $3+2$ carrier packet, the compression identities, and the boundary-polarized closed architecture are organized into

one tightly overdetermined program. Conversely, the manuscript does not claim novelty for standard benchmark conventions, nor does it claim that appendix-level information-theoretic or quasicrystal readings are part of the hard core.

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Layer or statement	Role	Proof source or remaining note	Main structural test or falsifier
Operational seed generation, minimal packaging, and boundary datum $\mathfrak{T}_\partial^{\min}$	Boundary datum	operational-seed theorem, minimal-packaging theorem, and Calderón polarization theorem	failure would collapse the operator-level starting point
Doubling data τ_{dbl}, boundary polarization ι_C, doubled scaffold, and seam	Theorem	doubling / deck-involution / Calderón polarization theorem in the main text	structural consistency of every later theorem
Canonical admissibility projector P_{adm}	Theorem	canonical admissibility complex and Hodge projector lemma	unified electroweak / QCD / CP / gravity selection must survive one common projector
Carrier decoder Y, bosonic rank-two closure, branch Yukawa rigidity, and one-family packet S^+	Theorem	algebraic normal-form lemma, primitive trace-balanced carrier lemma, compact Higgs index corollary, branch Yukawa rigidity theorem, exterior-packet decomposition, and derived $48 = 4 \cdot 12$ identity	group/matter mismatch would be global
Winding decoder $[u_\Sigma] = 1$, $\mathcal{H}_F \cong F \cong \mathbb{C}^3$, and $\Omega_{\mathrm{adm}} = 48$	Theorem	four-corner / four-puncture theorem, harmonic family-mode theorem, family-integrality-repair corollary, and occupancy corollary	flavor multiplicity must be fixed without extra family input
Determinant classes and unique Higgs doublet	Theorem	minimal clutching theorem and compact bosonic index on the seam normal sphere	any unavoidable extra light triplet would falsify the theorem
Electromagnetic fixed point, retained-seed decoder, exact charged-lepton ratios, and closed 41 chain	Theorem	stationary $U(1)$ effective action, retained-seed decoder proposition, mass-compression theorem, and family/Higgs specialization	failure would break the fixed-point equation or the exact seed compression
Geometric Hodge gravity sector, boundary spectral unit, absolute spectral Planck closure, and low-curvature spectral branch	Theorem	spectral Einstein functional, boundary spectral-unit rigidity, absolute spectral Planck closure, Schwarzschild–de Sitter corollary, sectorial Hamilton decomposition, constructive geometric measure theorem, and spectral UV completion theorem	inability to obtain one coupled gravity/Higgs closure with one boundary-normalized stationary root would break the branch
D_4 family minimization, rigid family holonomy, cubic transport decoder, and residue–winding transport closure	Theorem	variational family theorem, monodromy-rigidity theorem, algebraic transport-pole theorem, regularized transport-pole stability lemma, exact C_6 resolvent, hard holonomy theorem, and winding sum rule	hierarchy grammar and winding lock must agree on one packet set
Determinant-line phase and strong-CP closure	Theorem	positive polar decomposition, determinant-line phase theorem, and admissible vacuum positivity	a stable nonzero effective strong-CP angle would falsify the theorem
FRW reduction, seam transfer, and cosmology interface data	Theorem	seam-transfer theorem, boundary-winding theorem, FRW/interface theorem, determinant-line axion interface theorem, and leptogenesis-interface theorem	positive trace-class seam transfer and scalaron closure must persist
Canonical rigid object and uniqueness	Theorem	internal analytic closure, essential injectivity on the rigid image, and canonical-object corollary	existence of a second nonisomorphic rigid realization would falsify the theorem
Internal reduction on the weakened ambient class $\widehat{\mathrm{PhysAdm}}_1$	Theorem	weakened ambient admissibility category, internal carrier / family / transport reductions, canonical rigid closure, and uniqueness theorem	existence of a physically admissible candidate in $\widehat{\mathrm{PhysAdm}}_1$ not closing canonically to $\mathfrak{T}_\star$ would falsify the theorem
Renormalized observable hierarchy and scheme projection discipline	Theorem	admissible 1PI package, renormalized observable theorem, physical observable factorization, minimal observable basis, and appendix scheme conventions	existence of an admissible observable not factoring through $\Gamma^{\mathrm{ren}}_{\mathrm{TFPT}} \to \mathbf{O}^{\mathrm{TFPT}}_{\mathrm{phys}}$ or a hidden fit parameter outside **Sch** would falsify the theorem

Table 1. Closed theorem surface for the present version. Internal reduction and the passage to the renormalized and physical observable layers are theorem-level; only the final scheme projection remains an appendix-facing interface. Comparison conventions, threshold rows, and empirical ledgers stay isolated in the appendix-level readout.

Exterior component	SM representation	Hypercharge	Field content	Mult.
$\Lambda^0 E$	$(1,1)_0$	0	ν^c	1
$\Lambda^2 E_3$	$(\bar{3},1)_{-2/3}$	$-2/3$	u^c	3
$E_3 \otimes E_2$	$(3,2)_{1/6}$	$1/6$	Q_L	6
$\Lambda^2 E_2$	$(1,1)_1$	1	e^c	1
$\Lambda^3 E_3 \otimes E_2$	$(1,2)_{-1/2}$	$-1/2$	L_L	2
$\Lambda^2 E_3 \otimes \Lambda^2 E_2$	$(\bar{3},1)_{1/3}$	$1/3$	d^c	3
			Total multiplicity	16

Table 2. One chiral family from $S^+ = \Lambda^{\text{even}} E$, written in left-chiral notation with charge-conjugate singlets.

Residue term	Operator sector	Generated term	χ_{geo} scaling	Role
a_0	relative bulk trace	vacuum-energy bookkeeping	χ_{geo}^4	Hard
a_2	seam-even bosonic bulk term	$\chi_{\text{geo}}^2 R$, $\chi_{\text{geo}}^2 \Phi^\dagger \Phi$	χ_{geo}^2	Closure
a_4	local bosonic sector	$\lvert D\Phi\rvert^2$, $(\Phi^\dagger\Phi)^2$, $R\,\Phi^\dagger\Phi$, $\operatorname{Tr}\mathbb{F}^2$, R^2	χ_{geo}^0	Closure
boundary/defect term	seam correction and parity data	local defect counterterms, seam-even selection	Δ_Σ	Closure
η term	APS spectral asymmetry	sign and spectral-flow control	scale free	Closure

Table 3. Low-curvature residue map for the canonical spectral Einstein functional.

Sector	Admissible length data	Transport matrix element	Role	Status
charged leptons	$L_e^{\text{diag}} = (8,5,3)$	$\Lambda_e, \Lambda_\mu, \Lambda_\tau$	ladder, hierarchy, exact mass ratios	theorem
up quarks	$L_u^{\text{diag}} = (7,3,0)$	$\Lambda_u, \Lambda_c, \Lambda_t$	up-type ratio ladder	theorem
down quarks	$L_d^{\text{diag}} = (7,5,2)$	$\Lambda_d, \Lambda_s, \Lambda_b$	down-type ratio ladder	theorem
quark mixing	L_{ij}^q	Λ_{ij}^q	CKM transport entries	theorem
neutrino mixing	L_{ij}^ν	Λ_{ij}^ν	PMNS transport entries	theorem
hadron basis	center-neutral word lengths	block coefficients	meson and baryon transport basis	singlet closure on $\mathcal{H}_{\text{had}}$
strong CP	determinant spectrum of $\mathcal{Y}_f$	none on the positive branch	phase suppression	theorem

Table 4. Transport data on the closed branch. Word-length ladders, hard holonomy flavor closure, and strong-CP suppression now sit on the same theorem-level transport surface, while later appendix readout conventions remain separate.

Statement	What would falsify it	Sector affected	Local or global failure
Primitive relative spectral dynamics	ill-posed APS domain or no finite admissible spectral action on the retained branch	Hard / closure	global
Minimal carrier $E_3 \oplus E_2$	failure to recover the one-family packet or the internal carrier stabilizer from the carrier theorem	Hard	global
Topological family closure and $\Omega_{\mathrm{adm}} = 48$	failure of the topological family theorem to globalize the local count	Closure	local-to-structural
Bosonic relative index selecting Φ	no unique seam-even doublet or an unavoidable extra light triplet	Closure	local
local spectral-scale / gravity closure for $\bar{M}_{\mathrm{Pl}}$ and v_{geo}	inability to obtain one common closure for $\bar{M}_{\mathrm{Pl}}$, v_{geo}, the Einstein branch, and the R^2 branch	Closure	global
Transport / determinant-phase closure	nonzero determinant phase, failure of nontrivial topological support, or a transport-side counterexample to the admissible branch	Closure	local
Hadronic admissibility as singlet selection	necessity of physical states with $z(w) \neq 1$ in the low-energy spectrum	Closure	local
Nonperturbative admissible QFT closure	no admissible mass gap or no clustering on the retained branch	Closure	global

Table 5. Falsification matrix for the present main-paper claims.

ID	Obligation	Status	Recorded location
O_0	operational seed generation of the one-sided boundary datum	A	operational-seed subsection, collar-completion theorem, and minimal-seed corollary
O_1	boundary polarization and primitive reconstruction from the one-sided datum	A	primitive reconstruction section and admissibility complex
O_2	carrier closure through bosonic rank-two plus branch-Yukawa rigidity, rigid $3+2$ split, and physical gauge quotient	A	algebraic normal-form lemma, bosonic-rank corollary, branch-Yukawa rigidity theorem, and faithful physical gauge-group theorem
O_3	primitive winding balance and topological / harmonic family closure	A	family geometry and harmonic family-mode theorems
O_4	rigid admissible family holonomy class	A	D_4 monodromy rigidity / Riemann–Hilbert closure and canonical family holonomy theorem
O_5	minimal determinant classes and compact Higgs selection	A	determinant-class theorem and compact determinant / Higgs-index closure
O_{6a}	determinant-phase suppression from hard holonomy factorization	A	hard holonomy Yukawa factorization and determinant-line phase theorem
O_{6b}	topological sector positivity on the retained branch	A	retained γ_5-Hermiticity, topological-sector positivity, and nontrivial topological-support lemmas
O_{6c}	strong-CP closure	A	strong-CP positivity theorem
O_{7a}	CP-even vacuum, thermodynamic limit, and parent-gap stability	A	vacuum-closure block
O_{7b}	full Osterwalder–Schrader and CAR reconstruction on P_{adm}	A	reflection positivity theorem, tailored OS/CAR reconstruction appendix, and Lorentzian reconstruction theorem
O_{7c}	local Minkowski net and microcausality on $\mathcal{H}_{\mathrm{adm}}$	A	physical admissible local-net definition and admissible local-net theorem
$O_{7d,m}$	stable massive Haag–Ruelle and LSZ closure on the admissible branch	A	stable massive scattering theorem and LSZ appendix closure
$O_{7d,0}$	infrared-dressed massless scattering interface on the low-curvature branch	B	dressed-interface theorem and standard long-range dressing machinery
O_8	rigid reconstruction and the canonical rigid object	A	rigid reconstruction, essential injectivity, and uniqueness corollaries
O_9	internal reduction on the weakened ambient class $\widehat{\mathrm{PhysAdm}}_1$	A	internal reduction section
O_{10}	low-curvature geometric gravity branch, scalaron FRW reduction, and closed-branch cosmology interface data	A	geometric Hodge theorem, full FRW/interface appendix theorem, determinant-line axion theorem, and leptogenesis-interface theorem
O_{11}	rank-5 master compression of family count, occupancy, and the 41 chain	A	derived $48 = 4 \cdot 12$ identity and master-compression corollary
O_{12}	dual-root generation of the hypercharge packet and admissible cusp set	A	dual-root proposition and transport-closure theorem
O_{13}	exact admissible RG flow, 1PI graph expansion, and renormalized observable hierarchy	A	admissible effective-average-action definition, exact RG theorem, 1PI graph expansion, and observable-hierarchy theorem
O_{14}	exact charged-lepton source compression on the v_{geo} branch	A	charged-lepton source-compression theorem and exact diagonal kernel evaluation
O_{15}	single-winding compression of the transport hierarchy	A	canonical transport-kernel theorem and single-winding corollary
O_{16}	algebraic carrier normal form with downstream bosonic / Yukawa discharge of the former carrier-signature residue	A	algebraic normal-form lemma, bosonic-rank corollary, branch-Yukawa rigidity theorem, and weakened ambient class section
O_{17}	absolute spectral Planck closure, Planck metrology, and stationary vacuum geometry	A	main theorem stack: boundary spectral-unit rigidity, absolute spectral Planck closure, and Schwarzschild–de Sitter corollary
O_{18}	modular horizon thermality, Hawking readout, and compact-object thermodynamics on the vacuum branch	P	downstream-closure section together with the split-inclusion horizon appendix block
O_{19}	primordial perturbation closure from the seam state	P	downstream-closure section: reheating and seam-state mode theorems
O_{20}	CMB transfer hierarchy on the closed branch (Stage 1: spectra)	P	downstream-closure section: CMB transfer theorem and late-time spectrum corollary
O_{20a}	closed-branch operator algebra $(U_\Sigma^\star, P_{\mathrm{adm}}^\star, \tau_{\mathrm{dbl}}^\star, \iota_C^\star, U_5^\star, U_6^\star)$ on $V_{\mathrm{adm}}^\star$ from seam-transfer dynamics, replacing the present band-circulant proxies	P	Remark (L3), Remark
O_{20b}	canonical seam record state $u_\Sigma^\star$ closed in form on $\mathfrak{T}_\star$ (not a generator output)	P	Conjecture 11.11, Remark (L4)
O_{20c}	canonical trace word $\mathcal{W}_\Sigma^\star$, collapsing the variant family $\{A, B, C, D\}$ via D_4-seam equivariance and naturality	P	Conjecture 11.12, Remark (L4)
O_{20d}	closed-branch sky triad $\mathcal{O}^\star \in \mathrm{SO}(3)$ orienting the realised	P	Remark (L5), Remark (O2)

Closed statement	Input data	Output package	Role in rigid reconstruction	Principal structural falsifier (entry of $\mathsf{F}_{\mathrm{fals}}(\mathfrak{T}_\star)$)
Boundary polarization from one-sided datum	$\mathfrak{T}_\partial^{\min}$	$(\tau_{\mathrm{dbl}}, \iota_{\mathrm{C}}), \mathfrak{T}_{\mathrm{prim}}, \mathfrak{T}_{\min}^{\mathrm{cl}}$	establishes the doubled scaffold and reference/relative split	failure of the primitive scaffold
Carrier rigidity and physical gauge quotient	$\mathfrak{T}_{\mathrm{prim}}$	$X, E_3 \oplus E_2, G_{\mathrm{phys}}, S^+, b_1$	fixes the particle-theory kernel	group or matter mismatch
Two-stage projector and winding-balance family theorem	$P_{\mathrm{prim}}, P_{\mathrm{sing}} P_\Theta$, seam polarization	$X_f^\circ \cong \mathbb{P}^1 \setminus \mu_4$, F, $\Omega_{\mathrm{adm}} = 48$, $[\nabla_F^\star]$	closes multiplicity, harmonic family geometry, and admissible holonomy class	wrong family multiplicity, residual family modulus, or broken projector
Determinant line and geometric Hodge sector	P_{adm}, D_{geo}	Φ, a_Σ, $\bar{M}_{\mathrm{Pl}}$, v_{geo}, $\mathcal{H}_{\mathrm{phys}}$	closes Higgs, strong-CP seed, and low-curvature gravity	extra light triplet or broken gravity/Higgs closure
Transport, hadronic admissibility, and strong-CP closure	P_{adm}, U_6, family holonomy	Y_f, V_{CKM}, U_{PMNS}, δ_{ph}, $\mathcal{H}_{\mathrm{had}}$, $\bar{\theta} = 0$	closes flavor, hadrons, and the effective strong angle	broken sum rule, bad winding lock, or nonzero effective strong-CP angle
FRW reduction and cosmology interface data	Γ_{grav}, U_Σ	S_Σ, Λ_{IR}, axion/reheating inputs, leptogenesis input map	records closed-branch cosmology interface data	non-trace-class seam transfer or broken scalaron closure
Branch uniqueness, essential injectivity, and canonical rigid object	invariant tuple $\mathcal{I}(\mathfrak{T}_{\mathrm{pre}})$	$\mathfrak{T}_\star$ up to unique unitary equivalence	removes all remaining nonisomorphic rigid alternatives	existence of a second rigid realization with the same invariants, or failure of the vacuum-closure chain
Internal reduction on the weakened ambient class	$\widehat{\mathrm{PhysAdm}}_1$	every physically admissible candidate in $\widehat{\mathrm{PhysAdm}}_1$ closes canonically to $\mathfrak{T}_\star$	records the theorem-level input reduction	existence of a physically admissible candidate in $\widehat{\mathrm{PhysAdm}}_1$ not closing canonically to $\mathfrak{T}_\star$
Renormalized observable hierarchy	$\mathfrak{T}_\star, \Gamma_{\mathrm{TFPT}}^{\mathrm{ren}}, \mathbf{Sch}$	$\Gamma_{\mathrm{TFPT}}^{\mathrm{ren}}$, $\Gamma_{\mathrm{TFPT}}^{\mathrm{ren}}$, $\mathbf{O}_{\mathrm{phys}}^{\mathrm{TFPT}}$, and the scheme orbit $[\mathfrak{M}(\mathfrak{T}_\star)]_{\mathbf{Sch}}$	records the output-side physical and scheme stratification	existence of an admissible observable not factoring through $\Gamma_{\mathrm{TFPT}}^{\mathrm{ren}} \to \mathbf{O}_{\mathrm{phys}}^{\mathrm{TFPT}}$ or a hidden fit parameter outside $\mathbf{Sch}$

Table 7. Structural falsification map $\mathsf{F}_{\mathrm{fals}}(\mathfrak{T}_\star)$ in main-text form. Each row records a theorem arrow of the reconstruction together with the structural failure mode that would break it. Appendix-level empirical readout via $\mathcal{R}_{\mathrm{cmp}}(\mathfrak{T}_\star)$ is handled separately and does not appear as a theorem arrow in this table.

Obs.	TFPT formula	Metrology readout	TFPT value	Exp. value	Unc.	Residual	Scale / scheme	Status
$\alpha^{-1}(0)$	exact electromagnetic closure $\varphi_\Sigma(\alpha)$	$d_O = 0$ direct readout	137.035 999 2	137.035 999 18	2.1×10^{-8}	$+1.87\sigma$	Thomson on-shell	physical observable
$\tilde{\alpha}^{(5)}(M_Z)^{-1}$	$\mathcal{R}_{\mathrm{SM}}[\alpha(0)]$	$d_O = 0$ scheme image	127.940 5	127.93	0.008	$+1.32\sigma$	$\overline{\mathrm{MS}}$ at M_Z	scheme projection
λ_C	$s_{12}(V_{\mathrm{CKM}}^\star)$ from hard flavor closure	$d_O = 0$ direct readout	0.224 38	0.224 31	8.5×10^{-4}	$+0.08\sigma$	CKM physical angle	physical observable
$\sin^2\theta_{13}$	$\lvert(U_{\mathrm{PMNS}}^\star)_{e3}\rvert^2$ from neutrino closure	$d_O = 0$ direct readout	0.023 11	0.022 24	5.7×10^{-4}	$+1.524\sigma$	NuFIT NO	physical observable
Ω_b	late-time cosmology comparison from (T_R, η_B, Ω_a)	$d_O = 0$ $C_{\cos}$ readout	0.048 94	0.049 30	8.57×10^{-4}	-0.421σ	Planck 2018 present epoch from $\Omega_b h^2$	cosmology comparison
β	determinant-line / Chern–Simons response	$d_O = 0$ β_{rad}	0.242	0.35	0.14	-0.77σ	Planck 2018 Minami–Komatsu	physical observable

Table 8. Restricted appendix benchmark table. The rows are separated into physical observables, cosmology comparisons, and scheme projections under the declared interface. The former seed quartet is now split across flavor, neutrino, determinant-response, and cosmology sectors. The metrology column records the actual dimensionless TFPT readout, while the numeric comparison column records only the chosen interface representative. Supplementary rows and out-of-sample checks are collected in Appendix N.

Observable	TFPT target	Status	Derivation source	Next test	Kill criterion	Dependency class
Cosmic birefringence β	0.2424°	Physical observable	determinant-line / Chern–Simons response	calibrated CMB rotation analysis	calibrated $\beta = 0$ within $\pm 0.05^\circ$	determinant response
$\theta_{\rm eff} = 0$ / nEDM null test	$\theta_{\rm eff} = 0$	Theorem	determinant-line strong-CP closure	next-gen EDM searches	stable nonzero hadronic EDM signal	strong-CP closure
Axion haloscope window	$m_a \approx 65.19$ μeV $\nu_a \approx 15.764$ GHz	Cosmology readout target	determinant line / seam transfer / scalaron block	haloscope scan near 15.764 GHz	exclusion in 15.764 GHz ± 50 MHz	cosmology readout
$\delta_{\rm CKM}$	1.198 rad	Comparison quantity	exact holonomy phase / phase lattice	global flavor fit	exclusion at $\geq 3\sigma$	flavor readout
Rare kaons $K \to \pi\nu\bar{\nu}$	$\mathrm{BR}(K^+) = 9.40 \times 10^{-11}$ $\mathrm{BR}(K_L) = 3.47 \times 10^{-11}$	Comparison quantity	exact CKM point in short-distance amplitudes	NA62 current result / KOTO program	K^+ outside $[7, 12] \times 10^{-11}$ or K_L mismatch	flavor readout
PMNS phase / octant	$\delta_{\rm CP} = 240^\circ$ $\sin^2\theta_{23} = 0.4557$	Comparison quantity	neutrino closure / phase lattice	global oscillation fit	exclude 240° or lower octant at $\geq 3\sigma$	neutrino readout
Σm_ν	5.8764×10^{-2} eV	Comparison quantity	neutrino closure (NO)	cosmological mass-sum analyses	robust upper bound below the branch value	neutrino readout
$m_{\beta\beta}$	1.516×10^{-3} eV	Comparison quantity	Majorana lattice on the closed neutrino branch	light-Majorana $0\nu\beta\beta$ searches	detection implying $m_{\beta\beta} \gtrsim 1 \times 10^{-2}$ eV	neutrino readout

Table 9. Operational prediction set with explicit status and dependency classes.

Representative readout row(s)	Basis package in $\mathcal{B}_{\rm ind}(\mathfrak{T}_\star)$	Declared factorization map	Readout block
$\alpha^{-1}(0)$	$C_{\rm em}$	exact electromagnetic closure via $\varphi_\Sigma(\alpha)$	benchmark surface
$\varphi_0^{\rm ret}$ together with Corollary A.1	$K_{\rm UV}$	retained UV seed identities on the closed branch	UV bookkeeping
$(\bar{\alpha}^{(5)}(M_Z)^{-1}, \overline{\rm MS}$ masses, $\hat{s}_Z^2, \alpha_s)$	$T_{\rm phys}$	scheme projection of the physical electroweak layer	threshold readout rows
$(\lambda_{\rm C}, \delta_{\rm CKM}, K \to \pi\nu\bar{\nu})$	$F_{\rm fl}$	flavor-holonomy and short-distance flavor readout maps	benchmark / operational flavor rows
$(\sin^2\theta_{13}, \delta_{\rm CP}^\nu, \sin^2\theta_{23}, \Sigma m_\nu, m_{\beta\beta}, M_R)$	N_ν	neutrino-closure readout on the closed branch	benchmark / operational neutrino rows
β	$R_{\rm CS}$	determinant-line / Chern–Simons response map	benchmark / operational response rows
$\theta_{\rm eff}$ and nEDM / strong-CP rows	$C_{\rm CP}$	determinant-line strong-CP closure on the admissible branch	operational prediction set
$(\Omega_b, S_\Sigma, \Lambda_{\rm IR}, T_R, f_a, m_a, \nu_a, \Omega_{\rm DM}, \eta_B)$	$C_{\rm cos}$	seam-transfer / determinant-line / scalaron readout from the closed branch	cosmology and axion rows

Table 10. Basis factorization of the appendix-level readout rows. Each displayed row factors through exactly one basis package of $\mathcal{B}_{\rm ind}(\mathfrak{T}_\star)$ under the declared physical / scheme hierarchy.

Domain	Status	Output package	Source layer
Carrier packet	Theorem	$E_3 \oplus E_2$, G_{car}, S^+, γ, $b_1(N_{\mathrm{fam}}, N_\Phi)$	algebraic normal-form lemma, bosonic-rank corollary, branch-Yukawa rigidity theorem, and half-spinor decomposition
Kernel constants	Theorem	c_3, φ_{base}, φ_0^{ret}, δ_{top}	exact seam opening and carrier-form closure equation
Family / occupancy layer	Theorem	P_{adm}, F, $\Omega_{\mathrm{adm}} = 48$, δ_{top}	derived corner theorem, admissibility complex, and occupancy corollary
Geometric branch	Theorem	Φ, $\bar{M}_{\mathrm{Pl}}$, v_{geo}, R^2 channel	compact bosonic index plus canonical spectral Einstein functional
Renormalized low-energy layer	Theorem	$\Gamma^{\mathrm{ren}}_{\mathrm{TFPT}}$, charged-current point closure, v_{phys}, pole equations, determinant response, cosmology interface data	charged seam projector lemma, pole-level readout theorem, and renormalized observable hierarchy
Transport / flavor closure	Theorem	Y_f, CKM, PMNS	residue–winding closure, hard holonomy theorem, and neutrino closure
Hadronic / strong-CP closure	Theorem	$\mathcal{H}_{\mathrm{had}}$, anomaly inflow, $\arg\det M_u = \arg\det M_d = 0$	singlet admissibility plus hard-holonomy determinant suppression
Physical observables	Physical observable	$\mathbf{O}^{\mathrm{TFPT}}_{\mathrm{phys}}$ including pole masses, mixing data, determinant responses, and closed-branch interface packages before late-time cosmology continuation	renormalized observable hierarchy and sector factorization
Scheme projections	Scheme projection	$[\mathfrak{M}(\mathfrak{T}_\star)]_{\mathbf{Sch}}$ represented by $\mathcal{R}^{(s)}_{\mathrm{cmp}}(\mathfrak{T}_\star)$, threshold rows, and ledgers	declared scheme conventions and appendix comparison maps
Cosmology block	Appendix continuation	Λ_{IR}, θ_i, N_{DW}, T_R, and the cosmology interface block	seam transfer, determinant line, scalaron dynamics, and comparison-layer continuation
Record layer	Appendix continuation	$\mathcal{A}_{\mathrm{rec}}$, $\mathcal{A}_{\mathrm{obs}}$, Pred	appendix-level record and readout algebra elaboration
Appendix continuation	Appendix continuation	E_8 scale ratios and modular inner-seam language	appendix-level scale grammar and horizon extension

Table 11. Compact prediction ledger for the closed-form presentation after the main-text surface is split into theorem outputs, physical observables, scheme projections, and appendix continuations.

Relative object	Definition or schematic form	Role
D_{rel}	$D - D_{\mathrm{ref}}$ or a specified relative pair (D, D_{ref})	fermionic comparison operator
Γ_{grav}	$-6\chi_{\mathrm{geo}}^2 \mathcal{E}_D + \Gamma_D^{(4)}$ with $\mathcal{E}_D = \mathrm{Res}_{s=1}\, \mathcal{Z}_{\mathrm{rel}}$ and $\Gamma_D^{(4)} = \mathrm{FP}_{s=0}\, \mathcal{Z}_{\mathrm{rel}}$	canonical spectral Einstein functional from relative residues
Γ_{rel}	admissible relative action preserved by the sheet antiunitary involution	strong-CP and reflection-symmetry package
$\mathrm{Ind}_{\mathrm{rel}}$	APS-type index of a relative pair or superconnection package	carrier and bosonic counting
Z_{rel}	$Z_{\mathrm{adm}}[J, \eta, \bar{\eta}] / Z_{\mathrm{ref}}[0, 0, 0]$	normalized admissible generating functional
P_{adm}	$\text{s-lim}_{\beta \to \infty}\, e^{-\beta K_{\mathrm{adm}}}$; factorized form when the elementary projectors commute	operative admissibility selector on the closed branch
$\langle W(C) \rangle_{\mathrm{rel}}$	Wilson observable after subtraction of the declared reference sector	confinement diagnostics
$\mathcal{H}_{\mathrm{rel}}$	relative Hamiltonian used in admissible-sector minimization	optional record extension

Table 12. Conventions on relative objects.

Level	Objects	Claim role	Function in the present paper
Core datum	$\mathfrak{T}_\partial^{\min}$, $\widetilde{M} \to M$, Σ, τ_{dbl}, ι_{C}, D_{ref}, D_{rel}, χ_{seed}	Boundary datum / theorem	one-sided input plus minimal kinematic scaffold before carrier data and admissibility closure; S and $\mathcal{H}_{\mathrm{int}}$ refer only to a chosen realization
Post-carrier primitive kernel	$(\mathcal{A}, \mathcal{H}, D, J, \Gamma, \iota_{\mathrm{C}}, P_{\mathrm{prim}}, P_{\mathrm{adm}})$, $E_3 \oplus E_2$, Y, $[u_\Sigma]$, c_3	Theorem	primitive closure kernel carrying the admissibility, carrier, and seam-normalization data before the discrete/continuous branch split
Closure data	$(\mathsf{Adm}, K_{\mathrm{adm}}, P_{\mathrm{adm}})$, (F, T), $(D_{\mathrm{geo}}, \chi_{\mathrm{geo}}, \mathcal{B}_{\mathrm{rel}}, \mathbb{A}_\Sigma)$, $(U_6, D_y, \mathcal{Y}_y^{(\varepsilon)}, \epsilon_f)$, $(Z_{\mathrm{rel}}, \Theta, \mathcal{H}_{\mathrm{rel}}, \mathcal{V}_{\mathrm{adm}})$, $(\mathcal{A}_{\mathrm{rec}}, \mathcal{A}_{\mathrm{obs}}, \mathrm{Pred})$	Derived consequence	collects the admissibility, family, geometric, transport, record, and vacuum data used by the closure theorems
Comparison layer	$\mathcal{R}_{\mathrm{cmp}}$, $\mathcal{R}_{\mathrm{SM}}$, $\alpha(0)$, M_Z, $\{m_c, m_b, m_\tau, m_t\}$	Comparison convention	translates closed outputs into scheme-specified observables, with $\mathcal{R}_{\mathrm{SM}}$ only the practical numerical preconditioner

Table 13. Core datum versus theorem, derived-consequence, and comparison-convention structures.

Symbol	First-duty meaning in the theorem chain	Status class
$\mathfrak{T}_\partial^{\min}$	one-sided boundary datum $(\mathcal{A}_+, \mathcal{H}_+, D_+, J, \Gamma, B_\Sigma)$	Boundary datum
$\mathfrak{T}_{\min}^{\mathrm{cl}}$	doubled closed datum $(\mathcal{A}, \mathcal{H}, D, J, \Gamma, \tau_{\mathrm{dbl}}, \iota_{\mathrm{C}})$ reconstructed from $\mathfrak{T}_\partial^{\min}$	Theorem
$\mathfrak{T}_{\mathrm{prim}}^{\mathrm{kin}}$	primitive kinematic scaffold $(\widetilde{M} \to M, \Sigma, \tau_{\mathrm{dbl}}, \iota_{\mathrm{C}}, D_{\mathrm{ref}}, D_{\mathrm{rel}}, \chi_{\mathrm{seed}})$; when an almost-commutative realization is chosen, S and $\mathcal{H}_{\mathrm{int}}$ refer to that choice and are not additional reconstruction outputs	Theorem
$\mathcal{C}_{\min}$	minimal carrier kernel $E_3 \oplus E_2$ with one-family packet S^+	Theorem
χ_{seed}	primitive seam-even scalar response reconstructed on the primitive layer before geometric reconstruction	Theorem
σ (geometric), χ_{geo}	local Weyl conformal factor and derived local spectral scale $\chi_{\mathrm{geo}} = \Lambda e^{\sigma}$ on the geometric branch	Theorem
σ_{QCD}	QCD string tension $\sigma_{\mathrm{QCD}} = c_3^2 \chi_{\mathrm{QCD}}^2$ on the hadronic transport branch (lexically distinct from the Weyl conformal factor and from the standard-deviation σ in benchmark tables)	Theorem
$\mathrm{APS}_{\mathrm{dom}}^{\mathrm{prov}}$	provisional APS realization of the geometric anchor used in Theorem 3.9; an analytic starting representative	Boundary datum (analytic choice)
$\mathrm{APS}_{\mathrm{dom}}$	closed-branch APS domain obtained from $\mathrm{APS}_{\mathrm{dom}}^{\mathrm{prov}}$ by the Calderón upgrade Theorem 8.4	Theorem
$\mathrm{APS}_{\mathrm{fam}}$	family-pairing data induced canonically on the doubled admissible branch by the master closure roadmap	Theorem (derived)
P_{prim}	primitive admissibility projector built from ι_{C}, $\mathrm{APS}_{\mathrm{dom}}$, and D_{rel} alone (Definition 3.4)	Theorem
$P_{\mathrm{sing}}, P_\Theta$	color-center singlet projector and determinant-sector involution projector available only after carrier and bosonic-index closure	Theorem
$\mathsf{Adm}, K_{\mathrm{adm}}, P_{\mathrm{adm}}$	admissibility predicate, positive constraint operator, and full operative selector $P_{\mathrm{adm}} = P_{\mathrm{prim}} P_{\mathrm{sing}} P_\Theta$ on the admissible branch (Theorem 8.2)	Theorem
F, Ω_{adm}	topological family space and occupancy count after family closure; Ω_{adm} later feeds the closed reading of δ_{top}	Theorem
$[\nabla_F]$	$SU(3)_F$ conjugacy class of the rigid family local system used as the sole family-connection invariant in Definition 9.19	Theorem
δ_{ph}	unique algebraic transport root on the C_6 hexagon used on the closed branch	Theorem
Q_Θ	determinant-sector involution entering admissibility and the projector P_Θ	Theorem
$\mathcal{C}_\Sigma = \mathcal{C}_\Sigma$	*sheet-CP symmetry operator only*: $\mathcal{C}_\Sigma := CP \circ \tau_{\mathrm{dbl}}$ on the admissible branch	Theorem
T	transport generator on the family / holonomy side	Theorem
$\varpi_{\mathrm{spin}} = \varpi_{\mathrm{spin}}$	spin holonomy weight entering the geometric package	Theorem
$\mathfrak{T}_\star$	canonical rigid object of Corollary 9.23; endpoint of the theorem chain	Theorem
$\mathsf{F}_{\mathrm{fals}}(\mathfrak{T}_\star)$	structural falsification map of Section 12.3 (main text, hard-separated from $\mathcal{R}_{\mathrm{cmp}}$)	Theorem (side card)

Table 14. Symbol guide for theorem-level objects only. Comparison-layer and readout symbols are collected separately in Table 15. The previously colliding label $\mathcal{C}_\Sigma$ now denotes only the sheet-CP operator; seam cycle data are written $\mathfrak{C}_\Sigma$ (Table 15).

Symbol	First-duty meaning in the appendix layer	Status class
$\mathcal{R}_{\mathrm{cmp}}$	appendix-only comparison map $\mathcal{R}_{\mathrm{cmp}} : \mathfrak{T}_\star \mapsto \mathcal{R}_{\mathrm{cmp}}(\mathfrak{T}_\star)$ of Section A; never enters the theorem chain	Comparison convention
$\mathcal{R}_{\mathrm{SM}}$	practical finite-threshold numerical preconditioner used inside $\mathcal{R}_{\mathrm{cmp}}$	Comparison convention
$\mathcal{I}_{41}$	weighted abelian index $10\, b_1 = \Omega_{\mathrm{occ}} \gamma + N_\Phi$ before the closed 41 specialization (appendix bookkeeping)	Appendix bookkeeping
$\hat{m}_f,\ m_f^{\mathrm{obs}}$	internal source masses and their appendix-level comparison-surface images after matching	Comparison convention
$M_\Sigma,\ g_\Sigma$	seam matching scale read from the first positive seam boundary mode and the gauge coupling evaluated there (used inside $\mathcal{R}_{\mathrm{cmp}}$)	Comparison convention
$\mathfrak{C}_\Sigma = \mathfrak{C}_\Sigma$	seam cycle data used by the matching and holonomy package; lexically distinct from the sheet-CP operator $\mathcal{C}_\Sigma$ in Table 14	Appendix-level derived data
$\mathcal{A}_{\mathrm{rec}}, \mathcal{A}_{\mathrm{obs}}, \mathrm{Pred}$	stable record algebra, observer algebra, and prediction map on the admissible sector (appendix continuation only; not part of the main theorem chain)	Appendix continuation
σ (statistical)	standard-deviation symbol used in benchmark tables for residuals (lexically distinct from the geometric Weyl factor σ and from σ_{QCD})	Appendix bookkeeping

Table 15. Symbol guide for appendix-level comparison and readout objects. These symbols never participate in the main-text theorem chain (52) and are recorded here only because they appear in $\mathcal{R}_{\mathrm{cmp}}$ or in the appendix bookkeeping.

Symbol	Definition	Manuscript role	First origin	Reappears in	Physical meaning
θ	seam-loop angle	core datum	seam data	a_θ	geometric rotation variable
a_θ	$\theta/(4\pi)$	hard normalization	spin lift	c_3, phase grammar	normalized cover rotation
c_3	$1/(8\pi)$	hard kernel constant	seam normalization	δ_{top}, β, σ	seam normalization constant
φ_{base}	$1/(6\pi)$	hard kernel constant	carrier geometry	φ_0^{ret}	base opening
δ_{top}	$\frac{3}{256\pi^4}$; bridge reading $\Omega_{\mathrm{adm}} c_3^4$	hard kernel value with occupancy reinterpretation	retained numerical kernel	φ_0^{ret}, δ_2	topological surcharge
φ_0^{ret}	$\frac{1}{6\pi} + \frac{3}{256\pi^4}$	retained canonical seed	retained numerical kernel	λ_Y, β, flavor	minimal nontrivial opening
γ	$\mathrm{Tr}_E\, Y^2 = 5/6$	hard kernel invariant	carrier packet	κ_{E8}, δ_2	carrier norm
B	$\mathrm{rank}\, E_3 / \mathrm{rank}\, E_2 = 3/2$	hard kernel invariant	carrier packet	δ_2	carrier rank ratio
Ω_{adm}	$3 \times 16 = 48$	closure quantity with bridge reinterpretation	APS family closure	δ_{top}	admissible chiral-state count on the family-closed branch
b_1	$\frac{4}{3} N_{\mathrm{fam}} + \frac{1}{10} N_\Phi$	hard kernel output	abelian index	benchmark map	hypercharge trace coefficient
κ_{E8}	$\gamma / \ln(248/60)$	conditional grammar	Appendix C	scale ladder	optional scale spacing

Table 16. Constants atlas for the present manuscript.